```
root@pragma:~# ./init_oph --book
Fetching observer consistency protocols...
Connecting to floatingpragma.io...
```

REVERSE ENGINEERING REALITY

$$[\quad \mathcal{P} = \varphi + \alpha_{em}(\mathcal{P})\sqrt{\pi} \quad]$$

// WRITTEN BY

BERNHARD MUELLER

HTTPS://FLOATINGPRAGMA.IO | HTTPS://FLOATINGPRAGMA.IO/OPH

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Dedication

For my wife Noon.

Epigraph

> There is a theory which states that if ever anyone discovers exactly what the Universe is for and why it is here, it will instantly disappear and be replaced by something even more bizarre and inexplicable.
>
> There is another theory which states that this has already happened.
>
> Douglas Adams, *The Restaurant at the End of the Universe*, 1980

No Source Code, No Documentation

In 1632 Galileo published a book in the form of a conversation between three men and gave the best passage in it to the one who agreed with him. Shut yourself below decks on a large ship, Salviati says. Bring flies and butterflies. Bring a bowl with fish in it. Hang a bottle so that it drips into a wide vessel underneath. Watch how everything behaves while the ship is tied up at the dock, and take your time about it.

Then have the ship put to sea at whatever speed you like, provided the motion is smooth and does not rock you about.

The flies do not pile up against the stern wall. The fish swim as easily toward the front of the bowl as toward the back, without having to work harder in either direction. The drops fall into the vessel below and not one of them lands behind it. You can jump the same distance forward as back. Throwing something to a friend takes the same effort whichever way you face. And from none of these effects, Galileo writes, can you determine whether the ship is moving or standing at rest.

That experiment is harder than it looks, and different in kind from what everybody means by testing something. To test a thing from outside you at least get to stand next to it and poke it. Galileo's man is inside the thing he wants to measure. He is a part of it. What he is asking is whether a container gives any sign of itself to what it contains. The answer he gets back is that this one does not.

Negative results of that kind are the most informative things in physics. He has not found that the measurement is delicate. He has found that there is nothing to measure: the information does not exist inside the cabin, no quantity of butterflies will bring it into existence, and the four centuries since have consisted largely of people discovering that the same is true of the cabin they are actually in.

The book was on the Index of forbidden works within a year and its author spent the rest of his life under house arrest, which is a fair measure of how seriously the argument was taken.

Everything that follows is that experiment, run for longer with better equipment.

Six symptoms

What follows is six behaviors of the machine, for now without explanations which take the rest of the book.

Galileo's cabin has never been beaten. Four hundred years of trying, with instruments he could not have imagined. No experiment performed inside a smoothly moving laboratory has ever revealed that it was moving. The nineteenth century made a particular effort, looking for the laboratory's motion through the medium that light was supposed to be waving in, and came up empty to a precision that embarrassed everyone involved. The information is missing with a thoroughness that looks less like a limitation of instruments and more like a decision somebody made.

Clocks disagree. The disagreement is engineered around. The satellites your phone uses to find itself carry atomic clocks, wrong in two directions at once. They move quickly, which makes them tick slow by about seven microseconds a day. They sit high up where gravity is weaker, which makes them tick fast by about forty-five. The two effects do not cancel. Without a correction the position your phone reports would drift off by something like ten kilometers a day.

When the first atomic clock of this kind went into orbit in 1977, the correction was built with a switch on it, because not everyone involved was convinced the physicists were right. It launched switched off. They watched the clock run fast by very close to the predicted amount for about three weeks, and then turned it on.

Asking a question changes the answer. In 1909 a twenty-three-year-old at Cambridge named Geoffrey Taylor set a needle in front of a gas flame and photographed the pattern of light and dark bands around its shadow. Then he began dimming the flame with sheets of smoked glass. He dimmed it until the light crossing his apparatus was so feeble that particles arrived essentially one at a time, with nothing else present for them to interfere with. To collect enough of them he had to leave the plate in place for three months, which is a long time to wait to find out that nothing has changed. The bands came out looking exactly as they had at full brightness. Whatever each particle was interfering with, it was not the others, because there were no others.

The second half of that result is even stranger. Put a detector in the apparatus that can tell you which side of the needle each particle went, and the bands disappear. Delicacy does not rescue them. Any detector that succeeds in telling you the route destroys the pattern. Any detector that fails to tell you leaves the pattern intact, and how gently it fails makes no difference at all.

Two things far apart agree too often. In 2015 a team at Delft trapped a single electron in each of two laboratories 1.3 kilometers apart and measured both. The answers matched more often than they could have if each electron had been carrying its answer around with it. The laboratories were separated widely enough that no signal travelling at the speed of light could have crossed between them while the measurements were happening. That experiment closed the last of the escape routes physicists had been finding in this result for fifty years. The correlation is real, it cannot be used to send a message, and no account in which the particles had properties before anyone looked survives it.

Falling into a black hole appears to destroy the record of what fell. On 6 February 1997 Stephen Hawking and Kip Thorne signed a wager with John Preskill. Hawking and Thorne held that information swallowed by a black hole is erased from the world. Preskill held that it survives. The stake was an encyclopedia of the winner's choice, from which, as the bet specifies, information can be recovered at will. In July 2004, at a conference in Dublin, Hawking conceded and handed Preskill a baseball encyclopedia. Thorne did not concede. Hawking remarked afterwards that burning the encyclopedia and handing over the ashes would have made the losing case rather better than the book did.

The bet was worth making because both positions are unacceptable. If the record is erased, quantum mechanics breaks. If it survives, it has to escape from the one place in the universe that nothing escapes from. Three of the most capable physicists alive spent seven years on which of two impossibilities to prefer. At the end of it one of them changed his mind and one of them did not.

Gravity puts an area ceiling on what a region can hold. This last one is arithmetic rather than anecdote. A warehouse twice as wide in every direction holds eight times as much. A black-hole horizon behaves differently: double its radius and its entropy grows by four. More general gravitational bounds extend that lesson under their own causal conditions, though an ordinary empty region need not saturate the bound. The observer construction reaches a separate finite boundary count. Calling that count physical entropy or area requires a map to a realized screen and a calibrated exchange rate.

A universe without a manual

People who take apart hardware for a living meet this problem constantly: a sealed device, no documentation, no source, nobody to ask, and no way to open the case without destroying what is inside. There is a discipline for it. It comes down to a few habits.

Feed it inputs and record what comes out. Change one thing at a time. Watch the timing, because timing leaks structure that the outputs conceal: a system that answers one question quickly and a similar question slowly is telling you that those questions take different routes. Above all, watch what makes it fail. A device gives away more about its architecture in the ten seconds around a crash than in a week of ordinary operation, which is why the six items above are worth more than a century of things that worked.

And a third habit. Never assume a component you have not found on the board. If you have not located the clock, you do not get to reason about what the clock does. If you have not located the memory, you do not get to assume there is a place where the state is being kept.

This book runs that discipline under one aggravating condition, which Galileo's cabin introduced and which never goes away: the engineer is a part. There is no bench to put the machine on.

A reconstruction asks how much physics follows from a smaller starting point. OPH starts with bounded systems that read their boundaries, retain records and repair disagreements with neighbors. Agreement becomes a mathematical constraint. The test is how much familiar physics follows from those constraints, with every extra assumption visible.

The everyday picture of the world is that there is a three-dimensional container, time runs inside it at one rate for everybody, objects occupy positions in it, and physics describes how the objects move.

In 2002 the programmer Joel Spolsky wrote down a rule about descriptions of that shape. All non-trivial abstractions, to some degree, are leaky. An abstraction is a simplified account that lets you work without knowing what is underneath it. It holds until the thing underneath does something the simplification has no words for, at which point it leaks. The container picture of space and time is a **leaky abstraction** and it leaks in at least six places, which you have just finished reading. What comes back through each leak is not noise. It is a wrong answer, delivered confidently, and reproducible to as many decimal places as you care to measure.

One equation at a time

The framework has a name: Observer Patch Holography. Its case is the reach of one observer architecture. Under explicit probability assumptions, the algebra of readable questions supports quantum probabilities. Complete reversible boundary response and internal transport constrain the local symmetries of forces. A supplied matter law supports controlled motion of a real field in a continuum and, at fixed mesh, an interacting quantum state space. Scalar-density and magnetic readouts use the same field reconstruction in both descriptions; quantum probabilities come from the declared Hilbert representation. Objects that ordinary calculations introduce separately become parts of a connected argument.

The papers state the assumptions behind each result and supply proofs or calculations that can be repeated independently. Physical claims face a further demand: identify what an instrument would measure and what outcome would refute the claim. Appendix C points to that evidence. This book develops the argument in words, explaining what each step rests on.

There are equations in this book, not many, and every one of them is doing work. Each one is introduced by a sentence saying what it is about to say and followed by a sentence saying what it said. Every symbol in it is explained in words on the same page.

An equation is compressed prose. It decompresses under four questions.

What does it count? Every quantity in physics is a count of something: how many, how much, how often, how far. Find out what is being counted and you have read half the sentence.

What is being held fixed? An equation states that certain things change together. Nothing changes together unless something else is pinned in place. The pinned quantity is usually the interesting one, and usually not written down.

What is allowed to vary, and what happens at the edges? Set a term to zero and see what the equation collapses into. The collapsed version is generally a law you learned at school. Watching your old law fall out of the new one is how you check that you have understood either of them.

What do the units say? If one side of the equals sign is a length and the other is a time, then a speed is hiding inside the equals sign, or somebody has made a mistake. In 1999 NASA lost the Mars Climate Orbiter because one team's software reported thrust in pounds and the other team's software read it as newtons. The spacecraft went into the Martian atmosphere instead of around it. Units are the cheapest error detection ever invented.

One line

There is a single line that working physicists write down when somebody asks what the world is made of. It is short enough to fit on a coffee mug, which is where most of the people who have seen it encountered it. Four or five terms, depending on how you group them. Each term is a compression: unfold one and out comes a force, or a family of particles, or the reason anything has mass.

The Standard Model packages an extraordinary range of measurements in that line. Its symmetry and field choices constrain the terms, while experiments supply parameters. A deeper reconstruction must account for those choices as well as reproduce their successful predictions.

By the last third of this book, you will be able to read that line and ask which parts follow from observer consistency, which require further assumptions, and how the resulting claims can be tested.

That is the promise. This prologue makes no other.

Chapter 1

Why Is the Universe So Hard to Reverse Engineer?

In the summer of 1687, in the opening pages of the book that founded physics, Isaac Newton declined to define time. He had spent the previous section defining everything else, carefully, in the manner of a man who knows he is being read by people looking for a mistake. Mass, momentum, force, centripetal force, all set out with their units and their edges. Then he stopped and wrote that he would not define time, space, place and motion, on the grounds that these were well known to everybody.

Then he defined them anyway.

Absolute, true and mathematical time, he wrote, of itself and from its own nature, flows equably without regard to anything external. Absolute space, in its own nature and without regard to anything external, remains always similar and immovable. Two sentences, offered as clarifications of the obvious, and neither of them derived from anything. They are the load-bearing assumptions of the *Principia* and they were put in by hand, by a man who had just told you they were too obvious to need stating.

They held for two hundred and eighteen years.

They did not hold unopposed. Within thirty years Gottfried Leibniz was writing furious letters about them to Samuel Clarke, who was answering on Newton's behalf and, most historians think, with Newton reading over his shoulder. Leibniz's objection was not technical. It was that absolute space never does anything you could catch it at. Space, he said, is nothing but the order of things that coexist, and time nothing but the order of things that follow one another. Suppose God had built the entire universe three feet to the left of where it is, everything shifted together. What would be different? Nothing measurable, nothing observable, nothing at all. So the position of the universe in absolute space is a fact about a picture somebody drew around it, and about nothing else.

Clarke replied. Leibniz replied to the reply. The correspondence ran to five long letters each and stopped in November 1716, when Leibniz died, which is the only way that particular argument was ever going to end.

He was right, and it did not matter. Newton's mechanics worked, Leibniz's objection predicted nothing anybody could measure, and absolute space went into the textbooks for two centuries. The complaint survived in a thin line of people who were mostly ignored: George Berkeley restated it in 1721. Ernst Mach restated it in 1883, in a history of mechanics arguing that the inertia of a body has to come from its relation to all the other matter in the universe rather than from its relation to empty space. A young Einstein read Mach and said later that it had shaken him.

That line, the one saying that a quantity nobody can measure from inside is not a quantity, loses every argument it enters for two hundred years, and then wins.

It also leads to a question that sounds impolite. Three and a half centuries of work, by some of the most capable people who have ever lived, with instruments of absurd precision and budgets to match. The job is not finished. There is no theory of everything. There are two theories, they do not fit together, and the seam between them is where all six of the symptoms in the prologue live.

Why not?

The question is what a theory takes as input and what it explains. Different starting points make different parts of the world accessible to derivation. Comparing them means following that division all the way to a measurement.

1.1 Four parameters

In the spring of 1953 a young Freeman Dyson took a train to Chicago to show Enrico Fermi some calculations. Dyson's group had spent months computing how mesons scatter off protons. The numbers agreed with Fermi's measurements. Fermi was polite for about two minutes and then asked how many free parameters were in the calculation. Four, said Dyson. Fermi said that his friend Johnny von Neumann used to say that with four parameters he could fit an elephant, and with five he could make it wiggle its trunk. With that, Dyson wrote later, the conversation was over. He counted it among the luckiest things that ever happened to him, because it saved his group several more years of computing something that could not have meant anything.

Four parameters, and Fermi thought the answer was worthless.

Fermi's objection is a warning about flexibility. Suppose you have ten measurements and use a polynomial with ten independently adjustable coefficients to pass through them. At ten distinct input positions, interpolation guarantees a fit. Agreement becomes more informative when the same coefficients also predict measurements that played no part in choosing them. Counting parameters helps, but the allowed functions, measurement errors and independent tests matter too.

The knobs are called free parameters. Counting them is the single most useful thing you can do to an unfamiliar theory.

1.2 The log

What follows is a reading of the last three centuries as a crash log: what was assumed, what broke, and what got fixed. There are five entries.

Absolute space and time. Newton, 1687, by definition, as above. Everything in mechanics rests on it, everything in mechanics works, and for two centuries the assumption is invisible because it is also correct to within any measurement anybody can make.

A medium for light. The nineteenth century established that light is a wave. Waves wave in something: sound waves in air, water waves in water, and light waves in a substance nobody could detect but everyone was confident about, because the alternative was a wave in nothing, which was absurd. The ether had to be stiffer than steel to carry light at that speed, and thin enough that the planets swept through it without slowing. Physicists accepted both properties at once rather than give up the assumption.

In 1887, in a basement in Cleveland, Albert Michelson and Edward Morley floated a sandstone slab on a pool of mercury so it could be turned without vibrating, mounted an interferometer on it, and looked for the Earth's motion through the ether. The apparatus was sensitive enough to detect a fraction of the expected effect. They found nothing. Michelson described the result as decidedly negative and spent much of the next forty years repeating the experiment with better equipment, looking for the drift he was certain had to be there. He received the Nobel Prize in 1907, the first American to win one in the sciences, for the instruments rather than for what they had failed to find.

Physicists at the end of the nineteenth century are supposed to have thought their subject was finished. Lord Kelvin did not, and said so at the Royal Institution in London on 27 April 1900, speaking on what he called the two clouds hanging over the dynamical theory of heat and light. Cloud one was the Michelson-Morley result: nobody could find the Earth's motion through the ether. Cloud two was that the accepted theory of how energy distributes itself among the parts of a system gave manifestly wrong answers for radiation and for the heat capacity of gases. Kelvin was seventy-five and he had put his finger on the two loose threads in the building. Cloud one became relativity within five years. Cloud two became quantum mechanics within twenty-five. Between them they took down everything else in the lecture hall.

Absolute time, withdrawn. 1905. Einstein's move was to stop treating the null result as a failure and start treating it as data. If no experiment can find the ether, take that as the law, and see what else has to change.

What had to change was simultaneity. Two events that happen at the same moment for you happen at different moments for somebody moving past you. There is no fact about which of you is right.

Notice what survived. Space and time stopped being separate and stopped being absolute, and stayed a stage. There is a four-dimensional arena, events have locations in it, and physics consists of saying what happens at those locations. Ten years later general relativity made the arena bend, which is a far more radical move than it is usually given credit for, because a bending stage is a stage that participates. But it is a stage. Every equation Einstein wrote is a statement about a set of points. The set of points is put in at the start.

Leibniz would have recognized the situation exactly. The quantity that got deleted in 1905, absolute simultaneity, was deleted for his reason: nobody could measure it from inside. The arena survived because nobody had a version of the objection that reached that far.

Definite values, withdrawn. 1926. Quantum mechanics arrived in about eighteen months of extremely bad-tempered argument. It removed the assumption that a thing has a value before you measure it. The rule for converting the theory's central object into odds you can bet on entered physics as a footnote. Max Born wrote a paper on collision processes that summer, said the probability went as the wave function, and then added a correction in proof: it goes as the square of the wave function. That footnote is the Born rule, it is the only place in quantum mechanics where probabilities come from, and it won him the Nobel Prize twenty-eight years later.

The standard formulation supplies a space of states, a rule for evolution and a rule for probabilities. Reconstruction programs ask which of those ingredients follow from other assumptions. OPH belongs to that effort: its finite probability results state the assumptions under which the Born representation follows.

The Standard Model. Assembled between about 1954 and 1973, it combines symmetry principles with experimentally identified matter and measured parameters. Its successes span particle decays, collisions and precision measurements. The number of independent inputs depends on the neutrino model: the familiar counts run from nineteen to about twenty-six. Newton's constant and the cosmological constant belong to the gravitational description.

That is not the elephant problem. Fermi's complaint was that four knobs let you fit anything. The Standard Model's parameters are not knobs in that sense: they are measured quantities, each pinned down by many independent experiments, and turning any of them breaks a hundred other predictions. The Standard Model is not overfitted. Every one of those twenty-six numbers is an input. No part of the theory has an opinion about why any of them has the value it does. Ask why the electron weighs what it weighs and the correct answer from inside the Standard Model is that somebody measured it.

Nineteen numbers is few for a theory covering every non-gravitational phenomenon ever observed, so the trouble is not the count. The trouble is direction. Numbers go into the Standard Model and predictions come out. There is no arrangement of the theory in which any of those numbers comes out instead. A framework that produced even one of them, from structure, with nothing fed in, would be doing something categorically different from fitting.

The seam. A complete account must describe quantum matter and dynamical geometry together. Quantum field theory on a supplied spacetime and classical general relativity each explain a great deal; neither, by itself, specifies the whole quantum gravitational system.

Perturbative quantization of the Einstein action requires additional interaction terms at increasing orders. The original action therefore cannot define a theory at every energy through a fixed finite list of renormalized parameters. Effective field theory permits controlled low-energy predictions: Donoghue's calculation separates calculable long-distance quantum effects from unknown short-distance physics. A fundamental reconstruction must explain the additional structure.

The examples show how productive it can be to reconsider a starting assumption. Einstein did not add a component to Newton's machine; he deleted absolute time and watched the rest rearrange itself. Born did not add a mechanism to make particles have definite positions; he deleted the requirement and wrote a footnote about odds.

And every deletion was resisted, always for the same reason. An assumption you have been making since you were four years old does not present itself as an assumption. It presents itself as the world. The ether was not a wild idea somebody talked physicists into; it was the obvious consequence of the obvious fact that waves wave in something. Giving it up meant accepting a sentence that sounded like gibberish. Absolute simultaneity was not a theory. It was what the word "now" meant.

1.3 Four approaches to the problem

Four influential approaches show how differently that task can be posed. Each makes particular structures calculable, with its own starting assumptions.

Grand unification, 1974. Howard Georgi and Sheldon Glashow noticed that the three forces of the Standard Model would fit tidily inside a single larger symmetry, and published a paper called "Unity of All Elementary-Particle Forces", which is either a title or a dare. The idea was tidy in a way that felt like evidence. Quarks carry charges in thirds, which had never made sense and which suddenly did, because inside the larger symmetry a quark and a lepton are two faces of one object and the thirds are forced. The particles of one generation, which in the Standard Model are a list you write out, became a structure. And the three force strengths, which are not constants but change slowly with the energy you probe at, appeared to converge on a single value at an energy around ten to the sixteenth times that of a proton at rest. It was the most beautiful idea in particle physics and it made a prediction: if the forces are aspects of one symmetry, then quarks can turn into leptons, and the proton, which everybody had assumed was stable forever, decays.

The predicted lifetime was around ten to the thirty-first years. The universe is about fourteen billion years old, so you do not sit and watch one proton. You watch ten to the thirty-third of them for a year and wait for one to go. Which is what Super-Kamiokande is: fifty thousand tons of ultrapure water in a mine under a mountain in Japan, lined with photomultiplier tubes, watching since 1996 for a single flash of the right shape.

Nothing has happened. The floor on the proton's lifetime has been pushed past ten to the thirty-fourth years, three orders of magnitude beyond the original prediction. The simplest version of the theory has been dead for decades. The program did exactly what a scientific program should do. It made a specific claim, somebody built a swimming pool under a mountain to check, and the answer was no.

Lattice quantum chromodynamics, also 1974. In the same year, Kenneth Wilson made spacetime discrete, as a way of computing rather than as a claim about the world. The equations of the strong force cannot be solved with pen and paper, so put the world on a grid of points, let a computer grind through it, and shrink the spacing at the end. It works. It is how we know the proton's mass from first principles. The calculations run on some of the largest machines ever built.

The lattice is computational scaffolding. Its spacing is varied to estimate and control the error relative to the continuum theory. A finite calculation and a fundamentally finite world are different proposals. Lattice quantum chromodynamics supplies a precise way to calculate strong-interaction physics from an action and parameters, including quark masses, that the calculation takes as inputs.

String theory. Replacing point particles with extended strings gives a different framework for quantum gravity. In 1984 Michael Green and John Schwarz demonstrated cancellation of gauge and gravitational anomalies in a specific superstring theory. The extra dimensions in such constructions must be given a physical interpretation, often by compactifying them into a small internal geometry. Different compactifications can produce different low-energy particle spectra and couplings.

The program has substantial derivations. Andrew Strominger and Cumrun Vafa counted quantum states and recovered the area-entropy relation for a class of extremal black holes. Juan Maldacena's correspondence relates specified quantum field theories to strings and gravity in anti-de Sitter settings. It offers a route in which a gravitational bulk is reconstructed through a different description. Those results must be judged within their assumptions; selecting a description of the observed universe is a further physical question.

Background-independent quantum gravity. General relativity makes geometry dynamical. Loop quantum gravity takes that lesson into quantization without choosing a fixed background metric. Its quantum geometry includes discrete area and volume spectra. Abhay Ashtekar and Jerzy Lewandowski's account explains the construction and its starting structures. A quantized geometry and a complete account of observed matter are distinct requirements.

These programs do not share a single hidden assumption that OPH alone removes. Each changes the starting objects and derives consequences. The useful comparison asks how much structure follows, which inputs remain free, and which measurements test the resulting physical model.

1.4 The rule

A rewriting program supplies another explicit relational starting point. Stephen Wolfram's physics project starts with a **hypergraph**, a collection of relations whose elements carry no prescribed positions or distances. A rule replaces a matching pattern of relations with another pattern. Repeated replacements produce a causal record of which updates depend on which others.

The project's technical account develops proposed connections between these structures and geometry, relativity and quantum mechanics. Independence from the ordering of updates, called causal invariance, is central. Its treatment of observers also matters: they operate within the evolving system and have bounded computational resources. OPH shares the ambition to describe a world from inside; that ambition has more than one mathematical expression.

The scientific question is how the rule, observer model and limiting construction determine measurable physics. A candidate rule must do more than produce a suggestive picture. Its readouts must be identified with quantities an instrument can measure, with an error estimate that survives changes in the calculation.

OPH puts overlap consistency at the center of that selection problem. Its patches carry records and repair disagreement under explicit laws. The proposal earns its explanatory force where those conditions select a structure, and its physical force where the resulting readouts pass independent tests.

1.5 What is left

Different programs choose different starting objects: fields, quantum geometries, strings or discrete relations. A reconstruction has to show how its chosen objects support familiar physics. An assumption cannot also count as its own explanation.

OPH takes a specific route. Begin with finite observer patches and agreement constraints, then ask which structures can be reconstructed without supplying a physical spacetime. The observer architecture is an input. Space, clocks and measurable matter must be connected to it by further arguments.

A particle is a thing at a location, and there are no locations, so it goes. A field is worse: a field is an assignment of a value to every point, which presupposes the points and then adds bookkeeping on top. A wave is a disturbance in something. A string is extended, which means extended in something. A geometry is a statement about a set of points and how far apart they are. Even a spacetime event, the humblest object on the list, is a location with nothing at it.

The familiar spatial descriptions depend on a location structure. Relational descriptions offer a way to ask where that structure comes from.

The candidate developed here is a finite system that reads part of itself and its boundary, keeps a record of what it read, and can repair disagreement when its record and a neighbor's fail to match.

Note that nothing in this description mentions a location. Reading a neighbor does not require knowing where the neighbor is, only that there is one. Keeping a record does not require a place to keep it, only that the record can be read back. Disagreeing does not require a background, only two accounts of the same thing. The patch is specified through its accessible operations and relations, before assigning those relations a physical geometry.

That is a peculiar object to build a universe out of. Whether it works is the only question left. It has one property that recommends it immediately and one that ought to worry you. It assumes no arena, which is the entire point. And it has no obvious way of getting started, because to read a neighbor you need a neighbor, and to have a neighbor you need a world, and we have just thrown the world away.

That leaves a question older than physics. What could make a world built from such relations self-consistent, and why is there anything here?

Chapter 2

Why Is There Anything Rather Than Nothing?

Alfred North Whitehead and Bertrand Russell spent ten years trying to build arithmetic out of nothing, and they meant it literally: no numbers assumed, no quantities taken as given, no facts about counting smuggled in from the outside world. They began with a handful of logical marks and rules for shuffling them. The plan was to show that everything true about numbers followed from the shuffling alone.

The result, *Principia Mathematica*, ran to three volumes. The proposition that one plus one equals two turns up in volume two, at proposition 110.643, after several hundred pages of preparation. Underneath it the authors added a line of commentary, which reads in full: "The above proposition is occasionally useful."

Russell wrote later that his intellect never quite recovered from the effort, and that he had been definitely less capable of difficult abstraction ever since. He was thirty-eight when they started. Cambridge University Press estimated the work would lose money and asked the authors to contribute fifty pounds each toward publishing it, which they did. The book brought them no income.

It is easy to read that as an anecdote about eccentric mathematicians and miss what the project was. Whitehead and Russell were not being pedantic about two. They were testing whether a body of knowledge could stand on itself: whether you could put down marks and rules with no borrowed meaning, turn the handle, and have arithmetic come out. If it worked, then nothing about numbers had to be assumed, because everything about numbers would be a consequence. Ten years, three volumes, and fifty pounds each was what it cost to find out for arithmetic alone.

What they were building has a name. A **formal system** is a set of marks you are allowed to write, a starting position, and a set of moves that turn one arrangement of marks into another. Meaning is absent by construction. The marks stand for nothing, the moves have no reasons, and the system has no idea what it is about. Hand the rules to somebody who shares no language with you and they will apply them correctly.

Such a system makes its starting assumptions explicit. Chapter one introduced a relational observer architecture before assigning physical space or time to it. The question is what that architecture can support. Behind it stands an older question, posed by Leibniz in 1714: why is there something rather than nothing? Nothing, he pointed out, is simpler and easier.

2.1 Two answers that do not work

The traditional replies come in two shapes.

The first says: something outside made it. Something caused the world. That something is the reason there is anything at all. This is the oldest answer on record and a serious one. Aristotle got to it by asking what moves a thing, and then what moves that, and noticing that the question does not stop on its own.

The trouble it runs into is arithmetic. Whatever you nominate as the cause, you can ask the same question about it. There are two ways out, both bad. Either the chain runs backward forever, in which case you never arrive at a reason and have only postponed the problem an infinite number of times, or you stop at some point and declare that this one thing needs no cause. But if one thing can need no cause, the question is why that particular thing gets the exemption. No answer to that is available except that we stopped there. The item nominated as needing no cause has an almost perfect historical record of being the item the arguer arrived with.

A first cause is also an assumed component, put on the board because the argument needs it rather than because anybody found it, which is the one rule a reverse engineer cannot break.

There is a second objection, the one that does the damage here, because it does not depend on the regress at all.

Something outside. Something before.

Outside is a spatial word. It means separated by a distance, on the far side of a boundary, in a different region. Before is a temporal word. It means earlier along an ordering that both things sit in. Chapter one did not suggest that space and time might turn out to be built from something else. It took them off the starting inventory altogether, along with the arena and the parameter that things happen along. So there is no distance for a cause to sit at and no earlier for it to have acted in.

Which makes "something outside made it" a sentence assembled out of the two components a reverse engineer has not found on the board. The regress was never the real complaint. A regress is what turns up when a question has quietly helped itself to a place to stand and a moment to stand in, and then goes looking for somewhere to put the answer.

The second reply says it is just here. A brute fact, with nothing behind it, and asking why is a mistake. This has the virtue of stopping the regress immediately and it has one crippling defect. It explains nothing and it also forbids nothing. If the world is here for no reason, then a world with four spatial dimensions would also be here for no reason, and so would a world with none, and so would a world where the electron weighs twice what it does. You have not explained the world, you have declared that no explanation exists, and in the same motion you have thrown away any hope of understanding why it has the specific shape it has rather than one of the others. The twenty-six measured numbers of chapter one stop being a debt and become the natural state of affairs.

There is a way to feel the size of what that concedes. The mass of the electron is known to about eleven significant figures. Treat it as a brute fact and you have accepted that eleven digits arrived for no reason, and that the same is true of the other twenty-five numbers, and that the total quantity of unexplained precision in the world runs to a couple of hundred digits that simply are what they are. That is a great deal to put down to nothing in particular.

Both replies point away from the thing they are explaining, one to a cause and one to a shrug. Pointing away is the move that has just been withdrawn. With no outside and no before, there is nowhere for an account to come from except the thing itself. So what is left is not a third option chosen over two others. It is the only one still standing. It stands by elimination: a structure accounts for itself, holding together in a way that requires nothing to be added to it, so that asking what put it there has the same status as asking what makes a triangle have three sides.

That also does something to the question in the title. "Why is there anything" wants a because, and a because wants somewhere to come from. Take away the outside and the before and the why has no room to operate in, while the question underneath it survives intact and turns out to be a question about structure: what does a thing have to be like, to be the kind of thing that can hold itself up? What follows is the beginning of that answer. The beginning is small.

Stated like that it is a phrase rather than a proposal. Phrases of that shape have a long history of sounding like wisdom and meaning nothing. It gets edges from three things: what it means for a system to have consequences nobody chose, what it costs to contradict yourself, and what happens when you demand that something survive its own description.

2.2 Two rules and a pencil

Here is a formal system small enough to run by hand. You will need about ninety seconds.

The marks are the letters a and b. The starting position is the string `ab`. There are two moves:

Move one. Take any string you have and wrap it: put an a on the front and a b on the back. So `ab` becomes `aabb`.

Move two. Take any string you have and write it twice. So `ab` becomes `abab`.

That is the whole system. Start from `ab` and apply the moves in any order you like, as many times as you like. From `ab` you can reach `aabb` and `abab`. From `aabb` you can reach `aaabbb` and `aabbaabb`. From `abab` you can reach `aababb` and `abababab`. Go on for a while and you will generate a large and reasonably complicated family of strings, as many as you have patience for.

Here is a question about that family. Can you reach `aab`?

Try it for a minute and you will fail. By itself, failing to find something is not the same as showing it is not there. But look at the counts. The starting string has one a and one b. Move one adds one a and one b. Move two doubles both. So every string you can possibly reach has exactly as many a's as b's, in every case, forever, without anybody having written that down as a rule. `aab` has two a's and one b. It is unreachable. You know this without searching. You have proved something about an infinite collection of strings by looking at four lines of rules. The proof was sitting there before you began searching, which is the standard experience in this subject and does not become less irritating with practice.

The decisive quantity is the difference between the number of a's and the number of b's. It starts at zero. Neither move can change it: one move adds one to each side of the subtraction, the other doubles both. A quantity the rules cannot touch, however long you run them, is the most useful thing anybody can know about a system.

There is a second thing about that little system that's easy to overlook. The strings it makes are not about anything. `aabb` is not a description of some other, realer `aabb` sitting elsewhere in a warehouse of genuine strings. Ask whether the one in front of you is the authentic article or a copy and the question dies on the way out, because there is nothing available for it to be a copy of. You ran the rules, and what came out is what there is.

That is the first useful property of a system with no meaning in it. Its consequences are not limited to what its author had in mind. Whitehead and Russell wrote down their marks and their moves, and then spent a decade finding out what they had committed themselves to. The answer took three volumes because the answer was not up to them. Nothing was consulted. Nothing was measured. The counts were fixed the moment the rules were written. The ten years were spent reading them off.

2.3 The price of contradictions

Suppose a bank's ledger says that an account holds nothing and also says that the same account holds a hundred pounds. Both entries, on the same page, at the same time.

A clerk looking at that does something sensible. They decide one entry is a mistake, or they pay out the hundred and open an investigation, or they go and find the deposit slip. What a clerk does not do is conclude that the account holds nine million pounds. People reason around contradictions all day without much difficulty: two entries that cannot both hold get quarantined, while the rest of the ledger goes on working.

A formal system has no clerk. It has marks and moves, it applies every move that applies, and it has no faculty for deciding which of its own lines is the suspicious one. Applied honestly, the moves take it somewhere no clerk would follow.

Watch. The ledger says the account holds a hundred pounds. From that alone I may write down a weaker sentence: either the account holds a hundred pounds, or the moon is a biscuit. That step costs nothing, because attaching an alternative to a sentence you already have cannot make it less true. Anybody who accepts the hundred pounds is obliged to accept it.

But the ledger also says the account holds nothing, so the first half of my weaker sentence is out. An either-or with its first half ruled out leaves the second half standing.

The moon is a biscuit.

No step in that is a trick. A clerk would sign off on each one taken by itself. Run it again with a different second half and the account holds nine million pounds. Run it again and you get whatever you please, because the second half was never constrained by anything: it was chosen freely at the first step and it survived to the last. Two entries on one page, and the ledger certifies every sentence in the language.

An inconsistent description asserts every world at once, and a description that permits everything distinguishes nothing. It is not a false theory. It is not a theory.

Somebody found this out the expensive way, in June 1902. The letter that did it came from Russell. Gottlob Frege had spent two decades building arithmetic out of logic. The second volume of his *Grundgesetze der Arithmetik* was at the printer when Russell wrote to him on the sixteenth with one short observation about the class of all classes that are not members of themselves. Frege replied six days later. He had seen it at once. What he had seen was not that some theorem of his was wrong. It was that his system now proved everything, which is the same as proving nothing. He added an appendix while the volume was in press. It opens by saying that hardly anything more unwelcome can befall a scientific writer than to have one of the foundations of his edifice shaken after the work is finished.

Which is why *Principia Mathematica* runs to three volumes. The several hundred pages of preparation, and one plus one arriving at proposition 110.643, are not eccentricity. They are the price of a system that Russell's letter could not be written to.

The defense did not hold. Russell's letter worked by getting a system to talk about itself, and *Principia Mathematica* was built to make that impossible. In 1931 Kurt Gödel showed that it cannot be made impossible. Any system strong enough to carry out ordinary arithmetic is strong enough to describe its own formulas, and once a system can describe its own formulas, sentences about itself are among the things it can build. What Gödel assembled out of one was milder than Frege's disaster and worse to live with. Frege's system proved everything, which killed it outright. Gödel's sentence is true, the system it is written in cannot prove it, and no patch closes the gap, because the repaired system has a sentence of its own. Whitehead and Russell had bought a system that could not be attacked the way Frege's had been. What was not for sale was a system with nothing to say about itself, because a system with nothing to say about itself is too weak to be the foundation of anything. Chapter thirty-six is where the construction gets taken apart.

The result was read as a demolition, and as a demolition it was decisive: the programme of standing mathematics on logic did not recover. Read the same theorem as a specification and it says something the demolition reading discards. A structure rich enough to describe arithmetic is rich enough to describe itself. No arrangement of the axioms takes that capability away. It arrives with the strength.

Chapter one gave you the tool for seeing why that matters here. A theory with infinitely many free parameters fits anything and forbids nothing, which was the fatal problem with quantizing gravity by brute force. A contradiction is the same disease in its terminal form: not many knobs but all of them, not a poor prediction but every prediction at once. Consistency is what buys a description the right to rule anything out.

So the demand this book starts from can be stated. **A structure counts as existing exactly insofar as it accounts for itself with nothing left over.** No external cause is allowed, because a cause is one more thing standing in need of an account. No leftover assumptions are allowed, because an assumption is a thing you would have to explain. And it must not contradict itself, because a contradiction buys everything and therefore purchases nothing.

The obvious objection is that this demand is so severe that nothing could satisfy it. The answer to that objection is a number.

2.4 One demand, and a calculator

Take a calculator, or a phone, or a piece of paper if you are feeling nineteenth century. Pick any positive number at all. Any at all: your age, the year, seven.

Then do this. Divide one by your number, add one, and keep the answer. Then do it again to the answer. Then again.

Start with seven and you get 1.142857, then 1.875, then 1.533333, then 1.652174, then 1.605263, then 1.622951, then 1.616162, then 1.618750, then 1.617761. The numbers are hunting, overshooting one way and then the other, and closing in. Keep going and they settle on 1.6180339887.

Start with a thousand instead. Or with 0.001. Or with the number of stairs in your building. Same destination, every time, to as many decimal places as you have patience for.

There is a small bonus in the arithmetic if you look at the fractions rather than the decimals. Starting from seven you get 8/7, then 15/8, then 23/15, then 38/23, then 61/38, then 99/61, then 160/99. Read the numbers down the page: 7, 8, 15, 23, 38, 61, 99, 160. Each one is the sum of the two before it. You did not ask for that and the rule you were applying says nothing about addition.

What is 1.6180339887, and why does everything land on it?

It is the number that the operation leaves alone. If you feed in a number and get the same number back, then that number satisfies x equals one plus one over x. Only one positive number does. Multiply through by x and it reads x squared equals x plus one, which is a quadratic. Its positive root is one plus the square root of five, all over two. Which is 1.6180339887, and which people have called the golden ratio since long before anybody could write it down that way.

The hunting has a mechanism. When your number is too big, one over it is too small, so the answer comes out below the target. When your number is too small, one over it is large, so the answer overshoots above. Every step lands on the far side of the destination from where you started, and lands closer. The error does not merely shrink, it changes sign each time and shrinks, which is why the decimals settle from both directions at once.

A number that a process leaves alone is called a **fixed point**. This is the first of many in this book. The number itself turns up in enough seaside-gift-shop literature to have earned a bad reputation. Where it came from is the part that matters. Nothing was measured. No experiment was performed and no data existed at any point in the procedure. A single demand, that the operation return what it is given, reached into the continuum of real numbers and picked out exactly one.

Look at the shape of that demand rather than at the number it produced. The equation reads x equals one plus one over x. The unknown stands on both sides of the equals sign. What is being solved for appears inside its own specification, which in most arguments is the move that gets called circular and thrown out. Here it is the entire mechanism. Take the self-reference out and no equation is left, only a continuum with nothing in it to prefer.

That is the shape of the answer to the question in the title. A requirement severe enough to have exactly one solution.

That quadratic has two roots. The other is minus 0.618034. Check it: one divided by minus 0.618034 is minus 1.618034, and adding one gives you minus 0.618034 back. It is a fixed point, exactly as much as the other one is. But start your calculator anywhere near it and the numbers do not settle there, they flee. Being a point that a process leaves alone is one property. Being a point that a process moves toward is a different property. A structure can perfectly well have fixed points that nothing ever arrives at.

2.5 What one requirement is worth

Look at what the demand bought. Before it there was a continuum of positive numbers available, uncountably many, with no reason to prefer any one of them. After it, there was one. One number, to as many decimal places as anybody cares to compute.

That exchange rate between what goes in and what comes out is the engine of this book. It fails in both directions.

Require nothing and you get nothing. This is the part that sounds wrong, because requiring nothing sounds like permitting everything, and permitting everything sounds generous. It is empty. A structure with no requirements on it does not have many possible forms; it has no form, because a form is exactly a set of things that are not allowed. If every arrangement is as good as every other, then nothing distinguishes anything, and there is no fact about what the structure is.

Which is the same failure the bank ledger had, arrived at through the opposite door. A contradiction permits every conclusion and therefore asserts nothing. A structure with no constraints permits every configuration and therefore is nothing. Total inconsistency and total freedom are emptiness. A great deal of confused metaphysics comes from noticing only one of them.

Between those two failures is a window, far narrower than anybody expects. Put a small number of requirements on a structure and what survives is not a comfortable family of possible worlds to choose among. It is very nearly one world, with a definite number of directions in it, a definite list of things it is allowed to contain, and definite numbers attached to them, none of which anybody selected. The golden ratio is that phenomenon at the smallest scale it can occur at: one requirement, one survivor, out of a continuum.

This cuts both ways and the second way is the uncomfortable one. If adding a requirement collapses a continuum to a point, then a requirement that slips in without being noticed produces a world that arrives without being earned. Chapter one is three centuries of exactly that going wrong. So the constraints have to be counted, they have to be few, and every one of them has to be visible.

2.6 Three requirements

Everything here comes out of three requirements, taken one at a time over the next three chapters.

The first says that whatever the world is made of, each piece of it sees only a part and keeps a record of what it saw. The second says that where two pieces overlap, their records have to agree, and that any relabeling of a piece has to be a move the system can carry out on itself, with no hand reaching in. The third says that beyond what those two force, nothing is assumed: whatever is not pinned down stays as unconstrained as it is possible to be.

There is very little in there. Nothing about space, because there is none yet. Nothing about time, particles, forces or numbers. No mention of what the pieces are made of, how many there are, or what the records say. The first two are constraints on bookkeeping and the third is a refusal to add anything.

Three, and not more, and if that sounds too few to build a universe from, the history of the last comparable attempt is instructive.

Euclid began his geometry with five postulates. Four of them are the sort of thing nobody argues with: you can draw a line between two points, you can extend a line, you can draw a circle with any center and radius, all right angles are equal. The fifth is about parallel lines and it is longer and clumsier than the other four put together. Euclid appears to have been embarrassed by it, because he avoids using it for as long as he possibly can. For two thousand years mathematicians tried to prove it from the other four and get rid of it, and failed, and assumed they had merely failed rather than attempted something impossible.

In 1733 an Italian Jesuit named Giovanni Saccheri published a book with the confident title *Euclid Freed of Every Flaw*, in which he tried the last remaining tactic. Assume the fifth postulate is false, grind out the consequences, and wait for a contradiction. He ground out a great many consequences. Triangles whose angles sum to less than two right angles. Lines that approach each other forever without meeting. An entire coherent body of theorems, dozens of them, all following perfectly from four postulates and the denial of the fifth, and not one contradiction anywhere in it.

What Saccheri had in his hands was a complete and consistent geometry of curved space, a century before anybody else found one. What he concluded was that the results were repugnant to the nature of a straight line, which he offered as the contradiction he had spent the book hunting for. He published, and died that year, having discovered a new kind of space and declined it on the grounds that he did not care for the look of it.

A hundred and eighty years later Einstein went looking for a mathematics of gravity and found he needed geometry without the fifth postulate:

space whose triangles do not add up to two right angles, whose parallel lines do not stay parallel, and whose shape is settled by what is sitting in it. The postulate Saccheri was trying to rescue is false in the universe he wrote the book in.

Two things come out of that. Four postulates and a decision about a fifth had been quietly fixing the shape of space for two thousand years. Nobody found out by inspection, because you cannot see a postulate by staring at it. They found out by working: by taking the requirements seriously and grinding out what they force, which took twenty centuries and was being got wrong as late as 1733. And the consequences of a small set of requirements are not up for negotiation. Saccheri disliked his and they were true anyway.

That is the whole starting position. There is a hole in it.

Look at what the three tools above have in common.

A formal system produces consequences. Those consequences are strings: arrangements of marks that stand one way rather than another and can be read back. Consistency is a property only certain objects can fail to have. A rock cannot contradict itself. It sits there being whatever shape it is. No arrangement of atoms is a contradiction, because contradicting yourself means holding two claims about the same thing and a rock holds no claims. And a fixed point is whatever satisfies a condition, where a condition is a demand, and a demand has to be made of something and stated somewhere before it can do any demanding.

Every one of those is a property of a description rather than of a lump of world.

Which puts a clause into the founding demand that nobody wrote there. "Accounts for itself" has no application to a structure containing no accounts. Ask whether a universe with nothing in it that keeps a record is consistent, and the question slides off: there is nothing in such a place to be right or wrong about anything, so there is nothing for consistency to be a property of. The word does not become false. It stops reaching.

So the founding demand turns out to require something nobody put into it and nobody would have wanted. Somewhere in that structure there has to be a thing that holds a description, sets it against something, and is capable of getting it wrong. Otherwise the demand has nothing to bite on, and a structure the demand cannot bite on fails it, and by its own terms it does not exist.

One more thing follows. It does not get settled here. If the structure is everything, then whatever its record-keepers describe is inside it, so some of what they hold is a description of the thing they are part of. Gödel's theorem says a structure that strong can carry such a description. The demand above says it has to.

A calculator, two letters of the alphabet, and a bank ledger with two entries on it. What comes out at the bottom of them is that the universe has to contain something capable of being wrong.

Chapter 3

Why Would a Universe Need Anybody In It?

In 1883 a science teacher at the state normal school in Whitewater, Wisconsin, got tired of janitors. The heating in the building ran off a coal boiler in the basement. The only way to find out whether a classroom was too cold was for somebody to walk into it and look at the thermometer on the wall, which happened during lessons, repeatedly, all winter. Warren Johnson's solution was a device that would notice the temperature itself and ring a bell in the boiler room. He patented it, then built a company to sell it, and that company outlived him by more than a century, which is a generous return on being annoyed.

The thing he built is on your wall. It has a sensor, a number you set, and a switch. It does one thing forever: it looks at what the temperature is, looks at what you asked for, and if those two disagree it does something about it.

Chapter two ended by discovering that consistency is a property of descriptions, and that descriptions have to be held by something. The obvious next move is to reach for a person. It is the wrong move, because reaching for a person means assuming a brain, a body, a planet, several billion years of chemistry and a spacetime for all of it to happen in, which is every assumption chapter one deleted arriving back through the side door.

The right move is to ask what the job actually requires, and then find the cheapest object in the universe that can do it. The answer is the thermostat on the wall. Nothing has to be added to it.

3.1 What the job requires

Take the job seriously. Something has to hold a description of part of the world. What does it need, and what can be removed?

It has to see something, and not everything. This one runs the opposite way from intuition, so take it slowly. You might suppose that a better describer is one that sees more, and that a perfect describer would see everything.

Lewis Carroll got there first. In *Sylvie and Bruno Concluded*, published in 1893, a character called Mein Herr explains that his country's cartographers began with maps at six inches to the mile, then tried six yards to the mile, then a hundred yards, and finally arrived at the grandest idea of all, a map on the scale of a mile to the mile. It has never been spread out, he admits: the farmers objected that it would cover the whole country and shut out the sunlight. So we now use the country itself, as its own map, and I assure you it does nearly as well.

That is the whole argument. A description that leaves nothing out is a duplicate. A duplicate tells you precisely as much as the original does, which is to say it does the job nearly as well and to no purpose whatever. Description begins where something gets left out. A boundary is what makes a thing a describer rather than a copy. It belongs at the top of the list rather than in a footnote about limitations.

It has to write. A description exists where it is recorded and nowhere else, because an unrecorded one leaves the world in exactly the state it would have been in had nobody formed it. So there has to be a somewhere: a state, a mark, a position, a magnetisation, anything at all that can be in one configuration rather than another and stay that way.

It has to read what it holds, and the reading has to make a difference. Seal the thermostat's set number behind glass where the switch cannot get at it. Every part still works. The sensor senses, the number sits there at twenty degrees, the room is cold, and the boiler stays off, because nothing in the device ever consults the number for any purpose. What is left on the wall is a thermometer and a decoration.

So the requirement is not that a record be accurate. It is that something downstream depends on it. A mark nothing consults is a mark the world runs identically without. If the world runs identically without it then it is not describing anything, whatever it happens to say.

This is also the place where one thing gets to matter to another, worth being slow about, because everything later rests on it. Two systems can sit beside each other forever, each in whatever state it is in, with their states correlating as tightly as you like, and nothing whatever passing between them. What makes the second one matter to the first is that the first reads it and does something different for having read it. Remove that and there is no sense in which anything affects anything: there are states, and there is a tally of how often they coincide, and there is nothing else.

It is also where being wrong becomes possible, which chapter two asked for and did not have to. The mark could have been otherwise, so there is more than one state the thing might be in. It gets consulted, so what the thing does depends on which. And the view is bounded, so the mark does not contain the world, and there are always two ways the world can be that leave it saying the same thing. Two worlds, one mark, one response, and in at least one of them the response is not the one that world called for. Nothing was added anywhere to arrange that. It is what a bounded thing that writes, and then consults what it wrote, cannot avoid.

That is the list. Three items, and the third is doing most of the work.

3.2 Three for three

Score the thermostat.

A bounded view. It senses temperature, at one point, in one room. It knows nothing about the pressure, the humidity, the light level, the temperature two floors up, or the weather in the next country. Its window on the world is one number wide, which is about as bounded as a view gets. Passes.

Somewhere to write. Two places, in fact: the current reading and the number you set. Passes.

A reading that makes a difference. This is all it does. The whole device exists to hold two numbers next to each other, consult the pair, and throw a switch on what it finds. Nothing in it is decorative. Passes emphatically.

Three for three. It is screwed to the wall, so it holds each of its numbers from the writing of it to the reading of it, which is all the lasting the third item asks for.

Then score the thermometer beside it, because a list of requirements is worth nothing until something fails one. A mercury thermometer has a bounded view: one number, one place, nothing else. It writes, in the sense that matters, since the height of the column is a physical configuration that can be one thing rather than another, and rewrites whenever the room changes. It persists, being made largely of glass and screwed to the same wall.

Two for three. Nothing in the glass consults the column. The mercury stands where it stands and no part of the instrument is downstream of where it stands, so the height makes no difference to anything the thermometer does, there being nothing the thermometer does. Engrave a second number on the case and nothing changes, because a number on a case is not consulted either. A thermometer has a state. A thermostat has a state and an opinion about it. The gap between those two sentences is the gap between a thing and an observer.

That is a useful failure, because it shows which of the three is load-bearing. Take away the boundary and you get a duplicate. Take away the writing and nothing is retained. But take away the reading and you are left with something that looks entirely functional, sits on the wall for decades, correlates with the world to a fraction of a degree, and observes nothing whatsoever.

The thermostat's three out of three is the point rather than the joke. The bar is that low. A thermostat is an observer in the sense this book needs, in exactly the same sense that you are. The difference between the two of you is quantitative rather than categorical. You have a wider window, more places to write, and a great deal more that consults what is written there. Every one of those is a matter of degree. None of them is a new item on the list.

A sensor, a number you set and a switch meet all three in full. That is the whole of what anything needs to be an observer, thermostats and people included.

3.3 What nobody has to add

The list is short enough that the omissions need saying out loud, because people reliably try to put things back on it.

Understanding. The thermostat has no idea what temperature is, what a room is, or what it is doing. It has never formed a concept in its life. Nothing in the three mentions comprehension. Adding it would exclude every instrument in every laboratory on Earth, which would be an odd result for a theory of measurement.

Intention. The thermostat does not want the room to be warm. Something wants that, and it is not the thermostat, and the device works identically whether anybody wants anything.

A model of itself. Nothing on the list requires the record-keeper to have any representation of its own state, its own boundary, or its own existence. Warren Johnson's device never once represented Warren Johnson's device.

Complexity. Two numbers and a comparison. That is the entire specification. Any system that meets it is on the list regardless of what else it can do.

Life. Obviously not, and this cuts both ways, because it also means that the arrival of living things in the universe did not introduce a new kind of physical entity. It introduced a much better one of a kind that had been there all along.

The list has three things on it. Anything else that gets proposed describes how good an observer is rather than whether it is one.

3.4 The worst word in physics

Which brings up the vocabulary problem. It has to be handled here because a great deal of nonsense has been written on the other side of it.

The word "observer" arrived in physics through relativity, where it never meant a person and was never intended to. An observer in Einstein's sense is a coordinate frame: a convention for assigning three numbers and a time to an event. Textbooks anthropomorphise them for readability, and then have to explain that these observers are equipped with rulers laid out across all of space and clocks at every point, synchronised in advance, which is not a description of any person who has ever lived. Nobody was confused, because everybody knew the word was shorthand for bookkeeping.

Quantum mechanics then borrowed the same word for something that had to do the measuring. The shorthand stopped being harmless. If the theory says that an observation produces a definite outcome, and "observer" sounds like it means a person, then it sounds like the theory says people produce definite outcomes by paying attention. Which is not what any version of the mathematics says, and is nonetheless what a great many people have taken away from it, including some who should have known better.

Eugene Wigner is the honourable case, because he thought it through and then changed his mind. In a 1961 essay he argued seriously that consciousness has to enter the physical description, on the grounds that a measurement apparatus can be described as a quantum system and so can the physicist looking at it, and something has to break the chain. He later abandoned the position. What did it was solipsism: if it takes a mind to make an outcome definite, whose mind, and what were the other minds looking at meanwhile.

The confusion has a way of surviving its own refutation. Schrödinger invented the cat in 1935 as an attack. He was arguing that a theory which lets a radioactive atom smear a live animal across two states is producing something ridiculous. He used the word. The example was a weapon aimed at a position he rejected. It is the single most widely reproduced illustration of that position, printed on t-shirts by people explaining what quantum mechanics says. A joke about a theory has become the theory's best-known exhibit, which tells you roughly how carefully this part of physics gets read.

So, unambiguously, because the whole structure rests on it:

An observer here is any bounded system that keeps records. A thermostat is one. A photographic plate is one. A rock with a scratch on it is not one, and fails on the requirement people would think of last: nothing reads the scratch. Put the rock where the scratch works a lever and it joins the list. Consciousness has nothing to do with any of it. Conscious observers are a special case that appears very late, arriving without anything being added to the list of three.

3.5 Constraint-first

There is a second misreading, subtler than the first and more persistent, and the one that makes people suspicious of observer-centered physics for good reasons.

Putting observers at the bottom sounds like putting *us* at the bottom. That sounds like a demotion of the universe in favour of the people looking at it. It sounds like the world is shaped by our attention. An anthropic selection argument conditions on observers existing; it does not require their preferences to arrange the universe. The observer architecture here likewise concerns what systems can record and compare.

That is the opposite of what is being said. An observer in this book does not get what it wants. It gets what it can consistently record. The second is a much harsher master than the first. The thermostat cannot decide the room is warm. It can hold a number, and the number can be wrong, and being wrong has consequences it does not choose. Everything in the next thirty chapters comes from what a bounded record-keeper is *forbidden* to do: it cannot see everything, it cannot write without limit, it cannot compare what it has no access to, and it cannot disagree with its neighbors indefinitely without something giving way.

There is a cheap way to feel the asymmetry. Set the thermostat to twenty-five degrees in January and switch the boiler off at the wall. The device will hold the number twenty-five for as long as you care to leave it, and the room will be freezing, and nothing anywhere in the arrangement will bend. A record-keeper may write whatever it likes. What it cannot do is make the comparison come out in its favour. That gap, between what a thing may write and what it finds when it checks, is where every result in this book comes from.

Observer-first is constraint-first. Nothing anywhere in this book turns on anybody wanting anything.

3.6 What the starting point explains

A theory that supplies a spacetime can explain how its geometry changes and what matter does within it. Explaining why that starting structure is appropriate requires an additional argument. Chapter one's examples show several ways to move the boundary between what is assumed and what is derived.

OPH begins on the observer side of that boundary. Consider the sentence: *these two records agree.* It specifies a relation between bounded record-keepers without assigning them coordinates or a distance. The patch architecture makes such comparisons precise through its states, ports and repair laws.

That gives a vocabulary for reconstruction. Whether it describes physical spacetime depends on how the records, their relations and their dynamics connect to measured geometry. Agreement supplies the common constraint; the physical bridges must do the rest.

3.7 The bill

One thing has been quietly assumed for six pages and it comes due later.

A physical record uses memory and must be carried by a physical process. Its energy cost depends on that process. Copying a readable value into a prepared register can retain the input; erasing an unknown value removes information from the memory being reset. Under the thermal conditions developed later, that reset has a mean heat bound.

Finite memory and thermodynamic reset connect the observer description to physical resources. Counting records and counting erased bits are different operations.

That is the whole starting position: bounded record-keepers, comparing what they hold against what they see, at a cost.

Which produces the difficulty immediately. If every one of these things sees a bounded part, and there are many of them, and none of them can see what the others hold, then there is nothing in the setup that makes their records descriptions of the same thing. Each has its own numbers. Each is right about its own window. And no two of them have ever been in a position to check.

So what makes it one world?

Chapter 4

Why Do Two Descriptions Ever Agree?

Alice saw the food truck. Bob did not.

They were walking the same street in Barcelona, on opposite pavements, thirty meters apart, conducting the standard argument about where to eat. A delivery van had stopped on the corner of Carrer de Sepúlveda with its back doors open and a man unloading crates of bottled water. From where Bob was walking the van filled the gap between two buildings completely. From where Alice was walking the van was a van, and behind it, in the gap, was a truck selling grilled sandwiches with two people queueing and a third buying two at once.

At the corner they compared accounts. Alice said there had been a food truck. Bob said there had been no food truck, an empty stretch of pavement, and a man with a great many crates of water.

Both accounts were accurate. That is the difficulty.

Alice and Bob have been doing this since February 1978, when Rivest, Shamir and Adleman needed two people to exchange a message in a paper on public-key cryptography and preferred names to letters. They have since been assigned to nearly every problem in science that turns on two parties who cannot see what the other one knows, a category that takes in more of physics than anybody expected in 1978, and in close to fifty years of continuous employment neither of them has ever been told what the message says.

The comfortable picture of what agreement means says that there is one street, that the street has a definite content, that each of them writes down what the street contains, and that if both of them do the job properly the two descriptions come out the same. Two people did the job properly and the descriptions came out different. No amount of care by either of them would have fixed it, because the van is opaque and Bob was on the wrong side of it.

The question is harder than it looks. Take two record-keepers of the kind chapter three described, each seeing a bounded part, neither able to inspect the other's records, with no third party anywhere holding both. What would it even mean for their two accounts to agree? Something has to be demanded of them, or there is no shared world. Whatever is demanded has to be something they can check from inside their own windows, or it might as well be demanded of the weather.

4.1 What can be demanded

Nothing can require Alice and Bob to match on the food truck. Only one of them was in a position to see it. A rule that penalised them for that would be a rule against having a point of view. Every observer in this book has one by construction.

What can be required is that they match where they both looked. Both of them saw the van. Both of them saw the corner, the traffic, the time of day, the number of lanes. That shared region is called the **overlap**, the only place where a demand can land. Outside it, two accounts differing is two windows differing, which carries no information about anybody's competence.

Even inside the overlap the two accounts will fail to be identical, worth being precise about, because it is where the naive picture does its second piece of damage. Alice wrote down that the van was on her left. Bob wrote down that the van was on his right. Those are different sentences. They are not competing claims about the van. Nothing is wrong with either of them.

What holds between them is a rule. To turn Alice's account into Bob's, swap left and right, because they were facing opposite ways. Apply the rule to Alice's whole record and you get Bob's record for everything they both saw. So the demand that survives contact with two points of view asks for a rule, and asks two things of it. That it carry one account onto the other everywhere both of them looked. And that each of them be able to work out what it is without being told.

Where the rule fails to line up, there is a leftover. If Alice has the van two car-lengths from the corner and Bob has it four, no relabeling of left and right repairs that. The leftover is a quantity. You can measure it, you can add up all the leftovers across everything two parties both looked at, and you get a single number that is zero exactly when their accounts are compatible and positive when they are not.

That number is the most important quantity in the book. Physics is what happens when something tries to make it smaller.

4.2 Two detectors

Physics takes this problem seriously enough to spend money on it. The money is the argument.

On 4 July 2012, two collaborations at CERN announced that they had found a new particle. ATLAS gave its mass as 126.0 billion electron-volts, with a statistical uncertainty of 0.4 and a systematic uncertainty of 0.4, at a significance of 5.9 standard deviations. CMS gave 125.3 billion electron-volts, give or take 0.6, at five standard deviations. Five is the bar particle physics sets before anybody uses the word discovery. It means the odds of a fluctuation doing it are about one in three and a half million.

Those are different numbers. 126.0 is not 125.3. A reader who was told that physics had converged on a value might reasonably ask which one it converged on. Neither, is the answer. What was announced was that two intervals overlapped, and overlapping is what agreement between measurements has always meant. A measurement is a claim about a range. Two measurements agree when the ranges have points in common. The sharper the measurements, the more that agreement is worth, because the ranges are smaller and there is less room to overlap by luck.

The reason there were two of them is the whole point. ATLAS and CMS sit at different places on the same ring and were built by different people to different designs. The differences are not cosmetic. ATLAS bends particle tracks with a thin two-tesla solenoid and then bends them again in three enormous air-core toroids. CMS does it with one superconducting solenoid six meters across running at 3.8 tesla, which is a different way to spend the same money. Different magnets, different detector materials, different trigger electronics, different reconstruction software, different graduate students, different arguments in different corridors.

Each collaboration also blinded itself. The region of the data where the signal was expected to sit was masked while the analysis was being built, and unmasked only after the collaboration had committed to its procedures, so that nobody could adjust the method until the answer came out pleasing. Neither experiment tuned its analysis to the other's number.

Building the second detector cost roughly what building the first one cost, which is a remarkable sum of money to spend on a second opinion. It bought the one thing a single detector cannot produce at any price. One detector reporting 126 produces a number. Two detectors that share no components, no code and no staff, reporting 126.0 and 125.3 on the same afternoon, produce something that a number by itself is not: a fact about the world rather than a fact about an apparatus.

That is the street corner again, with a budget. Alice and Bob compared accounts on the overlap and found them compatible. The overlap of two collaborations is the particle, and the compatibility is the discovery.

4.3 A clock face

It is quarter past three in the afternoon. The clock on the wall says three. A railway timetable says 15:15. Nobody is confused, and nobody thinks the clock and the timetable disagree about what time it is, because everybody knows the rule: fifteen and three are the same thing on a clock face.

Look at what that sentence does. Fifteen and three are different numbers. They stay different numbers. The clock declares that for its purposes, two numbers count as one thing when they differ by twelve, and having made that declaration it stops carrying the difference. A clock is a machine for forgetting the difference between three and fifteen, and forgetting it is the entire service the clock provides.

Do that systematically and you get the tool. Start with a collection of things. Choose a rule that says which of them count as the same. The rule has to be sane, in a way that takes ten seconds to check: everything counts as the same as itself; if this counts as that, then that counts as this; and if the first counts as the second and the second as the third, then the first counts as the third. Sweep up each family of things that count as one another into a single bundle. Each bundle is called an equivalence class, the collection of bundles is called a **quotient**, and the operation of passing from the original things to the bundles is the single most useful operation in the book.

Twelve bundles come out of the clock. One of them contains 3 and 15 and 27 and 39. That bundle is what "three o'clock" names. There is no such thing as the true underlying number that three o'clock secretly is. There is a bundle, and 3 and 15 are two ways of writing it down.

Which brings the two friends back. Alice's record and Bob's record differ. Where they differ by something a permitted relabeling accounts for, they are two ways of writing down one thing, in the same sense that 3 and 15 are. Where they differ by more than that, the leftover is real and somebody's records have to change.

Everything hangs on the word permitted. The clock's rule is not a free choice made for convenience. If you declared that any two numbers count as the same you would get one bundle, which forgets everything, and a clock that says the same thing at every moment. The permitted relabelings are exactly the ones that no measurement made inside the system can detect, and no others. Working out which those are is a large part of the job of physics.

4.4 The rate has to be computable in the room

One clause on all of this carries the rest of the book. It is easiest to see with money.

Two firms keep books in different currencies and share an account. To settle it they need an exchange rate. In ordinary commercial life they look one up, which works because there is a market outside both firms that publishes rates all day.

Chapter three closed off that move. There is no outside. There is no venue where the correct description is kept, no referee holding both sets of books, and any procedure that requires one has smuggled in exactly the stage that chapter said cannot be assumed. The exchange rate has to be computable from inside the room, out of things both parties hold.

It can be. The two firms have both recorded the same transaction, one of them in each currency. The ratio of those two entries is the rate. It came out of the overlap, which is the one place both parties have data. Neither firm had to be told it by anybody.

Two properties come free with a rate obtained that way. It can be undone, because if the rate one way is known then the rate the other way is its reciprocal. And two of them can be chained, because a rate from the first currency to the second, followed by a rate from the second to the third, is a rate from the first to the third.

Chaining is where it stops being bookkeeping. Take three firms, each pair of which shares an account, so there are three rates. Convert from the first currency to the second, from the second to the third, and from the third back to the first. You have gone in a circle and arrived back where you started. The amount you are holding had better be the amount you set out with.

If it is not, somebody can go round the loop repeatedly and end up with more money than they started with, having bought and sold nothing. Currency traders call this triangular arbitrage, they have machines watching for it, and when it opens it closes in milliseconds. The condition that closes it is the requirement that going around a loop of translations brings you back to yourself.

That condition is the whole content of what physicists call gauge structure. Accountants make it obvious and field theory makes it look hard. It is doing the work behind electromagnetism twenty chapters from here. Descriptions related by a permitted relabeling say the same thing. The relabelings can be undone and chained. And every closed loop of them has to come back to the identity, or there is something to be extracted for free. The world does not offer that.

4.5 Ninety meters

In June 1792 two French astronomers left Paris in opposite directions to measure the world.

France wanted a unit of length that no king had defined. The Academy of Sciences had settled on one ten-millionth of the distance from the north pole to the equator. To get it, somebody had to survey the meridian arc from Dunkirk to Barcelona by triangulation and scale up. Jean-Baptiste Delambre took the northern leg, Dunkirk down to Rodez. Pierre Méchain took the south, Rodez to Barcelona. The plan allowed a year. The survey took seven, most of them conducted through a revolution and a war, with both men repeatedly arrested by people who found a stranger on a hilltop with brass instruments and a telescope to be exactly what a spy would look like.

Méchain finished his southern latitude work at the fortress on Montjuïc, the hill above Barcelona harbour, and had earlier measured the latitude at his lodgings in the city. Both were careful measurements of the same quantity by the same excellent astronomer with the same instrument. In March 1794 he compared them and they disagreed by three seconds of arc.

Three seconds of arc is about a six-hundredth of the width of the full moon. On the ground, it is a little over ninety meters. It is a very small number, and a fatal one, because the two figures were both his and there was no third figure to adjudicate. War with Spain had closed the border behind him and he could not get back to the hill to remeasure.

He concealed it. When the international commission met in Paris in 1799 to fix the meter, the numbers Méchain supplied had been adjusted to remove the discrepancy. He spent the rest of his life trying to make the disagreement go away, went back into Spain in 1803 to redo the work, contracted yellow fever, and died at Castellón de la Plana on 20 September 1804 at the age of sixty. Delambre, preparing the official record of the survey afterwards, worked through the notebooks and found the alterations.

Set aside the concealment, which belongs to Méchain. The structure underneath it is the one the food truck had. Two records of the same quantity failed to agree on their overlap, the leftover would not go to zero, and the one move that could have driven it to zero, going back to the hill with the instrument, had been made unavailable. He was in the position of Alice and Bob with the van permanently in the way. A mismatch that cannot be repaired does not resolve itself and does not fade. It sits there.

4.6 What the meter is

The meter they produced is wrong.

The quadrant they were measuring, from pole to equator, is 10,002,290 meters. They were trying to make it exactly ten million. Their meter is therefore short of its own definition by about a fifth of a millimeter, roughly the thickness of two sheets of paper. The error has nothing to do with Méchain's three seconds of arc. It comes from the flattening of the Earth, which they had to estimate and which was not then known well enough.

Nothing whatever has gone wrong.

Every meter stick in the world was made to match the standard, and every meter stick in the world therefore matches every other one. The definition was replaced in 1889 with a platinum-iridium bar, replaced in 1960 with a count of 1,650,763.73 wavelengths of the orange-red light of krypton-86, and replaced again in 1983 with the distance light travels in one 299,792,458th of a second. Three replacements, and the meter has never changed length, which is a peculiar thing to be able to say about a length.

It has never changed length because the meter was never the bar in the vault. The meter is the bundle. It is the equivalence class of every stick, wavelength and timing that counts as agreeing with every other one. A definition is a way of pointing at the bundle by holding up one member of it. Point at it with a bar, point at it with krypton, point at it with the speed of light and a clock, and you have pointed at the same bundle three times. A better representative buys precision. It has no power to change what is being represented, because it was never carrying the identity in the first place.

This is why the survey's error is not embarrassing and why French commerce did not collapse in 1799. The unit's job is to let two parties translate their measurements into each other's terms. It does that job through what everybody adopts, which fixes the translation. The pole is not consulted.

4.7 One world

The naive picture of agreement wanted one street with a definite content and two matching descriptions of it. A delivery van was enough to take it apart. There is no master copy that anybody holds a copy of. There are records held by parties who cannot see each other's, and there are overlaps, and on the overlaps there are translations, and the translations are constrained: they must be computable from inside, they must be undoable, they must chain, and every closed loop of them must come back to where it started.

What makes it one world is that the leftovers on the overlaps go to zero. That is the only sense in which Alice and Bob live in the same city, the only sense in which ATLAS and CMS found the same particle, and the sense that failed for Pierre Méchain on a hill above Barcelona.

Which raises the question the next chapter exists to answer. Alice and Bob compared notes and their accounts were compatible. Compatible accounts are two labelings of one thing, in the way that 3 and 15 are two labelings of one bundle. The labeling is thrown away by the comparison.

So something survived the comparison. It was not either of their descriptions.

They agree. On what?

Chapter 5

Why Do You and I See the Same Table?

Bob's kitchen table is oak, scratched, and a hundred and twelve centimeters along its long edge. At two in the morning he crosses the kitchen in the dark to get a glass of water. He does it by memory and by hand: four steps from the doorway, the cold edge of the worktop under his left palm, then the corner of the table, which he takes wide because he has caught his hip on it before. He never sees it. Nothing reaches his eyes at all.

Alice, coming in behind him, turns on the light and sees the whole thing at once.

Two accounts of the table. The physical signals behind them have nothing in common. Alice receives photons, scattered off the oak in a band of wavelengths a few hundred nanometers wide, focused onto a patch of light-sensitive cells at the back of each eye. Bob receives mechanical deformation of the skin of his palm, transduced by pressure receptors that would not respond to a laser shone directly at them. No signal either of them receives is available to the other. The two channels do not overlap anywhere in the physics.

They agree about the table instantly and without discussion. Neither of them experiences the agreement as an achievement.

Chapter four left them agreeing and did not say what they were agreeing about. The comparison threw away both descriptions. Something came through it anyway.

5.1 What survives

Start with what does not survive.

The oak is brown. That belongs to Alice. Bob has no version of it, because brown is what her visual system calls a certain distribution of wavelengths, and no amount of running your hand along a table delivers it. The oak is slightly waxy, and cooler near the window. That belongs to Bob. Neither of them is wrong, and neither of them can hand their version across.

What both of them have is that the edge runs straight for a hundred and twelve centimeters and then turns through a right angle. That the far corner stands a hundred and twelve by seventy from the near one. That the surface is rigid, so the distance between any two points on it stays what it was while you walk around it. That there is a hard flat obstruction between the doorway and the window, and that walking into the space it occupies will hurt.

Every item on that list is a relation between one part of the table and another. None of them is a property of the table on its own. The rigidity is a statement about pairs of points, the right angle is a statement about two edges, and the hundred and twelve centimeters is a comparison with a metal bar in a vault outside Paris, or with whatever has since taken over that bar's job. Two channels with no physics in common delivered the relations and dropped everything else.

A quantity that comes out the same under a change in how you look is called an **invariant** of that change.

The order matters here. It is the order most treatments get backwards. The invariant comes first. You find out what survives, and only then do you have any business asking what it survived.

5.2 A square and a pencil

Take a sheet of paper and draw a square on it, big enough to handle, five or six centimeters. Mark one corner with a pencil dot, and mark the same corner on the reverse side so you can find it after the square has been face down.

Cut it out. Then find every way to pick it up and put it back down so that the square occupies exactly the same patch of paper it did before. The outline has to land on the outline. Where the pencil dot ends up is your business.

Most people find four and stop. The four are the rotations: leave it alone, turn it a quarter turn, turn it a half turn, turn it three quarters. Then they remember they are allowed to turn it over. There are four more: one flip about the vertical line through the middle, one about the horizontal line, and one about each of the two diagonals.

Eight. There are exactly eight. Three things about those eight can be checked with the paper in your hand.

Do one move, then do another. Whatever you did, the result is one of the eight. A quarter turn counterclockwise followed by a flip about the diagonal that runs up to the right is a flip about the horizontal line. You can watch the dot arrive to confirm it. There was nothing else it could have been, because the square is back on its outline, with only eight ways to be back on your outline. The set is closed. Nothing you can do with these moves takes you outside them.

Every move can be undone by another one on the list. The quarter turn is undone by the three-quarter turn. Each of the four flips is undone by itself. No move in the set strands you.

Leaving it alone is on the list, too. It looks like a formality, but without it, the result of “undo the quarter turn” has no name.

Checking the closure by hand means checking sixty-four pairs of moves. An eight-by-eight grid holds them all at once, the eight moves along the top and the same eight down the left edge. Each cell answers one question: do the move on the left, then the move along the top, and write down which of the eight you have ended up with. Sixty of the sixty-four are filled in below.

Eight short names first, because writing “the flip about the diagonal that runs up to the right” sixty-four times is worse than learning a letter. Each rotation gets the number of quarter turns it is, counted counterclockwise: 0 for leaving the square alone, 1 for the quarter turn, 2 for the half turn, 3 for three quarters. Each flip gets the line it is a flip about: V for the vertical line down the middle, H for the horizontal line across it, S for the diagonal that leans like a slash, running from the bottom left corner up to the top right, and B for the diagonal leaning the other way. The four lines belong to the table rather than to the square, so they stay put while the square moves. Rows are the first move, columns the second.

then	0	1	2	3	V	H	S	B
0	0	1	2	3	V	H	S	B
1	1	2	3	0	S		H	V
2	2	3	0	1	H	V	B	S
3	3	0	1		B	S	V	H
V	V	B	H	S	0		3	1
H	H		V	B	2	0	1	3
S	S	V	B	H	1	3	0	2
B	B	H	S	V	3	1	2	0

Four cells are empty. Pick the square up and fill them in.

Two of the four hold the same pair of moves in opposite orders. Row 1 under column H is a quarter turn and then a flip about the horizontal line. Row H under column 1 is the flip first and the quarter turn after. Both answers are diagonal flips. They are flips about different diagonals, and the pencil dot settles which is which: one order leaves the dot in the corner it started from, and the other sends it to the corner diagonally opposite.

The other two blanks are quieter. Row 3 under column 3 is a three-quarter turn done twice: six quarter turns, which is a half turn. Row V under column H is a flip about the vertical line followed by a flip about the horizontal. The square, turned over twice and lying face up again, comes back rotated through a half turn. Two mirrors make a rotation.

Count along any row of the finished grid. Each of the eight moves appears exactly once, and the same holds down every column. The row for the quarter turn shows why. Two different second moves cannot land the square in the same place there, because a three-quarter turn stuck in front of both sequences cancels the quarter turn and leaves the second move standing on its own, so those two second moves would have to have been one move. Eight second moves, eight different answers, and only eight moves available for the answers to be. All eight are in the row, once each, and nothing anywhere in the grid produces a ninth thing.

Closed, undoable, and containing the do-nothing move. A collection of changes with those properties is called a **group**. Mathematicians have been calling it that since the 1830s. The word arrives last on purpose. The object was in your hands before it had a name. It would have been there if nobody had ever named it.

The three properties fit on one line of symbols. Write G for the collection of the eight moves, g and h for any two moves in it, and a dot between them for doing g and then doing h. Write e for leaving the square alone, and g with a raised minus one for the move that undoes g. The rounded E is read as "is one of". The conditions come in the order closure, do-nothing move, undo, because the undo is defined by what it gets you back to.

$$g \cdot h \in G \qquad e \cdot g = g \cdot e = g \qquad g \cdot g^{-1} = g^{-1} \cdot g = e$$

The first says the answer to any pair of moves is on the list. The second says that e changes nothing whichever side of another move it sits on, and the third that every move has an undo which works from either side. The do-nothing move is called the **identity**, the undo is the **inverse**, and one further condition is usually listed alongside them and costs nothing here: with three moves in a row, grouping the first two or the last two makes no difference, because the square gets picked up three times in the same order either way.

A group is a set with one operation obeying those conditions. Nothing in them mentions paper. The eight moves of the square are the **dihedral group of order eight**, order eight for the number of moves in it and dihedral from the Greek for two-faced, which here is a square with a front and a back. The rotations of an ammonia molecule satisfy the same three conditions, and so does every way of reordering a deck of cards, so anything proved from those conditions alone holds for the molecule, the deck and the paper square without being proved three times.

One more thing to see with the paper. The dot moves. Under the eight moves it visits all four corners, every corner reachable from every other. The set of places a thing can be sent by the moves in a group is its **orbit**, so the orbit of a corner is the four corners. The orbit of the midpoint of an edge is the four edge midpoints, a different set of four. No move carries a corner to an edge midpoint.

The center of the square has an orbit of one. It sits where it sits under every one of the eight. Nothing you can do to the square touches it.

That is an invariant, in your hand, on a scrap of paper. Something the moves leave alone.

5.3 Chapter four, from the other side

Look again at the three conditions.

Chapter four asked what has to hold between two record-keepers before they can be said to share a world. It came back with conditions on the translations between their records. A translation had to be computable from inside, using only what the observer holds. It had to be undoable, so that going from Alice's account to Bob's and back returns Alice's account. Translations had to chain, so that a route through a third party gives the same answer as the direct one. And every closed loop of them had to come back where it started.

Those are the conditions on the paper square. Undoable is the inverse. Chaining is closure. Coming back where you started is the do-nothing move sitting at the end of every loop.

The overlap rule of chapter four and the symmetry of a square are one thing seen from two directions. The translations between observers form a group, in the way the moves of a square do. The question "what do Alice and Bob agree about" is the question "what does that group leave alone".

Which gives a sentence that is absolutely essential to reality and, by extension, this book.

A symmetry is a change that leaves a named invariant alone, so a symmetry is the claim that part of your description was never physical.

Brown is not physical in this exact sense: it is what changes when you switch from Alice's channel to Bob's. The world does not change when you make that switch. The distance from corner to corner does not change. That one is the table.

5.4 Nought degrees

The clearest case of a quantity everybody treated as a fact about the world, until it was subtracted, is the one printed on every map.

Latitude has an origin the planet supplies. The equator is where the spin axis says it is. Any surveyor anywhere can find it without asking permission. Longitude has no such thing. The Earth is very close to symmetric about its spin axis, which means that turning the whole planet through some angle about that axis leaves every measurable relation between places exactly as it was. Rotating everything is a symmetry, so the number you attach to a meridian is a labeling. No experiment can read it off the ground.

For three centuries this was treated as an unsettled question rather than an empty one. The French measured from Paris, the Spanish from Cádiz or Tenerife, the Americans from Washington, and captains carried tables for converting between them. In October 1884 forty-one delegates from twenty-six nations sat down in Washington to fix it, and on the thirteenth they voted: twenty-two in favour of Greenwich, one against, which was San Domingo, and two abstentions, which were France and Brazil. France went on printing charts from Paris for another quarter of a century.

What the conference settled was a convention, and everybody in the room knew it. The tell is in the mechanism, because there is no other way to settle it. You cannot measure which meridian is the real zero, and you cannot be wrong about it either. Thanks to the vote, two charts could now be compared without arithmetic, which is worth having, and which is a different kind of thing from a fact about the planet.

Distances between places survived the vote unchanged. Distance is the invariant, the labeling was never physical, and the world's response to twenty-two votes was to carry on exactly as before.

Every symmetry anybody has found in physics is that same operation run on a bigger description. Here is a change you can make to how the world is written down. Here is what survives it. Whatever did not survive was never a fact about the world. That people wrote it down for two hundred years is a fact about people.

5.5 Vienna, orbit, and the Cretaceous

There is a symmetry so large that almost nobody notices they are assuming it.

A physicist in Vienna measures how fast a ball falls. A physicist on the International Space Station, four hundred kilometers up and moving at seven and a half kilometers a second, does the same experiment in a way that accounts for the local conditions. They get different readings, because the conditions differ, and the same law relating the readings to the conditions.

Do it again a hundred and fifty kilometers to the west. Same law. Do it in the Cretaceous, if you can arrange the transport: same law, and a version of this can actually be checked, because light arriving from a quasar eleven billion years old carries the spectral fingerprint of hydrogen behaving exactly as hydrogen behaves in a lamp in a basement in Vienna.

The laws do not depend on where you are. They do not depend on when you are. Those are two symmetries. Shifting everything sideways in space leaves the laws alone, and shifting everything along in time leaves the laws alone. Galileo's cabin is a third change of the same kind: put the whole laboratory into smooth motion and the laws do not notice that either, which is why the flies stayed where they were.

Strip them out and watch what happens to the idea of an experiment. If the law of falling depended on where you stood, a result obtained in Vienna would say nothing about a ball in Graz, and the Graz result would say nothing about the next street. If it depended on when, a measurement made on Tuesday would say nothing about Wednesday, which is to say it would say nothing at all, because a measurement is finished before anybody can use it. Repeating an experiment would be pointless. Publishing one would be a form of gossip. The word “law” would have no work to do.

Every experimental result anybody has ever quoted is leaning on those two symmetries. They are the reason a number obtained once is worth writing down.

And there’s more to it. Emmy Noether proved in 1918 that a continuous symmetry of the laws always comes with a conserved quantity, exactly one, determined by the symmetry. Shifting in space gives you a quantity that never changes: momentum. Shifting in time gives you another: energy.

Momentum is conserved because the laws do not care where you are. Energy is conserved because they do not care when. Those are two statements of one fact, about two directions. The procedure that turns a symmetry into its conserved quantity is exact and mechanical. Running it takes more mathematics than a paper square supplies.

5.6 The letter to Chevalier

The paper square is the whole of what Évariste Galois was working on, run on something harder.

He was asking which polynomial equations can be solved by a formula built from the coefficients using addition, multiplication and the extraction of roots. The quadratic has such a formula, which every schoolchild is made to memorize. The cubic and the quartic have them, found in Bologna in the 1500s, ugly and real. The fifth degree had resisted for two hundred and fifty years.

Galois stopped looking at the equation and looked instead at the symmetries of its solutions: the ways you can permute the roots among themselves without disturbing any relation between them that the coefficients can see. Those permutations form a group, in exactly the sense the square’s eight moves do. He then showed that whether a formula exists is a property of that group, and that from the fifth degree onward the group is the wrong shape.

The submissions went badly. Cauchy took the first memoir in 1829 and it never appeared. Fourier took the second in 1830, for the Academy’s Grand Prix, and died in May of that year with the manuscript among his papers, where it was not found. Poisson took the third in 1831 and reported in July that the reasoning was neither sufficiently clear nor sufficiently developed for the Academy to judge it.

On the night of 29 May 1832 Galois wrote to his friend Auguste Chevalier, setting out the results and asking him to make a public request: that Jacobi or Gauss be asked to give an opinion, not on whether the theorems were true, but on whether they were important. In the margin of one page, beside an argument he had not finished, he wrote that there was something to complete in the demonstration and that he did not have the time.

He was shot in a duel the following morning and died on 31 May, aged twenty. The letter was published in 1846. What was in it is taught to undergraduates in their first year, which is the ordinary fate of an idea that turns out to be the right one. It is why the eight moves of a piece of paper have a name.

5.7 The size of an atom

A notion that applies to everything explains nothing. Symmetry has a boundary, findable from where you are sitting.

Shift everything in space: symmetry. Shift everything in time: symmetry. Rotate everything: symmetry. Change the size of everything, so that every length in the universe is doubled and nothing else about the arrangement is altered: the world notices.

The evidence is in the room. Atoms have a definite size. A hydrogen atom is about one ångström across, a ten-billionth of a meter. That number is the same in Vienna, in orbit and in the Cretaceous. It is a length built into the physics. A world without scale symmetry is exactly a world in which such a length can exist. If doubling every length left the laws alone, no size could be preferred, and nothing would distinguish the atom you have from one twice as big. Atoms would come in every size, and chemistry, which depends on atoms of a given element being interchangeable, would not be available.

The consequences are structural rather than decorative. Galileo worked one of them out in 1638, in the *Two New Sciences*, and illustrated it with a drawing of a bone. Scale an animal up by a factor of ten in every direction and its weight goes up by a factor of a thousand, while the cross-section of the bones that carry the weight goes up by a hundred. The scaled animal is ten times worse off than the original, and its legs snap. Which is why you cannot design a bridge by building a small bridge and then being more ambitious. Every engineer who has tested a model in a wind tunnel has had to do arithmetic to correct for the fact that the model is not a small copy of the world. That arithmetic is a bill for a symmetry the world does not have.

So symmetry is a claim with content. The way you can tell is that it is false about scale.

5.8 What the table was

Alice and Bob agree about the table because there is a change, from her channel to his, that leaves something alone, and what it leaves alone is the geometry: the edges, the right angle, the hundred and twelve centimeters, the rigidity. Everything else in either account belongs to the observer that produced it.

That is the same operation as the paper square, where the eight moves leave the center alone. It is the same operation as chapter four's overlaps, where the translations between records leave the shared content alone. It is the operation physics has been running since the 1830s, and each time it has been run, something everybody had taken for a property of the world has turned out to be a property of the description, and has been subtracted.

There is a question left over. Alice and Bob cannot answer it between them, because two people who agree have very little to disagree about.

Put a third person in the room. Alice agrees with Bob. Bob agrees with Charlie. Charlie agrees with Alice. Every pair compares records, finds a translation, and comes away satisfied. Each pair shares one invariant, and each pair is content.

That has the sound of a settled matter. Three coins in three separate rooms are enough to take it apart.

Chapter 6

Why Isn't Agreeing With Your Neighbors Enough?

In 1785 the Marquis de Condorcet published an essay on applying analysis to decisions reached by majority vote, and buried in it is an observation that has been ruining committees ever since.

Take three people and three options. The first person ranks them A, B, C. The second ranks them B, C, A. The third ranks them C, A, B. Ask the committee to choose between A and B, and A wins by two votes to one. Ask it to choose between B and C, and B wins two to one. Ask it to choose between C and A, expecting a formality, and C wins two to one.

Every one of those votes is a real majority. Nobody was manipulated, nobody abstained, and each vote can be repeated as often as you like with the same result. There is no winner, and there is no winner in a way that no amount of further voting will fix, because the information that the three results are jointly impossible is not present in any of them.

Condorcet's committee is the first warning that pairwise agreement is a weaker thing than it sounds. Local data can be flawless, checkable and reproducible, and refuse to be assembled into anything. The voting version has no arithmetic attached to it that helps much. The version that does is simpler, checkable on a table.

Give Alice, Bob and Charlie a coin each and put them in three separate rooms. Each of them looks at their own coin and sees heads or tails. Nobody can see anybody else's.

They are allowed to meet in the corridor two at a time. When a pair meets they compare coins and write down one word: **same**, if the two coins match, or **different**, if they do not. The only restriction is that the pair must not write down which way either coin is facing. What gets recorded is the relation between two coins, which is the only thing either of them could check together.

Three meetings happen. Alice and Bob meet, and record different. Bob and Charlie meet, and record different. Charlie and Alice meet, and record different.

Set the coins.

Suppose Alice's is heads. Bob's has to be tails, because they recorded different. Charlie's has to be heads, because Bob and Charlie recorded different. And Charlie and Alice recorded different, so Charlie's has to be tails. It was heads two sentences ago.

Try the other branch. Suppose Alice's is tails. Bob's is heads, Charlie's is tails, and Charlie and Alice are both tails, having recorded that they differ.

That is all the branches there are. There are eight ways to set three coins. Every one of them contradicts at least one of the three records.

The obvious response is that somebody is wrong. Three coins have three faces. Records that no arrangement of faces can satisfy are records with a mistake in them, so the thing to do is find out whose. Look again at what the eight cases ruled out. Not the three records by themselves: the three records together with the assumption that there were three settled faces for them to be records of. The check kills the pair. It does not say which half to drop.

Nobody lied. No coin was misread. Each of the three meetings produced a record that is perfectly satisfiable on its own. Any two of the three can be satisfied together without the slightest difficulty. It is the third one that has nowhere to go. Which of the three you call the third one is your choice, because they are symmetric and none of them is the culprit.

Notice what the three of them can do about it, which is nothing. They can meet in pairs for the rest of their lives. Every meeting will confirm what the previous meeting found. No pairwise comparison anywhere in the arrangement contains the information that the three records are jointly impossible, because that information is not held by any pair. The three of them can go on being individually correct indefinitely.

Chapter five put a third person in the room and left the three of them agreeing. This is the bill.

6.1 "Same same but different" as arithmetic

The argument above is short enough to check by hand, but doing things by hand doesn't scale very well here. Twelve people generate sixty-six possible pairings. What is needed is a way of computing the answer.

Write **0** for same and **1** for different.

Then chain two records together. If Alice and Bob are the same, and Bob and Charlie are the same, Alice and Charlie are the same: 0 and 0 give 0. If Alice and Bob are the same and Bob and Charlie differ, then Alice and Charlie differ: 0 and 1 give 1. And if Alice and Bob differ and Bob and Charlie differ, then Alice and Charlie are back to being the same, because two flips return you to where you started: 1 and 1 give 0.

Ordinary addition says one plus one is two. Here there is no two, because there are only two states in the world being described. The count that matters is whether the number of flips is even or odd. So one plus one is zero.

This is called **arithmetic modulo 2**. It is the arithmetic of light switches: flick the switch twice and the room is as you found it. It is the arithmetic of turning a sock inside out. Every fact in it fits on two lines, the whole of what the coins require.

Take the three records again. Different, different, different: 1, 1, 1. Add them. One and one is zero, and zero and one is one.

The three records around the loop sum to 1.

For the coins to be settable at all, that sum has to be 0, because going all the way around the loop returns you to Alice's own coin, and Alice's coin does not differ from itself. An odd number of flips around a closed path is a demand that something be different from itself. The coins cannot arrange that.

Eight cases checked by hand, replaced by adding three ones.

6.2 Loops

The picture generalizes as soon as it is drawn.

Put a dot for each observer and a line between two dots whenever those two have compared records. A collection of dots and lines like that is a **graph**, one of the few pieces of mathematics that looks exactly like the thing it is about. The dots are called vertices, the lines edges, and two dots joined by a line are neighbors.

Walk along the edges from dot to dot and you have a **path**. Come back to the dot you started from without retracing your steps and you have a **cycle**. The triangle of Alice, Bob and Charlie is the smallest cycle there is.

Every edge carries a record, 0 or 1. Every cycle therefore carries a total, which is the sum of the records around it, computed in the arithmetic that has no two in it. The quantity you pick up by going all the way around a loop and arriving back where you started is called the **holonomy** of that loop.

The condition on the whole arrangement fits in one sentence.

A consistent global assignment exists exactly when every cycle has holonomy zero.

One direction of that is the coin argument again: if the coins can be set, then walking around any loop and coming home must find the coin you left, so the flips have to cancel.

The other direction is constructive: Suppose every loop sums to zero. Pick any dot and set its coin however you like. Walk outward. Every time you cross an edge you know whether to keep the value or flip it, because the edge says same or different, so every dot you can reach gets a value. The only way this can go wrong is if two different routes to the same dot disagree. Two routes to the same dot form a loop. Every loop sums to zero.

So they cannot disagree. The assignment is forced, everywhere, from one free choice at the start.

A route through the dots that reaches all of them without ever closing a loop is called a **spanning tree**, the skeleton of that construction. For any connected graph, a spanning tree has exactly one edge fewer than it has dots. Every remaining edge, the ones the tree did not use, closes exactly one independent loop when you put it back.

Which gives you the count of conditions, from nothing but arithmetic. Take the number of edges, subtract the number of dots, add one. That is how many independent loops the graph has, and therefore how many separate sums have to come out zero. The triangle has three edges and three dots: three minus three plus one is one. One loop, one condition, and it failed.

6.3 Ascending and descending

Roger Penrose went to the International Congress of Mathematicians in Amsterdam in 1954, where there was an exhibition of prints by M. C. Escher, and came away preoccupied. He and his father Lionel, a psychiatrist, spent some time afterward constructing figures that behave locally like ordinary drawings and cannot be assembled into an object, and published them in the *British Journal of Psychology* in 1958. One of them was a staircase.

Every flight goes up. Take any corner of the drawing, cover the rest, and there is nothing to object to: a step, then another step, each one higher than the last. Follow all four flights around and you arrive back at the landing you started from, having climbed the whole way.

The Penroses sent a copy to Escher, who put the staircase into a lithograph in March 1960 and populated it with a file of monks trudging around the roof of a monastery forever.

The staircase is the coin problem drawn. Each edge of the loop carries a record, in this case "up", every local record is fine, and the sum around the loop is not zero. What is impossible is not any of the steps. It is the closing.

6.4 A magnet that cannot decide

Physics met this in the 1950s and gave it a name.

A magnet is many small magnetic moments, each able to point one of two ways, each pushing its neighbors around. In an antiferromagnet, every pair wants to be opposite: if this one points up, the one beside it should point down. On a square grid that is easy to arrange: the result is the tidy alternating pattern of a chessboard.

Put the same rule on a triangular grid and it comes apart at the first triangle. Set the first moment up and the second down, which satisfies that pair. The third has to be opposite to both. There is no third direction. Whatever it does, one of its two bonds is unhappy, with no principle available to say which one to disappoint.

Gregory Wannier worked out the consequences for a whole triangular lattice in 1950 and found something that had no business being there: the system keeps a finite amount of disorder as the temperature goes to absolute zero. Ordinarily, cooling something to zero freezes it into its one best arrangement. Here there is no one best arrangement, only an enormous number of equally disappointing ones. The system goes on having a choice after the energy has been taken away.

Gerard Toulouse gave the phenomenon its name in 1977, while working on spin glasses, and gave it the same test the coins gave it. Go around a closed loop of bonds and multiply out what each one demands. If the loop comes back consistent, the region can settle. If it does not, the loop is **frustrated**. No arrangement of the moments inside it satisfies every bond, however long you wait.

Frustration is the coin problem in a material, and a measurable one. It changes the heat capacity, it shows up in neutron scattering, and it is the reason a class of magnets never orders properly no matter how carefully they are cooled. The obstruction lives in the loops, exactly where the arithmetic said it would.

6.5 Twelve patches

The triangle is small enough to be dismissed as a puzzle. So take the smallest thing that deserves to be called a universe and check every state of it.

Twelve observers. Wire them as the corners of an icosahedron, the twenty-sided solid, so that each observer has exactly five neighbors and there are thirty edges in all. Each edge is an overlap in chapter four's sense: two observers with something they can both check, and no way to reach each other except by checking it. Alice and Bob's overlap was the stretch of Barcelona street they could both see. Here it is one comparison, which is as small as an overlap can get and still be one. Twelve corners, thirty edges. The count above says the graph has thirty minus twelve plus one, which is nineteen independent loops, so nineteen conditions have to hold at once for the thing to be settable.

Each observer holds one bit. That is four thousand and ninety-six arrangements. Flipping all twelve at once changes no comparison anywhere, so there are two thousand and forty-eight distinct states.

Each of the thirty overlaps holds a record of the kind the corridor produced: same or different, the relation between two bits and nothing about either one. The thirty used here are a set whose nineteen loop sums all come out zero, so by the argument above the bits can satisfy them.

Two thousand and forty-eight is a number you can exhaust. So exhaust it.

First the machine needs a move. The one used here is the spanning tree of two sections ago, run as a procedure: take eleven of the thirty edges, enough to reach all twelve observers without closing a loop, and root them at one observer. An observer repairs by copying its parent's bit through the record on the edge between them. That is the whole move. The remaining nineteen edges are never repaired against. They are the conditions being checked.

That move is a demonstration engine rather than a law. Nothing so far says repair has to work this way, and chapter nine is where the move an observer actually makes gets derived from what a repair is allowed to cost. What this one buys is a machine small enough to enumerate.

It also stops. Repairs run outward from the root. An observer that has just been fixed can be unsettled again if its parent changes later, but its parent sits nearer the root, and nothing unsettles the root at all, so agreement sets from the root outward and every run runs out of work. A run can only stop when every observer agrees with its parent, and with the root pinned that leaves exactly one assignment.

Start in every one of the two thousand and forty-eight states and repair until nothing is left to do. That is one **run**: one starting state, one order of working through the observers, carried to a stop. Do it in every order. Sixteen schedules apiece, which for observers with no clock, no leader and no queue are sixteen different worlds as far as anyone inside can tell. Counted over every state, the machine has eleven thousand two hundred and sixty-four repairs available to it.

Two thousand and forty-eight states, sixteen schedules apiece, and eleven thousand two hundred and sixty-four repairs is not a sample. It is the whole space. Nothing here rests on a well-chosen starting configuration or a lucky order, because there was no choosing and no luck: every configuration this universe admits was started, and every order these observers could repair in was run.

Two results come out. The first is the boring one that has to be true before the second means anything.

Exactly one of the two thousand and forty-eight states is globally consistent. That one is arithmetic. The spanning tree argument already settled it: the nineteen loop sums are zero, so fix one observer's bit and every other bit is forced, which leaves one free choice, two assignments, and one distinct state once the all-over flip is divided out.

The second result is the machine's. Every single run lands on that state. Every starting point, every schedule, every order of repair. The answer does not depend on who went first. This machine's indifference is bought by the score, which falls whichever order you work in. Whether a repair rule has to behave that way in general is chapter ten's subject. The leftover at the end is zero, which is to say the observers end up in complete agreement and the agreement was never a matter of who spoke loudest.

6.6 One edge

Flip a single edge, one of the nineteen the tree did not use. Change one record, out of thirty, from same to different. Change nothing else. Every observer has five neighbors, every overlap works, and the machine is untouched.

Some loop now sums to 1. How many of the nineteen do depends on which nineteen you drew, since there is more than one way to pick them. What the choice does not affect is that they no longer all close.

Of the two thousand and forty-eight states, the number that are globally consistent is zero. There is no way to set twelve bits that satisfies all thirty records. No amount of repair produces one, because the obstruction is a property of the loops rather than of the bits.

What happens instead is the interesting part. The machine does not thrash, fail to halt, or give different answers on different days. It converges, exactly as before, from every one of the two thousand and forty-eight starting states and under every one of the sixteen schedules, onto one terminal arrangement.

That arrangement has a leftover of exactly one.

Every repair the observers can perform has been performed. Every disagreement that could be pushed somewhere else has been pushed somewhere else. And when the pushing stops, one unit of disagreement sits in the system. It cannot be removed by working harder, it cannot be removed by choosing a better order, and it cannot be pushed off the edge of the world, because a system of twelve observers wired into a closed surface has no edge to push it off.

You can move it. Repair in a different order and it sits somewhere else. What you cannot do is get rid of it. The total is one wherever it happens to be sitting.

6.7 The leftover

Look at what that leftover does.

It has a definite quantity, exactly one, not approximately one and not one on average. It has a location, in the sense that the loops carrying holonomy tell you where it is. It survives every process the world can apply to it. It is not a property of any single observer, because every observer is behaving correctly and repairing whatever it finds. And it is not a property of any pair, because every pair is satisfiable and always was.

It is a property of the way the whole thing is wired. It is as real as anything else in the arrangement, in the only sense of real that has been available since chapter four: it is what survives comparison between every point of view.

A conserved quantity, indivisible, located, indifferent to how you approach it, that no amount of local work will destroy.

The word for that costs twenty more chapters to earn, so it stays unsaid.

6.8 The order of the meetings

Alice, Bob and Charlie met in a corridor, two at a time, and the account above knows which meetings happened before which. The twelve observers ran sixteen schedules. Calling them schedules is a claim that there is an order to run things in. Every one of those sentences assumed that comparisons are events, that events happen one after another, and that there is a fact about which.

Take that away.

Three observers with no clock, no shared present, and no way of telling a neighbor who has gone quiet from one that has stopped existing.

They have to agree anyway.

Chapter 7

Why Is There No Such Thing As Now?

On 4 October 1971 two men boarded an eastbound commercial flight out of Washington with tickets made out for a party of four. Two of the four were listed as Mr. Clock.

Mr. Clock was four cesium-beam atomic clocks, Hewlett-Packard model 5061A, each about the size of a large suitcase and too delicate for the hold. Joseph Hafele was a physicist at Washington University in St. Louis who had worked out, while preparing a lecture in 1969, that an ordinary airliner and an ordinary atomic clock were enough between them to measure something people had been arguing about since 1905. He spent the following year failing to find anybody who would pay for it. After a talk he gave in 1970, Richard Keating, an astronomer at the United States Naval Observatory, came up and mentioned that he worked with atomic clocks for a living and that his employer owned several. The whole thing cost eight thousand dollars, of which seven thousand six hundred went on economy fares.

They flew east around the world, came back, and flew west from 13 to 17 October. Then they carried the clocks back into the Naval Observatory and compared them against the clocks that had never left the bench.

Every clock in the building had been running the entire time. None of them agreed.

Going east, the flying clocks had lost 59 nanoseconds against the ones on the ground, give or take 10, where the prediction had been a loss of 40 give or take 23. Going west they had gained 273 nanoseconds, give or take 7, against a prediction of 275 give or take 21. Both papers went into *Science* in July 1972, one titled “Around-the-World Atomic Clocks: Predicted Relativistic Time Gains” and the other, published immediately after it by the same two authors, “Around-the-World Atomic Clocks: Observed Relativistic Time Gains”.

Look at what those numbers are attached to. Each came out of subtracting one reading from another in one room, at the end, with the clocks side by side.

That is not the only comparison available, and why the others do not help is the whole of the point. Nothing stops you comparing clocks at a distance. Radio does it. A navigation satellite does nothing else all day:

the plane could have sent its reading down and the observatory could have written down what arrived.

The trouble is that the message takes time. The reading arrives stale, and correcting for that means knowing how long the trip took. You can time a round trip without difficulty: send a pulse up, wait for it to come back, read your own clock twice. Splitting that total into the trip out and the trip back is where it stops. Calling the two halves equal is the usual move and a perfectly good working rule, but nothing measures it, and checking it would mean comparing two clocks that are far apart, which is the thing you were trying to do.

What did the flying clock read at the instant the bench clock read noon on the second day out? Every route to an answer goes through an assumption somebody made rather than a measurement somebody took.

Everyone carries a picture around without inspecting it. There is a single present moment, everything in the universe is doing something in it, and clocks are devices for finding out what. On that picture the question about noon on the second day has an answer that Hafele and Keating merely failed to measure. It has none. The single present moment is a part of the description rather than a part of the world. The evidence is that no instrument reads it and no procedure produces one.

Agreement between observers has to be built out of messages: one observer sends, another receives, both write. Do messages need a shared present to happen in?

7.1 The messenger

Take away the clock and look at what is left, which is two observers and a channel.

Two armies sit on hills either side of a valley. The enemy is in the valley. Either army alone loses; both together win, provided they attack at the same time. The only way to communicate is to send a messenger down through the valley, where the messenger may be captured.

The first general sends a rider: attack at dawn. The rider may not arrive, so the first general cannot attack on the strength of having sent him. Suppose the rider does arrive. The second general reads the message and cannot attack either, because he knows that the first general does not know the message got through. A general who suspects he may be attacking alone stays put. So the second general sends the rider back with an acknowledgment. That rider may be captured, so the second general cannot rely on the acknowledgment having landed. The first general receives it, and cannot attack, because he knows the second general does not know the acknowledgment arrived.

Acknowledging the acknowledgment moves the problem one rider further down the road and changes nothing else.

E. A. Akkoyunlu, K. Ekanadham and R. V. Huber published the impossibility proof in 1975, in a paper called “Some Constraints and Tradeoffs in the Design of Network Communications”. The argument is three lines long. Any protocol that solves the problem delivers some finite number of messages. Consider the last one delivered. Its sender cannot know it arrived, so the sender’s decision cannot have depended on it, and the receiver’s decision cannot depend on a message that a working protocol has to be prepared to lose. Delete it. Both generals behave exactly as before. Delete the last message of what is left, and keep going, and you reduce any protocol whatever to one that sends no riders at all. Jim Gray named it the Two Generals Paradox in 1978, in a set of lecture notes on database operating systems. The name stuck to a result that was three years old.

The proof kills certainty, for any number of exchanges, on any channel that can lose anything. Anyone who has operated a network knows what that leaves. A participant that has gone quiet and a participant that has stopped existing send you exactly the same thing, which is nothing, and no amount of listening tells the two cases apart. You can wait longer, as every system built on this planet does, which is what a timeout is: a decision to call a slow neighbor dead at a moment somebody chose in advance and wrote into a configuration file.

The channel does something else to messages besides losing them. Send two messages one after the other across a network with more than one route through it, and the second can land first, because the routes are of different lengths and one of them had a queue on it. The receiver writes them down as they arrive. That is a real order, it belongs to the receiver, and the sending order it fails to match is held by somebody else on another hill. There is a fix, and everyone who builds these systems uses it: put a number in each message before sending it. The number works because both parties agreed on the numbering in advance, which makes it a piece of freight the message is carrying rather than a fact the receiver discovered by looking at the sky.

Every requirement chapter three put on an observer survives that. A bounded view, somewhere to write, and a reading that makes a difference, with the record lasting from the writing of it to the reading of it. Read the fifth one again. It asks that the thing outlive its own operations, which is a statement about order. It never mentions a duration, a rate or a second.

7.2 Water, pasta, cheese

A recipe carries the whole distinction. Most people have run one a thousand times without needing a name for it.

Five steps. Fill the pot with water. Bring it to the boil. Put the pasta in. Drain the pasta. Grate the cheese.

Four of those steps are wired together and one is loose. Boiling has to come after filling, since there is nothing to boil otherwise. The pasta goes in after the water boils, and putting it into cold water is a different act with a different result. Draining comes after the pasta goes in. The cheese can be grated at any point, before the pot comes out or while the pasta cooks or the previous evening. No arrangement of the kitchen makes that choice matter.

In how many different orders can this be cooked? The four wired steps have to happen in their one sequence. Grating can be dropped into any of five positions: before filling, between filling and boiling, between boiling and the pasta, between the pasta and draining, or after draining. Five orders, one dinner, and nothing distinguishes them afterward.

Draw it. Four dots in a row with arrows running along them, filling to boiling to pasta to draining, and a fifth dot sitting off to one side with no arrow touching it at all. There is no left-to-right axis in that picture and no place to put one, because putting one in means deciding where the fifth dot goes. Nothing in the kitchen decides that.

The structure underneath is a relation that says, of some pairs of steps, that one must come after the other. The relation has to behave: no step comes after itself, and if grating comes after boiling and boiling comes after filling then grating comes after filling. Beyond that it is permitted to be silent, and about grating and boiling it says nothing at all. A relation like that is a **partial order**. When it is silent about no pair, when every two items are related one way or the other, it is a **total order**.

A printed recipe is a total order. Step one, step two, step three, down the card, and somewhere between the cooking and the cardboard the one choice that never mattered got made and printed as though it had. A card has to be read in some order, being a card, so an order was invented for it.

Leslie Lamport wrote the distributed-systems version in July 1978, in the *Communications of the ACM*, in a paper called “Time, Clocks, and the Ordering of Events in a Distributed System” that runs from page 558 to page 565. He defined one event as happening before another when the first could have influenced the second, showed that this relation is a partial order and nothing stronger, and then gave an algorithm for extending it to a total order so that a computer system could number things. The algorithm needs an arbitrary rule for breaking ties between events that the partial order leaves unrelated. Any rule will serve. Choose a different one and you get a different total order, equally correct, over the same events.

Authenticated informational precedence is the partial order. A timeline is a total order. You can have the first without the second. The first is what a record system can hand out from certified read-from dependencies. The second is available to anybody willing to break ties, which can be done more than one way.

There is a physics program built around the same shape. Luca Bombelli, Joohan Lee, David Meyer and Rafael Sorkin called a locally finite partial order a **causal set** in 1987. Its order is meant to become the causal order of spacetime, while, after a sprinkling-density or equivalent count calibration, the number of elements in a region supplies a volume estimate. Under suitable conditions the causal order of a continuum fixes its light cones and conformal geometry; that calibrated volume datum supplies the missing scale. Standard causal-set approximations obtain discrete events by random Poisson sprinkling, which introduces no preferred lattice direction. Most partial orders do not resemble manifolds, so the hard part is proving that a given order has a faithful, four-dimensional spacetime approximation.

The observer construction generates an authenticated informational order: a child certifies a read, its parent wrote that register, and both the writer mark and value match. Its **source height** is the number of edges in a longest authenticated-parent chain ending at the event. Such a chain always exists. Height increases on every dependency and comes from provenance rather than execution-list position. It is an ordinal coordinate, not a physical clock. Independently, the response metric completes the rank-three source carrier to continuous three-space. Joining it to a real axis gives candidate event coordinates with one time direction and three spatial directions. Choosing which records populate that space requires an additional rule.

Two finite existence results separate representation from dynamics. Every finite strict partial order can be encoded exactly by authenticated reads, so the grammar does not force chains. Every finite log can also be placed injectively in the candidate space with its dependencies inside the forward cone. That auxiliary placement can add causal comparisons absent from the log. The supplied order and chosen placement do not establish a spacetime limit.

A more restrictive construction does have such a limit. Conservative records generate a growing population of distinct points in the three-dimensional source metric. Each point reads all points within a specified radius on the preceding layer, including itself. The read radius shrinks more slowly than the gaps between points, and the model clock assigns one radius divided by the chosen signal speed to each layer. The resulting order approaches the light-cone order of flat spacetime. Equally weighted event counts approach four-dimensional volume; inside an interior causal diamond, the fraction of comparable pairs approaches one tenth. The population, read rule and clock convention are explicit inputs.

A source-selected spatial readback and one explicit positive global scale τ place an event at $(\tau\, h, x)$ in that ambient target, where h is source height. If spatial changes satisfy the corresponding edge-speed bound, every authenticated dependency enters the future cone. Two concrete checks are enough for the reverse implication: positions separate events within every equal-height layer, and every increasing-height pair unsupported by ancestry is strictly spacelike. Order and cone then agree exactly. No separate no-collision assumption is needed: antisymmetry forces distinct events to occupy distinct points, and every source interval is exactly the corresponding cone interval on the placed event image. Neither τh nor the supplied spatial readback is thereby a physical clock or position, and no continuum limit follows. A manifoldlike refinement must preserve the source order and directions, calibrate event count to volume, recover topology and intrinsic dimension independently, and make the cone and physical signal order agree.

The declared continuum family supplies a retrospective clock from records. Choose fixed timelike intervals whose causal diamonds lie inside the spatial window. Count their events, including the endpoint records. Divide one count by the reference interval's count and take the fourth root. As the population becomes fine enough, the result approaches the ratio of the two proper durations. The event density cancels. An observer needs access to all the interval records, and the reference interval sets the unit. This reconstructs elapsed time after the second meeting; it does not turn the counting procedure into a running wristwatch.

The shape of the cone also has a precise conditional answer. Suppose some nonzero influence is possible and its allowed displacements form a closed convex cone containing no complete line: limits and positive combinations are allowed, but a nonzero displacement and its reverse cannot both occur. If the source's rotations and every boost along one axis preserve these possible influences, the cone must be the Lorentz cone, up to its time orientation. The symmetry must act on actual preparations, signals and readouts. A Lorentz transformation of a coordinate diagram alone does not establish that physical premise.

7.3 Caused, without earlier

Putting the pasta into unboiled water is a different event from putting it into boiling water. The arrow from boiling to pasta records that dependency and nothing else. It says the second act consumes what the first produced. Nobody consulted a clock to establish it. The arrow would be there in a kitchen with no clock in it.

Carry that across to observers. One record depends on another when its accepted commit certifies a read of a version the other commit actually wrote. The writer mark must name that parent, and the value read must equal the value the parent left. Assemble the authenticated dependencies and their chains and you have the informational partial order. Two records with no chain between them acquire no order merely because an executor happened to process one first. A clock, retry counter or queue position cannot supply the missing edge afterwards.

7.4 Eight things called time

Values in a program have types, meaning a declaration of what kind of thing each value is, and compilers convert quietly between types they consider close enough. Write a whole number where a decimal is expected and most languages hand you the decimal without comment. The convenience is real, and so is the class of defect where two quantities that were never the same kind of thing get added together because the machine was being helpful.

Eight separate kinds of thing go by the name of time here, held apart by exactly that mechanism, run in reverse.

There is the closure of the whole arrangement, the state that the repair operation returns unchanged. There is the repair order, a finite list of moves and the positions they occupy in the list. There is an observer's record order, which says of its own records which depend on which. There is a bare real parameter attached to a region's own internal transformation, carrying no unit and no duration. There is a worldline, a map that sends each of an observer's records to an event while preserving the order. There is a clock readout, a real number attached to each record that increases along that order. There is proper time, a nonnegative interval, which is what a wristwatch is for. And there is a global time function, one number attached to every event everywhere, increasing along every chain of dependency, for everybody at once.

The eighth is the one the everyday picture assumes. It is also the only one on the list that a given arrangement of events may simply fail to admit. No arrangement of observers keeping their own records hands one out.

"That repair took about a nanosecond" reads a position in a list as a duration. Accept it once and the positions can be added up, so a stretch of the world's history acquires an elapsed time in units, with no clock anywhere in the reasoning and no calibration performed by anybody. The sentence is humdrum, which is exactly why it gets through. Nothing about it sounds like a mistake.

Crossing between two of the eight requires a map and its assumptions. The count construction proves one such relation: interval counts recover proper-duration ratios in the specified effective geometry. Identifying that geometry and readout with physical clocks requires a physical calibration.

Take any clock readout and add seventeen to every reading. Every record reads later than the records it depends on, exactly as before, so the shifted readout is a perfectly legal clock for the same history. It is a different clock. Multiply every reading by any positive number instead and the same thing happens. The record order alone fixes neither the origin nor the rate. For positive readouts depending only on volume, additivity of successive inertial durations selects the fourth root up to its reference unit.

7.5 Two meetings

Einstein got to the same place from the opposite direction in 1905. He started from the speed of light being the same for every observer, which is a strong thing to be handed, and derived that two events simultaneous for one observer are not simultaneous for another. The demand was made by a light signal, and simultaneity came apart under it.

Coming the other way there is nothing to come apart. What each observer has is an order over its own records, built out of dependencies it can check. A shared present would be the eighth item on the list, and building one would mean assembling a single number covering every event held by everybody, consistent with every observer's order, which is an object nobody in the arrangement is in a position to construct. Simultaneity is not withheld. There is no quantity there to withhold, in the sense that Galileo's cabin holds no quantity saying whether the ship is under way.

Einstein's route needs the speed of light to come out the same for every observer, which is a measured fact about one particular kind of signal, handed over at the start and left unexplained. Coming from observers and their records needs no signal at all. What gets handed over is that each observer keeps an order over what it wrote. A shared present fails to turn up because nothing in the arrangement was ever asked to build one. Each observer ends up holding a list in its own memory. There is no second list anywhere saying how two of them interleave.

Which is why the only numbers Hafele and Keating could bring home were differences taken at a meeting. Two clocks compare where they meet. Each trip gave the flying clocks two meetings with the bench clocks, one on the way out of the Naval Observatory and one on the way back in. Everything between those two events belongs to a route rather than to a moment. The two routes, east and west, gave answers about a third of a microsecond apart.

7.6 Sixteen schedules

Go back to the twelve observers of chapter six, wired as an icosahedron, thirty overlaps between them, and every one of the two thousand and forty-eight states run under sixteen different repair orders.

Calling them schedules was generous. The sixteen are sixteen lists, each naming which repair move occupies which position, with no duration attached to any position and no rate at which positions go by. Two of those lists that differ only by swapping a pair of moves touching no overlap in common are not two histories of the world. They are one partial order, written down twice, and the ordering between those two moves is the numbering on the recipe card.

That is why all sixteen had to be run. Taking turns properly has nothing behind it here, so the agreement the twelve reach cannot be an artifact of a well-behaved order. The same answer comes out of all sixteen lists.

Which leaves the twelve of them in the position the generals were in, with better equipment and no better prospects. Each sees five neighbors and nothing else. Each holds an order over its own records and over nobody else's. A message from a neighbor lands after an unbounded number of intervening moves, out of the sequence it was sent in, and sometimes it does not land at all. A leader would have to be recognized as one by observers with no way of comparing what they hold, and a vote would have to be taken in a present that none of them share.

They converge anyway. Every state, every order, one answer, arrived at under exactly the conditions that a branch of engineering has been building expensive machinery to survive since 1975.

Chapter 8

Why Can Strangers Agree Without a Boss?

At 22:52 UTC on 21 October 2018, maintenance work on a failing piece of 100-gigabit optical equipment cut the link between GitHub's East Coast network hub and its primary East Coast data center. The link came back forty-three seconds later.

During that forty-three-second break, GitHub's database-topology software did what it was built to do. It runs a leader-election protocol called Raft. It could not reach the East Coast machines, and Raft carries no test that separates a machine which has stopped answering from one that has stopped. What it does instead is count. The nodes that could still hear one another formed a majority, agreed among themselves that the East Coast primary was gone, and promoted a cluster on the West Coast to accept writes in its place.

The East Coast primary had lost nothing except the ability to talk to the other coast. Applications in its own building went on sending it work, and it went on taking it. One of the busiest clusters accepted 954 writes during the window.

Then the link came back. Each coast was holding writes the other had never seen. Neither copy contained the other, so neither could be made the primary without throwing away data that had been acknowledged to somebody. GitHub ran for 24 hours and 11 minutes in a degraded state, showing information that was out of date and inconsistent, unable to publish pages or deliver webhooks. The divergent writes had to be reconciled by hand.

There is no bug in that story. Every component behaved as specified, on the information available to it. Forty-three seconds of missing fiber cost a day of the largest code host on the planet. The problem was that two groups of machines, neither able to see the other, both concluded they were in charge.

So how does a crowd of parties who cannot see each other's internal state, who share no clock, and who answer to nobody end up holding the same answer, and what does it cost? Engineers buy that agreement with hardware, messages and delay. Alice, Bob and Charlie get theirs for free.

8.1 Two ways to be broken

In 1982 Leslie Lamport, Robert Shostak and Marshall Pease published a paper in *ACM Transactions on Programming Languages and Systems* called “The Byzantine Generals Problem”, about a group of commanders surrounding a city who must agree on whether to attack, and some of whom are traitors.

Lamport wanted a nationality for the traitors that no reader would object to, and settled on Albania, which at the time was sealed tightly enough that he doubted any Albanian would ever see the paper. His colleague Jack Goldberg pointed out that there were Albanians outside Albania and that Albania might not stay sealed forever. Byzantium had the advantage of having been out of business since 1453.

A **crash fault** stops. The machine dies, the process is killed, the power goes, and what everybody else sees is silence.

A **Byzantine fault** keeps talking. It sends messages that are wrong, or that are inconsistent with each other, or that say one thing to one neighbor and something else to another, all of it with no outward sign that anything has happened. Malice is one way to get there and by far the least common. A memory cell that flips, a disk returning data from before the last write, a program that applied half of an update and then fell over: each of these produces a participant confidently answering questions with garbage. The confidence is the whole of the problem, because the other participants have no way to weigh it.

On 18 May 2003 a vote-counting machine in Schaerbeek, in Brussels, credited one candidate in the Belgian general election with 4,096 votes that nobody had cast. No fault was ever found in the software. The official finding was a spontaneous inversion of a single bit in memory at the thirteenth position, a position worth two to the twelfth, which is 4,096. The machine reported the result the way it reported every other result, with no error, no warning and no sign of distress. The reason anyone caught it was that the municipality had returned more votes than it had voters.

Crash faults are cheap to survive. You wait, you notice the silence, you carry on without the missing party. Byzantine faults are expensive. The exact price can be counted on a napkin.

8.2 Four dots

Draw four dots. They are four participants. You would like the group to survive one of them being Byzantine, without knowing which one.

Decide that any three of them, agreeing, may commit a decision on behalf of the group. Call a set that large a **quorum**.

Take any two quorums. Each has three members drawn from four, so the two of them together name six memberships among four participants, which means they share at least two members: three plus three minus four is two. At most one participant of the four is faulty. Therefore at least one member of the shared pair is sound. A sound participant does not vote for two different answers to the same question. So no two quorums can ratify conflicting decisions, whatever the faulty participant says to whom.

That is the entire safety argument. The same subtraction runs at any size. Surviving three broken participants out of ten takes quorums of seven. Any two quorums of seven out of ten share four members, at most three of them faulty. Surviving four out of thirteen takes quorums of nine, which share five, at most four of them faulty. Either way at least one sound participant sits in both. The property is called **quorum intersection**, the reason the ratio is two thirds rather than one half. Ordinary majority voting survives crashes. Surviving liars costs an extra third of the hardware.

Read the ratio as a purchase order. To survive one broken participant you buy four machines and use them to do the work of one. Every decision costs a round of messages from three of them. To survive two you buy seven. The bill is linear in the number of liars you are willing to tolerate and it never gets cheaper, because the subtraction that makes the argument work is doing so at the tightest possible margin: two quorums of three out of four share two members, and if either quorum shrank to two, they could share one, and that one could be the liar.

Notice also what the four dots require you to have before any of this starts: a list of who the participants are, a way to reach any of them, and a fixed count of how many there are. Take away the roster and the arithmetic has nothing left to subtract.

8.3 What Fischer, Lynch and Paterson proved

In April 1985 the *Journal of the ACM* published a paper by Michael Fischer, Nancy Lynch and Michael Paterson titled “Impossibility of Distributed Consensus with One Faulty Process”.

Suppose messages always arrive eventually, but with no bound on how long they take. Suppose one single participant may crash, and only crash: no lying, no corruption, one machine that stops. Then there is no deterministic protocol that guarantees both that everybody who decides decides the same thing and that everybody decides at all.

The result does not say that no efficient protocol exists. It says that no protocol exists. The shape of the proof is that any such system passes through configurations from which both answers are reachable, and that a schedule delivering one message at a time can always steer it out of one such configuration and into another. Every message gets delivered. Nobody is starved. The system is kept poised forever by nothing more aggressive than a choice about the order of the mail.

Faster hardware does not fix this. Better engineering does not fix it. What the result attacks is the combination of three things: asynchrony with no bound on delay, a deterministic rule, and the demand that agreement and termination both hold always. Give up any one of them and consensus comes back.

The industry actually made two of those surrenders.

The first gives up asynchrony. Assume that after some unknown point, message delays are bounded by some unknown constant, an assumption Cynthia Dwork, Nancy Lynch and Larry Stockmeyer formalized in the *Journal of the ACM* in 1988 and called partial synchrony. The protocol never violates agreement, whatever the network does; it promises to finish only during the stretches when the network behaves. Every leader-based system in production is standing on this hatch, which is why every one of them has a timeout in it, and why the timeout is the parameter that operators fight about.

The second gives up determinism. Let participants flip coins, which Michael Ben-Or showed in 1983 is enough: the protocol terminates with probability one, meaning that the runs in which it goes on forever have a total weight of zero, while carrying no promise about any particular deadline.

A leader with a timeout, or a coin. Every consensus protocol running in production is one of those two, or both.

The failover on 21 October 2018 was the first hatch working as designed. The nodes that could see each other formed a quorum, elected a new topology and never disagreed with one another about it, which is exactly what a protocol standing on partial synchrony promises: agreement always, progress when the network behaves. What diverged was the data underneath, because the databases accepting writes on the East Coast were not participants in the election and had made nobody any promises at all. Agreement held, about which machine was in charge rather than about the 954 writes.

8.4 A leader and a clock

Since September 2022 Ethereum has run its consensus on a schedule. Time is cut into slots of 12 seconds, one validator is drawn for each slot to propose a block, and a committee attests to what the proposer produced. Thirty-two slots make an epoch of 6.4 minutes. A checkpoint at an epoch boundary is finalized once the following epoch has been agreed, which puts final on the record 12.8 minutes after the fact and backs it with stakes that can be destroyed.

What that machinery buys is a single history that every participant agrees on, in one order, with a definite moment after which nothing in it can move. What it costs is on the face of the design. There is a leader for every slot. There is a wall clock that everybody reads, because a 12-second slot means nothing to a participant with no idea what time it is. Every full node holds the entire state, because checking the next block means having the previous one in full. And every transaction is placed in the one order relative to every other transaction, whether or not the two have anything whatsoever to do with each other. The system spends real work establishing which of them came first. Bitcoin buys the same single order with a lottery in place of a schedule, a race to find a number whose hash falls below a target that is retuned every 2,016 blocks to hold those 2,016 blocks to 1,209,600 seconds. Ten minutes a block is what that works out to, and 1,209,600 seconds is two weeks. Whoever wins a round is that round's leader. The chain the winner extends is one line.

8.5 Twenty strangers

In May 2018 a pseudonymous group calling itself Team Rocket posted a paper to a file sharing network, "Snowflake to Avalanche: A Novel Metastable Consensus Protocol Family for Cryptocurrencies", describing a way of reaching agreement in which nobody is in charge and nobody holds the tally.

Each participant keeps two things: its current preference, and a counter. That is the whole of its state. There is no vote count, no view of the network, no record of who said what.

Each round, a participant picks twenty others at random and asks each one what it prefers. If fifteen of the twenty give the same answer, the asker adopts that answer and increases its confidence in it. If twenty rounds in a row come back the same way, it decides, and the decision does not reverse. Twenty as the sample size, fifteen as the threshold, twenty as the number of consecutive rounds: those are the documented defaults on Avalanche's primary network, settings rather than laws. The protocol built out of the sample and the counter is called Snowball.

If a bag holds a hundred marbles and thirty of them are red, the probability of drawing a red one is thirty out of a hundred: the fraction of the cases that come out red, which is the fraction a long run of draws settles toward. That is all the sampling needs. When a sample of twenty comes back fifteen to five, that fraction is evidence about the fraction in the whole population. It can also mislead, because a sample of twenty drawn from a population split sixty to forty will occasionally come back fifteen for the minority, in the way that a fair coin occasionally lands heads six times running. The counter is what handles this. Being misled twenty rounds consecutively is enormously less likely than being misled once. The decision threshold is the dial that pushes the chance of a wrong decision below one in a billion, a bound the published parameters hold with a fifth of the network lying.

Each participant sends twenty questions per round regardless of how large the network is. Twenty when there are a hundred participants, twenty when there are a hundred thousand. The number of rounds needed grows like the logarithm of the population, which is the slowest growth on offer. There is no proposer, no miner, no round leader and no view change, so there is no single machine an attacker can take off the air to stop the thing.

And the dynamics are metastable. Start the network at an even split and it sits on a knife edge, where the sampling itself is the only thing capable of breaking the tie. As soon as one answer runs slightly ahead by accident, participants sampling the network get that answer slightly more often, adopt it, and become part of the majority that the next participant samples. The tilt feeds itself. Past a certain point the outcome is fixed. The only thing that broke the tie was the order in which twenty names came out of a hat.

8.6 Where the analogy inverts

A crowd of parties, none of them in charge, none of them holding the global picture, converging on one answer: that is what a network of overlapping observers does. The engineered version of it settles real payments in production.

Watch an Avalanche node take its sample. It picks twenty participants uniformly at random from the whole validator set, which requires it to hold a roster of every participant in the network and to be able to open a connection to any of them. That roster is a global object, and holding one is the price of the sample being uniform. The locality in that protocol lives in the size of the sample, twenty out of however many, rather than in who the twenty are allowed to be.

An observer in this world has neighbors, fixed, and communicates through those and through nothing else. The twelve of chapter six each had five neighbors. The five came with the wiring: the same five in every round of every schedule, with no round of repair ever producing a sixth. To reach a non-neighbor an observer has to go through a neighbor. There is no address book and no directory. A channel that bypassed the wiring would itself be an overlap, at which point it would be in the wiring.

So the engineered protocol is the loose version. What an observer can be affected by is settled by the diagram of who overlaps with whom. That diagram is finite and fixed.

Snowball's safety is probabilistic. It holds except with an error probability that the parameters drive down. One in a billion is a small number rather than zero. The order-independence of a repairing network of observers is of a different kind. Where it settles is fixed by the disagreements and the wiring, and running the repairs in a different order lands on the same place, in every run rather than in overwhelmingly many of them.

The twelve-observer sweep of chapter six is what that looks like from below, where every state and every schedule was enumerated rather than sampled. The count of exceptions was zero because there was nowhere for an exception to hide.

An engineer would take that guarantee over Snowball's in an instant and cannot have it, because getting it requires the thing engineers are never given, which is control over how the participants are wired to each other.

8.7 What a fixed neighbor list buys

Fix an observer and ask a question that the engineered protocols cannot even pose: what could the reading it holds possibly depend on?

Before any repair, its own state. After one repair, its own state and whatever its neighbors held, because a repair compares readings across a shared boundary and nothing else was consulted. After two repairs, the neighbors of those neighbors, and no further. The set of observers whose records could have contributed through a read chain grows by at most one step of the wiring per round of repair. At every stage it is a finite set you could write out by name.

The twelve-observer arrangement makes the counting easy, because the wiring is regular enough that everybody is in the same position. Take any observer of the twelve. Five of the others are its neighbors, five more are neighbors of those, and exactly one, the observer directly opposite it on the solid, is three steps away. So after one round of repair, six of the twelve can have touched what it holds. After two rounds, eleven. The twelfth arrives in the third round and not before. There is nothing it can do to arrive sooner.

Call the set the **dependency cone**: the observers within reach, where reach is counted in steps along overlaps rather than in kilometers.

Everything outside the cone is absent from the reading in the strongest sense available, which is that there exists no route by which it could have entered. A signal too weak to detect is a different situation. A better instrument fixes that one. If a distant observer repairs, or refuses to repair, or is deleted from the arrangement entirely, nothing about what this observer holds changes until the consequence has walked in along the overlaps, one step per round, the whole way.

Two distant readings in this world can be correlated, and chapter six built an arrangement where they are correlated by force, since the twelve observers had exactly one consistent state available to them and every one of them was pinned by it. Correlation is allowed and is ordinary. What no arrangement permits is a distant party choosing what you read. There is no dial they can turn, because the only channel between you is the wiring and the wiring has a pace, so whatever they do arrives as something that walked, or does not arrive.

Correlation without communication is the shape of the Delft result from the prologue, where two laboratories 1.3 kilometers apart got answers that matched too often and could not use the matching to send each other a single bit. In an arrangement whose only channels are its declared overlaps, that combination is arithmetic about which observers are within how many steps of which. The name for it is **no-signalling**, arriving here before there are any particles to attach it to.

8.8 Only the conflicts

One feature of the engineered protocol carries over exactly. It looks at first like an efficiency trick.

Avalanche does not order transactions. It organizes them into conflict sets, where a conflict set is the group of transactions that try to spend the same coin, and runs its sampling only within those sets. Two transactions that touch nothing in common are never compared, never ordered, and never made to wait for each other. There is no fact in the system about which of them came first, so no work is done to manufacture one.

The leader family does the opposite, and pays for it in the queue. Every transaction goes into one line whether or not it interacts with anything else in the line, which is why a payment in Lisbon waits behind a payment in Osaka that has nothing to do with it.

A network of observers repairing their overlaps is in the first camp, there by construction rather than by choice. A repair happens where a comparison disagrees. Two disagreements in unrelated parts of the arrangement are not competing for anything, do not touch a common quantity, and have no order between them. Sharing a register makes a dependency possible but does not create one by itself: an order edge appears only when a later authenticated commit certifies that it read a version written by the earlier commit. Conflicting writes without that read-from witness need not be ordered. The rest of the arrangement is under no obligation to have an opinion about when they happened.

The protocol is mostly a list of absences: no leader, no clock, no roster beyond the immediate neighbors, no global tally, and no order imposed on things that do not interact. What survives is a set of observers, a set of overlaps, and the repair each observer works out for itself wherever two readings of the same overlap fail to match. Nothing dispatches that repair. The observer proposes a move, checks it against what it holds, and keeps it only if the fit comes out better, which is the next chapter's subject.

Alice and Bob share an overlap. Alice reads twenty across it, Bob reads twenty-two, in whatever units that boundary is measured in. Both are careful, neither has malfunctioned, and both readings are of the same thing, which is why two numbers where there should be one is a problem rather than a curiosity. Something has to move. The protocol says nothing about which number moves, or by how much, or why that answer rather than one of the infinitely many others that would also leave the two of them matching.

Chapter 9

What Move Does Reality Actually Make?

David Kinnebrook arrived at the Royal Observatory at Greenwich on 23 May 1794 as assistant to the Astronomer Royal, which meant that most nights of the week he sat at the transit instrument and timed stars.

The method was eye and ear. You listen to the clock beating out seconds, you watch a star cross a wire in the eyepiece, and you split the interval between the last beat and the crossing into tenths by judgment. Nevil Maskelyne had been doing it for thirty years. Both men used the same instrument on the same stars. For a year their numbers matched.

From the beginning of August 1795, Maskelyne found that his assistant was setting the transits down half a second later than he should. He told him. Kinnebrook tried to correct it, and the gap widened. By January 1796 it stood at eight-tenths of a second. Maskelyne wrote that Kinnebrook did not seem to him likely ever to get over it, noted for the record that the man had been diligent and useful, and dismissed him. Kinnebrook left the observatory on 12 February 1796.

A star crossing a wire issues no receipt. No instrument at Greenwich held the true time of the transit, and none could have, because the true time of a transit was the thing the transit instrument existed to produce. Two men, one event, two numbers, and the single measurable quantity in the place was the difference between them: eight-tenths of a second, attached to nobody. Maskelyne resolved it by appointing one of the two readings correct. The reading he appointed was his own.

Twenty-seven years later, at Königsberg, Friedrich Bessel worked out what had happened. In 1823 he published the finding that these differences are involuntary, constant, and present between any two competent observers of the same event. He measured himself against his colleagues and published the results as pairs, because a pair was all there was to publish. Against Argelander he ran 1.223 seconds early. The corrections never attached to a person. They attached to a pair of people. A transit time belonging to nobody was not on offer at any price.

Every observer in this world sits where Maskelyne and Kinnebrook sat. Two of them share a boundary, each holds a reading of it, the readings disagree, no third reading exists, and nobody has been appointed correct.

The question is arithmetic. What does one observer physically do to its own state, and by how much?

9.1 Two readings and a seam

Take chapter six's wiring back out: twelve observers, each with exactly five neighbors, thirty shared boundaries between them. Call a shared boundary a **seam**.

In that universe each observer held one bit and each seam recorded one word, same or different, which is the coarsest reading a seam can carry. Widen it. Let each observer hold a quantity, so that what a seam compares is two accounts of one quantity and the record on a seam is a number rather than a word: how far apart the two ends are, and which way round. A seam is satisfied when the two readings at its ends are equal. The simplest such quantity is a count of units held.

A rule that acts on a small piece of a state and leaves the rest alone, applied over and over in no particular order, is a **rewriting system**. Chapter two had one with two rules and an alphabet of two letters. A state that no rule applies to is where the process stops. A stopping state is a **normal form**. Whether a state is one is a local question: an observer looks at its own five seams, finds nothing to do, and has its answer without knowing anything about the other six observers or about what any of them is doing at that moment.

So what is a repair allowed to be? Three requirements fix it.

First, it changes nothing outside the seam it repairs. The two readings at that seam's ends and the record between them are all it can touch, because an observer talks through its ports and nowhere else, and a move that reached further would need a fact about a place it cannot see.

Second, it leaves the total of those two readings alone. What the two of them are disagreeing about is one shared quantity described twice, so a repair re-attributes between two accounts rather than adding to them. The readings have no absolute level for a repair to correct them toward.

Third, when it is finished the two readings are equal. A move that leaves them disagreeing has repaired nothing.

Solve it. Call the two readings a and b. The third requirement says both end at one value, so call that value v. The second says twice v equals a plus b. Therefore v is half of a plus b. There is nothing further to determine, because two conditions have fixed two unknowns.

Both parties move to the midpoint. Split the difference.

The move carries no adjustable constant. It has no rate, no step size and no fraction of the gap to be closed per event, so there is nothing in it for anybody to have chosen or fitted to a measurement.

Three consequences arrive free. If a and b are equal to start with, the move does nothing, so the states where the law idles are exactly the consistent ones. Neither observer is corrected by the other: both move, by the same amount, in opposite directions. And Maskelyne's move, which was to overwrite the assistant's reading with the Astronomer Royal's, fails the second requirement, because it changes the total of the two numbers it touches, and needs somebody appointed before it can be carried out at all.

9.2 Why does the universe keep books?

Remove the second requirement and the answer disperses: both readings going to a, both going to b, both going to zero, and every weighted blend of the two are all legal moves. No information available at the seam picks between them. Restore it and one move survives.

The requirement belongs there because a rule that changed the total would have to say by how much, and every answer to that question is a fact about a quantity nothing in this world holds. The two observers can add their own two readings together. There is no register anywhere carrying the total of all twelve, no clearing house, and nobody to ask. A universe that repairs locally cannot help conserving things, because a local repair that changed a total would have to know the total. Nothing here knows a total.

The readings have no absolute level. Add the same amount to every observer's books and every seam in the arrangement reads exactly what it read before, which makes the level a convention of the kind chapter five found at Greenwich: unmeasurable from inside, impossible to be wrong about. The gap between the two numbers is the whole of what the pair carries. Now look again at the moves the requirement threw out. Both readings to zero, or to nine tenths of the midpoint, or to any weighted blend of a and b: every one of them needs a zero to move toward, and a level fixed by convention supplies none. Shift every book by the same amount and the midpoint shifts with it, while each of the alternatives lands somewhere else. Conservation is what survives when a move is forbidden to know where the origin is. Chapter twelve takes that freedom seriously enough to get a force out of it.

No admissible move alters the total of the two readings it touches, and reaches no reading beyond those two, so the total across all twelve observers is the same after one repair, after a thousand, in any order, forever. The total is an invariant, a quantity that comes out the same under a change, and the change it survives is every repair the world can perform.

A conserved quantity, in this world, is a quantity that no admissible local move can alter. What makes a move admissible is that it touches one seam and reduces disagreement. Newtonian mechanics conserves momentum and energy too, and nothing inside it says why the universe should keep accounts; the conservation goes in by hand and is then observed to hold. Here it falls out of the only move available.

Chapter five gave the standard correspondence, that momentum is conserved because the laws do not care where you are and energy because they do not care when. The correspondence is exact and it says which quantity goes with which symmetry. It takes the existence of the books for granted before it starts. The books come from the second requirement, which is there because no seam and no observer holds a total. Charge conservation, energy conservation, and the conservation of the quantities that turn out to label particles are all this same argument run on a different reading of a seam.

9.3 Take three numbers

Write three numbers at the corners of a triangle: 6 at the top, 0 at the bottom left, 3 at the bottom right. Every corner is joined to the other two. Take a pencil and work out, for each corner in turn, its own number minus the average of the two numbers it is joined to, and then double the answer, because each corner has two neighbors. Three subtractions and three doublings. Write the results beside the corners before reading the next paragraph.

The top corner sits at 6 and its two neighbors average 1.5, so it gets 9. The bottom left sits at 0 against an average of 4.5, so it gets minus 9. The bottom right sits at 3 against an average of 3, so it gets 0.

Three numbers in, three numbers out. Each output says how far above or below its own neighborhood that corner sits, scaled by how many neighbors it has. The three outputs sum to zero, because every gap in the list appears twice with opposite signs. The operation is called the **graph Laplacian**, it is defined on any graph whatever, and the sentence carried out three times with the pencil is all of it:

$$(Lx)_{\text{me}} \;=\; n_{\text{me}} \times (x_{\text{me}} - \text{average of } x \text{ over my neighbors})$$

Here x is the list of readings, one number per corner, x with the subscript me is the reading at my own corner, n with the subscript me is how many neighbors my corner has, and L is the name of the operation that turns one such list into another. Nothing on the right-hand side looks past the corners a corner is joined to, which is why an observer can work out its own entry without leaving its patch.

Repair a seam by hand. Take the 6 and the 3, whose midpoint is 4.5, and send both there: the triangle reads 4.5, 0, 4.5. Add them. Start over and repair the 0 and the 3 instead, which puts both at 1.5: the triangle reads 6, 1.5, 1.5. Add them. Three seams, six orders of working through them, and the running total is 9 at every point in every one of those histories, because a move that sends two numbers to their midpoint puts back exactly what it took out of them. No corner holds the 9, and no move available on the triangle can spend it.

The midpoint move is the graph Laplacian of a single seam, halved and subtracted: each reading moves by half the gap to its partner, toward the partner. Apply it to all thirty seams and weight the seams equally, and what acts on the twelve readings is one operator:

$$T = I - \frac{1}{60}L$$

Here T is the average effect of one repair on the twelve readings, I is the operation that leaves all of them alone, L is the graph Laplacian of the wiring, and the sixty is the thirty seams doubled, because a midpoint move closes half of a gap rather than all of it. So a reading moves toward the average of the five readings around it by a twelfth of the distance: the Laplacian counts that distance once per neighbor, and five sixtieths is a twelfth.

That operator is the one in the heat equation, written out on a graph instead of on a line. Heat in an iron bar obeys exactly this rule: the temperature at a point changes at a rate set by how far that point sits from the average of the temperatures next to it, which is why a bar heated at one end finishes up uniformly warm and why the warmth arrives at the far end late. The hot side and the cold side of a bar land on their midpoint for the reason two observers do. Repair is diffusion.

Which gives a way to prove the process stops. Score a state by the sum of the squares of the twelve readings, and call that its level. Replacing two readings by their mean drops the level by exactly half the square of the gap that was closed, so any repair that lands both readings on their exact midpoint strictly lowers it, and the level cannot go below zero.

Check it on the triangle. Square 6, 0 and 3 and add them: 45. Repair the seam between the 6 and the 0, which sends both of them to 3, then square and add again: 27. The level fell by 18. The gap that was closed was 6, and half of 6 squared is 18. The average was 3 throughout. A single move on the widest seam finished the triangle.

When the readings are counts, the level is a whole number. A whole number that strictly drops and cannot go below zero runs out of room after finitely many steps. That is the general recipe for showing a rewriting system terminates: find a quantity every rule strictly lowers, which cannot be lowered indefinitely.

Diffusion on this wiring carries a hard graph-local limit with it. A repair event touches one seam. A changed reading can enter a neighbor's certified support only through their shared seam, and a read-from chain can extend by at most one wiring step in one model event. That is the informational dependency cone of chapter eight, counted. It is a fact about the finite program, not yet a speed in meters per second. Identifying seams and model events with a physical ruler, a clock and signal reachability requires separate receipts; without them this bound is not the speed of light.

9.4 The rule that got stuck in 302 places

Let the readings be counts, and give the twelve observers twelve units to hold between them. Since the total is untouchable, the entire history of any run is confined to the arrangements of twelve units over twelve observers. There are 1,352,078 of those.

A space that size can be checked all the way through, so check it all the way through.

The obvious rule goes first. If a neighbor and I differ by two units or more, hand one unit across the seam. It reduces the gap, it preserves the total, it touches nothing but the two of us, and it is wrong.

Run every one of the 1,352,078 arrangements until the rule has nothing left to do. Three hundred and three of them have nothing to do at the outset. One is the balanced arrangement, everybody holding one unit, which is the answer. The other 302 are lopsided and the rule cannot move them.

They are simple to describe once you have seen one. Give one observer two units, give another observer none, place those two so they are not neighbors, and give everybody else one. Every seam in that arrangement has a gap of one or zero, so the rule has no move available anywhere. The arrangement is not the balanced one. Count them yourself. An observer has five neighbors out of the other eleven, which leaves six it is not joined to, and twelve places to put the full observer times six places to put the empty one comes to seventy-two. Two hundred and ten more have two observers holding two units and two holding none. Twenty have three holding two and three holding none. The three counts add to 302.

The defect is at a gap of exactly one. Handing that unit across reverses which observer is ahead without narrowing anything. A rule that accepts only moves which improve the score refuses it. So the process halts on a plateau where no move lowers the score, one observer keeps a spare unit and another keeps none.

The repair is to complete the move instead of nibbling at it. On a seam whose two readings add to an even number, the midpoint is a whole number and both sides go to it. On a seam whose readings add to an odd number, no pair of equal whole numbers has that total. The closest the two readings can come is half a unit either side of the midpoint, a pair that can be placed two ways round. Both placements are legal completions, including the one that reverses which observer is ahead. That one is legal because it changes nothing measurable.

Allowing that one move changes the audit. Under the completed law, exactly one arrangement out of 1,352,078 rests. It is the balanced one.

From every other arrangement there is a path of at most three events to a strictly lower level. Most of the stuck ones need two.

Watch one of them get out. An observer holding two units sits opposite an observer holding none, with the ten others holding one each. Every gap in the arrangement is one or zero, so nothing improves anywhere. The full observer and one of its neighbors hold two and one, which totals three, so the midpoint is one and a half, the two readings land half a unit either side of it, and they may land either way round: the surplus unit walks to the neighbor, and the level has not moved. It walks again, to a neighbor of that neighbor, and the level has not moved on that event either. On the third event the observer holding the surplus is finally adjacent to the observer holding none, their readings total two, the midpoint is a whole number, and both go to one. The level drops. Nothing was gained by the first two events except position, which is exactly what the one-unit rule refused to spend anything on.

Three hundred and two failures out of 1,352,078 arrangements is a failure rate of two in ten thousand, which is why the rule had to be run over every arrangement rather than over a sample of them.

9.5 Two tellers, one account

An account holds 1,000 and two tellers open it at the same moment. The first takes a deposit of 100, adds it to the 1,000 in front of her, and writes 1,100. The second pays out 200, subtracts it from the 1,000 in front of him, and writes 800. Both did correct arithmetic on the number they were given. The account ends at 800, the depositor's 100 is gone, and no entry anywhere records its disappearance, because every individual entry is right.

Two operations that touch nothing in common can be done in either order. No record afterwards can tell which happened first. Two operations that touch the same item cannot, because the order is part of the answer. Operations that conflict are operations whose order matters, which is the whole content of the word conflict.

In June 1981, H. T. Kung and John Robinson published "On Optimistic Methods for Concurrency Control" in ACM Transactions on Database Systems, proposing that a database should stop asking permission. Their transaction runs in three phases. During the read phase it works from the copy of the data it took when it started, keeping a list of everything it looked at, its **read set**, and a list of everything it intends to change, its **write set**, and writing nothing anybody else can see. During the validation phase it checks whether anything in its read set was changed by a transaction that committed while it was working. Only if that check passes does it reach the write phase and make its changes visible. A transaction that fails validation is discarded and run again from the start.

Locking says: nobody touches this while I work. Optimistic control says: work, then prove nobody touched it. Where conflicts are rare the second is cheaper, because nothing waits for anything. Where conflicts are common it is a disaster, since every collision costs the whole transaction and the work is done twice, which is the trade the designer of any such system is making whether or not anybody writes it down.

The acceptance layer of the repair law is that mechanism, item for item. A repair is proposed rather than performed. Its read set is the pair of readings at the ends of one seam, its write set is the same pair. It commits only if the fit on every overlap it touches is at least as good afterwards and strictly better on one of them. Otherwise it is discarded. The observer reads again. Nothing in that loop is a rule handed to the observer from outside. The three requirements say what a proposal has to satisfy; working out a candidate that satisfies them is the observer's own computation, and the check is what decides whether the candidate becomes the world. Because read set and write set are confined to a single seam, two repairs on seams with no observer in common cannot conflict, which is why nothing has to put them in an order. The check has a second effect that the tellers make obvious. A repair cannot buy agreement on one seam by wrecking it on another, because it is validated against every overlap it touches and refused if any of them comes out worse.

Put a dot for each transaction and draw an arrow between two dots whenever those transactions touched the same item in a way whose order matters. That is a **conflict graph**. A batch of interleaved transactions can be rearranged into a run where they went one at a time exactly when that graph contains no cycle. Whether local records assemble into one global account is settled around loops, which is the condition Alice, Bob and Charlie's coins failed, turning up here in a banking system.

9.6 Two budgets

The three requirements fix what the two readings end up holding. They say nothing about which register the repair gets written into, and a seam has two: the readings in the observers' own patches, and the record sitting between them. So a repair comes in two kinds. The difference is who pays for it.

In the first, an observer rewrites its own patch so that its reading matches what the seam says. The capacity spent is the observer's own, nothing outside its patch changes, and the boundary between it and its neighbor says afterwards what it said before.

In the second, the boundary record itself is rewritten, so that both sides read something new. The capacity spent is held jointly by the two observers on the overlap. Neither of them owns it alone.

The two budgets are not interchangeable. An observer with capacity to spend on itself can bring its own reading into line and leave the boundary saying what it always said. Nothing outside that patch registers that anything happened. Rewriting the boundary takes both of them, and afterwards both are holding something neither one held before.

9.7 Sixty ways to go next

Chapter six ran a twelve-observer universe under sixteen schedules and got the same answer every time, on a system holding one bit per observer and 2,048 distinguishable states. The machine with counts in it is larger: 1,352,078 arrangements, and at every point in a run there are sixty directed completions to choose from, one for each seam in each direction.

Nothing chooses between them. There is no clock, no queue, no leader, and no fact about which seam is repaired first, because the observers have none of those things and neither does anything else. Two runs from the same starting arrangement, repairing the same seams in a different order, write different numbers into different observers along the way. No register anywhere is keeping the list.

They land on the same arrangement. The three requirements do not say they have to: two rules on two letters are enough to build a rewriting system where the answer depends on which rule you reach for first, and where both answers are places at which nothing further can be done. So every schedule landing in the same place is either a lucky property of this particular wiring or a theorem. If it is luck, then what the twelve of them end up holding depends on which seam happened to be repaired first. There is no objective answer anywhere in the machine.

Chapter 10

Why Doesn't It Matter Who Goes First?

Among the crosswords in the May 1979 issue of *Dell Pencil Puzzles and Word Games* there was a nine by nine grid with some of its squares filled in and the rest blank. It was called Number Place. Every row, every column and each of the nine three by three boxes had to end up holding each digit from one to nine exactly once. The man who devised it was Howard Garns, a retired architect in Indianapolis, seventy-four years old, and Dell did not print his name on it. He died in October 1989. Five years before that, the Japanese publisher Nikoli had picked the puzzle up and given it the name it travels under, which is sudoku, and in 1997 a retired Hong Kong judge named Wayne Gould found one in a Tokyo bookshop, spent six years writing a program that could make them by the thousand, and talked The Times of London into printing one on 12 November 2004. Within weeks every British newspaper had a grid on its puzzle page.

Give the same grid to two people on the same train. One of them works the sevens first, because four of them are showing and they constrain each other. The other goes box by box from the bottom left. They do not speak, they do not look at each other's paper, and neither has the faintest idea what the other has written down. Forty minutes later both grids are full and all eighty-one squares match.

Nothing arranged that. Two different procedures ran in two different orders on different parts of the grid. The agreement at the end was neither negotiated nor checked. The two of them get off at different stops without ever finding out.

The reason is that the finished grid was fixed by the clues before either of them picked up a pen. There are 6,670,903,752,021,072,936,960 ways to fill a nine by nine grid legally, a figure Bertram Felgenhauer and Frazer Jarvis computed in 2005. A printed puzzle is a set of clues that cuts that number to one. In 2012 Gary McGuire, Bastian Tugemann and Gilles Civario finished an exhaustive search and established how few clues can manage it. Seventeen. Sixteen never suffices, on any grid, in any arrangement. Below seventeen the clues stop picking out a single completion. The two people on the train can hand in different grids and both be right.

The clues do half of the work and the pencils do the other half. Every move either of them makes is forced. A square gets a five because five is the only digit its row, its column and its box have left over for it, and once that is true it goes on being true whatever else gets filled in, because filling in another square removes possibilities and never restores one. A forced entry cannot be un-forced by a later forced entry. So the order the two of them work in decides which square gets filled at which minute. It has no way to reach the digit that square ends up with.

Repairs in the network of the last chapter happen when a mismatch is noticed. Which mismatch gets noticed first is whatever the world happens to hand over. There is no schedule, in the sense that there is no fact anywhere about the order. If the endpoint depends on that order, then the world an observer finds is a fact about scheduling. No observer inside can see a schedule.

10.1 Two librarians

Take a shelf of books that ought to be in order and are not. The rule is one line: if you find two neighboring books out of order, swap them. Any pair, whenever you like.

Photograph the shelf. Let one librarian sort it. Then put the books back exactly as the photograph shows and hand the shelf to a second librarian who has been out of the room. Both finish with the shelf in order, which is no surprise, because there is one way for a shelf to be in order. Count their swaps. Each of them will have made exactly the same number. Neither of them chose it.

Count, at any moment, every pair of books in the wrong relative order, wherever on the shelf the two of them happen to sit. A book that belongs before another one and stands to its right is one such pair, whether the two are touching or at opposite ends of the run. Call each of those a wrong pair. Swapping two neighbors that are out of order fixes exactly one wrong pair, because it reverses the relative order of those two books and leaves the relative order of every other pair untouched. So the count drops by exactly one at every swap. It drops to zero. The number of swaps is the number of wrong pairs the shelf started with, which is a property of the shelf and not of the librarian. This is the descent argument of the last chapter, on furniture: a quantity that strictly falls and cannot fall below zero cannot go on falling.

The shelf does something else on the way down. Take a shelf where two swaps are available: books three and four are out of order, and so are books seven and eight. Do the first one. Do the second one instead. The two shelves you get are different from each other. From the first, swap seven and eight; from the second, swap three and four. Both land on the same shelf.

Draw those four shelves with arrows between them and the picture is a diamond: one at the top, two down the sides, one at the bottom where the paths meet. The **diamond property** is the name for that shape. The property it names is that any two moves out of a state can each be followed by one move that brings them back together. Allow the branches any number of moves to find each other again and the condition is called **local confluence**. Allow the branches themselves to be any length, so that two long histories out of one starting state have to be reconcilable, and the name for that is **confluence**.

Three books show what the looser condition buys. Put Chesterton, Cervantes and Bunyan in positions three, four and five. The pair three and four is out of order, the pair four and five is out of order, and the two swaps share a book, so they are not independent the way the first example's were. Swap three and four and the shelf reads Cervantes, Chesterton, Bunyan. Swap four and five instead and it reads Chesterton, Bunyan, Cervantes. Those two shelves are further apart than the first example's pair. No single swap carries either one to the other. From the first, swap four and five, then three and four. From the second, swap three and four, then four and five. Both arrive at Bunyan, Cervantes, Chesterton. Two moves on each branch rather than one, and three swaps down each route, which is the number of wrong pairs those three books started with.

Those last two conditions are separate statements. Local confluence says any two first steps can be reconciled. Confluence says any two histories can be. A history is a hundred moves of drift, and knowing that its opening move was harmless says nothing about where the other ninety-nine took it.

10.2 A snowflake on Triple Divide Peak

Triple Divide Peak stands 8,020 feet up in Glacier National Park in Montana, on the one summit where two continental divides cross. Water leaving that summit on one face reaches the Pacific by way of the Columbia. On another face it reaches the Gulf of Mexico by way of the Missouri and the Mississippi. On the third it reaches Hudson Bay by way of the Saskatchewan. Two snowflakes landing a hand's width apart on the summit finish in different oceans.

Inside any one of those three basins the outcome stops depending on the route. A raindrop falling anywhere in the Columbia basin can take any of a billion routes down, through gullies and side creeks and a reservoir or two, and arrives at the same Pacific as every other drop in the basin. Different paths, one sea. That is confluence, drawn at the scale of half a continent.

Water is also guaranteed to stop. It goes downhill, its height is a quantity that strictly decreases, there is a floor, and every drop that lands on that mountain reaches sea level and stays. Termination promises that the process ends. It promises nothing about ending in one place. A mountain with three basins meeting on its summit is what the failure looks like.

The failure runs the other way as well. Loosen the librarians' rule so that any two neighboring books may be swapped whether or not they are out of order, and the branches go on being reconcilable, since enough swapping turns any shelf into any other shelf. What goes is the stopping. The two librarians can push Bunyan past Cervantes and back again forever, able to agree at every step and never arriving at a shelf to compare.

The mountain misleads in one place. Water runs downhill because there is a downhill, and it stops at the sea because the sea is sitting there waiting to be arrived at. The network has neither. Its settled configuration is the arrangement that satisfies every seam at once, picked out by the constraints in the way seventeen printed clues pick out one grid, before a single repair happens. Repairs delete the arrangements that fail. Nothing travels anywhere. Nobody who finishes a sudoku believes the grid was assembled by the pencil. The same restraint is owed to a network that ends up consistent.

Max Newman supplied the missing piece in 1942, in the *Annals of Mathematics*, in a paper on theories with a combinatorial definition of equivalence. He was a Cambridge topologist whose lectures had sent Alan Turing off to write "On Computable Numbers", and in the year that paper appeared he arrived at Bletchley Park, where he was given a section that everyone called the Newmanry and told to break the German high command's teleprinter cipher with machines. The first Colossus was delivered to him on 18 January 1944.

His result fits in a sentence. If a process always stops, and if every pair of single steps out of a state can be brought back together, then every pair of histories can be brought back together, and every starting state has exactly one endpoint. That is Newman's lemma.

The argument runs from the bottom of the descent upward. Take a state, and suppose that everything reachable below it has a single endpoint. That supposition is safe, because the process descends and cannot descend forever, so you can start at the bottom and work upward. Two moves lead out of your state. Local confluence brings those two moves back together somewhere below. The two states they led to both sit below the state you began at, so each of them has a single endpoint, and so does the state where they rejoined, which forces all three endpoints to be the same one. Your state inherits it. The induction runs over every level of the descent. The descent is finite because the disagreement it counts down is finite.

10.3 Two things to check

Newman's first condition, that the process stops, was settled in the last chapter. Every accepted repair strictly lowers the total disagreement across the network, that total is a sum of terms none of which can go negative, and so it has a floor at zero. A quantity that drops by a positive amount every time and cannot pass zero runs out. The network stops.

The second condition is the one that does the work. Two repairs happen at two different seams. If those seams share no patch, each repair reads and writes numbers the other never touches, and the two orders give identical results, because the repairs are not interacting in the first place. That is the shelf's first example, where books three and four had nothing to do with books seven and eight.

The case that has to be checked is the shelf's second one. Draw three patches in a row. There is a seam between the first and the second, another between the second and the third, and the middle patch sits on both of them. A repair on the left seam moves the middle patch's reading. A repair on the right seam moves the same reading. The middle patch has only the one.

A repair does one thing. It takes what the two sides of a seam expose to each other and moves them together. What each side exposes is overlap data, information about a region both patches can see, and assembling overlapping regions is associative: glue the first two and then the third, or glue the last two and then the first, and the object you finish holding is the same object. Two repairs sharing a patch are two such gluings. The order in which overlapping regions were glued is not recorded anywhere in the result. Whichever repair went first, one further move on each branch brings both branches to the same configuration.

The network has both of Newman's conditions. Every configuration of it settles into exactly one state. That state carries no trace of the order in which the repairs happened.

Schedule independence is what objectivity means here. The twelve-observer universe of chapter six settled onto one arrangement from all 2,048 of its states, under sixteen orders of repair apiece. Newman's two conditions cover the configurations nobody enumerated, and past twelve observers that is all of them.

In most of engineering the order does change the answer. It changes the answer here too, on the smallest arrangement that can show it, which takes two observers and one seam.

Give each of them a single bit and one rule: when you disagree with your neighbor, adopt your neighbor's value. Nothing about that is unreasonable. It is the politest repair rule available, it strictly reduces disagreement, and it touches only the seam it is applied at.

Start them disagreeing. The left one holds false and the right one holds true.

Let the left one move. It takes the right one's value, both hold true, and since they agree neither will ever move again. Go back and let the right one move instead. It takes the left one's value, both hold false, and neither will ever move again.

Two endings, both final, out of one start. Whoever moves first wins, permanently. The two worlds are easy to tell apart, and that is the trouble rather than the consolation: both holding true means the left one moved first, both holding false means the right one did, so the contents of the world are a transcript of a schedule that no rule chose and nothing required.

Confluence belongs to the repair rule rather than to locality. It is not a courtesy extended to whoever asks for it. Leave those two patches wired exactly as they are and change only the rule, from adopt your neighbor's value to something that defers to a fixed criterion, say that whichever of them holds true gives way. From the same disagreeing start there is one move available instead of two, both orders arrive at false, and the fork is gone. Courtesy was the whole problem: a rule in which each party defers to the other leaves nothing to settle the case.

The twelve-observer network has confluence, and chapter six settled it by starting every configuration the arrangement admits, rather than by arguing that the alternative sounded unphysical.

The engineered protocols of chapter eight buy order-independence with sampling and probability. The guarantee they buy is a guarantee about how often. What the observers get instead is two conditions and an induction over a finite descent, which is why their version has no failure rate attached to it.

10.4 What the network is made of

The arrangement in this chapter is one the OPH research team runs as a program, and the program is called OPH-FPE. It is not a numerical model of a continuous world. Every quantity inside it is a whole number or one of the six ways of permuting three labels, every move is exact, and no step anywhere rounds anything off. A run either meets a condition or fails to, and it returns the same verdict on every machine that executes it.

Each published run is archived entire: the configuration it was given, the seeds it drew from, the arrays it started with, the arrays it finished with, the full trace, and the state of all twelve ports on every patch. Beside them sits a verifier that borrows none of the simulator's own code. It builds the six permutations from scratch, recomputes the starting disagreements from the raw arrays, works out the finishing state, and recomputes every fingerprint. A reader with Python and a copy of the archive runs one command and either gets the same answer or finds out that they do not.

The run read here is the level-six rung of a tower built by subdividing an icosahedron over and over. It holds 81,920 patches and 122,880 seams between them, and every patch carries twelve local port slots, three of which route seams while nine stay exposed. At the start, 102,415 of those seams disagree, which is 83.3 percent of them.

10.5 Two cycles

The trace is kept in cycles. A cycle is a round in which every patch with something to repair gets a turn. That count belongs to whoever reads the trace afterward. Nothing in the network counts cycles, nothing waits for the next one, and there is no tick anywhere in the machinery.

A run of a process that failed to be confluent would be an anecdote: what happened to one arrangement on one pass, with no warrant to say anything about the next pass, which would take a different order and could finish somewhere else entirely. Newman's two conditions are what make the last line of this trace worth more than the run that produced it. The lines in between are the accidents of one order. The line at cycle 1 would have read zero under any of them.

Cycle 0 accepts 81,920 repairs and leaves 20,495 seams in disagreement. Cycle 1 accepts the remaining 20,495 and leaves nothing. Of 122,880 seams, every single one agrees, and the trace runs on to cycle 15 without another mismatch posted against it.

Each repair removes exactly one disagreement, which is the librarians' shelf at the scale of a network, and the number of moves the run accepts is 102,415, the number of disagreements it began with. Neither figure has anything in it about the order the repairs came in.

10.6 Sixteen shuffles

The last line of one trace is the last line of one trace. What settles the question is running the same arrangement sixteen times and shuffling the order before every pass.

All sixteen accept 102,415 moves. All sixteen finish with no seam in disagreement. All sixteen arrive at the same finishing state, and the check on that last point is a fingerprint: the whole terminal state is boiled down to one line of characters, and the sixteen lines are the same line.

Sixteen orders. One destination.

The same replay takes 512 places where two repairs share a patch and confirms that the two ways around the diamond meet, takes 512 places where two repairs have nothing in common and confirms that they commute, and relabels the local frames sixteen times over to confirm that nothing in the outcome was resting on the labels. None of those checks turns up a violation.

The records arrive after the arguing stops. Nothing is committed through cycle 6. Cycle 7 posts 47,412 of them, a little under three in five. Cycle 8 posts the rest, 81,920, one for every patch in the network, and from there to the end of the run nothing changes at all.

The spread of those records at the finish is 11.3134. The natural logarithm of 81,920 is 11.3135, and the whole of the gap between those two figures is the price of four coincidences: four pairs of patches hold matching readings, and each of the other 81,912 holds a reading that no other patch in the network holds. Spread counts how many different things a collection holds, on a scale where doubling the number of distinct items adds a fixed amount instead of doubling the figure.

10.7 Agreement first

Records lag agreement by six cycles, and they lag it because of what a record is. A record is something you can go back to and find unchanged. Write one into a neighborhood whose seams disagree, and a repair comes through and revises what you wrote, which makes what you wrote a draft.

The run puts a number on that. A patch's reading becomes a record when it has held still for eight cycles together. Anything in the middle of being repaired fails that test, because being repaired is what failing to hold still consists of.

Read the rule closely and notice what it does not say. It does not tell a patch to look at its neighborhood, or to wait for anybody, or to ask whether the argument nearby has finished. It asks one question, and the question is about the patch itself: has this reading stopped changing. The neighborhood enters only through the back door, because a patch whose seams are in dispute is a patch whose reading keeps getting revised out from under it.

Agreement comes first and memory second. The order is forced by what a record is rather than by any instruction to observe it, and it runs the opposite way to the picture most people carry, in which observers write down what they see and then compare notes and argue about the differences. Through six cycles of this run, every patch in the network is in the middle of a disagreement or fresh out of one, and not one of them has written anything down, and there is nothing anywhere to compare.

10.8 The last line

The last line of the trace reads like this. One hundred and twenty-two thousand eight hundred and eighty seams, every one of them in agreement. Eighty-one thousand nine hundred and twenty patches, each holding a record of its own. Whatever order the repairs ran in, that is where the network lands, and Newman's two conditions are the reason.

Everything settled. Nothing that settled is a number any two patches share. Each patch holds its own reading, taken through its own ports, of the piece of the world it can see. Walk up to two neighboring patches at the end of that run and ask each of them what the world is like, and you get two different answers from two observers in complete agreement.

The network has a settled state. It is written down in none of the eighty-one thousand nine hundred and twenty places where things get written down.

Chapter 11

What Does Everybody Actually End Up Holding?

Elizabeth Loftus and John Palmer sat forty-five students down at the University of Washington in 1974 and showed them seven short films of traffic accidents. Afterward each student was asked how fast the cars had been going. It was the same question every time except for one word. Some were asked how fast the cars were going when they smashed into each other, and the others got collided, bumped, hit, or contacted.

Everybody saw the same films. The smashed group put the cars at 40.8 miles an hour on average, the hit group at 34.0, and the contacted group at 31.8.

A second experiment used a hundred and fifty students and one film of a multiple-car crash. Fifty of them were asked about smashing, fifty about hitting, and fifty were asked nothing about speed at all. A week later, all of them were asked whether they had seen any broken glass. Sixteen of the fifty in the smashed group said they had, against seven in the hit group and six in the group that had never been asked about speed. There is no broken glass in the film.

Loftus and Palmer called the paper “Reconstruction of Automobile Destruction”. Nobody in those rooms was lying, and nobody was playing back a stored recording. Each of them was assembling an account out of the material available, and the material available included the verb.

Courts worked this out considerably earlier and built an institution around it. A trial does not ask what is inside a witness’s head, partly because there is no instrument for getting at it and partly because the contents turn out to depend on how you ask. What a trial produces instead is a record: the words actually spoken in the room, written down as they are spoken by somebody employed to write down words. Exhibits go into it. Private recollection does not, and neither does anything a juror happens to know from outside. Continental procedure has a maxim for the arrangement. Quod non est in actis, non est in mundo. What is not in the record is not in the world.

The record is a physical object made by a person under load. The Registered Professional Reporter certification, from the National Court Reporters Association, requires taking down two speakers at 225 words a minute with 95 percent accuracy, on a keyboard with 22 keys, which is fewer keys than there are letters and is the reason the output is phonetic rather than alphabetical. Everything a court later does with the case runs through what came off those 22 keys.

No authority outside the room supplies the verdict. Twelve people who sat through the trial compute it from the record. When they cannot agree about what a witness said they ask for the transcript and it is read back to them.

Chapter ten put eighty-two thousand patches through the repair process and the last of their 102,415 disagreements went at cycle 1, whatever order the repairs ran in. That settles the schedule. It says nothing about what any patch is holding at cycle 2. Each finished the run holding its own state, visible to nobody else, connected to the rest through what crosses its seams. So what do they end up holding in common, and how much of the world does it pin down?

11.1 The list everybody can check

When Alice and Bob met in the corridor with their coins, they wrote down one word, same or different, and never which way either coin was facing. That was the only restriction in the whole arrangement. Two observers cannot compare what only one of them holds; they can compare the relation between them. The relation is what gets written down.

Collect every one of those written comparisons into a single list. That list is what anybody can check, in the sense that any observer can walk up to the seam it names and find out whether the entry is right. Everything else in the network is somebody's private state.

The **record map** takes the full state of the network and returns the part of it that survives comparison: every seam entry, and nothing that only one observer could have known. Two observers who disagree about anything are disagreeing about an entry, because an entry is the whole of what either of them can put on the table.

What the record map deletes is real. Alice's coin is a feature of Alice, she can look at it whenever she likes, and it does not appear in the list because there is no procedure by which she and anybody else could jointly establish which way it faces. She has been deleting it at every meeting since the corridor, and so has everybody she has ever compared with.

A state of the network is **consistent** when every entry on that list is true of the patches it names. Twelve coins and thirty written records: consistent means all thirty records say what the pairs of coins they name actually do. The collection of consistent states is the **consistent set**.

Chapter six counted that set for the icosahedral net. Thirty records that pass the loop test, twelve coins, four thousand and ninety-six ways to set them, and exactly one setting satisfies every record, along with the setting you get by turning all twelve coins over at once.

11.2 Three verdicts

Fix a list. Ask which consistent states carry it.

The answer is a set: all the ways the world could be that would produce exactly this record and no other. In court it is the set of stories the testimony fits. The prosecution has one of them in mind, the defense usually has another, and the size of the set is what the argument is about. The set of consistent states carrying a given reading is called the **fiber** of that reading.

Wire three patches to each other and to nobody else. Three coins, three seams carrying one entry each, and eight ways to set the coins.

Read all three entries as same. Heads all round satisfies them and tails all round satisfies them, and those two differ by turning over every coin in the world, which no entry sees, so they count once. One world.

Read all three entries as different instead. Chapter six wrote different as 1, added the three of them in the arithmetic where one and one make zero, got one rather than zero, and no setting of three coins survives a loop that fails to close.

Let the third patch go quiet. Its two seams go unread, the record shrinks to the single entry between the first two, and that entry reads same. Count this one yourself. Four of the eight settings have the first two coins matching: heads heads heads, heads heads tails, tails tails heads, tails tails tails. All four satisfy the record, because the record has nothing to say about the third coin. Pair each setting with its all-over flip, the way the first reading paired heads all round with tails all round, and the four collapse to two: one world in which the third coin agrees with the other two, one in which it differs. The record names both and picks neither.

Three readings, one question put to each, and the fibers came back holding one world, none, and two. One recipe produced all three counts. It takes three letters to write down. Write r for a reading, s for a state of the network, and R for the record map that strips a state down to its entries. The fiber of r is then every consistent s whose R(s) comes out r: keep the states that satisfy the entries, discard the ones whose entries read differently, count what is left over. Fix a map, name one of its outputs, collect the inputs that land on it, and that operation is called taking a preimage. A fiber is a preimage.

A fiber can have three sizes, counting a state and its all-over flip as one. The three sizes are three different situations rather than three points on a scale.

It can be empty. No consistent state produces this record, so the testimony describes nothing that could have happened. No amount of further questioning will produce a story that fits. The reading is **unrealizable**.

It can hold exactly one. The record pins down the world completely. Any two observers who hold the record hold the same world whether or not they ever compare anything else. The reading is **reconstructing**.

It can hold several. Every state in it satisfies every record. Nothing anybody wrote distinguishes them. The reading is **ambiguous**. This is the fiber a jury is looking at when it acquits somebody it privately suspects. The acquittal is the correct output: an ambiguous fiber has more than one member and the record does not name one.

Advocates argue about which of the three they are in. The defense wants the fiber shown to hold more than one member, and needs only one alternative story that fits every piece of testimony. The prosecution wants it shown to hold exactly one. A witness caught in an unrealizable reading has produced an account that no arrangement of the world satisfies, which is a result rather than a doubt.

Only the middle case convicts.

The twelve-patch net supplies all three. Its consistent record list is reconstructing: two coin settings satisfy it, they differ by flipping every coin in the world, and no meeting anyone could arrange would tell them apart. Flip one edge of the record, changing a single entry out of thirty from same to different, and the reading goes unrealizable: two of the nineteen independent loops carry a residue, and the number of coin settings satisfying all thirty records drops from two to zero.

For the ambiguous case, take away a patch's ability to talk. Its five seams go unread, so the record shrinks to twenty-five entries. The eleven patches that go on comparing are pinned exactly as before, and the silent patch's coin is not pinned at all: four settings of the twelve coins satisfy every record that exists, in two pairs that differ by the all-over flip. The reading leaves one bit undetermined. The bit has an address.

Repair cannot fix that. Repair works by reducing disagreement. There is no disagreement anywhere: every record is satisfied. A rule that picked one of the two would be picking on grounds no record contains, which makes it a choice rather than a repair.

11.3 Four demands

Run the machine on the flipped-edge record anyway. Chapter six did, from all two thousand and forty-eight distinguishable starting states and under all sixteen schedules. It converged every time onto one terminal arrangement with a leftover of exactly one. The machine halts. It halts in the same place from everywhere. And the state it halts in violates a record, which is to say that halting and being consistent are two separate properties. The second one has to be checked by somebody.

The check is a procedure with a four-line specification. Take any state at all, read off its record, and return the consistent state that record names. It has to meet four demands. What comes back carries the record it was handed, since a procedure that revised the testimony on its way past would be answering a different question. A state that was consistent to begin with comes back untouched. Two states with the same record come back with the same answer. And when no consistent state carries the record at all, what comes back is a failure value, written bottom, rather than an answer. There is exactly one procedure meeting all four. It exists precisely when no reading is ambiguous.

The third demand is the one doing the work, and the one an ordinary repair procedure fails. Returning the same answer for two states with the same record forbids the procedure from consulting anything the record does not contain, including which state it was handed and which repairs it happened to run. Hand it an ambiguous reading and no procedure meets all four, because the third requires one answer and the second requires two.

Bottom is a proof that the fiber is empty, which is not what a search reports when it runs out of patience. On the twelve-patch net you can hold the proof in your hand. Add up the entries around each of the nineteen loops in the arithmetic where one and one make zero. Two of the sums come out one. A loop summing to one demands that some coin differ from itself. Nineteen sums settle four thousand and ninety-six settings without a coin being set, and they name the two loops the obstruction sits in, which no amount of running through the settings would.

A normal form in chapter nine's sense was what a process stops at, which made it a statement about the process. This one is a statement about the reading: the settled state is determined by the record rather than by the path, so it is an **observation-determined normal form**. Two observers who agree on the record agree on the world. They can be handed the record by anybody, in any order, having watched none of the repairs.

Which makes the settling two functions run one after the other. Write N for the whole procedure, the one handed a state and returning the world it settles into. The third demand says N consults what it was handed only through the record, so it comes apart into two steps:

$$N(s) \;=\; \mathrm{world}\big(R(s)\big)$$

Here world is the lookup that takes a reading to the consistent state that reading names, and returns bottom when the fiber is empty. The state s appears on the right inside the record map's brackets. The left-hand side reaches it through nothing else. Two networks holding different coins and writing down the same thirty entries settle into one world, and whatever separated them at the start is a difference neither can carry forward, because carrying it forward would take an entry, and the entries match.

Nineteen sums decided the twelve-patch case because its constraints are linear. For constraint systems at large, deciding whether any consistent state carries a given reading is NP-complete: hand somebody a proposed state and they can check it against the reading in seconds, and no known method of producing one beats searching. The ambiguity question comes out the same shape. Two consistent states sharing one record can be put in front of anybody in a moment, and ruling out every such pair is the expensive half. The record determines the world. Working out which world is a second problem, its cost set by the shape of the constraints rather than by the length of the record.

11.4 Flip all twelve coins

Take the settled state of the twelve-patch net and turn every coin over. Every same stays same. Every different stays different. All thirty records are untouched, so the record map returns exactly what it returned before, and the two states are indistinguishable to every observer in the net, to every observer who joins later, and to any sequence of meetings anybody cares to arrange. That is why chapter six counted two thousand and forty-eight states rather than four thousand and ninety-six: the two members of each pair are one thing.

The silent patch's undetermined bit looks similar and is not. Flipping that coin also leaves every existing record alone, but the moment the patch's seams come back into use, somebody writes down a comparison that comes out differently. It is a difference this record fails to see. The all-over flip is a difference nothing could see, under any arrangement of comparisons, ever.

Changes nothing could see form a group, in chapter five's sense: doing two of them in a row is another one, doing nothing is one of them, and every one of them can be undone. The group has a job here that chapter five could only describe. **The symmetry group is exactly the set of changes the record map cannot see.** On the twelve-patch net that group has two members: leave the coins alone, or turn all twelve over.

That fixes what physics is allowed to talk about. Only quantities that survive a change of description may be called physical. Chapter five arrived at that sentence by watching what two accounts of one table had in common, and the settled answer enforces it: relabel every patch's internal state however you like, run the repair, and the result is the relabeled version of the result you would have got. The answer is a statement about whole families of descriptions rather than about any one of them, which is called **presentation invariance**. It means no internal labeling choice anybody makes can reach the physics.

So the word objective has an exact referent here. It is a state. Objective reality is the observable normal form of the network: the consistent state that the shared record picks out, up to changes nothing can see. There is no further fact about the world hiding behind it, because a further fact would be a difference no record distinguishes. Those are exactly what the group deletes.

11.5 Paths and fibers

Schedule independence and reconstruction sound alike. They are statements about different objects. The first does not give the second.

Confluence is a property of paths. It takes one starting state, lets two repair sequences run out of it, and says the branches come back together. Reconstruction is a property of fibers. It takes two starting states that happen to carry the same record and says they settle to the same place. The first says nothing whatever about the second, because the first never mentions a second starting state.

The silent patch is the counterexample, small enough to check. Everything about it is confluent: every repair sequence terminates, every schedule from a given start lands in the same place, and Newman's lemma applies without amendment. Two observers who start from different coin settings and hold identical records land in different worlds anyway, because the silent coin is never touched by any repair and never recorded by anybody. Schedule independence in full, and a coin nobody can name.

Run the two together and you get a network that passes every confluence test there is and has no objective world in it. Any two observers can compare records forever without turning up a discrepancy. The thing they are converging on differs between them in a way no comparison reaches. The repairs are finished, the schedules agree, and there are two worlds.

The traffic runs the other way with a condition attached. A repair moves coins and leaves entries alone, so two branches out of one start carry that start's record all the way down, and if both branches stop, a record that pins down one consistent state hands them the same world. Schedule independence falls out, provided every state has somewhere to settle at all. Three properties, then: branches out of one state rejoin, endpoints from different states with one record agree, and every state has an endpoint to reach. The first and third leave the second free to fail, which is the silent patch. The second and third force the first.

11.6 One over n

The trichotomy counts the members of a fiber and says nothing about how far apart they are. The distance is what decides whether the answer survives a closer look.

Instruments quote a number for exactly that, called sensitivity: how much of a difference in the world it takes to move the dial by a readable amount. A thermometer whose scale marks crowd together at the top of its range reads temperatures perfectly well and separates two nearby ones badly. A buyer who checks only that each mark has its own temperature has checked the wrong thing. A reconstruction is an instrument pointed backward, from the dial to the world, and needs its sensitivity quoted in that direction.

A reading can be refined. Resolve more seams and the record gets longer, and throwing away the extra entries hands back the shorter record you started with. Stack those refinements and you have a tower. The question is whether the answers at each level are converging on anything.

Build one. At level n there are two consistent states a full unit apart, and the record separates them: it reads 0 on the first and 1 over n on the second. Two distinct readings picking out two distinct states is reconstruction, so every level of the tower reconstructs. Then watch the levels go by. The readings converge on each other, the states sit a unit apart wherever you look, and the factor by which a difference in the record gets magnified into a difference in the state is n. At level ten thousand the record separates two worlds by one part in ten thousand. The limit has one reading and two worlds.

So being determined by the record is not the same as being determined stably by the record. What separates them is the largest factor by which a difference in the record can be magnified into a difference in the state, which is the conditioning of the reconstruction, and a tower of readings has a limit worth speaking of exactly when the conditioning is bounded all the way up rather than level by level. Chapter ten's curve went to zero across all 122,880 seams in the network, and 122,880 seams is one level. The bound is what says a finer one would have gone to the same place.

This is what can make a continuum candidate stable. A limit taken through a tower with unbounded conditioning may be an artifact of whichever level you stopped at. A uniform bound, compatible refinement maps and convergence can make different solvers and orders land on one mathematical limit. They do not by themselves identify that limit with physical space or spacetime; that still needs the causal, dimensional, volume, ruler and clock attachments introduced later.

11.7 Hold the boundary fixed

An accepted repair leaves the record it was handed exactly as it found it, so a settled state carries the record it started with, and what it settles into depends on that record from the first move to the last.

The settled state is unique given the boundary data, which is the part of the record fixed from outside the region doing the repairing. Every start carrying that boundary data lands in the same world. Change the boundary data and the region settles into a different world, just as consistent, just as settled, and just as indifferent to who went first.

On the twelve-patch net the boundary data is easy to point at. Repair a region of the net and the entries on the seams that leave the region are handed to it from outside: the patches inside cannot change them, and every setting the region can settle into has to agree with them. Those entries are the region's boundary. The settled interior is a function of them. Two neighboring regions with different boundary entries settle into different interiors, and both are correct.

There is no single endpoint that the whole thing is heading toward. The terminal state is unique relative to its boundary record, and boundary records are not unique. Two different readings on the edge of a region produce two different settled interiors. Both of them are objective in the only sense the word has here, which is that every observer inside either one agrees with every other.

Which is the whole of what a jury does. It is handed a record it did not choose. The verdict follows from the record. Hand it a different record and a different verdict follows, with no less force. Twelve people, one transcript, and a verdict that belongs to the transcript rather than to any of them.

So look at what the settled world is made of. Thirty entries, each one a comparison between two patches, none of them a property of either patch on its own. The coins were the things in this world, in the sense that they were what the patches actually held, and the coins are the part that came out in the wash: two settings of them produce every entry identically, and the answer is stated about the pair rather than about either member.

Thirty entries survived the procedure. Twelve coins did not appear anywhere in the answer. The record map never once returned a coin.

Chapter 12

Why Is the World Made of Relations and Not of Things?

In the spring of 1918 Hermann Weyl sent Albert Einstein a paper arguing that the universe has no fixed unit of length. Einstein disagreed with it, presented it to the Prussian Academy of Sciences in Berlin anyway, and his objection was printed along with it, so that the proposal and the reason it could not be true went to press a few pages apart.

Let every point of space set its own unit of length, the way every workshop sets its own gauge blocks, and comparing two lengths at different places then needs a rule for carrying a unit from one to the other. Weyl's rule was the electromagnetic potential. Electricity and magnetism would be the bookkeeping of a freedom that had been sitting unexamined inside geometry, the freedom to recalibrate the ruler at every point, and the German word for setting the scale on an instrument is Eichung, calibration. The word that came back through the translation is gauge.

A workshop's gauge blocks are the object the word points at. In 1896 a Swedish machinist named Carl Edvard Johansson worked out that a set of 102 hardened steel blocks, finished flat enough that two of them cling together when wrung face to face, will combine to give any length between 0.5 and 300 millimeters in steps of a hundredth of a millimeter. He patented the set in 1901, and within twenty years it was what the world's engineering shops measured against. Every shop had its own box, every box was checked against a better box somewhere else, and Weyl's proposal was that the universe skips the last step: every point keeps its own box and no box anywhere is the master.

Einstein's objection was short. If carrying a unit of length depends on the route it takes, two identical objects sent by different routes arrive with different sizes. Atoms are objects of that kind and the light they emit is their scale marking, so hydrogen that spent a billion years in the arm of a distant galaxy would emit at frequencies displaced from hydrogen in a Berlin discharge tube by an amount depending on where it had been. Astronomy consists in large part of recognizing elements in starlight by matching their lines against laboratory samples. The lines match.

Weyl argued back for years and gave the scale geometry up in the early 1920s, and the word outlived it. In 1927 Fritz London noticed that the factor Weyl had written works exactly as intended if it is made imaginary, at which point it stops rescaling the size of anything and starts rotating the phase of a quantum wave function, which no instrument reads off. Weyl rewrote the theory in those terms in 1929, in a paper on the electron and gravitation. The phrase gauge invariance came through the rewrite with its modern meaning attached. What had to change was the quantity chosen freely at each point, which has to be one that no measurement reaches. The length of a ruler fails that condition in any laboratory with a spectrograph.

Twelve observers wired into thirty seams have a freedom of exactly that kind, for reasons with no geometry in them. Each keeps its state in its own labels, and nothing outside the twelve fixes the labeling, because there is nothing outside the twelve. Watch what the settling then produces. The repairs run, the disagreement drains out of the network, every one of the thirty seams reports itself content, and if you walk round the twelve afterwards and read off what each of them is holding, nothing in the run has pushed those numbers together. The seams converged and the labels did not. What is it that the twelve of them agree about?

12.1 Three teams, one algorithm

Three teams are handed the same specification and told to implement it. One writes Python, one writes Java, one writes Rust. When the programs are done, the only comparison anybody can perform is to feed all three the same inputs and check the outputs, because that is the only place the three implementations touch. Everything else about them, variable names, memory layout, which loop runs backwards, is invisible to that test. Each team can choose it without consulting the others.

Chapter five made a symmetry the claim that part of the description was never physical. Here the choice gets made three times, independently, by parties who never see each other's source. A symmetry with one global choice is a claim about the whole world. A symmetry whose choice can be made separately in every place is a **gauge** symmetry, and twelve observers make twelve of them.

The picture has a client in it. The client is the part to delete. Somebody outside the three teams wrote the specification and knows what the program is supposed to do. Nothing plays that role here. The input-output behavior of the twelve observers is the only thing about them that exists. Each of them wrote itself. What is left when the client goes is three implementations whose agreement has nothing to appeal to but each other, and no envelope arriving in the morning with the requirements in it.

Local choice has a price and it is charged at the seams. Two patches sharing a seam compare port readings written in two private conventions.

The comparison needs a translation. Chapter four supplied the shape of it, with two firms settling one account in two currencies and reading their exchange rate off the transaction they both recorded. Put a dictionary of that sort on every seam. The consistency condition stops being “my reading equals yours” and becomes “my reading equals this seam’s dictionary applied to yours.”

A rule that lives on an edge and translates one end into the other is called a **connection**. Chain the dictionaries around a closed loop and you come back holding a translation from your own conventions into your own conventions, which is chapter six’s holonomy, and the one thing it tells you is whether the trip returned you unchanged.

Notice where the information went. Each observer’s own labels are its private business. The thirty dictionaries are the only content the arrangement holds that anybody can check. In the large runs the freedom each patch holds is one of the six ways of permuting three labels, and every seam’s dictionary is one of the same six. If a patch decides to swap the names of two of its labels, the dictionary reading into it picks up that swap, the dictionary reading out of it picks up the reverse, and every round trip through the patch comes back with what it came back with before. Redenominate one city’s currency and each rate into and out of it changes, while the round-trip factor does not move.

12.2 One number for the whole arrangement

Take the wiring itself, and give each observer the freedom London’s rewrite left standing, an angle on a dial rather than one of six permutations. Twelve ports, thirty seams, five seams meeting at each port. A field on that graph is a number on each of the thirty seams. A relabeling is a number at each of the twelve ports, and what it does to a seam is add the difference between the numbers sitting at the seam’s two ends.

Some fields can be manufactured that way out of nothing and some cannot. Every field splits into those two parts. The part no relabeling could have manufactured is the collection of loop readings. That part is the field strength, because it is the part a measurement can reach.

The split shows up in what two observers can argue about. A seam value is something each of them wrote in private. The gap between their two numbers closes if either one changes a convention. A loop reading is the result of a round trip. No convention either of them can change will move it. Relabel one port and all five seam values touching it move, while the round trips through it come back exactly as before.

Then a source. Charge moves and the source has to move with it, so the source is a current: one number on each seam, saying how much runs along it and in which direction.

Score the arrangement with a single number: the energy held in the field strength, less the thirty products of each seam's value with the current running along it. A number scored over a whole arrangement like that is called the **action**.

Apply a relabeling and work out what the score does. The first term does nothing, since the field strength was built to ignore relabelings. The second term moves, and the amount it moves by is one line of arithmetic: for each of the twelve ports, take the number that port chose and multiply it by the imbalance in the current there, meaning what flows out along its five seams less what flows in, then add the twelve products.

Both directions of the theorem are sitting in that sum. If the current balances at all twelve ports, every product is zero and no choice of labels moves the score, whatever numbers the observers pick. If the current fails to balance at even one port, the observer at that port can pick a number that makes its product nonzero, and the score moves. Send three units out of one port along its five seams and two units back in. The imbalance there is one. Let that observer add seven to its own labels, and the score of the whole arrangement moves by seven while the eleven other products sit at zero.

The second direction is the one doing work. A current that fails to balance somewhere gives the arrangement a score that depends on conventions the twelve observers picked privately, and any one of the twelve could then move that score over lunch by rewriting its own labels, having changed nothing whatever about the world. An arrangement like that has no score at all.

Charge conservation and gauge symmetry are one statement read in two directions. Both directions came out of the same twelve products.

Look at what went into that and what came out. In went a wiring diagram of twelve ports and thirty seams, a rule saying which changes count as relabelings, and a source. Out came a restriction on what a source is allowed to be, with a conservation law as its content. Nothing about electricity was assumed anywhere in the argument. The only geometry in it is a list of which port is wired to which.

12.3 Nineteen loops

Count what the labels can reach. There are twelve dials to turn, one at each port, and turning all twelve by the same amount changes no seam, so eleven of the twelve directions do anything at all. Thirty seam values, eleven directions of pure convention, nineteen left over.

Nineteen turned up in chapter six as well, off this same wiring, out of a rule with three pieces of arithmetic in it: take the edges, subtract the dots, add one. Thirty seams, twelve ports, and thirty minus twelve plus one is nineteen.

Run the rule on two drawings small enough to check by eye. Sketch six observers in a row, which is five seams and six dots. Sketch two triangles sharing an edge, which is five lines and four dots. Do both subtractions in your head and keep the second answer.

Count the loops in the second drawing and the eye finds three: the left triangle, the right triangle, and the four-sided circuit around the outside. Walk the left triangle clockwise and then the right one clockwise. The shared edge gets crossed once in each direction and cancels. What survives of the two trips is the circuit around the outside. The third loop is the first two performed in sequence, two of the three are independent, and the drawing gives no hint as to which two. The eye overcounted by one on a figure with five lines in it, which is the argument for doing the arithmetic on a figure with thirty.

Three drawings, one rule, and the arithmetic for all of them fits on a line.

$$5 - 6 + 1 = 0 \qquad 5 - 4 + 1 = 2 \qquad 30 - 12 + 1 = 19$$

The row of six has nothing to walk around, the two triangles carry two independent loops where the eye found three, and twelve observers wired into thirty seams carry nineteen.

Nineteen does two jobs. It counts the sums that have to come out zero before the twelve can settle on anything, and it counts how much of the wiring is physics rather than paperwork.

The two nineteens are the same nineteen. Strip every label choice out of a field and what survives is nineteen numbers, one for each independent loop, and nothing else about the thirty seam values survives at all. Two fields with the same nineteen loop readings are one field written down twice.

Those nineteen numbers are also the only things any two of the twelve can both compute and both check. A round trip begins and ends at one port, so the observer sitting there can work it out from the dictionaries alone, without being told what anybody else calls their own state and without taking anybody's word for anything. The dictionaries sit on the seams where both ends can read them, and the shortest of the round trips chains three.

12.4 The laziest field

Give the numbers on the seams a different job. They stop being anybody's private labeling and become the field itself. A field costs energy: square each seam's number and add the thirty of them. The law tying charge to that field reads at the ports: the net outflow along a port's five seams equals the charge sitting there. Give it a list of twelve charges and it answers with a family of fields rather than one, because adding a pure circulation around any loop leaves every port's outflow alone. The family is nineteen-dimensional, one dimension for each independent loop.

A law that answers with a nineteen-dimensional family has not finished the job. Two further conditions each pick out one member of it. They pick the same member. Exactly one field in the family has zero reading on every loop. Exactly one has the least energy. They are one field for a single reason: the energy of any solution equals the energy of that one plus the energy of the difference between the two, and the difference is pure circulation, which costs energy unless there is none of it.

The field around a charge is the solution that circulates nowhere. Every other member of the family is that field with a circulation laid on top of it, which costs energy and transports nothing. The operator that selects against it is chapter nine's repair rule, my value against the average of my neighbors, running on the twelve ports of this same graph, and the only reading that rule leaves alone is the one that is the same at every port.

On a twelve-port graph all of this is exact arithmetic rather than approximation. Put one unit of positive charge at a port and one unit of negative charge at a neighbor. Send the whole unit straight down the seam that joins them and the arrangement costs one, one squared being one. Let it settle into the field that circulates nowhere and the same two charges cost eleven thirtieths, and the nineteen thirtieths that went missing was circulation delivering nothing to either port. Eleven over thirty and nineteen over thirty, off the same wiring that gave eleven directions of convention and nineteen loops.

12.5 One hundred and two thousand three hundred and thirty-three seams

The same behavior is visible in the large runs, where nothing about it was arranged. In the run of chapter ten, 81,920 observers wired into 122,880 seams, the raw labels began by disagreeing across 102,519 seams and ended disagreeing across 102,333, while the transport-corrected disagreement began at 102,415 and ended at zero. The dictionaries themselves were never touched on the way: a repair rewrites the label at one end of a seam and leaves the translation between the ends exactly as it found it.

At the finish, 102,333 of the 122,880 seams have two ends holding different labels, and no seam in the network is in disagreement. Both counts came off the same run.

Count the other way at the end of that run. The seams whose two ends held identical labels numbered 20,547, and the seams whose dictionary was the permutation that does nothing numbered 20,547, and they were the same seams. Two ends of a seam carry the same label exactly when the translation between them happens to be the do-nothing one, which is a coincidence of conventions with no content in it. The other 102,333 seams were in perfect agreement and said so in different words.

What settles is the relation, never the values. It could not have been the values. The labels are each observer's private business, no procedure anywhere in the arrangement reads them, and a process that drove them together would be enforcing a convention that nothing can check.

12.6 The residue in the loop

Chapter six ended with twelve observers, one flipped record, and a leftover of exactly one unit of disagreement that no repair removed and no ordering avoided. Repairing in a different order moved it somewhere else and left the total at one wherever it came to rest. It sat in the loops.

The nineteen loop readings are where it lives. A relabeling cannot reach it, because relabelings leave round trips alone, the way redenominating a currency leaves the arbitrage factor where it was. Repair cannot flush it, because repair works one seam at a time and no single seam holds it. A loop that comes back carrying a residue is a loop with something inside it. No rewriting of anybody's conventions disposes of it, because the conventions have been divided out of the reading before the residue is read.

That residue is charge. The quantity a current has to conserve for the arrangement to have a score at all, the quantity the law at the ports reads as an outflow, and the quantity trapped in a loop that fails to close are one quantity. A loop carrying a residue cannot be talked away as somebody's bookkeeping, because the bookkeeping was divided out of the reading before the reading was taken, and what survives that division has to be paid for in energy and balanced at every port. Weyl proposed in 1918 that electromagnetism is the bookkeeping of a freedom chosen separately at every point. He had the wrong freedom, and the bookkeeping was right.

Thirty seams. Twelve ports, five seams at each. Eleven directions of pure convention, nineteen loops, and one unit of charge sitting in the loops where no relabeling reaches it. Every number in that list was read off a wiring diagram. Nobody has opened one of the twelve observers to see what a port is.

Chapter 13

What Is One Piece of Reality Actually Made Of?

The standard way to test a circuit board used to be a bed of nails: a fixture with a spring-loaded probe for every pad, which pressed the board down onto several hundred needles at once and read off which connections had taken solder. It worked while the connections were where a needle could touch them.

Then the boards got dense. Traces went down into inner layers, and chip packages arrived with their contacts in a grid underneath the body of the part instead of along its edges, so the pins of a soldered-down chip sat sealed in a sandwich a probe cannot enter. The nails had nowhere to land.

In 1985 a group of electronics firms formed a committee about this and named it the Joint Test Action Group. Their answer gave up on reaching the pins and asked the chip to report itself. Put a small register cell beside every pin, chain those cells into one long shift register running around the boundary of the die, and add wires that let an outsider clock the chain out a bit at a time. In 1990 the arrangement became IEEE Standard 1149.1, titled Standard Test Access Port and Boundary-Scan Architecture. The port takes four wires: a test clock, a mode select, data in, data out, with an optional fifth for reset. Behind those four wires sit several million transistors about which the standard says nothing whatsoever.

Hand somebody an unfamiliar board and the first thing they look for is the port: four or five pads in a row on a tenth-of-an-inch grid, often with no header soldered into them. Those pads say the board expects to be interrogated and on how many lines, before anybody knows what a single chip on it is.

Chapter twelve left twelve observers wired into thirty seams, nineteen independent loops, and one unit of charge sitting where no relabeling reached it. All of that came off a wiring diagram. A wiring diagram says which port is joined to which port and says nothing whatever about what sits behind either one.

Open one. The question is what a piece of the world is made of when you get all the way down. The answer comes back as six parts and six moves.

13.1 The parts list

Start with the thing that holds a state. A **patch** is a machine small enough to be described completely: a bounded amount of internal state, a set of interfaces, and a finite list of moves it is allowed to make. Bounded has been in force since the definition of an observer. A patch that could hold unlimited state would have an unlimited amount to compare. The comparison would never finish.

A **port** is one place where a patch exposes part of itself to whatever is outside it. What appears at a port is a **packet**: a reading, drawn from a finite set of possible readings, that anything on the other side can pick up. The packet is not the patch's state. It is what the state looks like from the outside at that one interface, in the way that the cell beside a pin holds what that pin is doing rather than what the die is thinking.

Suppose a port could show any value on a continuous dial. Reading it to a hundred decimal places takes a hundred digits of storage in the reader, reading it exactly takes an infinite amount, and a machine with bounded state cannot hold either. A neighbor who writes down the first four digits and drops the rest has not read that port. It has read a different, coarser port with a finite list of settings, which is the port the architecture had all along.

Two ports routed to each other make a **seam**. Chapter six drew these as edges and counted thirty of them. A seam is where two patches read the same thing twice, once from each side, the only place in the architecture where a disagreement can be detected at all.

Everything any patch can ever learn about its neighbor arrives through the seams they share. Collect all of it, every reading available at every interface an observer has, and what you are holding is that observer's **screen**. A screen is where an observer's accessible information lives, in the same sense that the boundary-scan chain is where a chip's pin states live. It shows nothing to anybody, it faces no direction, and there is no seat in front of it.

An observer assembled out of several patches has a screen too, the readings on the ports facing outward from the assembly. The ports that patches inside the assembly share with each other are internal wiring to everything beyond it. So the size of a screen is a count of readings rather than a width in meters, and asking what an observer's screen looks like is asking how many distinct things its outward ports can be doing at once.

Last, a **record**. A patch that completes a move writes the outcome into the part of its state that later moves are required to leave alone. Chapter three priced this and chapter nine identified what gets written, which is the accepted repair. A record is a write that survives the next read, and survives it because the rules of the machine protect that region of the state rather than because writing is somehow permanent.

Six words, and every one of them names something you could point at on a board: the machine, the interface, the reading at the interface, the join between two interfaces, the set of all the readings, and the part of the state that stays put.

13.2 Six moves and no others

A patch does one thing, over and over.

It reads the packets exposed on its ports. It compares them against its own, through a declared rule saying which of its readings answers which of the neighbor's, since two patches keeping their state in private conventions have to be told what corresponds to what before a comparison means anything. That rule is chapter twelve's dictionary on the seam. The technical word for whether the comparison is possible at all is commensurability. Then, if the comparison came back unequal, the patch chooses a move from a finite menu of allowed local updates, one entry off a printed card. It writes a record of the move it made. It exposes the resulting packet at its ports, so the neighbor's next read sees the new state rather than the old one. And any claim about what it did is settled by an exact check run on the exposed data, by anything that has access to that data, with the patch's own opinion of its work carrying no vote.

Read, compare, choose, write, expose, verify. That is the entire operational repertoire of a piece of reality. No move reaches a patch on the far side of the network, none of them consults a clock, and the menu they are drawn from has an end to it.

Chapter twelve's large arrangement was running that loop at 122,880 seams at once. A patch reads what its neighbor has put on their shared seam. It applies the dictionary sitting on that seam, which tells it which of the neighbor's three labels answers which of its own. Suppose they do not answer each other. The menu on that patch holds the six ways of permuting three labels, and the entry that gets taken is the one the dictionary names: the neighbor's label carried across the seam and written down at this end as its own. The dictionary is not on the menu. It sits where it was put and no move in the run rewrites it. The patch takes one entry, writes down that it took it, and puts the resulting packet back on the port for the neighbor's next read. Nothing in that sequence required knowing anything at all about the far side of the network, which is the property that lets 81,920 of these run at once with no coordination between them.

The sixth move is the one engineers recognize on sight. A patch can be wrong. It can have run its comparison against a stale packet, or chosen a move that was not available, or written a record that does not follow from what it read. What settles the matter is a check performed on the readings themselves, which either passes or does not, and which anybody holding the same data performs identically. Anything that runs that check is a verifier. The data it runs on is chapter eleven's list everybody can check, the seam entries and nothing private to either end of a seam, so a verifier needs no access to the inside of the patch whose work it settles. Arithmetic has no opinion about whose readings it is run on.

13.3 Adding up the seams

Each seam can score itself. Take the two packets, apply the seam's dictionary to one of them, and produce a number that is zero when they answer each other and positive when they do not. That number is local: it involves the two ends of one seam and nothing else in the world.

Add the scores over every seam. What comes out is one number for the whole arrangement, the total mismatch. Every accepted repair lowers it. The score is built from finitely many seams each carrying finitely many possible values, so it takes values in a finite ordered set. A quantity that steps down and has a floor runs out of room.

Chapter ten watched that number come down. Of the 122,880 seams in that run, 102,415 were scoring above zero at the start, and at cycle 1 out of 16 none of them were. The settling curve counted the seams contributing anything at all. The quantity the machine was working against was their sum.

So every sequence of repairs stops. Wherever it stops, no patch has a move available: each one has read its ports, compared, and found nothing its menu can improve. That is a local normal form, exactly as much as the descent argument buys.

Everybody stopping and everybody stopping in the same place are two different claims. The score has paid for one of them.

13.4 What the second claim costs

Three more conditions, and each is a property the interface either has or lacks.

The first is that the repair menu is complete: for every mismatch a seam is able to report, the menu contains a move that clears it. Chapter nine showed what a missing entry looks like. Twelve observers passing units across seams, and a rule that acts whenever two neighbors differ by two or more. In 302 of the 1,352,078 arrangements, no seam has a gap of more than one, no rule can fire, and the arrangement sits lopsided. Every patch content, every local check passing, and the world not settled. The gap of one had no entry on the card.

The second is about the inside of a patch rather than the seam. When a repair changes what a patch exposes at a port, the patch has to have an interior state consistent with the new reading. If some admissible boundary value has no interior that produces it, then clearing the seam breaks something inside, repairing the inside changes what the port exposes, and the two repairs chase each other around the boundary forever. The condition that rules this out is a gluing condition: every value a port can legally be asked to display extends to a consistent state of the patch behind it.

A third requirement comes out of the relabeling freedom of chapter twelve. Two repairs leaving one state have to be reconcilable, which is chapter ten's diamond, and the diamond has to close in the description with private conventions divided out. Two patches that relabeled themselves differently have landed in the same place as far as anything checkable goes, so closing the diamond in the raw labels is the wrong test in both directions: it can fail where the world agrees, and pass where the world does not.

With the menu complete, the gluing condition holding, and the diamond closing where conventions have been divided out, the terminal state of the network is the same whatever order the repairs ran in. Chapter ten proved the shape of that argument on a shelf of books and a mountain with three basins. This is what it costs in parts: a scoring rule on every seam, a printed menu with no gaps in it, and interiors that can support every boundary reading their ports allow.

Two patches on one seam gave chapter ten two endings out of one start. The score in that arrangement runs down to zero, every move taken is an entry on the card applied to a mismatch that was there, and the sixth move's check passes on every reading either patch exposed. The descent supplies all of that and leaves both endings standing. A gap in the menu and an interior that cannot support its boundary are both printed into the card and the wiring before the first repair runs.

Those three requirements are the **synchronization contract**: obligations on the parts. Any assembly that meets them synchronizes. The descent gets the machine to stop. The contract is what makes the place it stops one place rather than a place per schedule, which is to say that whether there is a single world at the end of the repairs is a question about the connector.

13.5 Four things that have to travel

A host needs maintenance, so the live machine running on it is stopped, written down and started again on another host, with the programs inside carrying on through the move. Server operators do this on a schedule. What makes it work is a decision about which parts of the machine have to travel and which parts can be left behind and rebuilt, and getting the list wrong shows up as a program that resumes into a world it does not recognize.

A machine that is interrupted and resumed is the same machine when enough of it has been put back. Enough, for an observer, is four things. Its records, meaning the writes that later moves protect. Its accessible state, meaning the state of what it can actually reach. Its interfaces, meaning which ports it has and what is routed to them. And its future law, meaning which moves and which readings its machinery allows. Those four together are a **checkpoint**.

The fourth item is the one that gets forgotten. Restore an observer's records and its accessible state into a machine whose menu of moves is different, and the restored thing reads the same history and takes different steps from it. The law that says which moves are available has to travel with the records, or the records land somewhere that treats them differently.

Two checkpoints that agree exactly on all four give the same probabilities to everything the observer subsequently sees, for every event it is able to register. If the two restored states differ slightly, the two futures differ by at most as much, which is what makes a rebuilt observer usable when the copy is imperfect.

A great deal of what a running system holds is outside that list. Worker identifiers. Queue positions. Retry counters, repair-cycle indices, timestamps, and the latency of individual packets. All of it is real, all of it is needed to replay the machine's history, and none of it is part of the observer or of the observer's time.

13.6 A code with no distance in it

Look at what the seams do to the space of configurations. Each one demands that its two packets answer each other under its dictionary. A configuration that satisfies every one of those demands at once is legal, and one that fails any of them is not. Chapter twelve's arrangement had 81,920 patches holding one of six labelings apiece, which is finitely many possibilities in the sense that writing out the number of them takes 63,747 digits, and that is before any seam dictionary is counted.

An object of that description has a standard name in engineering: it is a **constraint code**, the same kind of thing as the codes that protect a hard drive or a deep-space link, where a handful of parity checks pick the legal words out of all the possible words. This carrier is a finite constraint code.

What a code usually arrives with is a distance. Send every bit three times and the legal words are 000 and 111, which differ in three places. Corrupt one symbol and you hold 010, which is one step from 000 and two steps from 111, so the nearer legal word is the one that was sent and the error is corrected rather than merely noticed. Corrupt two symbols and 011 is nearer to 111, and the correction confidently produces the wrong bit.

The number three is the distance of that code, and the distance is what says how much the code can fix: a distance of three corrects one error, a distance of five corrects two, and a distance of two catches an error without being able to say which symbol moved.

The architecture supplies the checks and does not supply the distance. Nor does it supply a bottleneck across which information has to squeeze, a rate at which disagreement mixes through the network, or a bound on how many seconds settling takes. Each of those is a property of a particular wiring diagram, and has to be computed for that diagram, from the diagram. The six moves say which configurations are legal. How far apart two legal configurations sit depends on which port is joined to which port.

13.7 What one boundary can hold

Take a patch and ask what anything outside it can determine about it.

Everything arrives through the ports. What arrives at a port is one packet from a finite set. So two interior states that expose the same packets on every port are the same state to every neighbor, to every neighbor's neighbor, and to the network entire. They can differ internally in any way you like. The difference has nowhere to be read.

Which turns the amount that can be known about a region into a count of arrangements of its boundary. The number of values one port can show, multiplied by itself once for every port the patch has. In the runs of chapter twelve the freedom each patch held was one of the six ways of permuting three labels, so a reading in that arrangement is one of six things, and the bound is six multiplied by itself once per port.

Assemble many patches into one observer and the same argument applies to the assembly. Adding a patch deep inside adds state and adds nothing whatever to what an outsider can read, because every port it has is routed to another patch inside the assembly. The count moves only when a patch puts a port on the outside. Two assemblies with the same outward-facing ports and wildly different interiors are one object to everything beyond them, with the same list of possible readings and no way for anybody to tell.

The interior does not appear in that product anywhere. However much a patch holds inside, and however finely it holds it, what an outsider can distinguish is bounded by the readings on its boundary. This is a finite boundary-accessibility law. The ports are not square meters, and their arrangement count is not a physical entropy. Connecting it to a gravitational area law requires a realized physical screen, an entropy identification and a calibrated conversion from boundary records to area. Getting this far took a machine with a bounded interior and finitely many ports, and nothing about gravity at all.

Six, multiplied by itself once per port. The six is an accident of one arrangement's three labels. The exponent is a count of holes in a connector. The six moves do not fix it.

The connector is a closed surface, its ports have to be arranged so that every one of them sees the same thing, and the seams have to pair up with nothing left over. Impose them all at once and the arithmetic permits one count and no other.

Count the ports.

Chapter 14

Why Twelve?

In the Egyptian galleries of the Metropolitan Museum of Art there is a lump of serpentinite the size of a plum, 3.2 centimeters tall, cut into twenty triangular faces with a Greek letter carved into each one. A missionary named Chauncey Murch picked it up in Egypt at some point between 1883 and 1906, the museum bought it from his family in 1910 with money Helen Miller Gould put up, and the label allows the object itself anywhere from the second century BC to the fourth century AD. Nobody knows what it was for. The museum's own entry on it offers an alphabet oracle, a text with one line written out against each letter, where the letter that lands face up picks the line that is yours.

Whoever cut it was working to one number. The number was twenty. Twenty faces, twenty letters, twenty answers.

Count its corners. There are twelve, and five faces close around every one of them. Nobody in Ptolemaic Egypt sat down and picked twelve. Twelve is what happens when twenty triangles close up with five meeting at each corner, and any craftsman in any century who set out to make that object would have handed you the same twelve whether he wanted them or not. The same object sits on shelves in a good many houses in injection-molded plastic with the numbers one to twenty on its faces. Its corners come to twelve as well.

Chapter thirteen opened one piece of the world and found a boundary with ports on it, packets going out through the ports, seams where two patches meet, and a verifier deciding claims. The question the boundary leaves standing is how many ports it has. That count is settled the same way the twelve corners in the museum case are settled, by four conditions on how the pieces meet and some arithmetic anyone can do on paper.

The conditions are these. The boundary is cut into triangular faces, since three is the smallest number of patches whose comparisons close into a loop. Every seam has exactly two faces along it, because a seam is where two patches meet and a third would have nothing to meet across. The faces around any port close into a single ring with no loose end anywhere, because there is no edge of the world for a loose end to hang off. That ring is five faces long. And any loop of comparisons you can walk on the boundary is the rim of some collection of faces, with none left over that the faces cannot reach.

14.1 Count it twice

A theater with thirty rows of twenty-four seats has seven hundred and twenty seats, and counting the same room column by column, twenty-four columns of thirty, gives seven hundred and twenty again. The two answers agree because they are answers about the same seats. Counting one collection two different ways and setting the two results equal is what settles the port count.

The collection to count is seam-ends. Every port has five seams running out of it, so the number of seam-ends is five times the number of ports. Every seam has two ends, so the number of seam-ends is twice the number of seams. Five times the ports equals twice the seams. On the object in the museum case that reads sixty seam-ends counted corner by corner, and thirty edges with two ends apiece, which is the same sixty.

The second collection is face-sides. Every face is a triangle and has three sides, so the number of face-sides is three times the number of faces. Every seam has a face on each side of it and supplies exactly two, so the number of face-sides is twice the number of seams. Three times the faces equals twice the seams.

Two equations, and feeding one into the other leaves everything standing on the port count. If the ports number P, the seams number five P over two and the faces number five P over three. Both of those have to come out as whole numbers, since half a seam is not a thing the boundary can have, so P is even and P is divisible by three. Before any question of shape arises, the number of ports is a multiple of six.

14.2 Nineteen loops, twenty faces

Chapter six counted the independent loops in a wiring diagram of ports and seams. Lay a route that reaches every port and never closes on itself, which uses exactly one seam fewer than there are ports, and then put back the seams the route did not use: each one closes exactly one new loop. Seams, minus ports, plus one. On the twelve-port wiring that came to nineteen, and chapter twelve found the same nineteen again from the other end, as the nineteen loop readings that survive when every observer's private labeling is divided out.

Count the faces of the same object. Twenty.

Nineteen and twenty sit one apart. The reason they do is the entire content of the word spherical.

Take the twenty faces and add up their boundaries in chapter six's arithmetic, where two of anything cancel. Each seam lies along exactly two faces, so each seam gets counted twice and drops out, and the grand total is nothing at all. So the twenty face boundaries carry exactly one relation between them: any nineteen of them determine the twentieth, and no smaller collection of faces adds up to nothing, since a collection that stopped short would have a seam with a face on one side of it and nothing on the other, and that seam has nothing to cancel against. Twenty faces supply nineteen independent loops.

The wiring has nineteen independent loops. The faces supply nineteen independent loops. Those two nineteens are the same nineteen. Every closed walk across the ports is a sum of triangles. There is nothing else in the boundary for a loop to be.

That is what being a sphere consists of, checkable from inside the object without stepping back to look at it. Cut the surface of a doughnut into triangles and two loops on it are the boundary of nothing at all: the loop that goes around the hole and the loop that goes through it. Take any collection of triangles on the doughnut and its boundary misses both of them. On a doughnut the faces fall two short of the loops. The shortfall runs two for every handle.

So for a boundary that closes into a sphere, faces minus one equals seams minus ports plus one. Move the terms about and the same statement reads: ports, minus seams, plus faces, equals two. Cut a sphere into a million triangles of wildly unequal size and the combination comes out two. Cut a doughnut any way at all and it comes out zero.

Each of the three counts on its own depends entirely on how the cutting was done. Drop a new port into the middle of one face and join it to that face's three corners: the ports go up by one, the seams by three, the faces by two, and one minus three plus two is nothing. The combination sits where it sat. That is an invariant in chapter five's sense, a quantity left alone by a family of changes. The family here is every way of cutting a surface into triangles. What survives all of them is one small whole number that counts holes. The name it goes by is the Euler characteristic of the surface.

14.3 Twelve, thirty, twenty

Two independent relations on the same three counts, and they meet. Take the port counts the whole-number test left standing, fill in the seams and the faces from the two double counts, and do ports minus seams plus faces on paper. Six ports carry fifteen seams and ten faces, and six minus fifteen plus ten is one. Twelve ports carry thirty seams and twenty faces, and twelve minus thirty plus twenty is two. Eighteen carry forty-five and thirty, and eighteen minus forty-five plus thirty is three. One, two and three, against six, twelve and eighteen.

The letters say the same thing and say why it happens: ports minus seams plus faces is P minus five P over two plus five P over three, which over a common denominator of six is six P minus fifteen P plus ten P, all over six, and that is P over six. The counting alone, with nothing about shape anywhere in it, ties the port count to the one combination a sphere pins down.

$$\text{ports} - \text{seams} + \text{faces} = \frac{\text{ports}}{6}$$

The left side is two for anything that closes into a sphere, so the right side is two, and the ports number twelve. Feed twelve back through the two double counts: the seams number thirty and the faces number twenty. That is the icosahedron chapters six through twelve have been running on, arrived at from four conditions and a page of arithmetic.

Run the argument backwards. Twelve ports with five faces around each of them make ports minus seams plus faces equal to a sixth of twelve, which is two, and among closed surfaces the value two belongs to the sphere alone. Two is the largest value any closed surface reaches. The only other positive value is one, belonging to a one-sided surface that cannot sit in ordinary space without cutting through itself. Twelve ports force a sphere, and a sphere forces twelve ports.

14.4 Four around a point, six around a point

Every condition in that argument does work. Three of the four can be broken and leave a boundary behind. The fourth cannot: put a third face along a seam and the seam has no two sides, so there is nothing left to run a count on. Break the other three one at a time and watch the twelve go.

Change the ring around a port from five faces to four. Run the identical double counting: four times the ports equals twice the seams, three times the faces equals twice the seams, and ports minus seams plus faces comes out at a third of the ports. A sphere sets that combination to two, so the ports number six. That object exists, has six ports, twelve seams and eight faces, and sits in every mineralogy display case as an octahedron, the shape a fluorite crystal cleaves into. Three faces around a port gives four ports, which is the shape of the smallest die in the box.

Six faces around a port drops the port count out of the arithmetic entirely. The double counting gives six times the ports equals twice the seams, and the combination collapses to zero times the ports, which is zero however many ports there are. Zero is the doughnut's number and the sphere's is two. Six triangles around every point is the flat triangular grid, which runs across a tabletop forever and closes up into a doughnut of any size you like, and never once closes into a sphere.

Break the spherical condition instead. Take the twenty-faced object and glue every point of it to the point diametrically opposite, so that walking off one side brings you back on the other. Ports pair up and there are six of them left, seams pair up and there are fifteen, faces pair up and there are ten. Six minus fifteen plus ten is one. Five faces ring every port. The surface has one side and cannot be a sphere. Its port count is six, and a sixth of six is one.

Break the triangles. Pick up the six-sided die from the same shelf and try to find a triangle in its wiring: start at a corner, walk along an edge, walk along another, and see whether a third edge takes you home. It never does. On a cube, no two neighbors of a corner are neighbors of each other, so there is no face anywhere in it whose three corners are pairwise joined. A cube does not fail the count. It fails to admit the structure at all. No amount of relabeling its eight corners repairs that.

14.5 Sixty rotations

Chapter five had you cut a paper square and find every way of putting it back on its own outline, and eight came out. Do the same thing to the twenty-faced object, turning it without ever reflecting it, and count the ways it lands on itself.

Every one of those turns leaves the object standing where it stood and shuffles its parts underneath. A turn carries every port to a port, every seam to a seam and every face to a face, and a collection of moves that shuffles a set of things among themselves in that way is a **group action**. One collection of turns is acting on three different sets here. The same sixty turns are doing all three at once. Chapter five also left a word for the set of places a single thing can be sent to under such an action, which is its orbit. Every port here can be carried to every other, so the orbit of a port is all twelve. Ask the opposite question. Which turns leave one chosen port sitting exactly where it is? Only the ones that spin the object about the line through that port, and since five faces ring it, five such turns bring the ring onto itself, counting the do-nothing turn. The set of moves that leave a chosen thing where it is is its **stabilizer**. The stabilizer of a port has five members.

Every turn in the whole collection is accounted for exactly once: send the chosen port to one of its twelve destinations, then spin about the destination in one of five ways. Twelve times five is sixty. That multiplication is the general fact, and works from any starting point: the size of an orbit times the size of a stabilizer is the size of the group.

Which means the count can be run backwards through any feature of the object, and run with the object in your hand.

Pinch a twenty-sided die between finger and thumb at the centers of two opposite faces and turn it about that pinch. A third of the way round it fills its own outline again, because the triangle under your thumb has three corners and takes each of them in turn, which is three positions in a full turn counting the one you started in. Move the pinch to two opposite corners, where five triangles meet under each fingertip, and turn it again. Five positions. Take the awkward grip, at the midpoints of two opposite edges, and there are two, the second being the half turn about the line through those midpoints.

Three, five and two, read off with a hand rather than argued. Every face can be carried to every other face by some turn in the collection, and every edge to every other edge, so each of those three numbers is a stabilizer sitting under a full orbit. Multiply each one by the number of things of its kind on the solid.

$$20 \times 3 = 12 \times 5 = 30 \times 2 = 60$$

Twenty faces held three ways apiece, twelve corners held five ways, thirty edges held two ways. Three counts that share nothing except the lump they were taken from, and one answer. The solid reports the size of its own symmetry three times over. The port count comes back out of it as sixty divided by five.

Nobody picked those sixty turns. Four conditions on how comparisons meet forced twelve ports, and the twelve came with sixty turns attached, as much a consequence of the conditions as the corner count is. The symmetry belongs to the boundary the way the corner count belongs to it, arrived at rather than assumed, and every rule that can be written on such a boundary has to come back to itself under all sixty turns. Most rules do not.

Sixty moves carrying twelve ports around means no port is distinguished from any other by anything the wiring holds. Any port can be sent to any other, and having sent it there, its ring of five neighbors can be spun into any of five positions, so a description of the boundary that starts at one port and works outward could have started anywhere and come out the same. Two patches comparing readings through a seam are comparing quantities of the same kind. No calibration step anywhere in the machine had to establish it.

Fifteen of the sixty are half turns, one for each pair of opposite seams. They fall into five families of three, each family being three half turns about three lines at right angles to each other, and every turn in the group shuffles those five families among themselves. Follow the shuffling and you get sixty permutations of five objects. That is the group behind chapter five's fifth-degree result, the one whose shape rules out a solution formula of the sort the quadratic has, and it is sitting inside a gaming die.

14.6 Ten thousand three hundred and ninety-five ways to pair up twelve things

Twelve ports. Nothing in a list of twelve labels says which of them has anything to do with which.

Pair them up, six pairs with every port used once. Take the first port and choose its partner from the remaining eleven. Take the lowest unpaired port and choose from the nine left. Then seven, then five, then three, then one. Eleven times nine times seven times five times three is ten thousand three hundred and ninety-five. That is how many pairings twelve labeled things admit. Every one of them is as good as every other, so long as the twelve are only labels.

Put the wiring back and all but one of the pairings dies. Start at any port and count how far the others are, in seams, along the shortest route. Five ports are one seam away, which is the ring around it. Five more are two seams away. That accounts for eleven of the twelve, counting the port you started from. The twelfth is three seams off. Chapter nine leaned on that three when it put a ceiling on how far news has to travel, and every port has exactly one port at distance three, never two.

One pairing out of ten thousand three hundred and ninety-five survives that, the one matching each port with the single port farthest from it. It uses every port once and gives six axes.

The same six pairs can be found without counting a single distance. Take any two ports and count the ports adjacent to both of them. Sixty of the sixty-six pairs come back with two ports in common. The other six come back with none, the six axes again, each one a pair of ports that share no neighbor anywhere in the wiring.

Any relabeling that preserves the wiring preserves distances, so it carries the far port of a port to the far port of its image. None of the sixty turns can scramble the pairing. Count the group against the axes and the arithmetic closes again: six axes, and the moves holding one axis in place number ten.

The pairing itself is a move, sending every port to the port opposite. Include it and the moves that preserve the wiring number a hundred and twenty rather than sixty, the extra sixty being mirror moves that no amount of turning the solid in your hands reproduces. That single move commutes with every other one in the collection: whatever else happens to the ports, opposite stays opposite.

Pick up a modern twenty-sided die and check what the manufacturer did with the numbers. One is opposite twenty, two is opposite nineteen, three is opposite eighteen, and every pair of opposite faces sums to twenty-one. The reason is mechanical. A die pressed slightly flat on one pair of faces lands on those two more often than it should, and if the pair reads nineteen and twenty, the die runs high. Pairs summing to twenty-one cancel that bias. The numbering is a decision. The pairing it uses is the only one the solid offers.

14.7 Thirteen ports

A boundary with thirteen ports on it, each with five seams running out, would have sixty-five seam-ends. Seams have two ends. Sixty-five is odd. There is no such object, in this world or any other, for reasons that never get as far as physics. Fourteen ports pass that test and fail the other one: they give thirty-five seams, thirty-five seams give seventy face-sides, and seventy face-sides will not divide into triangles. Between them the two double counts dispose of every count that is not a multiple of six. The multiples of six need the surface to settle them. Eighteen ports would need ports minus seams plus faces to come out at three, twenty-four would need four, and no closed surface reaches either, since two is the ceiling and the sphere owns it. Six ports would need one, and one belongs to the surface you get by gluing every point to the point opposite, where a port and its far partner are the same port and a patch carried once around comes back mirrored. Twelve is the whole of what the conditions permit.

The one line working physicists write down when somebody asks what the world is made of carries somewhere between nineteen and twenty-six numbers that were measured in laboratories and entered by hand. The number of particle families in it is three because three is how many the detectors reported. Twelve came by the other route. Four conditions on how comparisons meet, every one of them checkable by something sitting inside the boundary with no view of it, leave exactly one count standing, and the count is twelve, with thirty seams and twenty faces attached.

None of which puts a small solid anywhere. The object in the case in New York is 3.2 centimeters of serpentinite lying on a shelf in a gallery. A screen is where an observer's accessible information lives. Being spherical was a statement about loops closing. An observer checks it by counting its own comparisons without ever standing outside anything. On the declared operational-cost branch, twelve readings build an exact rank-three candidate position carrier. Identifying that carrier with the three physical directions of the gallery requires the source-causal spacetime attachment. The count that produced the twelve used no direction anywhere in it.

Twelve readings, one per port, is everything one piece of the world exposes about itself.

Twelve numbers. Their slow-mode candidate position carrier has exactly rank three. On a source-selected physical refinement, that carrier is the proposed origin of the three spatial directions an observer sees; the finite rank theorem alone does not make it physical space. Every physics ever written has taken the observed three as a given and moved on.

Chapter 15

Why Does the Position Carrier Have Three Directions?

In September 1879 a man left his house in Essex carrying a rack of brass resonators and walked to the parish church to hit a bell.

The house was Terling Place, where John William Strutt, third Baron Rayleigh, had a wing fitted out as a laboratory. The bell was the second of the five in the village tower, counting down from the highest, cast by a founder named Gardiner in 1723. The brass was a set of Helmholtz resonators made in Paris by Rudolph Koenig, who had been apprenticed to the violin maker Vuillaume in 1851, set up on his own account in 1858, and spent the next forty years turning Helmholtz's ideas into apparatus a physicist could order by post. Each resonator is tuned to a single frequency. Hold one to your ear inside a noise and that frequency comes up out of the mess loud enough to name, while everything else drops away. Rayleigh's verdict on them, printed a decade later, was that without the security the resonators afford, deciding which octave a bell is sounding in is very uncertain.

He published in January 1890, in the *Philosophical Magazine*, under the title "On Bells". Much of the paper is a list of what he heard, bell by bell, with the founder and the casting date beside each one, going back to a Terling bell cast by Graye in 1623.

One entry is a hemispherical bell of about three hundredweight, a little over 150 kilograms of bronze, lent to him by the London founders Mears and Stainbank. Four tones came out of it plainly and he named all four. Then this: "The gravest tone has a long duration. When the bell is struck by a hard body, the higher tones are at first predominant, but after a time they die away, and leave e-flat in possession of the field."

For a fraction of a second the bell has four voices. The loudest of them are the high ones. After that it has one. Which of those two descriptions is the bell depends on how fast the listener is. Give the job to an instrument that cannot register anything shorter than a few seconds, and the four voices never happen. It reports one note, the lowest, and it reports that note as the whole bell.

You can walk forward, walk sideways, and climb a ladder, and any motion you are capable of is some combination of those three. Three numbers pin a body down and two will not: an air traffic controller is given latitude, longitude and altitude, and losing any one of the three leaves a whole line of places the aircraft could be sitting. A fourth independent number has never been available to anybody. General relativity generalizes the shape of space in every respect except the count of directions, which it takes as a starting datum. Nothing in its field equations prefers three to five or to ten. The same equations take it for granted that the three run smooth, with a place between any two places and no smallest step anywhere in them.

The slow band gives an abstract continuous three-dimensional Euclidean carrier under the stated repair rule and response metric. Its integer records can populate finite regions at a specified resolution. A supplied rule for interactions between nearby records then supports a scalar field whose motion approaches an ordinary continuum wave equation, with an explicit error bound. This construction needs no inserted Cartesian grid. Its population rule and action remain assumptions, and its model time needs a physical interpretation. Chapter seven's authenticated event order is a separate object; matching records to physical signals requires more than adjoining a time axis. The pages below derive the carrier and explain what finite resolution can accomplish.

15.1 Twelve readings and a grid of weights

Chapter fourteen left the carrier with twelve ports, thirty seams, five seams meeting at each port, six axes of paired ports, and sixty rotations carrying the whole arrangement onto itself. Chapter nine left a rule running on it: each port's reading becomes itself, less a sixtieth of the total gap between it and the five readings around it. Sixty because there are thirty seams and a midpoint move closes half of a gap rather than all of it.

Write the twelve readings as a column of twelve numbers, top to bottom, one line per port. A column like that is a **vector**, and the word carries no more than that here: an ordered list, kept in order so that the third entry always means the third port.

The rule turns one such column into another. Look at what a single output line is made of. Port three's new reading is eleven twelfths of port three's old reading, plus a sixtieth of each of its five neighbors' old readings. Every output line has that shape, with different neighbors in it. Each new number is a weighted sum of the old numbers, and the weights can be laid out in a square grid: twelve rows for the twelve outputs, twelve columns for the twelve inputs, and in row three, the number eleven twelfths in column three, a sixtieth in the five columns belonging to its neighbors, and zero in the other six. A square grid of weights used this way is a **matrix**. This one has 144 entries in it, of which 72 are zero, and eleven twelfths plus five sixtieths is one, so every row of it adds to one and so does every column. That is the arithmetic statement of something chapter nine established by a different route: repair moves the total around the twelve observers without creating or destroying any of it.

Run it once on something. Put 12 units at one port and nothing anywhere else. That port keeps eleven twelfths of what it had, or 11, and each of its five neighbors picks up a sixtieth of 12, which is 0.2. Eleven plus five lots of 0.2 is 12, so the total sat where it was, and the gap between that port and each of its neighbors shrank from 12 to 10.8. That is one pass.

The question the bell asks is what the grid does after many passes. That means applying it a hundred times, which by hand means a hundred rounds of multiplying and adding across 144 entries, and after all of it a column of twelve numbers that has to be interpreted.

15.2 Two observers and a number raised to a power

Alice holds 10 and Bob holds 0. The only rule in operation is that each of them moves a tenth of the way toward the other.

One pass: Alice holds 9, Bob holds 1. Second pass: 8.2 and 1.8. Third: 7.56 and 2.44. Do the fourth on paper. Nine tenths of 7.56 plus a tenth of 2.44 gives Alice 7.048, and the same sum with the tenths swapped gives Bob 2.952. The numbers are unpleasant and getting worse, and their sum is 10 at every stage, and their difference goes 10, then 8, then 6.4, then 5.12, then the 4.096 you just produced.

The arithmetic behind that is two lines. Let Alice hold A and Bob hold B. After one pass Alice holds nine tenths of A plus a tenth of B, and Bob holds a tenth of A plus nine tenths of B. Add those two expressions together and the tenths cancel, leaving A plus B, which is what the pair started with. Subtract the second from the first and the tenths reinforce, leaving eight tenths of A minus B.

Look at what the rule did to those two combinations. It multiplied the sum by 1 and the difference by 0.8, exactly, and it will do the same on the next pass and on every pass after that, from any starting values whatever, because the arithmetic that does it never refers to what the numbers are. The whole two-by-two grid, read in those two combinations, is two ordinary numbers.

A direction that a grid of weights merely scales, without swinging it round to point somewhere else, is called an **eigenvector**, and the number it gets scaled by is its **eigenvalue**. The prefix is the German for “own”, and these are the grid’s own directions, the ones it was built out of whether anybody noticed or not.

What is left of the gap between Alice and Bob after a hundred passes is 0.8 raised to the hundredth power of what it was, which is one button on a calculator rather than a hundred rounds of arithmetic. Ten passes leave 0.8 to the tenth, which is 0.107, so a gap of 10 has closed to a little over 1. The same substitution works with twelve observers instead of two, on one condition.

The weight carrying port three into port seven has to equal the weight carrying port seven into port three. A grid where that holds everywhere is **symmetric**. When it holds, a full set of such directions always exists: as many of them as the grid is wide, sitting at right angles to one another, and every possible column of readings is a sum of pieces along them. That statement is the **spectral theorem**. It holds for every symmetric grid of numbers of any size. The repair grid qualifies, because a seam pushes both of its ends toward the middle by the same amount.

So the twelve-port rule is twelve numbers wearing a costume. Getting them out of the 144 by force means solving an equation of the twelfth degree.

15.3 Sixty rotations and four numbers

The wiring hands them over instead.

The rule cannot tell one port from another. It is written entirely in terms of which port shares a seam with which, and a rotation of the arrangement carries seams to seams, so rotating the readings and then repairing gives the same twelve numbers as repairing and then rotating. Either order, same answer, all sixty rotations.

That is a heavy constraint. It turns into a count. The sixty rotations shuffle the twelve readings around among themselves, and under that shuffling the twelve directions of reading-space fall into pieces: collections of directions that the rotations mix internally and never mix with anything outside. Some such pieces can be cut into smaller pieces of the same kind. Some cannot. The ones that cannot are where the argument lands. For twelve readings on this wiring the uncuttable pieces have sizes one, three, three and five, and one plus three plus three plus five is twelve.

A rule that commutes with all sixty rotations acts on each uncuttable piece as multiplication by a single number. Suppose it did something else, treating one direction inside a piece differently from another. The rotations carry the first direction to the second, so the rule would have to agree with itself about two directions it was distinguishing, and it does not get to do both. That argument is **Schur's lemma**, which Issai Schur published in Berlin in 1905 to put the character theory of finite groups on a new footing. It is chapter five's symmetry doing arithmetic on 144 unknown weights.

So a rule respecting this wiring, however elaborate its 144 entries look, has at most four distinct answers in it. One for each uncuttable piece.

The two pieces of size three are not copies of each other. The sixty rotations move the directions inside one of them along a different pattern from the other, which is why the two get separate numbers and why the count is four rather than three. Two pieces of size three went into the wiring. The declared carrier branch uses one of them.

15.4 The three rates

The piece of size one is the direction in which all twelve ports shift together. Nobody disagrees with anybody, there is nothing for a repair to close, and the rule leaves it exactly where it is: eigenvalue one, forever. It also carries no comparison, which means no observer inside the arrangement can read it, and it drops out. Eleven directions are left.

The other three numbers are the fraction of a disagreement that survives one pass of repair in each of the three remaining pieces. Two of the three are irrational.

$$\frac{55+\sqrt{5}}{60} \approx 0.9539 \qquad \frac{9}{10} = 0.9000 \qquad \frac{55-\sqrt{5}}{60} \approx 0.8794$$

Three directions keep 0.9539 of whatever they were holding, five keep exactly nine tenths, three keep 0.8794, and those three numbers, with the 1 that no observer can read, are the entire content of a twelve-by-twelve grid of weights. The 55 over 60 in the outer two is the eleven twelfths a port keeps of its own reading, and the square root of five over sixty is what its five neighbors put back or take away. Slow band, middle band and fast band, named for the rates rather than the sizes.

The square root of five is the same square root of five a reader met in chapter two, in one plus the square root of five all over two, which a calculator kept landing on from any starting number. It is here because the solid carrying the twelve ports is built out of it, and because a fraction cannot be its own reciprocal plus one. The distance between the top rate and the bottom one is the square root of five divided by thirty, which comes to 0.0745 per pass, and thirty is the number of seams. One of the two pieces of size three fades slower than the other by that margin, which is a small number to have a body in.

Both of them can be written down. Set the arrangement on a table, choose any flat surface to measure heights above, and give each port a reading equal to its own height above that surface. Those twelve numbers fade at 0.9539 a pass and at no other rate. There are three independent versions of the pattern because there are three independent directions to measure a height along. The other three-dimensional piece is the same recipe run on a different solid. Chapter two's calculator settled on 1.618034 and the quadratic behind it had a second solution, minus 0.618034, a fixed point that nothing ever arrives at. Build the twelve corners using minus

0.618034 everywhere the golden ratio stood, read the heights off those, and the patterns that come back are the ones that fade at 0.8794.

15.5 Slow ears

Strike the arrangement, meaning disturb it away from agreement in no particular direction, and the disagreement lands across all eleven directions at once. Repair works on all three bands simultaneously, at the three rates above.

Follow it. A disturbance in the fastest band is down to half in 5.4 passes. The middle band takes 6.6 passes to halve, and the slow band takes 14.7. Spread the initial disagreement evenly across the eleven directions and three elevenths of it, 27 percent, sits in the slow band at the start, which is a minority position. After ten passes the slow band holds 42 percent of what is left. After thirty passes, 73 percent. After sixty, 94 percent. After a hundred, 99.5 percent.

Those are shares of a total that is itself collapsing. After a hundred passes the slow band holds nearly nine parts in a thousand of what it was handed, the fast band holds under three parts in a million, and the ratio between them is 3,414. Nothing was tuned to produce that. Change the rates and you would have to change the wiring. The wiring is what chapter fourteen showed there is only one of.

Which is Rayleigh's hemispherical bell, with the founder's name filed off. Four voices at the strike, one voice a moment later, and e-flat left in possession of the field.

Observers here are built out of repair records. A repair record is not made in one pass. Chapter nine's graph-local dependency bound is one seam per model event, and the two most distant ports on this wiring sit three seams apart, so a round-trip comparison needs at least six seam events on the declared operational-cost branch. Six passes is longer than the fast band's half-life of 5.4, so more than half the fast content is gone before that comparison can return. This selects the slow candidate carrier band within the finite model; it is not a physical signal-speed derivation.

On the declared operational-cost branch, that settles which content can carry the candidate position readback. Such a readback is a quantity two separated observers can compare and both get the same answer for. The comparison takes rounds to run. Content that has faded to nothing before the comparison completes cannot anchor it: by the time the second observer reports, the quantity it was reporting on has relaxed away. Only the slowest band lasts long enough. Its size is three.

The five-dimensional piece is the largest one on the wiring and it loses anyway. Its content halves in 6.6 passes against the slow band's 14.7, and after sixty passes the five directions are holding about a twentieth of what the three are holding. Whatever an observer writes into those five directions has drained most of the way out before a second observer can be told about it. A quantity like that describes nothing that has a place.

That is where the carrier's three comes from. It is the number of independent directions in the one band that outlives every other band, on the declared wiring.

15.6 The table of angles

The slow band is three directions wide. Look at what the twelve ports are doing inside it.

Each port leaves a shadow in the slow band, a direction with a length. To compare two directions, take the number that says how much they point the same way: one when they coincide, zero when they are at right angles, minus one when they point exactly opposite. It is the cosine of the angle between them, the only thing about two directions that survives forgetting where they are.

Do that for all 144 pairs of port shadows and the table has exactly four different numbers in it. A port against itself gives 1. Two ports sharing a seam give one divided by the square root of five, which is 0.4472, an angle of 63.43 degrees. Two ports that are two seams apart give minus that. A port and the port directly across the arrangement from it give minus 1.

Those are the angles the twelve ports of chapter fourteen's arrangement stand at, arriving here out of a rule about disagreement rather than out of anybody having measured anything. What went into the calculation was a list of which port shares a seam with which. What came back was a solid.

The count of independent directions in a table like that is called its **rank**: how many of the twelve shadows you have to be given before the other shadows are forced. Two will not do it. Four is more than necessary. The rank of that table is three. Every entry in it is 1, or minus 1, or one over the square root of five with a sign in front, so the whole calculation runs in arithmetic built out of the square root of five and nothing anywhere in it is rounded.

That three is the slow band's size arriving a second time, since shadows lying in a three-directional band cannot supply more independent rows than the band has directions. A four-directional carrier built by this mechanism would need a slowest band of size four. The sizes available on this wiring are one, three, three and five.

15.7 Six counters

A three-dimensional carrier space in the abstract has nowhere in it to stand, and getting into it takes whole numbers.

What an observer holds about where it is, at any moment, is a tally. Every comparison it completes through a port adds one to a count. The twelve ports come in six pairs facing opposite ways, and a comparison recorded out through one port is the same event as a comparison recorded inward through the port facing the other way, so the two counts are one count with a sign on it. Twelve ports, six pairs, six running totals, and every one of them a whole number, because a comparison either completed or it did not. Nobody ever made three and a half comparisons. An observer that has completed forty comparisons out through one port and thirty-seven inward through the port facing it holds a 3 on that axis, and five more numbers like it, and that list is everything it has ever known about where it is.

Six whole numbers, dropped into a space with three directions in it. That looks like it should lose information, and on ordinary coordinates it does: six real coordinates fed through the table of angles collapse onto three. On whole numbers it loses nothing. For two different tallies to name the same point, some nonzero collection of whole numbers would have to be flattened by the table, which would make the square root of five rational. It is not. Distinct tallies name distinct points. Different histories can finish with the same tally.

The allowed conservative tallies are dense. Take any point of the three-dimensional band and any positive tolerance, and some finite signed tally lands within it. Some points are reached exactly; every point can be approximated. These are statements about the available records. Which sequences of comparisons the repair dynamics accepts is a separate question.

15.8 Nobody ever holds pi

Write down 3. Then 3.1. Then 3.14, then 3.141, then 3.1416.

Every one of those is a ratio of two whole numbers: the fifth is 31,416 over 10,000. The list crowds in on something, and the something it crowds in on is not on the list and never will be. Nobody has ever written pi down. Every circle any engineer has ever cut was cut to a finite number of digits, and the number the digits are converging on is not a thing anybody has held.

The argument that a list like that is a problem is older than the number line. Zeno of Elea made it in the fifth century BC, in a book that is lost. It survives because Aristotle set it out in the *Physics* in order to answer it: the first argument “asserts the non-existence of motion on the ground that that which is in locomotion must arrive at the half-way stage before it arrives at the goal.” Cross a room and you cross half of it first. Then half of what is left, then half of what is left after that. The halves do not run out, so the crossing is a list of tasks with no last entry on it, and Zeno took that to mean the crossing never happens.

Add the list up by hand. A half. A half and a quarter is three quarters. Put in an eighth and you have seven eighths, and the gap left over is an eighth, the very piece you just put in. That happens at every stage: each total falls short of 1 by exactly the piece last added, which is also exactly what all the pieces after it come to. Writing n for the number of pieces added, the pattern is one line.

$$\frac{1}{2}+\frac{1}{4}+\cdots+\frac{1}{2^n}=1-\frac{1}{2^n}$$

At n equal to three the left side is the three pieces you just added, seven eighths, and the right side is 1 minus an eighth, which is the same number. Name a tolerance and the line says which stage gets you inside it. A thousandth needs ten pieces, because 1 over 2 to the tenth is 1 over 1024, and every total after the tenth is closer than that one.

That is the whole of what "the sum is 1" says. The surprise is in what it does not say. The limit is not the last term. There is no last term. None of the totals is 1, no amount of adding produces one, and the statement is about where the totals pile up rather than about any total on the list. Calling it a trick is fair, provided the trick gets named accurately: it replaces an act nobody can perform with a test on finite quantities that anybody can run. For any tolerance somebody names, some finite stage gets inside it and every stage after that stays inside. That is what an infinite sum is, all of it. Cauchy had pushed the argument into inequalities in the *Cours d'analyse* of 1821 and stopped short of putting it that way; Weierstrass put it that way to a differential calculus class in Berlin in the summer of 1861, twenty-three centuries after Zeno.

The fix for pi is a single operation and it is the operation that built the number line. Take the ratios of whole numbers, which have gaps in them wherever a sequence crowds together without arriving, and add in the limit of every such sequence. What comes out has no gaps left. It is the real numbers: the fractions together with every destination the fractions were pointing at. The name for the operation is **completion**.

Run that operation on the record points inside the three-dimensional band, using the distances the table of angles supplies, and what comes out is an abstract continuous three-dimensional Euclidean completion. The original record points belong to it, just as fractions belong to the real line. Every other point is the limit of a sequence of increasingly accurate records. Physical position, scale and the matching of neighboring observers' frames remain separate identifications.

The sixty rotations come along. Each of them permutes the twelve ports, so each leaves every entry of the table of angles precisely where it was, so each preserves every distance in the completed space. A transformation that preserves all distances is a **rigid motion**: what a machinist does picking up a part and setting it down the other way round, with nothing stretched. The group chapter fourteen counted at sixty elements acts on the space it built as the turns of a rigid body.

15.9 What holds at a finite resolution

The two limits answer different questions: how strongly the slow band dominates, and how finely records can sample its space.

Run the repair grid any finite number of passes and count the independent directions in what it does. The answer is eleven. Every time, at ten passes and at ten million, because all three bands are present, none of them has reached zero, and a number smaller than another number is not zero. Three appears when the overall size is divided out and the number of passes is allowed to grow: the fast and middle bands vanish against the slow one, and the rank of what survives is three. At any actual number of passes it is eleven.

For spatial sampling, choose a bounded region and a minimum separation between records. Add available records until none can be added without breaking that separation. With a sufficiently dense initial menu, the selected records cover the region with a bounded maximum gap. Their number is bounded above and below by fixed multiples of the inverse cube of the resolution. That familiar cubic scaling follows from the three-dimensional distance metric. The chosen separation and region are additional inputs.

These finite populations support a controlled field model. Give each point a real field value, weight it by the volume nearest to it, and supply a positive interaction between nearby values. One stationary action determines the motion. With sufficiently rapid refinement of the population relative to the shrinking interaction range, the field approaches a continuum wave. Errors in smooth detector readings vanish on a fixed time interval before boundary effects arrive. The theorem concerns this supplied scalar action; it does not identify arbitrary repair traffic with field propagation.

An effective continuum therefore asks for a finite error small enough for the observer's question. It does not require a finite list to contain every real point. Record availability, accepted operations and elapsed physical time are distinct: a tiny displacement obtained by canceling a long sequence need not be quick to execute. Zeno's runner crosses the room because crossing it was never the same act as finishing a list with no last entry.

Which leaves one word doing more work than it has been paid for. Every rate here is per pass. Nine tenths per pass, half in 5.4 passes, 94 percent after sixty. A pass is an event that has finished happening. The shares only mean anything if the events fall into an order. What a pass is, why passes fall into an order at all, and why that order has a preferred way round are three questions a wiring diagram does not answer. Rayleigh's bell rang down and did not ring up.

Chapter 16

Why Does Time Run, and Only One Way?

A latex bead a little over six thousandths of a millimeter across sat in a dish of water in a laboratory in Canberra, held in place by a focused laser. A trap of that kind pulls the bead toward the brightest point with a force that grows the further out it strays, which is how a spring behaves. The dish was then dragged sideways past the trap at 1.25 micrometers per second while a quadrant photodiode read the bead's position a thousand times a second, for as long as ten seconds at a stretch.

Dragging a bead through water takes work. The work finishes up in the water as heat. Scored across a few seconds and longer, every one of the five hundred and forty recorded runs did what the second law of thermodynamics requires: the bead lagged behind the trap, the water took up the difference, and the whole business ran the way anybody would expect it to. Score the same recordings over a hundredth of a second instead, and the stretches that run backwards come out about as numerous as the stretches that run forwards. The water hands energy to the bead rather than taking it. The entropy of the arrangement goes down. Widen the window to two seconds and the backwards stretches thin out; widen it past a few seconds and they stop occurring.

G. M. Wang, E. M. Sevick, E. Mittag, D. J. Searles and D. J. Evans published that in *Physical Review Letters* on 29 July 2002, under the title "Experimental Demonstration of Violations of the Second Law of Thermodynamics for Small Systems and Short Time Scales", seventeen words that leave the abstract with nothing to announce.

A rule that fails on short recordings and holds on long ones is a statement about an average. So some quantity is being averaged, the average comes out with a sign, and something has to say where the sign comes from. That is one of the three questions hiding inside the word time, and the other two are older.

16.1 Three jobs, one word

Ask anybody what time is and you get a description with four separate claims folded into it: events fall in a single line, the line runs forward and cannot be run backward, the line is the same for everybody, and it goes at one rate. Chapter seven took away the third one. Two clocks in different places have no shared present in which to disagree.

The other three claims are three different objects with one word stretched over them.

The first is **order**: which of two things happened before which. Order is a relation between events. It has no units, no rate and no duration in it. Chapter seven's recipe carried the whole of it, four steps wired in a line and a fifth that floats.

The second is **direction**: why the relation runs one way round and never the other. That is a property of an operation rather than of a relation, and the one the bead in Canberra was testing.

The third is **flow**: why anything seems to be moving, and why there is a rate at all. Order tells you which of two events came first, where there is a fact about it at all. Direction says why the pair never runs the other way round. Between them they say nothing whatever about passing. You could be handed the complete order of every event in the universe and the direction of every strand in it and have, out of that, no motion, no duration, and nothing that goes by. Chapter seven's recipe card carries an order, and printing it down a card draws an arrow along that order, and the card lies on the counter going at no rate whatever. The answer to flow is a separate answer, arrived at by a different route. It is not the arrow of time experienced from the inside. The state an observer holds determines its own clock, the way a piece of music determines a tempo without a metronome anywhere in the room. Five chapters from here, that stops being an image.

16.2 What gets ordered

Chapter seven taught partial orders on a kitchen counter, before this world contained anything to order, and chapter nine supplied the missing ingredient. Events are authenticated semantic commits. An accepted repair commit is one possible event when its reads and writes are recorded; record, readback and feedback commits can also be events. The archived source-history receipt uses the latter instrumentation carrier and excludes recurrent seam-repair transactions from its physical-event interpretation.

Take the twelve observers of chapter six, wired as an icosahedron, thirty seams between them, every observer with five neighbors. A repair reads the two readings at the ends of one seam, writes the two readings at the ends of that same seam, and touches nothing else in the world.

Sharing an observer makes two repairs possible competitors for the same registers. It does not by itself order them. An informational edge appears when the later accepted commit certifies that it read a register version written by the earlier one. If it merely follows the other repair in an executor's list, there is no edge. Repairs whose certified resources are disjoint have no such fact, and no clock or queue position can add one afterwards.

Two pairs, concretely. A repair on the seam between the first observer and the second may write a version that a later repair between the second and third certifies reading. If it does, the writer mark and read certificate establish the dependency. Against that, a repair between the first and second observers and one between the fourth and fifth touch disjoint registers. Ask which came first and the record system has nothing to interrogate.

Count the possible conflicts. Thirty seams make four hundred and thirty-five pairs of seams. A given seam has two ends, and at each end four other seams arrive, so eight seams share an observer with it and twenty-one do not. That gives one hundred and twenty pairs that may compete for a register and three hundred and fifteen that are structurally disjoint. These are topology counts, not counts of informational precedence pairs. The actual order comes from the versions read in one execution.

That is a great deal less order than a line would carry. A line puts everything in single file and answers all four hundred and thirty-five pairwise questions before a single repair has run. The observers instead produce a web after the fact. Every strand is a certified read-after-write dependency, and unrelated commits remain incomparable. The wiring says where dependencies are possible; the provenance says which ones occurred.

The everyday picture flattens that web into a timeline. The difference shows up in what you are allowed to ask. "Which of these two happened first" has an answer for some pairs and none for others. The wiring constrains which dependencies are possible, while the recorded provenance fixes which comparabilities actually occur.

16.3 Seven pairs go in, one comes out

Chapter nine derived the repair move from three requirements and got the midpoint. Two readings at a seam. Both of them go to the average.

Watch what that does to counts. A seam whose two ends hold 6 and 0 finishes with both ends holding 3. So does a seam holding 5 and 1. So does 4 and 2, and 2 and 4, and 1 and 5, and 0 and 6, and 3 and 3, which was sitting on the answer to begin with. Seven arrangements of two whole numbers go into that move and one arrangement comes out, and every one of the seven is consistent with what the seam says afterwards.

Run it backwards. You have two readings of 3. The question is what they were. Those readings do not say. Without another record, all seven inputs remain compatible with the output. Averaging has removed the distinction from this description of the seam. Whether another part of the system retains it is a separate question.

This repair write leaves the total alone and satisfies the seam while discarding which input it received from the two-reading description. A semantic commit can record such a write, but commits can also retain readback and history. The acceptance rule restricts which repairs are allowed; it does not decide what information survives elsewhere.

Those two restrictions differ. The acceptance test is a rule about available moves. Seven into one is a fact about the averaging map. Relaxing the rule cannot recover an input from the output alone.

A write can be made undoable: retain the input in an initially blank register. The combined output and record then distinguish all seven cases. No erasure is forced by keeping that copy. Reusing finite memory cyclically may require resetting it later, and the cost belongs to that reset under its physical operating conditions. Logical erasure and reversible record retention are different operations.

Order is a relation between two records, checkable by whoever holds both and carrying no units. Direction is a property of a single move, showing up as an arithmetic loss inside that move. Drawing one axis and putting an arrow along it welds the two into a single object. The weld is why people go looking for the arrow in the first instant of the universe.

Boltzmann's account concerns a different arrow. High-entropy arrangements vastly outnumber low-entropy ones, and an explanation of their prevalence must also account for the initial conditions. David Albert called that initial-condition assumption the past hypothesis in *Time and Chance* in 2000. The seven-into-one map establishes an information loss in a reduced description. Connecting it to sustained physical irreversibility requires a state ensemble, dynamics and an environment; the arithmetic of averaging does not settle the cosmological initial-condition question.

16.4 Counting what got lost

The seven arrangements that collapse into one are a quantity of missing information. Ask how many yes-or-no questions it takes to pin down one item out of a list. Two items, one question. Four items, two questions, since the first splits the four into halves and the second splits a half. Eight items, three. Sixteen, four. Each doubling of the list costs one more question, which is the defining habit of the **logarithm**: it converts multiplying into adding. Counting questions this way makes the logarithm base two. The unit of the answer is the **bit**.

The list is rarely flat. If one item out of the four is likely and the other three are long shots, a well-chosen first question separates the likely one from the rest, and most of the time you are finished after one question rather than two. So the count wanted is an average over the odds. Claude Shannon wrote it down at Bell Labs and published it in the *Bell System Technical Journal* in July and October of 1948, in two halves of a paper called "A Mathematical Theory of Communication".

$$H = -\sum_i p_i \log_2 p_i$$

Here H is the average information in one outcome, measured in bits; the index i runs over the possible outcomes, and p sub i is the probability of outcome i. The minus sign compensates for the negative logarithms of probabilities below one.

Three cases, by hand. A fair coin gives 1 bit. A coin biased nine to one gives 0.469 bits: heads contributes 0.137 and tails 0.332 to the average. A fair eight-sided die gives 3 bits. Assign equal probabilities to the seven inputs at the seam and their entropy is log base two of seven, or 2.807 bits. Mapping that ensemble to one output removes that much entropy from the seam description. Unequal probabilities change the amount, and a retained input record preserves the distinction in the larger system.

Shannon told Myron Tribus that he had gone to John von Neumann for a name, and that von Neumann told him to call it entropy, on the grounds that the same expression appeared in statistical mechanics and that nobody knows what entropy is, so in an argument he would have the advantage. Tribus and Edward McIrvine printed the story in *Scientific American* in September 1971.

The entropy belongs to the description rather than to the thing described. The seam holds 3 and 3 whatever anybody knows. The 2.807 bits are a fact about somebody's odds over the seven arrangements. A different set of odds gives a different number for the same seam. Chapter ten met this quantity under another word: the run that finished with 81,920 records had a spread of 11.3134, which is the same count in a different unit, the way a length is the same length in inches or in centimeters. In bits it is 16.32.

Which is where the schoolroom gloss goes wrong. Repair makes the world tidier and more determinate, and a seam that goes from 6 and 0 to 3 and 3 has had its uncertainty cut, so anybody reading entropy as a measure of how messy the seam is will get the sign backwards. The seam is as messy after the repair as before it: two whole numbers, sitting at two ends of one overlap. What changed is how many questions it takes to say which two.

16.5 Two messages down one channel

The quantity that runs one way is a comparison rather than a count. Take a settled arrangement of the sort the repair law leaves alone, and call the odds it assigns to each state the reference. Take your own odds over states, which need not be the reference. Ask how many extra questions per state your odds cost you compared with the reference, and the answer is a second quantity, the **relative entropy** of your odds against the reference. It is zero exactly when the two agree, positive otherwise, and measures how far your description sits from the settled one.

Pass both descriptions through the same process, whatever the process is, and they get harder to tell apart. Send two distinguishable messages through the same noisy channel and the outputs are closer together than the inputs were; nothing you do to both of them afterwards can make them further apart again. Relative entropy is the number that quantifies that, and can only fall under further processing.

For a stochastic repair channel that preserves a supplied reference, relative entropy to that reference cannot increase. This is the finite second-law inequality. The deterministic averaging rule and the equilibrium resampling channel are separate constructions; applying the inequality requires the stated reference-preservation condition.

16.6 The identity underneath the inequality

The falling is an average of something that does not always fall. Take one repair step. Before it, the world is in some state and your odds give that state a probability, and the reference gives it a probability too. The logarithm of the ratio between them is your surprise at finding that state. After the step, the world is in another state, and there is a second surprise, computed the same way against the same reference. The **entropy produced** by that transition is the first surprise minus the second.

Some transitions have positive entropy production and others negative. For strictly positive normalized initial and reference laws, a reference-preserving stochastic kernel, and a positive output law, the average exponential of minus that quantity is exactly one. Detailed balance is not needed for this integral identity. The positivity assumptions keep the logarithmic ratios defined on every required state.

One further condition is checkable at the seam: measured against the reference, the weight the step carries from one state to a second matches the weight it carries back. A repair in that balance satisfies the following at every pair of states, exactly.

$$p(x)\, K(x, y) = e^{\sigma(x,y)}\, q(y)\, K(y, x)$$

Here p of x is the probability your description gives the state before, K of x y is the chance the repair takes that state to the state after, q of y is the probability of the state after, K of y x is the chance the same repair runs the other way, and sigma is the entropy produced. The equation says the forward weight of any transition exceeds the reverse weight of the same transition by exactly the exponential of the entropy produced, at every pair of states, with nothing approximated. Gavin Crooks published the general relation of this form in *Physical Review E* in 1999, the relation the Canberra bead was measured against.

Two consequences follow. Collect all the transitions that produce the same amount of entropy and add up their weights, and the forward total is that same exponential times the reverse total. Count the entropy produced in bits and the exponential turns into a doubling per bit, which is the same statement in the unit the coins were counted in. A transition producing ten bits runs forwards about a thousand times as often as backwards. One producing forty bits, about a million million times as often. The arrow is a ratio rather than a prohibition, which is why a small enough system watched for a short enough time can be caught running the wrong way, and why nobody has ever caught a cup of coffee doing it.

The mean entropy production equals the fall in relative entropy exactly. Convexity of the exponential gives Jensen's inequality: its average is at least the exponential of the average exponent. Since the first average is one, mean entropy production is nonnegative. This is an exact inequality between real-valued quantities, with no approximation.

Detailed balance also makes the equilibrium two-time correlation symmetric: measure one quantity before a step and another after it, then swap their positions. The averages agree. This is the finite analogue of the reciprocity associated with Lars Onsager, whose work connected paired transport processes such as heat and electric currents. Reading these finite correlations as physical transport coefficients additionally requires identified currents and a calibrated time scale.

16.7 Faking a past

If the seam no longer distinguishes its inputs, a retained record can preserve access to them. The question is whether a claimed record can be forged.

Set the machine one task. A chain of records claims a history worth twenty bits, meaning twenty yes-or-no questions' worth of detail about what happened. A competing chain claims the same history and has no such history behind it, so the twenty bits it would need to produce are twenty bits it can only guess. A blind guess at twenty bits succeeds one time in two to the twentieth, which is one time in 1,048,576, or 9.5 times in ten million.

The machine ran a hundred thousand attempts. The expected number of successes under that blind-guess model is about a tenth, and the number obtained was zero. The per-attempt probability assumes independent uniform hidden bits and no usable side information. More attempts increase the chance of a success; a retained copy or predictable history would change the guessing problem.

The recorded past is the retained history of commits and their certified read dependencies. Its authentication rests on those records and checks, not on an assumption that every write destroyed its input. Unresolved repairs have no committed outcome in that history yet.

Chapter six left one unit of disagreement in the declared twelve-observer model. One flipped record, thirty seams, nineteen loops, and a leftover of exactly one that no allowed repair removes. Repair order changes where it sits. A retained history can record the writes that moved it and their certified dependencies, whether or not an input remains recoverable from the local readings.

Every strand in that web was laid by one commit reading what another commit wrote. Chapter nine's local rule bounds how many seams that dependency can cross in one model event. This is a graph-local propagation bound. Turning seams and events into meters and seconds requires a physical ruler, a clock calibration and proof that the informational order agrees with physical signal causality. Without those bridges it is not a derivation of the speed of light.

Chapter 17

Why Is There a Speed Limit?

On 3 December 1725, in a house on Kew Green, James Bradley pointed a telescope straight up through a hole in the ceiling and a second hole in the roof.

The instrument belonged to Samuel Molyneux, who had ordered it from the clockmaker George Graham. Twenty-four feet from lens to eyepiece, hung from an iron frame bolted to the chimney stack, because a chimney is the most rigid thing in an English house. Pointing it at the zenith and nowhere else was the design: a star directly overhead is seen through the least air, so any displacement left over belongs to the sky rather than the weather.

The star was gamma Draconis, which passes almost exactly overhead at London. What the two of them wanted from it was parallax. If the Earth goes around the Sun, a nearby star should shift against the distant ones, one way in June and the other in December, and nobody had ever caught it happening.

The star moved. It moved the wrong way, it moved too far, and it moved at the wrong time of year.

Parallax keeps a schedule set by where the Earth is. Gamma Draconis kept one set by which way the Earth is going, and those two run three months apart. By March 1726 the star sat about twenty seconds of arc south of its December position, by June it had come back, and by September it was twenty seconds north.

Every star Bradley checked answered to the same number, which settled it. Parallax has to shrink with distance, being a triangle with the Earth's orbit for a base and the star at the far vertex, and this displacement took no interest in distance whatever.

The displacement is a ratio instead. Divide the speed of the Earth along its orbit by the speed of light, thirty kilometers a second against three hundred thousand, and you get one part in ten thousand, which as an angle is twenty and a half seconds of arc.

The effect is called **aberration**, what motion does to a direction. The number Bradley divided by has to be a speed light has, the same in June as in December, when he and his ceiling were moving in opposite directions at sixty kilometers a second. Why is there a top speed at all, and why does everybody who measures it come back with the same figure?

17.1 Two mirrors

Build a clock out of light. Two mirrors facing each other one meter apart, and a pulse of light bouncing between them. One tick is one round trip: two meters of travel, about six and two-thirds billionths of a second.

Put the clock on a train and stand beside the track.

From the platform the pulse does not go straight up and straight back down, because the mirrors slide sideways while it is in the air. It leaves the bottom mirror, travels along a slant, and arrives at a top mirror that has moved along the track since it set out. Then it slants back to a bottom mirror that has moved again. The path from the platform is a pair of slants. A slant is longer than the gap.

The speed is the same on both paths, by the fact Bradley's ratio needs. A longer path at the same speed takes longer, so one tick of the clock on the train occupies more time as read from the platform than it does as read on the train.

How much more is a right-angled triangle and nothing else. The gap between the mirrors is one side. The distance the train carries the mirrors while the pulse is crossing is the second side, at right angles to the first. The slant the pulse actually travels is the hypotenuse, and Pythagoras fixes it: the square on the hypotenuse is the sum of the squares on the other two. Divide through by the gap and the stretch factor turns on one thing, the train's speed as a fraction of the pulse's, and on nothing about the mirrors or the clock. Put the train at eighty percent of light speed and the triangle comes out whole: call the slant five, the train covers eight tenths of it while the pulse is in the air, which is four, and twenty-five take away sixteen leaves nine, so the gap is three. The slant is five thirds of the gap, and so is the tick. At ninety-eight percent it is a shade over five.

Cosmic rays supply the check. A muon is a heavy relative of the electron, manufactured in quantity when a cosmic ray hits the upper atmosphere ten or fifteen kilometers up. Left alone it survives 2.2 millionths of a second on average, which at almost the speed of light carries it about 660 meters, so almost none of them should reach the ground. In 1941 Bruno Rossi and David Hall counted muons at Denver and again at Echo Lake, 1,624 meters higher, sorted them by how fast they were traveling, and found the slower ones thinning out over that drop faster than the quicker ones did. Read backwards, their counts gave the muon a lifetime of its own of about 2.4 millionths of a second, against the 2.2 that later measurements settled on. The fast ones were not sturdier. They were running on longer ticks.

17.2 The train and the lightning

Lightning strikes both ends of the train at once. A guard stands on the platform at the midpoint between the two strike marks, and a passenger sits at the midpoint of the train, and the passenger is traveling toward the front.

Light from the two strikes reaches the guard together. Both flashes crossed equal distances at one speed, so for the guard the strikes happened at the same moment. The passenger is moving toward the flash from the front and away from the flash behind, so the front flash reaches the passenger first, and the passenger, doing the same arithmetic with the same speed, concludes that the front of the train was hit first.

There is no third party who can break the tie, because breaking it would take a comparison that is not the arrival of light at a place, and nobody has one. The two of them hold different facts about the order of two events. Both sets of records are correct.

Chapter seven reached the same absence from the other end, with no signals in the argument at all: no observer in the arrangement was in a position to assemble a shared present out of what it held. Here it fails for a narrower reason, that light has one speed, handed over as a measured fact about one kind of signal and left standing.

17.3 Two events, one subtraction

The guard and the passenger disagree about the order of two events, about how far apart they were, and about how long the gap between them lasted. Chapter four's overlap rule says something has to come through a comparison like that intact, or the two of them are not describing one world.

Take two events. Take the time between them on one observer's clock, multiply by the speed of light so that it becomes a length, and square it. Then square each of the three distances between the events along that observer's three directions, and subtract all three.

$$s^2 = c^2t^2 - x^2 - y^2 - z^2$$

Here t is the time between the two events on one observer's clock, x, y and z are the distances between them along that observer's three directions, and c is the speed of light, present so that a duration and a length can appear in the same subtraction. Two observers in relative motion assign different values to every letter on the right and the same value to s^2, making the quantity a fact about the pair of events rather than about anybody's account of them.

That quantity is the **interval**. Its sign sorts every pair of events in the world into three classes.

When the interval is positive, the events are separated by more time than distance. One can influence the other, some observer sees them happening in the same place at different moments, and every observer agrees on which came first. When it is negative, they are separated by more distance than time. Influence runs in neither direction, some observer sees them happening at the same moment, and observers disagree about the order with no fact anywhere to settle it. When it is exactly zero, the separation is one that light can cross and nothing slower can.

Draw the zero case at a single event and it is a cone opening into the future and another into the past. That cone is the **light cone**. Every pair of events in the universe sits inside one, on one, or outside.

All of which runs on a single input, that light comes out at one speed for everybody. Einstein put that in by hand in 1905 and said he was putting it in by hand. A null result in a Cleveland basement eighteen years earlier had said the same thing, and promoting a null result to a law is a way of writing physics down. It is not a way of explaining it.

17.4 The sky is the data

So take the arrangement apart and ask what an observer physically holds.

The exact OPH result in this chapter is Lorentz-frame kinematics on the declared geometric modular branch. Independently of event order, a real axis and the rank-three source carrier define an ambient product target of real dimension four with the Lorentz sign pattern, while the carrier's unit sphere labels its future-null directions. The authenticated direct-parent order computes source height as the exact longest parent-chain length ending at each event and proves that it increases strictly along generated ancestry. Every finite log has an auxiliary injective forward-causal placement in the target, obtained from an event enumeration and a sufficiently large height scale. This constructs a finite one-plus-three precursor without assigning an intrinsic four-dimensional geometry to the event poset. Physical light cones, event locations and a universal signal speed additionally require a source-selected spatial placement, a calibrated ruler and clock, equal-height spatial separation and spacelike increasing-height incomparable pairs through a manifoldlike refinement; scaled source height alone is not that clock.

An observer has a bounded interface. What reaches it reaches it through a port. Everything it can read is indexed by which way the thing came in. At the resolution of one carrier that is twelve directions, and refining the carrier fills them in, and the object the directions fill in is a sphere. That is the screen with its geometry showing: the sky, in the plain sense that the sky is where you see everything from where you are standing.

Two observers in relative motion look at the same stars and disagree about where they are. Bradley measured the disagreement at one part in ten thousand. It grows with speed. His stars did not swing round together the way the constellations swing round on a globe you turn. They were all pulled toward one point, the point the Earth was heading for that month, and a pull toward a point is a different operation from a turn: stars near that point crowd together, stars out to the side barely change their spacing at all, and the amount depends on where on the sky you look. So angular separation between two stars does not survive: your sky and mine put the same pair of stars at different distances apart, and neither of us is wrong. Overall size does not survive either. There was never anything to measure it against, since a sphere of directions has no radius that anybody in the arrangement holds.

What does survive is the angle at a crossing. Two families of directions that cut across each other at right angles on my sky cut across each other at right angles on yours. A set of directions that closes into a circle on mine closes into a circle on yours, at whatever size it comes out. Those relations hold up under every comparison two observers can run. Chapter five called that an invariant.

Name the invariant and the group follows.

The maps of a sphere onto itself that preserve every crossing angle have a picture. The picture is a map projection. Set the sphere on a flat table so it touches at the south pole. From the north pole run a straight line through any other point of the sphere and carry it on until it hits the table. Every point except the north pole lands somewhere. The entire infinite table gets used. Navigators have bought angle-preserving charts for centuries and they all buy them for the same property: two roads crossing at forty degrees on the globe cross at forty degrees on the chart. What a chart like that destroys is size, which is why Greenland is a catastrophe on a sea chart and why no navigator has ever complained, because nobody steers by area.

It does one further thing. Every circle on the sphere arrives on the table as a circle, except for circles through the north pole, which arrive as straight lines, a straight line being what a circle of unlimited size looks like.

Which turns the question into a question about the flat table. The table has an arithmetic on it.

Write a point two units right of the origin and three units up as the single number $2 + 3i$, where i is the instruction "turn a quarter turn counterclockwise". Turn twice and you are facing backwards, so i times i is -1, and that is the entire content of the symbol. Numbers of this shape are **complex numbers**. They add the way arrows add, tip to tail. The useful part is that they multiply. Multiply $2 + 3i$ by i using the ordinary rules of school algebra and you get $2i + 3i^2$, which is $-3 + 2i$: the point three left and two up, the arrow you started with swung a quarter turn to the left. Nothing was arranged to make that come out. It follows from two quarter turns making a half turn, and from the rule for expanding brackets.

Multiplying the whole table by one complex number turns it through an angle and stretches it by a factor, both at once, everywhere at the same time.

Turning keeps angles. Stretching keeps angles. Sliding keeps angles, and so does the operation that undoes multiplying. Build a map of the table out of those and it keeps angles, and the most general such map that covers the whole table once and sends every circle to a circle sends a point w to $(aw + b)/(pw + q)$, with a, b, p and q four complex numbers. Maps of that shape are called Möbius transformations. Every angle-preserving map of the sphere onto itself is one of them.

Count the knobs. Four complex numbers is eight real ones. Multiplying all four by the same thing moves no point anywhere, which deletes two. Six knobs.

Turn them and watch the sky. Three of the six rotate it rigidly, the way you turn a globe: the constellations swing round and no star changes its distance from any other. The remaining three do something no rotation does. They gather the whole sky toward one point and thin it away from the opposite point, so the constellations crowd toward the direction you are heading and open out behind you, and each of the three is labeled by a speed. Bradley's twenty seconds of arc is that motion run at one part in ten thousand, and the speed shows through the fact that the twenty is not quite a constant: the displacement runs about 20.8 seconds of arc in January, when the Earth is nearest the Sun and traveling at 30.3 kilometers a second, and about 20.2 in July, when it has slowed to 29.3.

Those six, three rotations and three crowdings, are the Lorentz group, named for Hendrik Lorentz, who wrote the transformations down in 1904 as a way of keeping the ether. They are the transformations relating observers in relative motion, the ones Einstein reached from two postulates, and the sphere carries them as the only thing it can do to itself without disturbing an angle.

Textbooks from 1905 onward drew a fast-moving sphere as an ellipsoid squashed along its direction of travel. The question of what a camera would record went fifty years without a general answer. In 1959 Roger Penrose published three pages in the Proceedings of the Cambridge Philosophical Society pointing out that a camera records no such thing. The outline of a sphere is a set of directions, angle-preserving maps take circles to circles, and a photograph of a sphere at any speed you like comes out round. James Terrell got there independently the same year in the Physical Review, under the title "Invisibility of the Lorentz Contraction".

17.5 Four numbers in a square

The sky has six knobs on it and the interval has four signs in it, and one square of four numbers carries both. Chapter fifteen laid the repair weights out in a twelve-by-twelve grid and took its eigenvalues. Take the smallest block worth anything at all: two by two, four entries in a square. Every such block carries a single number of its own: multiply the top left by the bottom right, multiply the top right by the bottom left, and subtract the second product from the first. What is left over is the block's **determinant**.

Fill the square with a particular pattern. Real numbers down the diagonal, a complex number in one off-diagonal corner, and in the other corner its **conjugate**, meaning the same number with the sign of its i part flipped, so that the conjugate of $x + iy$ is $x - iy$. A square filled that way is called Hermitian, after Charles Hermite, and the pattern means it takes four real numbers to write one down rather than eight. Call the four t, x, y and z, measuring time in the distance light covers in it, and lay them out.

$$\begin{pmatrix} t+z & x-iy \\ x+iy & t-z \end{pmatrix}$$

The four numbers on the diagonal and in the corners are the same t, x, y and z that the interval was built from, one time and three distances, written into a square instead of a row.

Do one. Take five for the time, and two, one and two for the three distances. The diagonal reads $5 + 2$ and $5 - 2$, and the corners read $2 - i$ and $2 + i$.

$$\begin{pmatrix} 7 & 2-i \\ 2+i & 3 \end{pmatrix}$$

Seven times three is twenty-one. The other product is $2 - i$ times $2 + i$, which expands to $4 + 2i - 2i - i^2$, where the middle terms cancel and $-i^2$ is $+1$, so it comes to five. Twenty-one take away five is sixteen. Then do the interval on the four numbers you started with, twenty-five minus four minus one minus four, and it is sixteen as well.

Take the determinant in letters. Top left times bottom right is $t^2 - z^2$. The other product is $x - iy$ times $x + iy$, which multiplies out to $x^2 + y^2$, the cross terms canceling and i squared turning the last one from a minus into a plus. Subtract, and the determinant is $t^2 - x^2 - y^2 - z^2$. That is the interval, with the units chosen so light covers one unit of distance in one unit of time.

Nothing in that multiplication knew about time or distance.

Count the signs. One plus and three minuses, and nobody chose a single one of them. They come from the shape of the square. The diagonal is real, so its product is a difference of two squares, and the corners are conjugates, so their product is a sum of two squares that the subtraction then takes away whole. Einstein reached that pattern of signs from two postulates about light. Subtracting one product from the other in a two-by-two produces it in a line of school algebra.

Set the determinant to zero and you have the algebraic null cone. A two-by-two block with determinant zero has one of its rows a multiple of the other, so it carries one independent row rather than two. Chapter fifteen called that count the rank. Algebraic null directions are rank one, and generic non-null blocks are rank two. Calling those directions physical light rays requires a separate identification with signal propagation.

Keep five for the time and take three, nothing and four for the distances, which are the three, four and five of the right-angled triangle the light clock ran on. The square comes out as nine and one down the diagonal with a three in each corner. Nine times one is nine, three times three is nine, and the determinant is zero. The bottom row, three and one, is the top row divided by three, so there is one independent row and not two.

Take the rank-one blocks with t positive, which is the future half of the algebraic cone, and agree to ignore overall size, since scaling a direction by a positive number does not change which way it points. What is left is an algebraic sphere of directions. The match is exact both ways inside this module: each future null ray names one point of the sphere, and each point names one ray. This S^2 is not yet the measured sky or the boundary of physical reachability. On a realization that separately attaches the module to physical events and identifies its null cone with signal propagation, the sky and that boundary become one object described twice. Only then do the angle-preserving motions of the first acquire the physical interpretation of the determinant-preserving motions of the second.

Chapter fourteen counted sixty rotations carrying a twelve-port carrier onto itself, and no port came out distinguished, because some rotation among the sixty carries any port to any other. The motions that leave a determinant where they found it do that work one level up. They close under doing two in a row and each has one that puts the square back, so they are a group, and they are the six knobs on the sky: the group Einstein's two postulates about light were describing.

Pick a direction into the future with determinant one and call it an observer at rest. Ask which vectors are perpendicular to it, using the interval to decide perpendicular rather than ordinary distance. The answer is a three-dimensional local rest space inside the Lorentz module. On that space the interval with its sign reversed is positive for everything except zero, ordinary Euclidean distance in three dimensions. Identifying this frame space with the rank-three carrier readback, and attaching both to physical events, are separate branch receipts.

One more count comes from the same subtraction. A light ray picks a direction inside that three-dimensional rest space, the way it is going. The directions left over, the ones perpendicular to it, make up a space of three minus one dimensions. Two. A light wave has exactly two independent ways of waving, which is why polarized sunglasses cut glare and why they stop cutting it when you tilt your head far enough.

17.6 Cleveland, answered

Michelson and Morley built an instrument to display one number: the speed of their basement through the ether.

That number is one of the six knobs. Turning it moves the stars around on their sky and changes no crossing angle, no circle and no interval, which is to say it changes nothing either of them could compare against anything. The apparatus was not too insensitive. There was no quantity in the basement for it to be sensitive to. No instrument built since has found one.

The six motions shuffle the algebraic sphere as hard as you like and leave the algebraic cone exactly where it was, because the cone is where a determinant vanishes and the six motions preserve determinants. They therefore preserve the causal type of separation vectors supplied in this module. They do not by themselves attach finite records to physical events, identify determinant zero with signals, or prove that informational reachability is Lorentzian causality. On a physical realization with those separate conditions, every frame sorts every attached event pair into the same three classes: influenceable, not influenceable, and the boundary. Chapter seven's source-derived order remains informational until a placement makes reachability and cone order agree both ways; event separation then follows from antisymmetry.

On a physical realization, the speed of light converts the ruler and clock units attached to the null boundary. Seam counts and committed repairs do not fix that conversion. What the Lorentz branch supplies is the invariant cone in the frame module; identifying it with signal propagation and calibrating its units are independent receipts.

Metrologists conceded the point in 1983. At the seventeenth General Conference on Weights and Measures the meter was redefined as the distance light covers in one 299,792,458th of a second, which fixes the speed of light at 299,792,458 meters per second exactly. Nobody has measured it since, and nobody can, because measuring it would consist of measuring the meter.

A separation that light can cover is rank one: one independent row, nothing held behind it. Everything that travels slower is rank two. The second row is the reason it travels slower. That row has to be re-asserted every round or the thing it belongs to stops being there, which means something is being spent to hold it, at a rate, against a clock, and it is being spent right through the chair you are sitting in.

Chapter 18

Why Does Anything Have Energy?

Willem 's Gravesande, who taught mathematics and astronomy at Leiden, laid a tray of soft clay on the floor in 1722 and dropped brass balls into it from measured heights.

He was settling an argument about which multiplication to perform. Both sides agreed that a moving ball carries something, and that the something grows with the weight of the ball and with its speed. Descartes' followers held that the quantity worth tracking is the weight multiplied by the speed. Leibniz's held that it is the weight multiplied by the square of the speed, and called it vis viva, living force. The dispute had run for decades, mostly in Latin, mostly in print, and neither side had proposed anything you could put on a floor.

A ball released from four times the height arrives at twice the speed, so a measured drop hands the clay a chosen speed. A ball arriving at twice the speed sank four times as deep. At three times the speed, nine times as deep. Two balls of the same size and different weights, released from heights in inverse proportion to those weights, left the same dent.

The square won, in mud, and neither side could say what the square was a quantity of.

Émilie du Châtelet took the argument up in France. She had entered the Academy of Sciences prize competition on the nature of fire in 1737, anonymously, as every entrant was required to do; the result was announced on 16 April 1738 and the prize was divided among Leonhard Euler and two others, and her essay was printed with the winning memoirs the following year, the first scientific paper the Academy published by a woman. Her *Institutions de physique* came out anonymously in Paris in 1740. Its twenty-first chapter, on the force of bodies, put the clay in front of readers who had been treating the question as a matter of definition. She came down on the side of the square. In 1745 she began translating the *Principia* into French, with a commentary running alongside Newton's Latin that supplies the derivations he had left out. She was carrying a fourth child while she finished it, working eighteen-hour days, and she dated the commentary on the last day she worked on it. She died at Lunéville on 10 September 1749, six days after the birth, at forty-two. The complete text appeared in 1759, the only complete *Principia* in French that exists.

Thomas Young gave the number its modern name in 1807: the term energy, he wrote, may be applied with great propriety to the product of the mass or weight of a body into the square of the number expressing its velocity. That sentence assigns a word to a product and assigns nothing else. Newton's *Principia*, which defines mass, momentum, force, centripetal force and half a dozen other things with the care of a man expecting hostile readers, has no energy in it anywhere. Textbooks define energy as the capacity to do work and define work as force through a distance, which tells you how to compute the number and leaves open what the number counts. Chapter five gave the classical answer: energy is the quantity conserved when the laws do not care when. That correspondence is exact. It wants two things before it starts: a background time to shift and a stage for the shifting to happen on.

18.1 The rate a region is worked through

Chapter nine left this world with one move. Two neighbors share a seam, each holds a reading of it, the readings disagree, and both readings go to their midpoint. Nothing else happens anywhere. The disagreement outstanding across a set of seams is a sum of squares of gaps, every repair strictly lowers it, and the lowest it goes is zero, which is what the seams read when every pair of neighbors agrees.

The machine holds one move, one kind of seam and two readings at each seam. The descent of the outstanding disagreement is the only quantitative thing any of it does, so energy is made out of that descent, there being nothing else to make it out of.

Take a screwdriver to a screw. One eighth of a turn, repeated, drives the screw into the wood. The whole trip of the screw is that one small move done again and again.

A smooth continuous change is a flow. The small move whose repetition produces the flow is its **generator**, and the pairing runs both ways: a flow determines its generator, and a generator run out determines the flow. The trip of the screw is a long list of positions. The one turn is a single instruction. The list follows from it.

Every region in this world advances along a clock of its own. Something generates that advance. That something is the region's energy. The energy of a region is the rate at which its records are worked through by repair, counted against that region's own clock.

The word behaves identically wherever physics uses it. What generates turning about an axis is the angular momentum about that axis, and what generates sliding sideways is the momentum in that direction. Advance along a clock has the energy for its generator, and quantum mechanics has run on exactly that definition since the 1920s without ever saying what does the advancing.

A clock of a region's own is the only kind this world has. Chapter ten ran eighty-two thousand patches with no tick anywhere in the machinery, nothing waiting for the next round and nothing counting rounds. What a region has instead is a clock its own state manufactures, out of nothing but what it holds. Two regions worked through at different rates differ in a fact about themselves, since no clock outside them has a setting they could differ in.

A region whose records agree everywhere generates no flow and reports nothing above the floor. A region holding standing disagreement drives its clock hard. The driving rate is the number a calorimeter or a particle detector hands you.

In ordinary quantum mechanics the energy of a system is the operator somebody writes down for it. You pick an operator called a Hamiltonian to match the system in front of you and fit its constants to measurements. The fitting is done by a person with a pencil. Here the operator is manufactured by the state the region holds. The region holds its state whether or not anybody has written anything about it. Two regions in the same state have the same energy. The pencil never enters.

18.2 Positive, and additive

The disagreement outstanding across a region's seams is a sum of squares, which puts a floor under it at zero, and zero is total agreement. Every process available in this world is a descent toward that floor. No region can spend past it. So the rate is never negative. There is no arrangement of records that goes on falling forever.

Arthur Wightman wrote that into the axioms of quantum field theory in the 1950s and called it the spectrum condition. It has to be assumed there, because without it every system has somewhere lower to drop into and hydrogen radiates without limit. Here it is the floor of a sum of squares.

Additivity comes off the seams. Two regions with no seam between them have no comparison to make across the gap, so they have no repair to do across it, and neither one's rate says anything about the other's. The rates add. That is why energy accounting works at all: a calorimeter that stops two beams together reports the sum of what it reports for each of them alone, an assumption so ordinary that nobody writes it down, and holds because separated regions share no seam.

That sum has a condition on it. The condition breaks the moment the two regions touch. Run a seam between them and their rates stop adding, since the seam carries a disagreement belonging to neither of them and repair at that seam changes both. The gap between the joint rate and the sum of the separate rates is the quantity a chemist charges for a bond. The sign of that gap is why two hydrogen atoms end up as a molecule.

18.3 Both sides of every entry

In November 1494 the printer Paganino de Paganini finished a large book in Venice for Luca Pacioli, called the *Summa de arithmetica*. Twenty-seven pages of it set out what the merchants of the city had been doing with their books, the first printed account of the practice.

Every transaction is written twice, once as a debit in one account and once as a credit in another. The books balance, and they balance for a reason that has nothing to do with anybody adding them up at the end of the day: a single-sided entry cannot be written at all. A clerk who wants to steal has to falsify two accounts that other entries also touch.

The repair move has that shape by construction. Chapter nine's second requirement was that a repair leaves the total of the two readings it touches alone, which makes every repair two entries, what comes off one reading going onto the other, the pair of them one event. The requirement is there because a move that changed a total would have to know the total, and no register in this world holds one.

Energy conservation is that fact read at the level of rates. A region is worked through at some rate, the working happens at seams, and every seam is shared with a neighbor, so what leaves one region's account arrives in the next one's, entered twice. The rule performing the entries is one rule, the same at every seam and at every event, carrying no parameter that could take a different value somewhere else. There is a general procedure for turning a symmetry into its conserved number. It is not needed to see this one. Chapter five's correspondence, that energy is what you get when the laws do not care when, is this seen from the other end: the repair rule has no when in it to care about.

In 1775 the Royal Academy of Sciences in Paris resolved to examine no further proposals for perpetual motion, and put them on a list with squaring the circle, trisecting the angle and doubling the cube, none of which anybody had proved impossible either. Every one of those designs is a proposal to make an entry on one side of a boundary and not the other. The obstacle is the list of available moves. Repair acts on one seam, it touches the two readings at that seam's ends, and it leaves their total where it found it, so a wheel that gains what nothing else loses is asking for a move that has never been on the list.

18.4 A wave that would not spread out

In August 1834 John Scott Russell was riding along the Union Canal near Edinburgh when a boat drawn by two horses stopped suddenly. The water it had been pushing rolled forward off the bow as a single rounded heap, some thirty feet long and a foot to a foot and a half high, and went on down the channel without spreading and without slowing. He followed it on horseback at eight or nine miles an hour, for a mile or two, and lost it in the windings of the canal.

What he chased was a shape. The water in the heap changed the entire way: each stretch of canal rose, moved forward a little and settled back down, while the heap went past it. The thing that traveled was an arrangement re-establishing itself out of new material at every point along the route. It held together because two effects that each ruin an ordinary wave were canceling. A tall crest runs faster than the water beneath it and steepens the front. Spreading pulls the crest apart into a train of ripples. Balance the two against each other and the heap keeps the shape it left the bow with.

The seam network holds patterns of that kind. A bounded set of records reproduces itself after every round of repair, because flattening it would open larger disagreements at the boundary of the set than it closes inside. Repair works on such a pattern at every event and gets nowhere with it, so holding the pattern costs repair work at every event, and what the work buys is the pattern being there for the next one.

Mass is that standing cost: the rate a pattern demands while its records go nowhere. Nothing about a stone on a table is idle. The arrangement that makes it a stone is being re-asserted at every event, at a rate the stone's own clock sets. That rate is what a balance reads off when you put the stone on one.

A pattern of that kind turns its material over, exactly as the heap in the canal did, and the charge falls on the arrangement rather than on whichever records are holding the arrangement at a given event. So the cost is steady while the material under it is not, which is why the proton's mass comes out at 1.0072764665789 atomic mass units, with the uncertainty sitting in the last two of those digits, in every laboratory that has weighed one.

18.5 Four numbers

Chapter seventeen put four real numbers into a two-by-two Hermitian matrix and found the interval of special relativity sitting in its determinant: the first number squared, minus the squares of the other three. The transformations that carry one observer's description into another's are exactly the ones that leave a determinant alone, which is why every observer computes that combination and gets the same answer.

A region's entry is four numbers of that kind on the declared physical carrier branch. The first is the rate the region is worked through against that region's own clock, which is its energy. The other three are the drift of its records across the seam network in the rank-three carrier directions from chapter fifteen. Identifying those coordinates and that clock with physical momentum and energy requires the separate realization receipts.

Put the four numbers into the matrix and take the determinant. Every observer agrees on it. What they agree on is the square of the mass.

Move the terms to the other side of the equals sign and the statement about the determinant becomes the relation everybody writes down, which says how tightly a region's energy, its mass and its drift are locked to one another.

$$E^2 = m^2 + p^2$$

Here E is the region's energy, m is its mass, and p is the size of the drift, with distances counted in seams and durations in repair events so that the conversion between the two is one. Fix the mass and fix the drift and the energy has one value available to it, which is why a detector measuring two of the three is measuring all three.

Set the three drift numbers to zero and the equation reads E = m. Run it the other way: if the energy equals the mass, then p squared is zero, and a sum of three squares comes to zero only when each of the three is zero, so the drift vanishes in every direction at once. Energy equals mass exactly when a pattern is going nowhere. The exactly-when is a two-way statement about a matrix rather than an approximation for slow objects.

On the physical Lorentz/action branch, the zero-drift mass shell gives $E = mc^2$. Chapter nine's one-seam-per-event statement is only a graph-local dependency bound; it neither fixes c nor identifies the standing cost with physical mass. Those identifications require the ruler, clock, causal and action receipts carried by the branch.

The corner is worth something because the conversion is squared. A helium-4 nucleus falls short of the two protons and two neutrons that make it by 0.0304 atomic mass units, and the energy released when those four combine, measured in the laboratory, is 28.3 million electronvolts, which is that shortfall multiplied by the square of the speed of light.

18.6 The beam and the calorimeter

Positivity, additivity and conservation went into classical physics as three things the world was observed to do, and Newton, Young and du Châtelet all worked with a quantity whose properties had to be granted before the arithmetic could start. They are three readings of one move: a sum of squares has a floor, separated regions share no seam, and a single-sided entry cannot be written.

Every one of the four numbers is something a region hands over when it is asked for it. A calorimeter reads a beam's energy by stopping it and measuring the heat, which is one question put to a region and answered. Put another question at the same moment, about where the beam is, and the two do not have a joint answer, and a better calorimeter does nothing whatever about that. Which pairs of questions have a joint answer is a fact about the questions, and no instrument gets a vote. Ask the calorimeter its one question, beam after beam, and the answers arrive in stable proportions, and nothing in the four numbers and nothing written at any seam sets those proportions.

Chapter 19

Why Can't You Ask Everything At Once?

At Petco Park in San Diego on the night of 24 September 2010, Aroldis Chapman of the Cincinnati Reds released a fastball that the stadium's tracking system clocked at 105.1 miles an hour, which is 169.1 kilometers an hour.

Two instruments were pointed at that ball. A radar gun read its speed off the shift in a returned microwave pulse. A camera read its position off the light coming back from it. Both worked, at the same moment, on the same object, and neither one spoiled the other's answer. Run the radar first and the camera second, or the camera first and the radar second, and the pair of numbers comes out the same. Nobody at the ballpark has to decide which instrument goes first, because the ball has a position and a speed and both are sitting there waiting to be read off.

That is two questions about one object, answered together, in either order, with neither answer costing the other anything. Every measurement made anywhere before 1925 was assumed to work that way.

The world declines to extend that courtesy to every pair of questions. Which pairs it allows, and what asking anyway costs the answer already in hand, are settled before any instrument is switched on.

19.1 Socks then shoes

Put on your socks, then your shoes. Then try it the other way.

The two orders are not the same operation. The difference is visible from across the room. Other pairs are indifferent: left sock then right sock, or right sock then left sock, and you finish in the same state either way. Dressing contains both kinds of pair.

So take the operations themselves as the objects worth studying, rather than the states they act on. You can do one operation and then another, which composes them. You can scale an operation, or add two of them, in any setting where addition means something. A collection of operations that admits all three of those moves is an **algebra**. The word carries no more than that. When every pair in an algebra composes the same way in both orders, the algebra is **commutative**. When some pair does not, the algebra is non-commutative. Adding up a shopping list is commutative. Getting dressed is not.

Ordinary numbers are the commutative case. Add them, scale them, multiply them in whichever order suits you, and the product comes out the same. Relax that one clause, leave every other rule of the arithmetic where it was, and you have crossed the whole distance between the physics of the nineteenth century and the physics of an atom.

The failure has a size. The size is what you get by doing both orders and comparing. Write A and B for two operations, write AB for doing B and then A, and the difference is the **commutator**:

$$[A, B] = AB - BA$$

That expression is zero exactly when the two operations can be swapped freely. Its size when it is not zero says how much difference the swap makes. Socks and shoes have a large one. Left sock and right sock have zero.

Chapter nine had a version of this with money in it: two edits landing on one account, each of them correct arithmetic, and the balance depending on which one was written first. Order mattering is a property that operations can have in a world with no quantum mechanics anywhere near it. The questions you can put to a physical system are operations of exactly this kind, most pairs of them fail to commute, and the failure belongs to the pair of questions rather than to anybody's equipment.

The radar gun and the camera commute. **Two questions can be answered together exactly when their commutator is zero.** A pair whose commutator is something other than zero is a pair you have to choose between.

19.2 Ten days on Helgoland

In June 1925 Werner Heisenberg was twenty-three and his hay fever had swollen his face badly enough that he left Göttingen for Helgoland, a red sandstone island in the North Sea with almost nothing growing on it and therefore almost no pollen. He spent about ten days there.

The problem he took with him was the hydrogen spectrum. Heat hydrogen and it emits light at a set of sharp wavelengths, the same set every time, catalogued since Johann Balmer fitted a formula to four of them in 1885. The standard picture had electrons on orbits, which reproduced those wavelengths and collapsed for every atom with two electrons in it.

What Heisenberg did was to throw out the orbits. Nobody had ever seen one. What a laboratory actually produces is a table: for each pair of atomic states, how bright the transition between them is and at what frequency. So he built his scheme out of the tables, with one entry for each pair of states, and no statement anywhere about where an electron is.

Then he had to multiply two tables together, because the energy of a system involves the square of a quantity, and squares need multiplication. The rule he was forced into combines a row of one table with a column of the other. It produced an answer. It also produced a different answer when he did the two tables in the other order.

He sent the work to Max Born, who read the multiplication rule and recognized it, recalling it from student days as the way you multiply matrices. Heisenberg's paper reached the *Zeitschrift für Physik* on 29 July 1925. Born and Pascual Jordan submitted the sequel that rewrote the whole thing in matrix language on 27 September. The story that gets told about a lone twenty-three-year-old on a rock at dawn leaves out that the theory arrived through three people over four months, one of whom had to remember an undergraduate course to see what the other one had written down.

Position and momentum came out of that scheme as tables that fail to commute. Measure a particle's position, then its momentum, and you have performed a different physical operation from measuring its momentum and then its position, and the difference is a fixed quantity with Planck's constant in it. It does not go to zero as the apparatus improves, because there is no apparatus in the statement.

The uncertainty relation is that fixed quantity, read as a floor. The spread in position multiplied by the spread in momentum cannot go below it, and where the floor sits is set by the size of the commutator of the two questions. Confine an electron to a region the size of an atom, about a tenth of a nanometer across, and the spread in its speed cannot be smaller than roughly six hundred kilometers a second, against the two thousand two hundred kilometers a second at which the electron in a ground-state hydrogen atom actually travels. The electron is not being jostled by the measurement into a speed it would otherwise not have had. There is no state of the electron with a sharp position and a sharp speed for the measurement to disturb.

19.3 Three magnets in a row

On the night of 7 February 1922, in the building of the Physikalischer Verein in Frankfurt, Otto Stern and Walther Gerlach sent a beam of silver atoms from an oven through a magnetic field arranged to be much stronger at one edge of the gap than at the other, and let the beam land on a glass plate.

A magnet in an uneven field gets pushed toward the strong side or the weak side depending on which way it points. Silver atoms come out of an oven pointing every which way, so the deposit should have been a smear.

It was two spots.

The deposit was too thin to see at all until Stern leaned over the plate to look, and the sulfur in his cheap cigars turned the silver into black silver sulfide, which developed the trace like a photograph. Two spots, nothing in between, and the beam had been split into precisely two by a question that could have been asked along any direction they chose to point the magnet.

That is what **spin** is, operationally. Point a magnet along a direction and you get a two-outcome measurement: the atom goes one way or the other, and the only thing you get to choose is the direction. There is no third outcome and no continuous dial. Nothing rotates, either. The name arrived in 1925, when two graduate students in Leiden, George Uhlenbeck and Samuel Goudsmit, proposed that the electron was a small spinning ball of charge, and the word outlived the picture by a century.

Chain three of these and the arrangement stops behaving like a set of properties being read off.

Send the beam through a magnet pointing up, and keep only the atoms that went up. Send those through a second magnet pointing sideways. They split fifty-fifty. Every atom in that beam had been certified up a moment earlier. Keep the ones that went sideways-left. Send those through a third magnet pointing up again, and ask the question whose answer is on record.

Half of them go down.

The atoms were up. The sideways question was asked. Half the certified-up atoms come out down. Skip the middle magnet and every atom comes out up, every time, so the middle magnet is doing it.

The temptation is to say that the sideways magnet knocked the atoms about, and that a gentler magnet would knock them about less. Gentleness has been tried, at every scale anybody can build. The answer never improves. What is going on is that the up-down question and the sideways question do not commute. The second answer overwrote the first because the two of them were never jointly available to be written.

Where the beam sits between the second magnet and the third is a **superposition**: the sideways-left beam, described in up-down terms, carries two numbers, one for up and one for down. Those numbers are **amplitudes** rather than probabilities. Probabilities are their squares. The difference is that amplitudes can cancel and probabilities cannot. The dark bands in the prologue's three-month photograph are places where two amplitudes of opposite sign met and left nothing. A beam that was secretly half up atoms and half down atoms, with nobody having looked, produces no cancellation anywhere, and a laboratory tells the two situations apart by bringing the split paths back together and seeing whether the original beam comes back.

19.4 Four multiplications

Heisenberg's tables carry one entry for each pair of states, and a two-outcome question has two states, so its table is four numbers in a square, which is a size a person can multiply on the back of an envelope.

Write U for the up-down question and S for the sideways one, both laid out in up-down terms, with the physical scale stripped off so the entries come out as plain numbers.

$$U = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \qquad S = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$$

U is diagonal because up and down are the answers it sorts into, plus one for the atoms that go up and minus one for the ones that go down, with zeros in the corners that would mix them. S has its entries in exactly those corners, since it exchanges up with down.

The rule is the one Heisenberg was forced into. For the entry in a given row and column of the product, run along that row of the left table and down that column of the right one, multiply the pairs and add.

Do US, which is the sideways question asked first, and then do SU. Both products have zeros down the diagonal. US comes out with a one in the top right and a minus one in the bottom left, and SU arrives with those two entries carrying the other sign. Subtract, and nothing cancels: a two in the top right, a minus two in the bottom left. The two orders are exact opposites, entry for entry.

Those four numbers are the middle magnet. Half of the atoms certified up came out down because a subtraction anybody can do in a minute refuses to vanish. There is nothing in those four entries to be gentler with.

Try a pair that behaves. Keep U, and write a second table with any two numbers you like down the diagonal, five in the top left and two in the bottom right, zeros in the other corners. Multiply it both ways round with U. Both products have five in the top left, minus two in the bottom right and zeros in the corners, so the difference is zero in all four places, whichever two numbers you picked. Tables that sort the atom into the same two answers commute, always, and the radar gun and the camera are a pair of that kind, which is why nobody at Petco Park had to decide which went first.

19.5 A question that squares to itself

Take the smallest well-behaved kind of question: one that admits yes or no and nothing else. Ask it, get an answer, and ask it again straight away. The second answer agrees with the first. Whatever operation the asking performs, doing it twice does what doing it once does.

An operation with that property is a **projector**. It has a name from geometry, because the operation that flattens an arrow onto a plane has the same property: flatten a flat thing again and it does not get flatter. Every yes-or-no question about a physical system is one of these, and every projector is a question. Its opposite is what is left of the identity operation when you remove it, so a question and its opposite add up to certainty.

A complete list of mutually exclusive alternatives is a set of projectors that never overlap and that add to the operation that does nothing at all. The three magnets gave two such lists on the same atom, one for up-down and one for sideways, and neither list is a refinement of the other. The list a question is asked inside of is its **context**. In richer systems than a single atom the same question sits inside many different complete lists.

To turn questions into odds you need one more instrument, and it is arithmetic. Add up the diagonal entries of a square array of numbers. That sum is the **trace**, and has the one property that makes it usable: rewriting the array in a different set of directions leaves the trace alone. The trace of a projector counts how many independent directions the question says yes to.

A **state** is the thing that assigns odds. It is a rule that hands every question a number between zero and one, hands the question with the certain answer a one, and adds up correctly across a complete list of exclusive alternatives. Probability enters this world as an additive weight on yes-or-no questions rather than as a count of how often something happened in a long run of trials. Every such rule can be written as an array of its own, called a **density matrix**, paired with the question through the trace:

$$p = \mathrm{Tr}(\rho P)$$

Here P is the question written as a projector, the Greek letter rho is the density matrix holding the state of whatever you are asking about, and p is the probability of the answer yes.

Run it on the beam between the second magnet and the third. In up-down terms that beam is the array with a half in all four places. Asking whether the atom went down is the array with a one in the bottom right and zeros elsewhere. Multiply the two, add the diagonal entries of the product, and out comes one half, which is the number the third magnet reported. Then set the two corner halves to zero, which turns the beam into one that really is half up atoms and half down atoms, and run the same multiplication. One half again. Every up-down question hands the two arrays the same number. The corners are the only place their difference lives.

The arrows those arrays act on, with lengths and angles supplied by the state, form a **Hilbert space**. Physics courses usually start there, with the space as the given object and the algebra of questions built afterward out of operations on it. The order runs the other way. What an observer has is a bounded list of questions with a composition rule, and the space is a way of writing that list down: pick a state, and the algebra plus the state manufacture a space of arrows on which the questions act as operations. Different observers with different question lists manufacture different spaces. They agree wherever their lists overlap, which is the only sense in which the wave function is a public object.

The manufacturing takes a few steps. None of them begins with a space. The algebra acts on itself, so that every question is both an operation and a thing operated on. The state supplies the geometry: it hands a number to every question, so it hands one to each pair of them, and a rule that gives a number to each pair of things is what lengths and angles are made of. Throw away whatever comes out with zero length, and what is left is a collection of arrows carrying the algebra as operations on it.

Two questions can be answered together exactly when a single complete list of alternatives is fine enough to answer both. That condition is chapter five's, wearing different clothes: compatible measurements are measurements that leave the same thing alone.

19.6 Averaging over what you do not hold

An observer holds a patch. The questions it can put are the ones on its side of the seam. The full description of the world involves everything past it.

So the state it holds is manufactured by averaging: take the description of the whole thing, and for every question you can actually ask, work out the odds and throw away everything that only a question from the far side could have detected. The operation is called the **partial trace**. Its output is a density matrix, which is why density matrices exist at all. A density matrix is what a description looks like after somebody has been cut out of the picture.

The averaging cannot be run backward. Two different worldwide descriptions can average to the same patch-level density matrix, and once you hold the density matrix there is nothing on your side of the seam that distinguishes them. It is a shadow. No cleverness turns it back into the object, because the object had features the averaging was built to delete.

A courtroom artist settles the same question every working day, with a few minutes, a partial view, and a drawing to hand in. The drawing has to be consistent with what was seen, and it must not commit to a detail nobody saw, because a detail that happens to be wrong is worse than a vague one. Edwin Jaynes wrote the rule down for physics in the *Physical Review* in 1957: among all the descriptions consistent with what has actually been measured, take the one with the largest entropy, which is the one that adds nothing else. The averaged description a patch holds is exactly that: the maximum-entropy state consistent with the questions it can put and the answers it has recorded, and every feature it lacks is a feature nothing on its side could have established.

19.7 Sorting the mail

Watch what writing a record does to an algebra.

Take a complete list of exclusive alternatives, and take any question at all. Cut the question into the pieces that live inside each alternative, keep those, and delete every piece that straddles two of them. That operation is called **pinching**, a mail room: letters keep their contents, they go into labeled bins, and what is destroyed is every comparison that only made sense between two bins.

Pinching is what writing a record does. The usual name in physics for the process is decoherence. The name describes the deletion rather than explaining it. What comes out of the mail room is an algebra with a property the original did not have: every surviving question commutes with every alternative on the list. The record cannot be disturbed by anything else the observer can ask, which is the whole of what makes it a record. You can read it without changing it, it reads the same the second time, and a neighbor who reads it gets your answer.

The set of things in an algebra that commute with everything in it is the **center**. Something is in there whatever the algebra is. Take the table with ones down the diagonal and zeros in the corners, the operation that does nothing, multiply either magnet by it in either order, and the magnet comes back with its four entries where they were. Doing nothing commutes with everything. Records live in the center.

Everything in the center commutes with everything else in it, so the center is a commutative algebra. A commutative algebra of this kind has one shape available to it. Strip a table of its off-diagonal entries and what is left is a list of numbers, one for each alternative: a value attached to each way things could have turned out, which is a function on a set of outcomes. A classical description has that in it and nothing else. The reason the public world wears that shape is a theorem about multiplication rather than a story about what a measurement does to a particle.

Look at what an observer finds when it looks at its own center. Questions with definite answers. Answers that survive being read. Answers that survive being copied to a neighbor, and that come back the same. Order that never matters, because everything in the center commutes with everything. Which is a complete description of the world of tables and chairs and pointer readings, arriving as the commuting part of a structure that is not commuting, rather than as a separate layer of reality sitting underneath.

The center is also the invariant part. Chapter five defined a symmetry as a change that leaves a named quantity alone, and said that a symmetry is the claim that part of your description was never physical. Chapter eleven built the record map, which takes the state of a network and returns the part that survives comparison. The center of an observer's algebra is what every internal change leaves alone. Symmetry, the record map, and the classicality of anything anybody can write down are one structure looked at three ways.

The relation between the two layers is sharper than a metaphor about shadows, and runs one way only. Upward, the record layer sits inside the operator layer perfectly: there is a map from records into questions that is faithful, that preserves sums and products and opposites, that lands region by region exactly inside each region's own algebra, and that gives every record the expectation value it had. Records are questions, of a particularly docile kind.

Downward there is no such map at all. A map that respected products would have to hand a single number to a whole block of questions that do not commute with each other, and no single number does that job, because the block contains pairs whose two orders of composition differ and a number multiplied by a number does not care about order. The best available downward map is an average. It preserves positivity and totals, it leaves records alone, and it fails to respect multiplication: there is an explicit projector in the block whose average of its square differs from the square of its average.

So the classical description is a subalgebra plus an average, the embedding is exact, and the average loses multiplication. There is no projection of the world onto the world of records. The reason is arithmetic rather than epistemology.

Condition on a record after it is written, which is the operation that makes records usable at all. The state you hold afterward is the one that gives that question probability one and keeps every compatible question's odds unchanged, which is the update rule Gerhart Lüders wrote down in 1951. Apply it twice and the second application does nothing. The operation is a retraction onto certainty: one step takes you into the set of states that agree with the reading, and further reading leaves you where you are. That is why a committed record is a stable classical fact inside a structure whose questions mostly refuse to commute.

Which inverts the usual complaint about quantum mechanics. **Non-commutativity is what creates constraint.** If every question could be answered alongside every other, every observer's answers would be one row of an enormous spreadsheet, any two rows could be filled in independently, and nothing would connect what one observer holds to what another one holds. A world of that kind has no seams in it, nothing to reverse engineer, and no structure to find. The limits are the content.

19.8 The observer, upgraded

Chapter three gave three items, arrived at by asking what the job requires and finding the cheapest object that does it: a bounded view, somewhere to write, and a reading that makes a difference. The third unpacks into three of its own, since a reading that makes a difference needs a way to update, a way to compare, and enough persistence to still be there at the next comparison. Those five become four objects. First, a local algebra of questions, which is the bounded view, bounded because the interface is. Second, a record algebra sitting in the center of that one, which is the somewhere to write, and centrality is what makes the writing stay put. Third, an interface that reads and updates the same way each time, so that comparison gives the same verdict on Tuesday as it gave on Monday. Fourth, enough checkpoint data to fix the odds on every question the observer can put next. The thermostat on the wall has all four: a question list one item long, a record of the setting, a comparison it performs identically every time, and a state that says what it will do next. The five-item version tells you what to look for on a bench. The four-object version can be handed to somebody else, because each of the four is a mathematical object with a name, and two observers can be asked whether they have the same one.

19.9 One generator

Chapter eighteen's screwdriver turned an eighth at a time and drove a screw the whole way into the wood, one small move repeated being what a smooth change is made of. Do the same to an algebra. A change that is reversible and that preserves the structure, meaning it respects sums, products and opposites, is the algebraic version of a rigid motion, and a continuous family of them, starting from the change that does nothing, is a flow: for every real number t, one such change, with the change at t plus s equal to doing the one at t and then the one at s.

On a full block of non-commuting questions, every flow of that kind is generated by a single time-independent quantity: one operator, fixed once at the start, whose repeated infinitesimal application produces every member of the family. Marshall Stone proved the correspondence in the *Annals of Mathematics* in 1932. The generator is what physics calls the Hamiltonian. Chapter eighteen's energy is that generator. A world with continuous reversible change in it has a Hamiltonian whether anybody wrote one down or not.

Ask how the family changes over a single instant rather than over a stretch and what comes back is the equation of motion: applied to the state it is the **Schrödinger equation**, applied to the questions it is Heisenberg's, the same statement written from opposite ends. That is the second of chapter one's three axioms, arriving here as a consequence of continuity and reversibility.

The flow fixes its generator to within one real number. Add a fixed amount to the Hamiltonian everywhere at once, which is to say add a multiple of the operation that does nothing, and that piece commutes with every question in the block and moves none of them, so the family of changes comes back exactly as it was. Two generators producing the same flow differ by that one amount and by nothing else. The gaps between the energies, the ratios of the gaps, the whole spectrum up to a rigid shift, all of it is pinned by the motion. Which is why a laboratory reads differences in energy. Balmer's four wavelengths are differences. The one number the flow declines to fix cancels out of every one of them.

The mirror statement is stranger and it is about the record layer. Every change of a record algebra that preserves the structure is a permutation of the finitely many labels the records can take. There are finitely many such permutations. A continuous family of them that starts at the identity has nowhere to go, since it would have to jump, so it sits at the identity forever. A purely classical public record cannot flow continuously and reversibly at all.

Put those two together. The world of pointer readings has no continuous reversible dynamics of its own, in the exact sense that its only structure-preserving motions are discrete relabelings. Whatever flows is flowing in the part that refuses to commute. The record layer changes by relabeling or it changes not at all.

19.10 A rule that fits every count

Max Born wrote the odds rule down in 1926, in a paper on collisions, putting it in the body of the text as the amplitude and correcting it to the square of the amplitude in a footnote. That footnote is the **Born rule**, a postulate in the standard formulation of quantum mechanics. Reconstruction theorems ask which other assumptions force the same probability rule.

Andrew Gleason derived it in 1957, in the *Journal of Mathematics and Mechanics*, from almost nothing: any assignment of numbers to yes-or-no questions that lands in the unit interval, gives the certain question one, and adds up across every complete list of exclusive alternatives is the trace against exactly one density matrix. The argument assumes no continuity anywhere and no shape whatever for the rule. There was one hole. Gleason's argument needs three dimensions or more, it fails at two, and two is where a spin lives. Paul Busch closed it in *Physical Review Letters* in 2003 by widening the questions to include blunt ones, which answer yes with a weight instead of sorting the world cleanly in two. The wider version covers every finite dimension including two.

A theorem that short, reaching that far, gets waved through and not believed, since it appears to conjure the whole of quantum probability out of bookkeeping.

Run a two-outcome instrument through its settings and count what comes out: eight settings, a pair of exclusive answers at each, and a frequency for each answer. Then go looking for any odds rule that fits those frequencies, with the trace formula given no privileges in the search.

One exists, and it is explicit. It reproduces every counted frequency exactly, as an identity between whole-number ratios rather than as a fit. Every question it scores lands between zero and one. The question with the certain answer gets one. A question and its opposite get numbers that add to one. And on every sum of exclusive questions that the run's own counts contain, it adds up correctly.

It is the odds rule of no state whatsoever. There is no density matrix at all whose trace against those questions returns those numbers.

Where it breaks is a pair of questions that nobody asked. Take the question this rule scores at 143/512, blunt enough that two copies of it side by side are again a legal question. Adding up demands that the pair score twice 143/512, so double it: 143/256. The rule scores the pair 35/64, which over the same denominator is 140/256. The gap is exactly three parts in two hundred and fifty-six. That single pair is enough to prove the rule belongs to nobody's state.

That locates the boundary. Additivity restricted to the sums the counted data happens to contain amounts to nothing more than the numbers in each setting adding to one. A demand that weak leaves the fake alive. The work is done by requiring the adding-up to hold for compatible questions the instrument was never pointed at. Impose that, and the same counts leave exactly one state standing, without a continuity assumption and without any appeal to long runs.

So the counted frequencies do not select the probability rule. What does select it reaches past every count anybody took. The demand is legitimate because two observers whose patches overlap have to agree about a question they share, and the weight either of them gives it cannot depend on which other questions they chose to ask alongside it, because the other observer made a different choice and the seam has to close. Context independence is the overlap condition, applied to odds.

19.11 Constraint

The commutator says how badly two questions refuse to be asked together, and the uncertainty relation prices that refusal in meters and kilograms and seconds. Questions that refuse nothing collect in the center. What collects there is the world of tables and pointer readings. Once the refusals hold across every context at once, the odds have one form available to them. It is the trace formula.

An observer holding an algebra of questions and a state has, in that pair, everything there is to have. Constraint is the part of the pair a commuting world could not supply, and constraint is what makes an undocumented machine readable from a seat inside it.

Constraint has a size. Put two observers far enough apart that no signal can cross between them while they work, hand each of them a pair of questions that do not commute, and compare the two sets of answers afterward. The agreement runs past anything a sealed envelope of instructions could have produced, and then it stops. Where it stops is one fixed number, 2.8284. No apparatus anybody has built has ever gone past it.

Chapter 20

Why Does Nature Cheat, But Only By Exactly $2\sqrt{2}$?

Early in 1981 the editor of *Foundations of Physics* sent a manuscript to Asher Peres at the Technion in Haifa and asked whether it ought to be published. It came from Nick Herbert, it was called FLASH, for first laser-amplified superluminal hookup, and it described a telegraph for sending messages faster than light.

The design was clean enough that saying which part of it failed took three separate proofs. Prepare a pair of photons together and send them in opposite directions. Alice, at one end, chooses between measuring plane polarization and measuring circular polarization, and her choice is the message: one question for a zero, the other for a one. At the other end Bob does nothing subtle. He drops his photon into a laser gain tube, which turns one photon into a shower of photons like it, and inspects the shower. Alice's choice fixes which mixture Bob's half is drawn from, a shower of copies would show which mixture that was, and the distance between the two of them does not enter the timing anywhere.

Peres knew the paper was wrong for the reason anybody would: a working version breaks relativity, and relativity does not break. Nothing in Herbert's argument touched relativity anywhere, though, so the error had to sit somewhere else, and Peres could not find it. He recommended publication, telling the editor that the paper was obviously wrong, that it would draw considerable interest, and that whoever located the error would push physics forward. He kept his own part in it quiet for twenty years. "I was the referee who approved the publication of Nick Herbert's FLASH paper, knowing perfectly well that it was wrong," he wrote in 2002, in a note titled *How the no-cloning theorem got its name*.

FLASH appeared in 1982, in volume 12 of the journal, running from page 1171 to page 1179. The refutations beat it into print. Bill Wootters and Wojciech Zurek published theirs in *Nature* that October, volume 299, page 802, under a title John Wheeler contributed, "A single quantum cannot be cloned," and Dennis Dieks published the same result independently in *Physics Letters*, volume 92A, page 271. The second referee, GianCarlo

Ghirardi, had put a special case of the argument into his report and voted to reject; he published it the following year in *Il Nuovo Cimento*.

Every component of the telegraph works except one. The correlations Herbert wanted to exploit are real, they are stronger than any set of instructions the two photons could have carried with them, and they are measured routinely. The gain tube is the piece that fails, for a reason that turns out to be the same reason the correlations stop where they do. How much better than a sealed answer sheet can two separated people agree, and what is it about an unmeasured photon that no machine can copy?

20.1 Four questions and a referee

Put the correlation in the form of a game, with a referee who keeps score.

Alice and Bob may confer for as long as they like beforehand and may carry anything they can build. Once the game starts they are separated and no signal passes between them. The referee tosses two coins and hands each of them one question, labeled 0 or 1, and neither is told which question the other received. Each answers with one bit, written here as plus one or minus one because it shortens the arithmetic. They win the round if their two answers differ when both were handed question 1, and match otherwise.

Three of the four question pairs reward matching and the fourth rewards differing. The referee's coins deliver all four equally often. So the obvious strategy is to match always: agree in advance that both will answer plus one, whatever they are asked. That wins three rounds in four. Seventy-five percent.

Nothing sealed does better. Write Alice's answer sheet as two numbers, one for each question she might be handed, and Bob's the same way. Score the round by multiplying the two answers together, which gives plus one when they match and minus one when they differ. Over many rounds each of the four question pairs earns an average of those products, a number between minus one and plus one that runs from perfect agreement to perfect disagreement. That average is the **correlation** for that pair. Take the four correlations, add the three that reward matching, subtract the one that rewards differing, and call the total the score. A strategy that satisfied all four demands at once would score four.

For a sealed answer sheet the score is Alice's first answer times the sum of Bob's two, plus her second answer times the difference of Bob's two. Pick four numbers of your own, each plus one or minus one, and add up the two pieces. Bob's two answers are each plus or minus one, so they are equal or they are opposite. If they are equal, the difference is zero and the sum is plus or minus two. If they are opposite, the sum is zero and the difference is plus or minus two. Either way the score is exactly two in magnitude, every time, for all sixteen sheets, and averaging over any mixture of sheets keeps it there, because an average of numbers no bigger than two is no bigger than two. The ceiling on sealed instructions is John Bell's, from 1964. The four-setting form is the one John Clauser, Michael Horne, Abner Shimony and Richard Holt wrote down in 1969.

20.2 Fifty-two cards

To do better, Alice and Bob need one description of the pair rather than two descriptions of the parts.

A playing card carries a suit and a rank. Four suits, thirteen ranks, fifty-two cards. Describing a card means fixing both. The count of descriptions is the product of the two counts rather than their sum. The operation that builds the descriptions of a pair out of the descriptions of its parts is the **tensor product**. Its first property is that arithmetic: dimensions multiply.

"Hearts" and "seven" is a description that comes apart into a fact about the suit and a fact about the rank. "The red seven or the black eight" does not come apart into anything. It fixes how color and rank go together while leaving both of them unsettled, and learning either one hands you the other. Most joint descriptions are of that kind. The ones that factor are a thin special case.

Alice's questions and Bob's are far enough apart that no light ray crosses between them during a round, so every question on her side commutes with every question on his: the bracket across the gap is zero, and the order in which the two of them ask changes nothing. What does not commute is Alice's first question with her second.

20.3 Four lines

Write Alice's two questions as A0 and A1, and Bob's as B0 and B1, each one an operation on the pair's joint description that returns plus one or minus one. Asking a plus-or-minus question twice returns you to where you started, so each of the four squares to the do-nothing operation. Build the score as an operator: A0 times the sum of Bob's two, plus A1 times the difference of Bob's two. The score of any state is the average of that operator in that state.

Square it. Sixteen terms come out. Four of them are squares, and each question squares to the do-nothing operation, so those four add to four times it. Eight more cancel in pairs, plus against minus, once you use the fact that everything on Alice's side commutes with everything on Bob's. The last four collapse into a single product of brackets.

$$(\text{score operator})^2 = 4 + [A_0, A_1]\,[B_1, B_0]$$

The square bracket is chapter nineteen's commutator, the amount by which the order of two questions matters, and is zero exactly when the two can be answered together. The four is a sealed answer sheet's two, squared. The second term is the whole of the quantum contribution, a product of two commutators, one from each side.

Put a number into that line. Suppose Alice can answer both of her questions at once and Bob both of his, which makes both brackets zero. Anything multiplied by zero is zero, so the second term goes. The square of the score operator is four times the do-nothing operation. Something that squares to four times the do-nothing operation reads out as two or as minus two. An average of twos and minus twos lands between them. The sealed answer sheet's ceiling comes back out of a line with no envelope anywhere in it. The 1969 inequality is this identity with its brackets switched off.

How big can that term be? Every operator has a size, the largest factor by which it can stretch anything it is applied to. The name for it is the **operator norm**. A question returning plus or minus one stretches nothing, so its norm is one. A commutator of two such questions is a difference of two products of things of norm one, so its norm is at most two. Multiply Alice's bracket by Bob's and the product has norm at most four. The square of the score operator therefore has norm at most eight, so the score operator has norm at most the square root of eight. An average can never exceed the largest value being averaged. So no state, no choice of questions, and no amount of prior conferring gets a score above $2\sqrt{2}$, which is 2.8284.

Work the square root of eight out by hand rather than reading it off a machine. Eight is four times two, and the square root of a product is the product of the square roots, so the square root of eight is twice the square root of two. The square root of two is 1.41421, the length of the diagonal of a square one unit on a side, and the first irrational number most people are shown. Double it. 2.82842. Multiply 2.8284 by itself and you get 7.9998. The two ten-thousandths short of eight are the digits dropped off the end.

One line read at its two ends gives both ceilings. With the brackets at zero it says four, and the score tops out at two. With them opened as wide as they go it says eight, and the score tops out at 2.8284. A bracket reaches two only where doing Alice's two questions in one order comes out as the exact negative of doing them in the other. A pair that refuses each other by less has a ceiling below 2.8284. The four in the middle does not move, which leaves the 0.8284 of score between the two answers held entirely inside a product of two commutators.

Divide one ceiling by the other. 2.8284 over 2 is 1.4142, the diagonal again: questions that refuse to be asked together beat a sealed envelope by the same factor a square's diagonal beats its side.

Carry that back to the referee. A score of S wins a round half the time plus S divided by eight, which turns the sealed ceiling of two into seventy-five rounds in a hundred and the ceiling of $2\sqrt{2}$ into 85.36. Just over ten extra rounds in a hundred is the whole of the quantum advantage in this game. Stuart Freedman and John Clauser went looking for those ten rounds at Berkeley with photons from a calcium cascade, found the margin where quantum mechanics says it is, and published on 3 April 1972.

Count what went into that ceiling. Four things that square to one, and two of them commute with the other two. Geometry never appeared. There is no space in it, no distance, no light cone, no wave, and nothing that could be called a particle. The ceiling holds anywhere four questions behave that way and sizes can be compared. This world's questions behave that way. Boris Tsirelson found the number in 1980 and published it in *Letters in Mathematical Physics* under the title "Quantum generalizations of Bell's inequality".

The gap between two and 2.8284 is that commutator term. Non-commuting questions are why joint knowledge has limits at all, and the same brackets, multiplied together, are why separated agreement has a ceiling. What stops Alice from answering both her own questions at once is what lets her agree with Bob more often than any answer sheet permits. Take the brackets to zero and the score falls back to two, and the world becomes a world of answer sheets.

20.4 Real numbers all the way

Set up two slots, each with two outcomes, so the pair has four dimensions. Put them in the state whose two answers are perfectly opposed along whichever direction you ask. Have Alice ask along two perpendicular directions and Bob along the two directions bisecting hers, an eighth of a turn from each. The score comes to $2\sqrt{2}$ on the nose. Every matrix in that arrangement is real: the state, the four questions, the whole construction. The most strongly correlated thing two separated systems can be put into needs no imaginary numbers whatsoever.

Change one thing about it. Take a state diagonal in the record basis, which is to say a state that has been written down, an outcome list with weights sitting in chapter nineteen's commuting center, and read it with four readouts diagonal in that same basis. Operators diagonal in one basis commute, so both brackets vanish, the second term in the square is zero, and the score is at most two, for every such state and every such readout. The written layer of the world plays this game exactly as well as an answer sheet does, and for the same reason: a committed record is an answer sheet, filled in and handed over. Everything an observer can point at, quote or file has come through that door, which is why the public world behaves like a world of objects with properties, and why the extra ten rounds in a hundred have to be hunted for with equipment.

20.5 What the gain tube would have needed

A copier is one fixed machine, chosen in advance, that takes an unknown state in one slot and a blank in the other and hands back the unknown state in both. A machine you have to configure for the state you are about to copy is useless on a photon arriving from a laboratory you cannot see.

Two states have an overlap, a number from zero to one, which is how often one of them passes a test built for the other: zero for perfectly distinguishable, one for identical. Reversible operations preserve overlaps, because preserving the structure overlaps are computed from is what reversible means here. Run the copier on two states and compare the outputs. Each slot contributes the original overlap, so the output overlap is the input overlap multiplied by itself. Preserved and squared at once, so the overlap equals its own square, and exactly two numbers do that: zero and one.

The machine copies states that are perfectly distinguishable, or states that are identical, and nothing in between. Herbert's gain tube was asked to copy a photon whose polarization nobody had measured, which is the in-between case and the only case the telegraph needed. Two lines, and the design is gone.

Look at which states the copier does handle. Overlap zero, perfect distinguishability, which is what a record is: an outcome that has been committed, sitting in the part of the algebra that commutes with everything, as distinguishable from another record as one word on a page is from a different word. Copying is forbidden for unwritten states and permitted for written ones, without limit and without disturbance, which is why a photocopier is a machine anybody can buy.

What is copyable is exactly what has been written down as a record, and that is why redundancy works: why a backup is a backup, why a thousand people read one printed page and report the same words, why a measurement copied into a hundred notebooks is the same measurement in all hundred.

20.6 Twelve directions

Alice and Bob have been free to point their equipment any way they like. Chapter six wired twelve observers into an icosahedron, and chapter fifteen read off which way each of the twelve ports points; a universe with twelve ports has no thirteenth direction to offer, so the four settings have to come from those twelve. One description of a pair of slots survives every rotation of the solid without being a mixture of others, the perfectly opposed one.

Four settings drawn from twelve, in order, is 20,736 choices, which is few enough to take all of them. The best score is reached by 960 of the 20,736, and all 960 run the same way. The twelve ports pair off into six axes, corner to opposite corner, and any two of those six meet at 63.43

degrees. Each of the 960 uses three of the six, one axis shared between Alice and Bob. Three of the four pairings put her axis at 63.43 degrees to his, where the correlation has size one over the square root of five, 0.4472. The fourth puts her on his own axis, where the size is one. The four line up rather than cancel, so the score is one plus three-fifths of the square root of five in magnitude. Multiply the square root of five, 2.2360679775, by six and move the point one place left: 1.3416407865. The one on top of that makes 2.3416407865. Sealed instructions on the same four settings land on plus two in eight of their sixteen patterns and on minus two in the other eight, and nothing outside that.

The icosahedron's score clears two, so twelve observers on a closed surface do the thing the telegraph was built to exploit: they agree in a pattern that no answer sheet reproduces. It stops short of 2.8284 because of the corners. Reaching the ceiling wants Bob's directions an eighth of a turn from Alice's, and an icosahedron's twelve corners offer exactly four angles between them, zero, 63.43 degrees, 116.57 degrees and 180. An eighth of a turn is not among them. Twelve fixed directions cheat, and take two fifths of the distance from two to 2.8284, and stop.

The ceiling came out of two objects and nothing else: an algebra of questions, carrying its brackets, and a state assigning odds to them. That pair is the entire inventory of an observer. Go through it looking for a clock and you come back holding a list of questions and a list of numbers, and nothing in either list moves. Something takes time anyway.

Chapter 21

Why Does Time Feel Like It's Moving?

Augustine was in his forties and a year or two into the bishopric of Hippo Regius, a port on the North African coast that is Annaba in Algeria today, when he wrote the eleventh book of the *Confessions*. The thirteen books took him from about 397 to about 400. Twelve of them are about his life and his God. The eleventh is about time, and in its fourteenth chapter it comes apart in his hands.

"What then is time? If no one asks me, I know; if I want to explain it to a questioner, I do not know."

The line gets quoted as a piece of wisdom, which flatters it. Augustine was after an instrument. He works through the possibilities in the chapters on either side of that sentence and discards each one: the past has gone and cannot be laid alongside a ruler, the future has not arrived and cannot either, and the present has no extent, because anything with extent divides into a part that has gone and a part that has not come. He ends the chapter holding a ruler with nothing in front of it.

What he settles on, having run out of things in the world to measure, is that the measuring happens in the mind: the soul stretched across a memory, an attention and an expectation, so that the quantity being compared against a ruler is an impression rather than an interval of anything. He gives it a name, *distentio animi*. The name has kept philosophers busy for sixteen hundred years. It moves the ruler indoors and leaves the question of what is being laid against it exactly where it was.

Sixteen centuries of instrument-building have not improved his position. In 1967 the thirteenth General Conference on Weights and Measures redefined the second as the duration of 9,192,631,770 periods of the radiation from a particular transition in a cesium-133 atom, which is the most carefully specified duration in the history of the species and is a duration. A cesium clock counts periods and reports a count. Subtract two counts and you have an interval. Every timepiece anybody has built, from a notched candle to the fountain clocks that keep international atomic time to sixteen digits, reports a difference between two events, and none of them has an output for the present moment, because there is no part of the mechanism that could produce one.

Chapter sixteen separated order, direction and duration. Informational order records which commit certified reading a version written by another. The wiring constrains possible conflicts; authenticated provenance fixes the actual edges. Averaging seven inputs into one illustrates a loss from a reduced seam description, while retaining an input record can preserve the distinction elsewhere. Neither this arithmetic nor the dependency graph gives an elapsed time. A clock requires an additional dynamical rate and a calibration.

21.1 An algebra and a state

An observer holds an algebra of questions, which is everything it can ask about its own patch, carrying the brackets that record which pairs can be asked together and which cannot. And it holds a state, which is the odds it puts on every one of them.

That pair fixed the floor under position and momentum, and put the ceiling on separated agreement at $2\sqrt{2}$ and not a thousandth higher. It also decided which part of the world is classical, by collecting the questions that commute with everything into a center and letting the records live there.

Take the pair apart looking for a clock.

The algebra lists questions and how they compose. The state assigns their probabilities. Neither object comes with seconds written on it. Together they can define a modular flow, a mathematical family of reversible changes. Interpreting its parameter as a clock reading requires a physical dynamics and an instrument.

Chapter ten's run makes the distinction concrete. Eighty-two thousand patches worked all 122,880 seams into agreement, but the reported cycles came from the executor. Such bookkeeping does not identify a physical tick. A clock needs a changing degree of freedom, a readable record and a calibration.

21.2 A room left alone

Leave a room alone for long enough and everything in it arrives at one temperature. The coffee, the mug, the table, the air above the table. The description of that room has collapsed to a single number. The collapse discards almost everything that was ever true about it.

A temperature helps determine the direction of heat exchange when two systems meet. Its rate also depends on their materials and contact. The equilibrium description and the dynamics leading to it are different pieces of information.

Run that argument backwards.

Chapter sixteen supplied a reference law preserved by a specified resampling channel. Invariance names an equilibrium only relative to that dynamics. For the modular construction, the question is which reversible flow has the chosen state as its normalized thermal reference.

So work out what it would have to be.

A change that qualifies is a reshuffling of the observer's questions among themselves that respects the algebra: questions go to questions, sums to sums, products to products, and the whole thing is reversible. Line those shuffles up, one for every real number, and the family is what chapter eighteen called a flow, driven by the move it repeats, its generator.

Under the modular theorem's algebra and faithfulness hypotheses, two demands characterize the normalized flow.

The first is that the state's odds are unchanged by it. Every question is worth after the shuffle exactly what it was worth before. That is what being settled means, written out.

The second is where the non-commuting part earns its keep. In a world where every pair of questions could be answered together, the state's weight for asking A and then B would equal its weight for asking B and then A, because there would be nothing to tell the two orders apart. Chapter nineteen's entire subject is that they differ. So the state carries a gap between the two orders, one gap for every pair of questions. The gap is a fixed feature of the state rather than a nuisance to be argued away. The second demand is that swapping the order costs exactly one unit of the shuffle, the same one unit for every pair of questions, with the amount of shuffling measured along a direction at right angles to the flow's own parameter. That last clause is complex analysis, the one step in the whole construction that cannot be done on a table.

For a faithful state in the modular theorem's setting, those demands select a unique flow. In the finite matrix example, faithfulness means that every eigenvalue of the density matrix is positive. The resulting parameter is modular time. A nontrivial flow, a suitable prepared readout and a physical calibration are further ingredients of an operational clock.

Minoru Tomita, who lost his hearing at the age of two and published very little in a long career, wrote the construction down in 1967 in a manuscript that the people who received it found close to unreadable. Masamichi Takesaki worked out what it said, passed the results to Jacques Dixmier in the summer of that year, and published a usable version in 1970 as a thin volume of Springer lecture notes. The thing is called **modular flow**, after the algebraic machinery it drops out of rather than after anything it does, and its generator, in chapter eighteen's sense of the word, is called the modular Hamiltonian.

In the finite matrix case the recipe is short: take the matrix logarithm of the strictly positive density matrix and change its sign. This dimensionless operator generates modular flow in a specified sign convention.

$$K = -\log \rho$$

Here rho is the density matrix and K the dimensionless modular Hamiltonian. The logarithm is natural: a doubling contributes 0.693. To apply it, first diagonalize rho, take the logarithm of each positive eigenvalue, and transform back. The formula is a finite matrix realization; general operator algebras use the modular operator.

21.3 One system, two outcomes

That recipe can be run by hand on the smallest system there is.

Take a two-level system with supplied energies zero and E. Put it in a Gibbs state at a supplied temperature T. The upper-to-lower probability ratio is the exponential of minus E divided by T when temperature is written in energy units. These energy and temperature inputs give the state its thermal interpretation.

Write the state as its array. Take the logarithm, which acts on the two diagonal entries separately. Put a minus sign in front. What comes out has E divided by T in the upper slot, zero in the lower, and a constant added to both, contributed by the requirement that the odds add up to one.

The constant does nothing whatsoever. A generator shifted by a constant produces the same flow, since the constant piece commutes with the whole algebra and shuffles nothing. Delete it.

What survives is the energy operator divided by the temperature in energy units. Its modular flow follows the same operator orbits as the supplied Hamiltonian evolution, with the parameter scale and orientation fixed by the temperature, time unit and sign convention.

Put numbers on this diagonal thermal state. Let the lower level have probability 20 in 21 and the upper level 1 in 21. The ratio determines the two diagonal probabilities. Off-diagonal coherences are absent on this specified Gibbs branch.

The lower slot of the generator holds zero, so the whole generator is whatever stands in the upper slot. That is the logarithm of the ratio between the two odds.

$$K = \log \frac{\text{odds of the lower answer}}{\text{odds of the upper answer}} = \log 20 = 2.996$$

Add it up yourself. Twenty is two times ten, so its logarithm is the sum of theirs: write 0.693 for the two, write 2.303 for the ten, and the twenty comes to 2.996, four thousandths short of 3. The check runs from the other end, since the exponent 3 gives 20.0855, a shade over the twenty on the slip.

The dimensionless generator gap is 2.996. On the supplied thermal branch, that equals the energy gap divided by temperature in energy units. It does not determine either quantity separately.

The slip of probabilities therefore determines a modular Hamiltonian. To identify a physical Hamiltonian H, supply a positive inverse temperature beta and the Gibbs relation: K equals beta times H plus log Z times the identity. Scaling H and the temperature by the same positive factor preserves the slip.

21.4 Whose clock

The flow belongs to the algebra–state pair.

Different faithful states can define different modular flows, although some distinct states give the same flow. A tracial state, which weights a finite matrix block uniformly, gives a trivial flow even on a noncommutative algebra. Comparing two physical clocks additionally requires their dynamics, readouts and a relation between their time units.

The two-level example shows the parameter dependence. Keep H fixed and halve the temperature. The probability ratio squares from twenty to one to four hundred to one, and the modular generator gap doubles from 2.996 to 5.99. The two modular parameters then use different scales for the same Hamiltonian evolution. A prediction about two clock instruments requires their physical readout models.

Chapter eighteen approached energy through generators. Here the state supplies a dimensionless modular generator. Joules enter through the separate energy identification and calibration, rather than through a count of unsettled records.

The state dependence here is precise: a normalized modular flow is determined by the specified algebra and faithful state. That mathematical fact does not by itself identify a subjective experience, a laboratory duration or a rate in seconds.

21.5 The center does not move

Records live in the center of the algebra: the questions that commute with everything, which is what makes a record readable without disturbance, copyable to a neighbor, and the same on the second reading as on the first. Run the flow on the center and it does nothing at all. Every element of the center is left exactly where it is by every transformation in the family, at every value of the parameter.

The flow was built to close the gap between asking A and then B and asking B and then A. An element of the center has no gap with anything, because it commutes with everything in the algebra. So there is nothing for the flow to fix about it, and it sits untouched. The price of commuting with everything is having no clock of your own.

Chapter nineteen proved the harsher version, which reaches past this flow to every continuous motion a record layer could have had. The only structure-preserving moves a record algebra has are permutations of its finitely many labels, and a continuous family of permutations would have to jump to get anywhere, so it sits where it started. The classical layer changes one discrete relabeling at a time, or it holds.

These facts locate possible modular motion outside the center. They do not prove a mechanism for experienced time. Even there, a tracial state has trivial flow. To build a clock, one must specify a nontrivial dynamics, a preparation and readout that reveal change, and a physical time scale.

21.6 The flow has no arrow

The flow is reversible. Every transformation in the family has its inverse in the family, which is the transformation at minus the same parameter. That was settled before either demand was written down: what qualified as a change at all was a reshuffling of the questions that loses nothing, and a shuffle that loses nothing can be undone.

This reversible flow does not itself supply a dissipative arrow. Chapter sixteen's averaging example loses distinctions from the seam description, while a retained record can preserve them elsewhere. The conditional thermal resampling construction gives a separate entropy-production inequality under its stated reference and energy assumptions.

The constructions answer different questions. An authenticated history gives event dependencies. A specified dissipative channel gives an entropy-production direction, while reversible dynamics gives a rate in its own parameter. A physical clock combines a repeatable dynamical process, readable records and a calibration. Recording its readings does not make every commit a logically irreversible erasure.

21.7 The region's own clock

Chapter eighteen defined energy and left an unpaid bill inside the definition.

A region's dynamics can have an energy generator once its time parameter and energy convention are fixed. A repair-event count alone supplies neither that generator nor an elapsed duration.

For a finite region with a faithful density matrix, the modular construction supplies K = −log rho and its dimensionless parameter. A physical clock adds a changing preparation and readable records. Comparing rates between regions requires a common physical interpretation of those readouts.

On a supplied Gibbs branch, K = beta H + log Z times the identity, with beta = 1/(k_B T) for T in kelvin. The constant disappears from the flow. Identifying H in joules and relating its evolution to seconds require the energy, temperature and time dictionary. Modular reconstruction establishes the relation once those inputs are supplied.

21.8 The ship that light never catches

Now supply a Minkowski quantum field theory in its vacuum state, a wedge algebra satisfying the Bisognano–Wichmann hypotheses, and an ideal observer with constant proper acceleration. This is a physical continuum example of the modular construction.

Chapter seventeen constructed a Lorentz kinematic carrier, with three rotation and three boost directions. In the present supplied spacetime, boosts act geometrically on a wedge, and a uniformly accelerating trajectory follows one boost orbit.

In Minkowski spacetime, an eternally uniformly accelerating observer has a causal horizon. Signals from beyond the relevant horizon never reach that trajectory. This uses the spacetime's physical causal structure; the finite kinematic carrier by itself does not identify an observer's physical light cone.

Use the quantum field theory's wedge algebra and the vacuum restricted to it. Locality, the vacuum and the hypotheses linking the algebra to spacetime are supplied here. The geometric boundary alone does not construct that field theory or state.

For this vacuum–wedge pair, the Bisognano–Wichmann theorem identifies modular flow with Lorentz boosts at the fixed normalization. Along a specified uniformly accelerating orbit, boost parameter is proportional to proper time, with the conversion set by the acceleration and physical units.

Joseph Bisognano and Eyvind Wichmann proved it for quantum field theory in two papers in the *Journal of Mathematical Physics*, "On the duality condition for a Hermitian scalar field" in volume 16 in 1975 and "On the duality condition for quantum fields" in volume 17 in 1976. They were not working on a theory of time. The question in front of them was technical, whether the operators available inside a region exhaust the operators that commute with everything outside it. The flow of the region turned out to be a Lorentz transformation.

Here the algebraic and geometric flows can be compared because the field theory, vacuum, wedge and proper-time convention have all been specified. Their agreement gives a concrete physical realization of modular flow under those hypotheses.

21.9 Warm

The physical identification also gives a thermal detector relation.

The vacuum restricted to the wedge satisfies the thermal KMS condition for boosts. Converting boost parameter to the uniformly accelerated observer's proper time gives the Unruh temperature. A detector reads that relation through its coupling to the field; it is not a statement that every thermometer or finite observation behaves as an equilibrium bath.

Bill Unruh calculated an ideal detector response. In the stationary, long-duration limit, a ground-state inertial detector in the Minkowski vacuum has zero excitation rate. A uniformly accelerated detector has a thermal excitation-to-de-excitation ratio at the Unruh temperature. Finite switching and the chosen coupling require their own response calculation.

He did that calculation in 1976, in a paper called "Notes on black-hole evaporation" in *Physical Review D*. The temperature is proportional to the acceleration. The constant of proportionality is about four parts in ten to the twenty-first of a degree for every meter per second squared.

Which is why nobody has run into this by accident. Warming yourself by a single degree calls for an acceleration around twenty-five billion billion times the strength of gravity at the Earth's surface, sustained, with you in the vehicle. Every acceleration a human body has ever survived leaves the reading far below what any thermometer built resolves, and the passengers have complaints about the other effects.

The algebra and faithful state define a modular parameter and a normalized KMS relation. The supplied Minkowski field theory and vacuum identify that flow with boosts; acceleration and the physical units convert it to proper time and temperature. The laboratory readings depend on these additional identifications.

In this physical example, duration and temperature are related through one specified dynamics. The general modular theorem supplies the mathematical relation, while a clock and a thermal instrument supply its calibrated readings. What does the temperature reading measure?

Chapter 22

Why Does Writing Something Down Cost Heat?

In 1824 a French military engineer of twenty-eight paid a Paris publisher named Bachelier to print six hundred copies of a book about steam engines, and then waited for a scientific community that did not read it. Sadi Carnot's *Reflections on the Motive Power of Fire* runs to a hundred and eighteen pages and asks the question every engine builder in Europe wanted answered, which is how much work you can get out of a fire.

His answer has no pistons in it. The most work any engine whatever can deliver, per unit of heat it swallows, is fixed by two numbers: the temperature of the thing it takes the heat from and the temperature of the thing it dumps the remainder into.

$$\eta = 1 - \frac{T_{\text{cold}}}{T_{\text{hot}}}$$

The Greek letter names the efficiency, the fraction of the heat that comes out as work. The two temperatures are the hot source and the cold sink, both counted from absolute zero. Brass or iron, steam or air, a better valve, a cleverer linkage, a bigger boiler: the ceiling sits where those two temperatures put it and no engineering moves it.

Carnot got that while believing heat was a weightless fluid called caloric, which fell from hot to cold the way water falls down a millrace and did as much work on the way. Caloric does not exist. Every heat engine built since has come in under the number he got out of it.

Cholera reached Paris in the spring of 1832 and killed something near twenty thousand people in the city that year. Carnot caught it on 24 August and was dead within the day, at thirty-six. His belongings were burned as a precaution against contagion, and nearly all of his papers went with them. What survived was a set of working notes his brother Hippolyte published in 1878, in which the fluid had been quietly dropped.

Look at what the surviving formula says. An engine's ceiling depends on the temperatures and on nothing else. That is a peculiar thing for a fact about brass and steam to be. Chapter twenty-one reached the same quantity from the other end, out of an algebra and a state, with no furnace anywhere in the argument. So what is a temperature a count of?

The everyday version sits on your desk. A drive with no moving parts gets warm while it is being written to. Its heat depends on the device and protocol. The universal erasure bound concerns a more specific operation than writing a record.

22.1 The doorkeeper

On 11 December 1867 James Clerk Maxwell wrote to Peter Guthrie Tait describing a way to break the second law of thermodynamics using a very small employee.

Take a box of gas at one uniform temperature, divide it with a wall, and put a door in the wall. Temperature is an average: some molecules in there are moving fast and some slow. Station a being at the door with good eyes and a light touch. When a fast molecule approaches from the left, it opens the door and lets it through to the right. When a slow one approaches from the right, it opens the door and lets it through to the left. The door is weightless and frictionless, so opening and closing it costs nothing.

Wait. The right side heats up and the left side cools down, out of a box that started uniform. Then run Carnot's engine off the difference, and run it again, forever, in a box that nobody has done any work on.

Maxwell called his employee a finite being and later grumbled that it was really more of a valve. William Thomson named it a demon in *Nature* in 1874, meaning the word in its Greek sense, a spirit working quietly in the background.

The demon survived every attempt to kill it for sixty-two years, because every attempt went looking in the wrong place. People examined the door, the hinges, the light the demon needed to see by. Leo Szilard moved the search in 1929, in a paper in *Zeitschrift für Physik* whose title says where he was looking: on the decrease of entropy in a thermodynamic system by the intervention of intelligent beings. He stripped the gas down to a single molecule in a box, with a partition dropped in the middle. The demon looks, records which side the molecule is on, and uses that one recorded fact to let the molecule push the partition outward and lift a weight. The work per cycle comes out as the temperature times the logarithm of two, out of a box at one temperature, from one bit of recorded information.

Rolf Landauer, at IBM, identified the relevant operation in 1961: erasure. Resetting an unknown memory state to a fixed value reduces its entropy. In the standard thermal setup, the memory initially has no correlations with a heat reservoir at temperature T, and the combined evolution preserves information. Entropy lost by the memory must then be accounted for in the reservoir and their correlations. For one equally probable bit, the mean heat delivered to that reservoir obeys

$$E \geq k_B T \ln 2$$

E is the mean heat delivered to the initially thermal reservoir, and T is its temperature. The constant converts kelvin to joules. The logarithm of two is the entropy decrease of an equiprobable bit reset to a definite value, with no usable side record reducing the erasure task.

Put a room temperature into it, three hundred kelvin, and the bill for erasing one bit comes to 2.87 divided by ten to the twenty-first, in joules. A single photon of green light at a wavelength of 550 nanometers carries 3.61 divided by ten to the nineteenth, in joules, which is enough to pay off about a hundred and twenty-six bits.

Charles Bennett finished the demon in a review of the thermodynamics of computation in 1982. The demon's memory is finite. It can run a few cycles free, filling that memory with records of which side each molecule was on, and then it is full, and to take another measurement it must reuse or extend that memory. In the ideal thermal cycle, resetting each independent fair bit costs at least the work that bit enabled it to extract. The demon was never an engine. It was a machine borrowing against its own memory. The loan came due at the same rate every time.

Antoine Bérut and colleagues put the number on a bench in 2012 and published it in *Nature*. A silica bead about two micrometers across, held in an optical trap shaped into two wells, one well for zero and one for one. Push the bead into a known well and you have erased a bit. Measure the heat that leaves. As the erasure cycles were slowed down, the mean heat per erasure came down onto that value and did not go below it.

Chapter three asked what a physical record costs. Landauer answers the erasure part of that question, once the memory, reservoir and entropy change are specified.

22.2 Four laws, numbered out of order

Thermodynamics has four laws. They were not discovered in the order they are taught.

Start with the one everybody uses without noticing. Nobody has ever established that a bath and a bowl of soup are at the same temperature by putting the soup in the bath. You put a thermometer in the soup, wait, read it, then put the same thermometer in the bath, wait, read it, and compare two numbers taken minutes apart with an instrument that has been in contact with each and never with both at once.

That procedure is a bet on transitivity. If the thermometer settles with the soup, and the same thermometer settles with the bath, then the soup and the bath will settle with each other. The bet is a law: two systems each in equilibrium with a third are in equilibrium with each other. Ralph

Fowler and Edward Guggenheim wrote in 1939 that it “could with advantage be known as the zeroth law of thermodynamics”, the other three having been numbered decades before anybody noticed this one was doing work underneath them.

Transitivity is what makes temperature a number. A relation that is transitive, and symmetric, and holds between a thing and itself, sorts everything in the world into classes with nothing left over, and classes in a row can be labeled by numbers on a scale. Without transitivity you could say that this is hotter than that, pair by pair, and there would be no scale to write either of them on, no degrees, no thermometer, and no sentence of the form “the water is at 47 degrees”.

The first law separates changes of energy into heat and work. Work changes the available energies at fixed probabilities; heat changes probabilities at fixed energies. For a finite update that changes both, comparing only the two endpoints adds the product of those changes. It vanishes for an ordered stroke holding one quantity fixed. In the differential law the product is second order, so the usual two-term formula is exact even when both vary continuously.

The second law arrived in chapter sixteen, in a laboratory in Canberra with a latex bead in water. In the finite model, choose a reference distribution and a stochastic channel that preserves it. Relative entropy to that reference cannot increase under the channel.

The third law is the one popular accounts skip. Walther Nernst put it to the Göttingen Academy in December 1905: as a system is cooled toward absolute zero, its entropy approaches a fixed floor. The floor is set by how many arrangements the system has at its lowest energy. One such arrangement and the floor is zero. Many, and it is the logarithm of how many, and sits there down to the last fraction of a degree.

Ordinary ice does this. Each oxygen atom in ice has four hydrogens near it, two close and two further off, and there are many ways to satisfy that rule across a whole crystal without ever violating it locally, which is chapter six’s triangle of frustrated neighbors wearing a different mineral. Linus Pauling counted the arrangements in 1935 and got a floor of 0.805 calories per degree per mole. William Giauque and J. W. Stout measured the heat capacity of ice down to fifteen degrees above absolute zero and reported 0.82, give or take 0.05, in a paper of 1936. That is 3.4 joules per kelvin per mole of entropy that stays in a block of ice at the bottom of the temperature scale.

The other half of the third law is a prohibition. You cannot reach absolute zero in a finite number of steps, however good your refrigerator, because each step takes a fraction of what is left rather than a fixed amount. A sodium gas at the Massachusetts Institute of Technology was cooled in 2003 to 450 picokelvin, give or take 80, which is four hundred and fifty trillionths of a degree above a floor nobody reaches.

22.3 One rule, read twice

Four laws, arrived at over eighty years by people working on engines, gases, chemical affinities and cold. A finite observer model recovers their conditional counterparts from a supplied reference, a compatible resampling rule, and an identification of energy and temperature.

Read it first as a rule about descriptions. Jaynes's instruction from chapter nineteen applies unchanged: among all the descriptions consistent with what has actually been measured, take the one with the largest entropy, the one that adds nothing else. Apply that with one quantity held fixed, the average energy, and the answer is forced. The odds fall off exponentially with energy, cheap arrangements common and expensive ones rare, with a single multiplier in the exponent setting how fast the fall is. Chapter twenty-one ran into that exponential from the far side, in a two-outcome system whose flow came out as the energy divided by the temperature.

The positive multiplier becomes inverse temperature after the energy scale and thermal reference have been identified: beta equals one over k_B T. A probability distribution alone supplies no temperature in kelvin. On that thermal branch, erasing an equiprobable bit has a minimum mean heat cost of k_B T times the logarithm of two, or 2.87 divided by ten to the twenty-first joules at three hundred kelvin.

On a specified spectrum with two distinct energies, equal Gibbs distributions have equal inverse-temperature multipliers. Equality is transitive, which gives the finite zeroth-law identification. A spectrum with only one energy cannot distinguish temperatures from its probabilities. Interpreting equal multipliers as equilibrium between a thermometer, soup and bath additionally requires a physical contact model.

Read the rule the second way, as a rule about transitions. A repair step is handed some things that neighbors have settled between them and some things nobody has settled. What should it do? The same instruction applies: change nothing you do not have to, and assume nothing you have not been given. Leave every settled quantity exactly as it stands, and redraw everything else from the reference, inside the set of arrangements that agree with what is settled.

That instruction picks out one map. Applying it twice does nothing that applying it once did not do, because the second application finds the same settled facts and the same reference. It leaves the reference where it is, since redrawing from the reference cannot move it. And it preserves the average of every quantity that can be read off the settled part, which is the first law in the form chapter nine derived it: a repair that changed a total would have to know the total. The heat and the work come off the same map. Shifting the energies of the arrangements without touching the settled odds is the work channel, redrawing the odds at fixed energies is the heat channel, and the cross term is what a step that does both at once picks up on the way through.

For the declared resampling map, distance to its preserved reference cannot increase. That is the second-law inequality of chapter sixteen. It compares two probability descriptions under one channel; it does not identify the reference with physical truth or identify every source repair with that channel.

Full resampling settles the unresolved probabilities in one step. Local repair can take longer while preserving the same public record. An exact eight-state example keeps one recorded bit fixed and updates two other bits from their conditional probabilities. Their correlations decay toward equilibrium; the recorded bit remains unchanged. This supplies a finite model of memory and relaxation under a specified transition law. Identifying that law and its clock with a physical system requires further evidence.

Both constructions preserve every initially positive state weight. A zero-temperature Gibbs state has zero excited-state weights when excited states exist, so finitely many such steps cannot reach it from full support. Its physical temperature reading uses the specified energy and thermal reference.

Landauer's bound follows on the identified thermal branch. If a reference-preserving heat stroke lowers entropy by c nats at positive inverse temperature beta, it expels at least c divided by beta of mean energy. Converting that energy decrease into reservoir heat uses the physical heat-stroke model. Setting c to the logarithm of two gives the one-bit bound.

22.4 A one-way loop that obeys the law

Chapter sixteen's pointwise fluctuation relation uses an extra condition: measured against the reference, the weight a step carries from one state to a second matches the weight it carries back. This is **detailed balance**. The integral identity requires stationarity and its positivity assumptions, not detailed balance.

The second law does not need it. The cheapest way to see that is a machine with three states and a permanent circulation in it.

Label three states one, two and three. From each state, the rule is: stay where you are with probability one half, or step clockwise to the next state with probability one half. There is no counterclockwise move at all. Nothing in this machine can go from state two to state one, ever, except the long way through state three, which takes four steps on average.

Check what it does to the flat description that puts a third of the weight on each state. State one receives half of its own third, from staying, and half of state three's third, from the clockwise step. That is a third. So is every other state, by the same count. The flat description does not move, which makes it the reference.

Detailed balance fails outright. The traffic from state one to state two is a third of a half, one sixth per step. The traffic from state two to state one is zero, because that move does not exist. There is a current going around the loop that never dies down.

Watch the second law hold anyway. Start with all the weight on state one, a description sitting the logarithm of three away from the flat one, 1.585 bits. One step spreads it into halves on states one and two, and the distance falls to 0.585 bits. A second step gives a quarter, a half and a quarter, at 0.085 bits. A third step brings it to 0.024. The descent is exact, it never reverses, and it happens inside a machine with a one-way street in it.

Stationarity is what the second law asks for: that the reference be left alone. Detailed balance is a stronger and separate demand. The extra content it carries buys the detailed fluctuation relations of chapter sixteen and Lars Onsager's paired transport coefficients. A world could have the second law with none of those and this loop is what it would look like.

22.5 The price of settling

Take a memory with two equally likely inputs and reset both to the same output. Its entropy falls from one bit to zero. The same count applies to a reduced seam description only when those are its input probabilities and the operation actually identifies the two inputs.

A comparison or a retained readback need not perform that erasure. Keeping the input in another register preserves the distinction in the combined system. Counting commits alone therefore does not count erased bits.

Bare Shannon entropy and relative entropy are different quantities. The stationary-reference inequality controls the latter. Deterministic settling can reduce the former, but its information loss alone supplies neither a thermal reference nor a heat measurement.

For a concrete thermal example, take the seven equiprobable inputs of chapter sixteen and erase their distinction, with no usable side record. If a physical memory performs that reset with an initially uncorrelated thermal reservoir at three hundred kelvin under the standard Landauer conditions, the mean heat delivered to the reservoir is at least k_B T ln 7: about 8.06 divided by ten to the twenty-first joules. This is a bound per such erasure. A semantic commit supplies neither that entropy decrease nor that reservoir temperature.

22.6 Five entropy readings

The same entropy mathematics appears in five kinds of reading, though the physical dictionaries connecting them have to be supplied and tested.

An engine reads entropy through heat and temperature. A gas reads it through Boltzmann's count of arrangements. A memory reads a Shannon count of questions, with Landauer giving a lower heat cost for physical erasure. A black-hole horizon carries the Bekenstein-Hawking entropy fixed by its area. An erasure loses alternatives from an informational record. These quantities share formal relations, but the finite record count becomes horizon entropy only after a physical carrier, temperature, energy and entropy map identifies the two readings.

A logically irreversible physical write has a Landauer lower cost when its device and thermal environment satisfy the law's premises. The finite observer model counts discarded alternatives and boundary channels without supplying that laboratory realization. A record count comes back in square meters only on a separate horizon-record branch that calibrates the count against physical entropy and area.

Chapter 23

Why Does a Region's Memory Scale With Its Surface?

John Archibald Wheeler confessed to a crime in his office at Princeton, and the crime was tea. He would set a cup of hot tea next to a glass of iced tea and let the two come to a common temperature. Energy is conserved in that operation, every joule of it accounted for. Something else goes up and does not come back down, because the difference between hot and cold, once mixed, cannot be unmixed by any procedure anybody has found in two centuries of looking. The evidence of what he had done stays in the world afterwards, spread too thin for any instrument to gather it back up.

Then the twist, which he put in 1970 to Jacob Bekenstein, the graduate student sitting in the room: "Jacob, if a black hole comes by, I can drop both tea cups into the black hole and conceal the evidence of my crime."

Chapter twenty-two priced a deleted bit in joules and derived the law the crime obeys. The total entropy of the world does not fall. Wheeler was describing a way to make it fall, using the one object in the universe that swallows things without giving anything back. The argument bites because nobody outside can be told afterwards what was in the cups.

A black hole is a place rather than an object. The place is defined by a failure. Pack enough mass into a small enough region and there is a surface around it across which signals travel inward and never outward, made of no material whatever. A light ray sent outward from just beyond it gets away. One sent outward from the surface itself stays on the surface forever. Outside that surface, the entire object is described by three numbers, which are its mass, its electric charge and how fast it spins. Two holes agreeing on those three are the same hole to everything in the universe beyond them, however different the histories that made them.

A teacup crossing that surface is not destroyed at the crossing. For a large enough hole the cup goes through without anything local happening to it at all. The crossing is a statement about which signals can be sent back rather than a statement about the cup. What it costs is the return path. Every question anybody outside might have asked about the tea has lost its channel.

Wheeler's crime comes down to accounting. Entropy counts what somebody would have to be told to know the rest, the tea had a great deal of it, and what the tea fell into is described by three numbers.

Bekenstein was twenty-five, a National Science Foundation predoctoral fellow at Princeton's Joseph Henry Laboratories, and he came back with an address for the missing count. The entropy stayed in the world and moved onto the hole. The area of the horizon is fixed by those three numbers, and behaves the way entropy behaves, which is to say it goes one way and only one way. Hawking had proved in 1971 that in any classical process the area of a horizon never decreases. Throw something in and the area grows. Two holes merge and the area of the result exceeds the sum of the areas that went in. Bekenstein's paper, "Black holes and entropy" in *Physical Review D* volume 7 in 1973, says that the entropy of a black hole is proportional to the area of its horizon, measured in units of a fundamental length squared, and that the sum of the entropy outside the hole and the entropy of the hole never decreases. The teacups are covered. The horizon grows by enough to pay for them.

23.1 A quarter of the area

Hawking thought this was wrong, and he had a good reason: anything with an entropy and an energy has a temperature, anything with a temperature radiates, and a black hole was the one object in physics defined by letting nothing out. He set out to show that the numbers could not be made to work. What his calculation produced instead was a temperature, a thermal spectrum, and a rate of emission, published in *Nature* on 1 March 1974 in two pages titled "Black hole explosions?". The temperature fixed the constant Bekenstein had been unable to pin down, and turned a proportionality into an equation.

Take the area of the horizon, count it in units of a very small square, and take a quarter of the count.

$$S = \frac{A}{4\ell_P^2}$$

Here S is the entropy of the hole, A is the area of its horizon, and the length written ℓ_P is the Planck length, which is what you get by combining Newton's constant, Planck's constant and the speed of light into a length, about 1.6 times ten to the minus thirty-fifth of a meter. Nothing on the right-hand side of that equation refers to anything inside.

Storage scales with volume in every warehouse ever built. Double a horizon's radius and the volume inside it goes up by eight while its memory goes up by four. Every further doubling widens that gap by another factor of two. Bekenstein worked the arithmetic into a comparison in *Scientific American* in August 2003: a black hole one centimeter in diameter has an entropy of about ten to the sixty-sixth bits, which is roughly the thermodynamic entropy of a cube of water ten billion kilometers on a side.

For a black hole, the number of distinguishable states is fixed by the horizon area. More general gravitational entropy bounds apply to suitably defined light-sheets and physical states; an ordinary room need not saturate a black-hole equality. Doubling the shelving does not evade the gravitational ceiling, but the ceiling and the room's actual entropy are different quantities.

Black holes are where the sharp equality was found. Pack enough energy and entropy into a region and gravitational collapse can close a horizon around it. Gerard 't Hooft's 1993 proposal, "Dimensional reduction in quantum gravity," made the area-scaling principle general. Its precise use outside horizons requires the causal and energetic conditions of an appropriate entropy bound.

Chapter thirteen got a related boundary-accessibility statement out of a wiring diagram, with no gravity anywhere in the argument. Everything an outsider can learn about a patch arrives through its ports, so interior states exposing the same boundary readings are indistinguishable to the network. This gives a finite count of boundary arrangements. It does not prove that the count is physical entropy, that each port carries equal area, or that a general region saturates a horizon law. Those identifications require a realized physical screen and an independently calibrated exchange rate.

23.2 What a cut costs

Chapter nineteen said what an observer holds and how it comes to hold it. The description on your own side of a seam is manufactured by averaging over everything only the far side could have detected, and chapter sixteen said what the averaging costs: the description that comes back has an entropy, a count of what its holder would have to be told.

Take two patches together in a state pinned down as far as anything can be pinned down, where every question that can be settled about the pair at once has been settled. Cut it in two and average over one half. What comes back is vague. The definiteness was living in the relation between the two halves. The cut went through it. That content is **entanglement**. The number measuring how much of it the cut severed is the entropy of the half left behind.

Of either half. Two sides of a cut through a definite whole come back with exactly equal entropies, however lopsided the two pieces are. A thimble of water cut out of an ocean returns the same number as the ocean it was cut from. That means the smaller side sets the ceiling for both, and it means the number was never a property of what is inside either piece. It belongs to the cut.

Chapter six's coins hold a record of that shape, with no quantum mechanics in the building. Alice and Bob write one word, same or different, which states the relation between two coins and says nothing about either coin: complete about the pair, silent about the halves. Read the entry and you know what will be on Bob's coin the moment you have seen Alice's, and you know nothing whatever about either one before that. The quantum version differs in one respect, which chapter twenty measured. There are no faces sitting underneath the entry for it to be a description of. The correlations run further than any set of faces could account for. The entry is the object. There are no coins under it.

A cut through the finite patch model passes through seams, because seams are the only places where two patches hold anything in common. The severed seam registers therefore bound the shared information across the cut. The entropy equals a bare seam count only under an additional uniform-capacity and state condition. Adding a patch deep inside changes no boundary register, while adding a crossing seam changes the available boundary carrier.

23.3 The doorway

Two rooms and a doorway between them. You are in the near room with instruments, and every signal from the far room comes through the doorway, so anything you can learn about what is happening over there arrives as something happening in the doorway. Tell somebody exactly what the doorway is doing, in full detail. What does the far room then add?

If the answer is nothing, the doorway **screens** the two rooms. Given the doorway, the two rooms are independent: they can correlate with each other only through what passes between them, and you have been handed that.

Run a cable through the wall and the screening breaks. The two rooms share something the doorway never carried, the far room goes on telling you things about the near one after the doorway has been fully described, and no amount of watching the doorway recovers what the cable is doing. Whether a boundary screens is a fact about how the rooms are wired. The cable is the thing to go and look for.

Ask how much two systems tell you about each other once a third is given in full, and the answer is the **conditional mutual information** of that arrangement. Elliott Lieb and Mary Beth Ruskai proved in 1973 that the quantity is never negative, a way of saying that no doorway leaves you knowing less than nothing about the relation between its rooms. Zero is the interesting value. Zero means the doorway carries the whole relation.

Chop a network into three pieces of exactly this shape: a region, which is the **cap**; a thin band of patches around its edge, which is the **collar**; and everything else, which is the exterior. When the collar screens the cap from the exterior, the state splits: fix what the collar is doing and the inside and the outside become two independent descriptions, with nothing running between them that the band has not accounted for.

23.4 Rebuilding from the collar

If the collar screens exactly, the exterior can be rebuilt from the collar alone. Reconstructed rather than estimated: there is a map that takes the collar's state and returns the state of everything outside, exactly as it was, and the same map runs in the other direction and rebuilds the cap. The operation is called a **recovery map**, and Dénes Petz wrote down the canonical one, which takes a process and a reference state and runs the process backwards, returning a description some system could actually be in.

Which is what a repair is. A patch that disagrees with its neighbor rewrites itself from the interface data the two of them share, touching nothing beyond its own collar, because a rule that reached further would have to read parts of the network no patch has access to. The repair law of chapter nine is the recovery map of the collar, applied locally. Nobody picked it off a list of plausible dynamics. Given a cap, a collar and an exterior, it is the map that exists.

Chapter ten ran eighty-two thousand patches into agreement across 122,880 seams with no patch consulting anything beyond its own neighbors. That run is this property in hardware. Each rewrite was a reconstruction from a collar. What made it legitimate to rewrite a patch out of its edge data, rather than merely convenient, is that the edge data carried the entire relation between that patch and the rest of the world.

Real collars screen approximately rather than exactly. There is a theorem covering the difference. Omar Fawzi and Renato Renner proved in 2015, in *Communications in Mathematical Physics* volume 340, that when screening fails by a small amount the recovery map fails by a correspondingly small amount, with the conditional mutual information of the arrangement bounding how far the rebuilt state can sit from the original. A patch rebuilding itself out of its own edge data is wrong by no more than that number.

A cap, a collar and an exterior are the three pieces a local rule needs. The collar is a surface one patch thick wrapped around the cap. The seams crossing it are what a repair reads, the same seams a cut through the boundary severs.

23.5 Hamming's Monday

At Bell Telephone Laboratories in 1947 a relay computer in New York spent a weekend on Richard Hamming's jobs. He was junior enough that the weekend was his entire allocation: start the run at five on Friday evening, collect the output at eight on Monday morning, with nobody in the building in between. The machine hit an error early on, dropped the work, and moved on to the next job in the queue. Two Mondays running he came back to a blank reel of tape and had nothing to hand his colleagues in Murray Hill. What he wanted to know was why a machine clever enough to notice that a relay had gone wrong could not work out which relay and set it back.

He published the answer in 1950, in the *Bell System Technical Journal*, volume 29. Four bits of data go in and seven come out, the extra three computed from the others and arranged so that a receiver holding a word with one bit flipped can name the position that moved instead of merely reporting that something did. Chapter thirteen took this world's carrier to be a code of that family, with the seams doing the checking, and settled there that a connector hands over its checks and hands over nothing whatever about how far apart two legal configurations sit.

What Hamming had beyond his list of checks was seven relays bolted in a row on a frame. A row has positions and the positions can be counted off with a finger. A surface made of seams has no row and no frame. What it has is the number of seams a cut through it severs. That number arrives without units.

23.6 A conditional exchange rate

Bekenstein's proportionality, with Hawking's quarter in it, prices a physical horizon: hand it an area and it returns an entropy. Reading the equation in reverse does not price an abstract seam by itself. That step assumes a selected horizon cellulation, identifies one correctable boundary-record unit with one unit of horizon entropy, and assigns equal weight to the selected seams. Under those conditions one such unit corresponds to four Planck squares. The finite connector supplies neither the physical horizon nor the equality of record and entropy units.

On that calibrated branch, the selected record count and horizon area are one quantity in two units. Four Planck squares comes to 1.04 times ten to the minus sixty-ninth of a square meter, corresponding to about 9.6 times ten to the sixty-eighth selected units per square meter. The conversion is a physical adapter, not a consequence of seam combinatorics.

If a realized horizon carries that cellulation and the adapter is local and additive, its selected units reproduce the horizon area. This does not imply that every physical surface has an exactly quantized area, that every abstract seam has the same geometric weight, or that counting seams on an arbitrary region reproduces a ruler reading.

23.7 Draw a square on a horizon

On a selected finite horizon cellulation, draw a closed curve dividing the carrier into two pieces. A curve routed between the selected units partitions their count, so the two counts add to the count for the whole.

Entropy divides this way only if the horizon-record adapter is local, additive and uniform on the selected state. Under those premises, equal-area pieces carry equal record capacity. The finite seam partition by itself proves additivity of the count, not equality with physical entropy or area.

The conditional conversion puts about ten to the sixty-seventh selected entropy units in a ten-centimeter square on a saturated horizon. It is a scale illustration, not a derived count for any boundary in the world.

A patch reads its collar registers one at a time. A ruler reads the same carrier only after the physical placement and area calibration identify those registers with locations and area elements.

Wheeler's two teacups went in carrying entropy nobody could gather back afterwards. The surface around the hole grew by enough to cover them. That surface is made of no material at all. The only property it has to offer is a size. A count of what somebody would have to be told came out balanced, exactly, against a shape.

Chapter 24

Why Does Gravity Look Like Geometry?

The gravity argument in this chapter is conditional. The source metric gives a continuous three-dimensional carrier. A specified population of conservative records and complete local reads produces orders and event counts approaching flat spacetime. The read radius shrinks more slowly than the population gaps. For fixed interior timelike intervals, fourth roots of count ratios recover proper-duration ratios, using a reference interval as the unit. Identifying these events and readouts physically, and showing that matter uses the same cone, are further conditions.

A field can be calculated on those same prepared addresses. Nearest neighbors exchange a scalar field through one specified action, which also defines its quantum states and detectors. In a finite calculation, a disturbance confined to one slab produces a response at a detector in a separate slab; the entire certified error stays smaller than that response throughout the observation window. The comparison is with a massive field in a cube with fixed walls. The addresses are stored separately from the changing field values. A separate, coarser execution records every field read and write across 64 sites and bounds its time-stepping error against the same finite action. Its recorded dependencies and the earlier counting construction are different. The action and model time are supplied. Identifying that time with a physical clock requires an additional argument.

Curved spacetime and Einstein's equation require more. One physical refinement must carry a common metric and conserved stress, compatible clocks and volume, stable geometry, and controlled curvature. Exact order reflection is one route; vanishing causal error with refinement is another. The gravity theorem also assumes the modular and null bridges, fixed-cap entropy stationarity, the small-ball formulas, universal coupling, a vacuum reference and physical scale. Smooth tensor curvature must be identified on that family, or through the separate small-ball argument. Scalar-curvature convergence alone does not identify the Einstein tensor.

In the autumn of 1907 Albert Einstein was examining patents in Bern and owed Johannes Stark a review article on relativity for a yearbook of radioactivity and electronics. He was assembling it at his desk in the office when the thought arrived that he described fifteen years afterward, in a lecture in Kyoto on 14 December 1922, as the happiest of his life. If a person falls freely, he will not feel his own weight.

Every other influence in physics discriminates. A magnet lifts a steel paperclip and does nothing at all to an aluminum one of the same mass, because the response depends on what the object is made of. Gravity pulls harder on a heavier object by exactly the factor that makes a heavier object harder to shift, the two cancel, and everything falls at one rate, which is why David Scott could stand on the Moon in August 1971 with a hammer in one hand and a falcon feather in the other and drop them together for the camera. Write the description in a frame that falls along with everything else and gravity drops out of it, leaving nothing anywhere in the equations for it to do.

The sharpest test of that cancellation flew between 2016 and 2018 in a French satellite called MICROSCOPE, which carried two hollow concentric cylinders, one of a titanium alloy and one of a platinum alloy, around the Earth in an orbit continuously corrected so that nothing but gravity acted on the pair. Electrostatic forces held both cylinders in place. The size of those forces reports any difference in how the two of them fall. Five months of usable free-fall data came out of two and a half years in orbit. The difference in acceleration between titanium and platinum came out at one and a half parts in ten to the fifteenth, with a statistical uncertainty of two and three tenths in the same units, which is to say a null result at the level of one part in a thousand million million.

So gravity belongs to the place rather than to the thing. Which leaves two questions. Why does an influence that treats everything identically turn out to be a shape, and what decides how much shape a given lump of matter makes?

24.1 Two walkers on the equator

Two people stand on the equator, a kilometer apart. Both face due north. Both are told to walk straight ahead and never turn, and both are conscientious about it: no course corrections, no navigating, nothing but one foot in front of the other in the direction they set off in.

They collide at the North Pole, about ten thousand kilometers later.

Both of them walked straight the whole way. The gap between them closed from a kilometer to nothing, at an accelerating rate over the last stretch, and no force acted on either walker at any point in the trip. A cord tied between them would have gone slack rather than taut. The convergence is a fact about the surface they crossed. They can establish it without ever seeing the Earth from outside, because they have their own two pairs of boots and a tape measure. Send a third walker off from a point between them and the same thing happens to each of the three gaps at once. That is what makes the effect a property of the ground rather than an agreement between two people.

The path you get by going straight ahead is called a **geodesic**. On a flat floor it is a straight line, on a sphere it is a great circle, which is why the flight from London to Tokyo goes over the Arctic and why that route looks bent on a wall map. Two geodesics that set off parallel and end up crossing is what curvature means. The walkers measure it from the inside.

Free fall is walking straight ahead. Two ball bearings released side by side above the Earth converge, and the amount they converge by is a fact about where they are rather than about what they are made of, which is what MICROSCOPE measured to one part in a thousand million million. Nothing in the account calls for a sheet with a weight on it, and nothing bends into a further dimension for the bending to happen in.

Release a small ball of particles, all at rest with respect to each other, and let them fall. The ball's volume changes. John Baez and Emory Bunn, in the *American Journal of Physics* in 2005, wrote Einstein's equation as a sentence about that ball: the rate at which it begins to shrink is proportional to its volume times the energy density at its center, plus the pressure in each of the three directions at that point.

24.2 The equation, read rather than solved

The whole of that, in the notation the textbooks use, takes one line.

$$G_{ab} + \Lambda\, g_{ab} = 8\pi G\, T_{ab}$$

That says the shape of the world at a point is fixed by what is sitting at that point.

The letters a and b each stand for one of the four directions, one of time and three of space. Four directions taken two at a time gives sixteen pairs, and swapping the two directions of a pair changes nothing, leaving ten of the sixteen independent, so that single line is ten equations. The object g_{ab} is the **metric**: ten numbers at every point of the world that turn a pair of directions into their contribution to chapter seventeen's interval, which makes the metric the ruler, the thing that says how much time and how much distance separate two nearby events. The object G_{ab} is a particular combination of the rates at which those ten numbers change from point to point, assembled so that its own bookkeeping closes automatically, with nothing leaking in or out. On the right, T_{ab} is the stress: energy density, momentum flow and pressure, everything that is there. And $8\pi G$ carries Newton's gravitational constant, which fixes how much bending a given amount of energy buys.

The one term that spoils the arrangement is Λ, the cosmological constant, which multiplies the metric itself and has nothing on the other side of the equals sign to account for it.

Einstein reached the equation without that term in November 1915, by requiring that it reduce to the old law of gravity for weak fields, that it not depend on the coordinates anybody chose, and that both sides conserve what has to be conserved. Those requirements narrow the possibilities to one. They do not say why geometry should respond to energy in the first place.

24.3 Four pages in 1995

Ted Jacobson's paper occupies pages 1260 to 1263 of volume 75 of *Physical Review Letters*, titled "Thermodynamics of Spacetime: The Einstein Equation of State".

He needed three things. An observer who accelerates steadily has a horizon behind it, a boundary past which signals never arrive. A horizon carries an entropy equal to a quarter of its area. And an accelerating observer finds empty space warm, at a temperature set by its acceleration. Then Clausius's relation, which is the first thing anybody learns about heat: the heat flowing into a system, divided by the temperature it arrives at, equals the entropy that heat buys.

Jacobson's move was to demand that this hold on every one of those horizons. A black hole has one. So does the air above your kitchen table, and so does every other point of the world, one horizon for every direction an observer passing through that point might accelerate in.

Matter crossing such a horizon carries energy, so it delivers heat. The horizon's entropy is a quarter of its area, so the entropy change is an area change. The temperature is the one the accelerating observer reads. Insist that heat over temperature equals the area change at every point and in every direction at once, and the geometry has no freedom left: it has to distort in exactly the way that keeps the accounts straight everywhere, and that distortion is Einstein's equation, coupling constant included.

Jacobson had to bring the area law and the temperature in from outside. Chapter twenty-three's collar gives a finite boundary-capacity candidate, and chapter twenty-one gives an algebraic flow candidate. Reading the first as physical area entropy and the second as an accelerated observer's measured temperature requires the corresponding geometric, modular and unit attachments; those identifications are branch premises, not consequences of the informational order.

The physical branch identifies a settled repair state with a maximum of generalized entropy, the entropy of a small region's contents plus one quarter of its boundary area in calibrated units. The first-order change of anything at a maximum is zero. Turning that stationarity statement into

Clausius's relation requires the supplied entropy, temperature, heat-flow and physical-area identifications. The finite repair theorem alone does not establish those thermodynamic premises.

24.4 Where the eight and the pi come from

Take a small ball around a point, in the rest frame of one observer, small enough that the geometry inside it is nearly flat, and run the stationarity condition on it.

The contents side first. Chapter twenty-one's generator, restricted to this ball, weights the energy density by a factor that falls from a maximum at the center to zero at the edge, namely the difference between the squared radius of the ball and the squared distance from the center, divided by twice the radius. That shape is what a boost centered on the ball does. Integrate that weight against the change in energy density over the whole ball and out comes four pi times the fourth power of the radius, divided by fifteen. That is an integral a first-year student can do in a line. The fifteen comes out of it.

One factor is owed, and chapter twenty-one supplied it. Entropy is heat divided by temperature. The temperature an accelerated observer reads is its acceleration divided by two pi. Dividing by that temperature multiplies the heat by two pi, taking the matter side to eight pi squared times the fourth power of the radius, divided by fifteen.

The geometry side. Alfred Gray and Lieven Vanhecke, in *Acta Mathematica* in 1979, worked out how the size of a small geodesic sphere depends on the curvature around it. The piece of that needed here says that if you hold the volume of a small ball fixed and raise the curvature, the area of its boundary falls by four pi times the fourth power of the radius, divided by fifteen, times the change in exactly the curvature combination that stands on the left of Einstein's equation. Curving the surroundings costs boundary area.

Set the sum of the two to zero, remembering that the area enters the total entropy divided by four times Newton's constant. Every appearance of the radius cancels. Both fifteens cancel. What is left is eight pi squared on one side against pi over Newton's constant on the other, and the ratio of those two is eight pi times Newton's constant, which is the coupling that stands in front of the stress in the field equation. Conditional on the imported small-ball, temperature, entropy-area and physical-scale normalizations, the coefficient is fixed by the ratio of two elementary integrals rather than fitted as an additional parameter. The finite source-order theorem does not derive those normalizations.

On the branch just named, that gives one of the ten equations in one observer's rest frame. The other nine follow when the effective continuum supplies observers through the same event in an open set of timelike directions and the overlap law carries the same relation between their frames. Written out in terms of velocity, each relation is a polynomial of degree two. A polynomial that vanishes for every velocity in an open range has every coefficient equal to zero. Ten components then follow from the conditional family of agreeing observers.

24.5 A gas law without molecules

Jacobson ended his abstract with a sentence that has been irritating people for thirty years: it may be no more appropriate to canonically quantize the Einstein equation than it would be to quantize the wave equation for sound in air.

Pressure times volume equals a constant times temperature. Émile Clapeyron put it in that form in 1834. It held up in every laboratory in Europe through the rest of the century without containing a single molecule. In 1908 Jean Perrin put grains of gamboge resin in water under a microscope, counted how the population thinned out with height, and got Avogadro's number out of the count, which finished an argument about whether matter came in pieces at all. Wilhelm Ostwald, who had spent a career arguing that atoms were a bookkeeping convenience, gave that position up in the preface to the fourth edition of his own textbook in 1909, citing Perrin by name. The gas law did not change a symbol, because it was never a claim about molecules. It is a claim about a gas that has settled down, and stays true whatever the gas turns out to be made of.

On the physical branch stated at the start of the chapter, Einstein's equation sits on that shelf. It is the effective condition that the supplied observer, geometry, stress and thermodynamic data have to satisfy at a settled state. The finite informational network alone does not establish that identification. Quantizing the metric in the hope of finding the atom of space is then analogous to quantizing the pressure of a gas in the hope of finding the molecule. The pressure is real, it is measurable, it will lift a piston, and it belongs to a settled state rather than to a constituent.

On that same branch, the 1907 thought is recovered as a conditional theorem. Every observer uses the supplied Lorentz-frame construction, so the local inertial reading of a falling observer and that of an observer floating in empty space have the same form. For pressureless matter, the supplied stress interpretation and conservation law then imply geodesic motion. These are consequences of the effective geometric branch, not of the read-from order by itself.

24.6 Nine balances in a supplied frame

On the conditional smooth branch, a horizon is traced by null directions and a symmetric stress tensor can be evaluated twice on any one of them. That is the physical picture behind null tomography. The exact finite theorem does not begin with nine observed light rays.

It begins in a separately supplied three-plus-one tensor interface indexed by the same finite event type used by the source-derived placement: matrix-valued geometry and stress fields, four step maps and the Lorentz form. A coordinate frame of nine null vectors is fixed inside that tensor interface. The exact rank-three unit sphere gives nine algebraic inverse images of that frame. Neither the dependency order nor the source dynamics chooses this distinguished nine-set, and no theorem identifies tensor-coordinate differences with coordinate differences in the constructed carrier. The set is not invariant under every rotation or Lorentz transformation.

The theorem asks for nine balance equalities along those fixed algebraic representatives. Symmetry, discrete conservation and connectedness make them imply balance on every null vector, followed by an Einstein-shaped tensor identity with one constant metric term in that supplied tensor interface on the finite event type. The dependency order shares those events but does no work in the tensor calculation. No dependency edge chooses a direction, a finite-difference operator or a tensor. The vacuum reference, Newton scale and physical normalization belong to the separate gravitational interpretation.

Why can nine equalities suffice? Take the six coordinate vectors pointing forward and backward along one spatial axis at a time, then three tilted vectors with the required sign patterns. They form one coordinate tomography frame. Its determinant is exactly eight thousand one hundred and ninety-two over twenty-seven, or two to the thirteenth over three cubed, so the linear system inverts. This is exact algebra and the inversion is stable. Calling these nine supplied equalities nine measurements would add a physical direction-selection and instrument theorem that the finite result does not contain. Turning the matrix fields into smooth curvature and physical stress requires the physical causal-order, link-direction, count-to-volume, topology, refinement and curvature-convergence conditions named at the start of the chapter.

24.7 The vacuum and the tenth number

The vacuum has a stress. Whatever is in empty space when nothing is in it, the same everywhere, looking identical to every observer no matter how fast they are moving, has to be built out of the only object with that property, and the only object with that property is the metric. So the stress of the vacuum is a number times the metric.

Contract it with a light ray. The metric evaluated on a light-ray direction twice is the interval of a light ray with itself. That is zero: chapter seventeen defined a light ray as a separation whose interval is exactly zero. So the vacuum's contribution to what a light ray reports is a number times zero.

Exactly zero, for any vacuum energy density whatever, however enormous, and zero by the definition of the probe rather than by a cancellation between large terms.

The single direction the light rays cannot see is exactly the direction the vacuum's energy points in. That is why the cosmological constant survives in the field equation as a term nobody derived: the small-ball calculation that produced everything else on the left-hand side is built out of horizons, horizons are made of rays, and rays are blind to it.

Add up the zero-point energies of the quantum fields of empty space, cut the sum off at the shortest length anybody takes seriously, and you get a vacuum energy density around a hundred and twenty orders of magnitude larger than the one the sky reports, the sky's value of the cosmological constant being about 1.1 times ten to the minus fifty-second per square meter. That is the worst prediction in the history of physics, and being wrong by that factor takes a kind of commitment a merely careless calculation cannot reach. In the null-response reconstruction considered here, local light-cone data do not determine the trace part of the metric response. A value for the cosmological constant therefore needs information beyond this reconstruction.

The OPH capacity branch asks whether the capacity of public records associated with an observer horizon can fix the remaining constant. That requires a physical map from record capacity to horizon entropy and volume on the same realized spacetime branch. A large capacity gives a small constant under this map; it does not explain agreement with measurement. The finite informational order and the nine-direction theorem supply no cosmological value by themselves.

24.8 When Newton's constant is an area scale

On the named physical-scale branch, the area law says the entropy of a boundary is a quarter of its area divided by Newton's constant. Reading that from right to left makes Newton's constant the conversion between a physical area and an identified record entropy. The conversion is not supplied by record count alone: it requires the independent area, entropy and unit attachment. With that attachment, one square meter of boundary corresponds to about ten to the sixty-ninth records.

Chapter thirteen said the size of a screen is a count of readings and not a width in meters. This is where the two get joined. Chapter twenty-three's collar says what the readings are: the seams a cut along a region's boundary severs. Take one cell of that screen. Whatever area the cell covers, that area is a whole number of one small unit, the same unit everywhere, and on the wiring this world has the cell carries a quarter of that same whole number in severed seams. Divide the cell's area by four times the entropy it carries and the whole number cancels, top and bottom, exactly, leaving the unit standing on its own.

The branch identifies Newton's constant with that calibrated unit of area, one square of the physical screen. The finite carrier does not select the square-meter calibration or prove that its combinatorial cell is the physical area unit.

Under that physical calibration the area is 2.61228 times ten to the minus seventy square meters, a square 1.6163 times ten to the minus thirty-five meters on a side. Multiply it by the cube of the speed of light and divide by Planck's constant over two pi, which is the conversion any area has to go through to be quoted in the units gravity is measured in, and out comes 6.67430 times ten to the minus eleven cubic meters per kilogram per second squared. The laboratory value is 6.67430 times ten to the minus eleven, with an uncertainty of 22 parts per million. This numerical agreement tests the supplied calibration; it is not a scale derived from the informational poset.

That uncertainty is where the comparison stops being informative, and worth being plain about which side it sits on. Newton's constant is the least precisely known of the fundamental constants by a wide margin, measured to five figures where the electron's magnetic moment is known to twelve, because gravity is feeble and everything in the laboratory has mass. A derived value landing inside a window that wide is a weak test passed, not a decimal place matched. MICROSCOPE measured gravity's indifference to what falls at one part in a thousand million million, and the strength of the thing being indifferent is known to twenty-two parts per million, ten orders of magnitude worse, because a difference between two cylinders nulls out everything the two have in common and a magnitude has to be assembled out of a kilogram, a meter and a second.

On a branch carrying all of the physical and continuum inputs, the effective geometric account has ten numbers at every point, a rule fixing how they answer to what is there, the small-ball coefficient and a calibrated area scale. The finite theorems construct a four-dimensional ambient Lorentzian carrier and its cone, while provenance independently generates the event order. The declared record family supplies a flat order-and-volume limit. Identifying the tensor field with smooth curvature and selecting the physical scale require the common physical branch. Conditional on the common refinement and those attachments, curvature is what a settled network of observers looks like from a distance, free fall is going straight ahead, and the two walkers on the equator were doing physics the whole way to the pole.

And nothing is standing in it. A geometry can be told how much energy and momentum is present at a point, which is all the field equation ever asks. It cannot be told why what is present comes in exactly three kinds of interaction, one of them with eight varieties, or why the twelve ports of chapter fourteen have room for those and for nothing else.

Chapter 25

Why Exactly These Forces?

The French army sent a twenty-seven-year-old engineer named Charles-Augustin de Coulomb to Martinique in February 1764 to build a fort. Eight years on Fort Bourbon got him a good deal of what is taught as soil mechanics and a succession of tropical illnesses that damaged his health for the rest of his life. He came home a captain in June 1772.

The Academy of Sciences in Paris wanted a better mariner's compass, and Coulomb's entry, which took a share of the grand prize in 1777, hung the needle from a fine thread instead of resting it on a pivot, because a pivot has friction in it and a thread has almost none. That left him needing to know exactly how hard a twisted thread pulls back, so he measured it, at length, and published the numbers in 1784: the pull rises with the fourth power of the wire's diameter, falls off with its length, and holds exact proportion to the angle of twist, which makes a long thin one as weak a spring as an experimenter could want and an exact one.

In 1785 he hung a light horizontal arm from one, mounted a small ball on its end, charged it, and brought a second charged ball up beside it. The two pushed apart, the arm swung, and the twist in the thread read off how hard they pushed.

Three readings from that first memoir on electricity carry the whole argument. Left alone, the thread let the balls settle thirty-six degrees apart and carried thirty-six degrees of twist. He turned its top through a hundred and twenty-six degrees, forcing them closer, and they came to rest eighteen degrees apart, held by a hundred and twenty-six plus eighteen, a hundred and forty-four degrees of twist. Half the separation, four times the push. He wound the top through five hundred and sixty-seven degrees, better than a full revolution, and the balls held out at eight and a half degrees, with five hundred and seventy-five and a half degrees of twist in the thread. The separation had come down by a factor of a little over four, and sixteen times thirty-six is five hundred and seventy-six.

Which is a fact about distance: every version of the law written down since 1785 says only how far apart two things are. What does a force law look like on a carrier that has no distances in it, and how many forces will that carrier hold? The twelve observers of chapters six through fourteen have a wiring diagram, thirty pairings saying which port is joined to which, and not one number anywhere saying how far apart anything is.

Two centuries of better instruments have left the exponent where Coulomb put it. Williams, Faller and Hill, writing in *Physical Review Letters* in 1971, wrote it as two plus an unknown small amount and then measured the amount. It came to two point seven parts in ten million billion, give or take three point one, which pins the exponent at two out to the fifteenth decimal place. Fifteen decimal places of pure distance. The machine has no ruler in it.

25.1 A value at every place

Between 1698 and 1700 Edmond Halley took a small Royal Navy vessel called the Paramore twice across the Atlantic, on two voyages fitted out for the single purpose of measuring the Earth's magnetism, and recorded the angle between where his compass pointed and where north actually was. The angle changes as you sail. In 1701 he published the results as a chart, and rather than print a table of positions and readings he drew curves through the places where the deviation came out the same. Sailors called them Halleyan lines for the next hundred years.

What that chart holds is a number attached to every place on it. A captain anywhere in the Atlantic could read his local deviation off the paper. The reading changes from place to place, the rule for finding it does not, and a ship crossing the ocean experiences the chart as a sequence of small differences between one position and the next.

That is a **field**, one value per place, and the word covers nothing else. The temperature in a room is a field. So is the wind over Halley's Atlantic, which hangs a speed and a heading at each place rather than a single number, and so is the deviation he charted, which needs only one.

On the twelve-port carrier the places are the ports. Give every port a number and each seam reads the gap between its two ends, thirty readings nobody had to supply separately. Chapter twelve had the same twelve numbers standing for something else, the private labeling each observer keeps at its own port. Twelve values and thirty differences: no more furniture than Halley put on his chart, and considerably less ocean.

25.2 Twelve numbers that add to nothing

Chapter twelve left a law sitting at the ports: what a port pushes out along its five seams is the charge it holds. That is Gauss's law with the geometry taken out of it. By itself that law picks no field, since a current going round in a circle leaves every port's books balanced, so one list of charges comes back with a whole family. One member of the family has the least energy, the one with no circulation in it.

Energy on this carrier is a sum over the thirty seams. Each seam contributes the square of the number sitting on it. Squaring is what makes circulation expensive: a current running around a loop delivers nothing to any port and puts a value on every seam it crosses, and every one of those values gets squared and added in. So the arrangement of least energy is the one that has stopped moving anything it does not have to move. On a wiring diagram with nineteen independent loops there is exactly one such arrangement for any list of charges.

Adding the same amount at all twelve ports changes no difference anywhere, so the twelve numbers are pinned only up to a common shift, exactly as chapter five's meridians were pinned only up to a choice of zero. The shift is fixed here by requiring the twelve values to add to nothing.

And the surface is closed. Twelve ports wired into a sphere have no edge, no outside and nowhere for a net outflow to go, so the twelve charges have to sum to zero or the law has no solution at all. A single unit of source at one port therefore arrives with its own balance: one twelfth taken off every port, the same subtraction everywhere, including at the port the source is sitting on. That subtraction is the only balance that treats the twelve ports alike.

Put one unit of source at one port, then, and ask what the other eleven read.

25.3 Four fractions

Walk the wiring outward from the source and count hops. The eleven others sort into three groups: a ring of five, a second ring of five past it, and the port directly opposite, three hops out and the only one chapter fourteen leaves at that distance.

Four distances, therefore, and four numbers. Here they are, exact, in the order the walk produced them:

$$\frac{7}{36}, \qquad \frac{1}{90}, \qquad -\frac{7}{180}, \qquad -\frac{1}{18}$$

The first is the potential at the source itself, the second is what every port in the first ring reads, the third is what every port in the second ring reads, and the fourth is the reading at the far port.

Put them over a common denominator of a hundred and eighty and they read thirty-five, two, minus seven and minus ten. Add all twelve of them up, taking five at the second value and five at the third: thirty-five and five twos on one side, five sevens and a ten on the other, forty-five against forty-five. The twelve values add to nothing, which the closed surface required of them.

They fall. Each ring reads lower than the ring inside it. The ordering was not put in anywhere. The hop counts came out of walking the wiring.

The four values came out of the charge law and the least-energy condition. One of those was a walk and the other was a piece of algebra. They agree about which port lies further out.

Subtract each value from the one inside it, in one-hundred-and-eightieths: thirty-five take two is thirty-three, two take minus seven is nine, and minus seven take minus ten is three. Over sixty those are eleven, three and one. The fall is steep near the source and shallow far from it, the shape of the thing Coulomb's thread was weighing.

The potential crosses zero somewhere between the first ring and the second. A potential around an isolated charge in open space never does that, and this one has no way to avoid it: twelve numbers that must sum to nothing, with the largest of them sitting at the source, have to go negative somewhere further out.

Compare what the same law costs in the ordinary setting. To write down the potential of a point charge in a laboratory you need the distance between two points, and a distance needs a coordinate system to measure in, and a coordinate system needs a space to be laid over. Three assumptions, stacked, before the first symbol goes on the paper, and not one of the three is a component anybody has found on the board.

The input to that calculation was a list of thirty pairs saying which port touches which. Nothing else went in. No coordinate, no distance, no direction, no angle, no space for anything to be at, and the hop counts that organize the answer were themselves read off the same list of thirty pairs. Coulomb's shape came out of a table of who touches whom, in exact fractions with denominators small enough to check by hand.

One force, worked to the last digit, out of a wiring diagram with thirty entries in it. The same thirty entries have to say how many forces there can be.

25.4 Dials

Chapter five's group was eight moves of a paper square, and eight is a menu. You pick one.

Take instead a dial on the front of an amplifier. Turn it through any angle you like, including a millionth of a degree, and then turn it through another. The result is a setting of the dial, so the moves close. Every turn is undone by turning back, and leaving the dial alone is on the list. Closed, undoable, do-nothing included: a group, by chapter five's three conditions, with the difference that the moves come in a continuous supply rather than in a list of eight.

One thing about a dial that a menu of eight moves does not have: you can turn it by almost nothing. Set it a hair off the do-nothing position and you have a nudge, and nudges behave far better than the moves they generate. Two nudges in different directions can be added. A nudge can be doubled or halved. Every large turn is a small nudge repeated, so the whole continuous supply of moves is determined by the nudges available at the do-nothing setting, which is a much simpler object: a flat space of directions, with as many independent directions as the dials it came from.

The useful number about a collection of dials is how many independent dials it has. That number is its **dimension**. The amplifier has one. So does a sheet of paper spun about a pin stuck through it: every rotation of a plane is that one dial at some setting. A rigid object turned in three dimensions has three, which is why aircraft have exactly three words for it: pitch, roll and yaw.

Counting dimensions is a way of settling questions, and chapter six ran one without calling it that. The independent loops of a wiring diagram are a collection with a dimension of its own, and edges less dots plus one reads it off: thirty less twelve plus one, nineteen. Nineteen was then the number of conditions that had to hold at once for the twelve observers to agree. Two of them refused, both facts settled before anybody drew a field on the wiring.

Two more collections, the ones physicists reach for. Take an object with two slots in it, each holding a complex number, and ask for every reversible change that leaves the total size alone. Then ask the same of an object with three slots. Both counts are short enough to do on your fingers.

Each slot has a dial of its own, which turns that slot and leaves the others where they were. Each pair of slots has two more, which lean the two members of the pair against each other. Two slots: two single dials, one pair, two pair dials, four. Three slots: three single dials, three pairs, six pair dials, nine. Singles and pairs together come to the number of slots multiplied by itself, every time.

One dial in each collection turns every slot together by the same amount, a change of overall convention that says nothing about how the slots stand relative to each other. Divide it out. Four less one is three, nine less one is eight, and the count that survives needs nothing but how many slots there are:

$$n^2 - 1$$

Here n is the number of slots and n squared is that number multiplied by itself. Four slots give fifteen. Five give twenty-four. Between eight and fifteen this family gives nothing at all.

25.5 A book on a table

Take a hardback off the shelf and lay it flat, cover upward, spine to your left.

Turn it a quarter turn clockwise on the tabletop. Then flip it away from you, end over end. Note where the cover faces and where the spine points.

Put it back the way it started and do the same two moves in the other order: flip it away from you end over end first, then turn it a quarter turn clockwise on the tabletop.

The cover is face down both times. The spine is nearest you in one case and furthest from you in the other. The two results are a half turn apart.

A group in which every pair of moves gives the same result in either order is **abelian**, after Niels Henrik Abel, and one containing even a single pair that does not is non-abelian. The amplifier dial is abelian: thirty degrees then fifty is fifty degrees then thirty, and there is no third way for it to come out. Turns of a solid object in three dimensions are non-abelian. The book on the table is the demonstration. The three-dial and eight-dial collections are non-abelian for the same reason, because three of the dials in each of them are turns of exactly that kind. The single dial is the only abelian collection here.

25.6 Twelve dials on a closed box

Back to the carrier, a box with twelve ports and no lid.

Each port holds one reading. A response is whatever the box does when the readings are disturbed: twelve numbers go in, twelve come back, and the box is the thing in between. Reversible means the twelve that came back are enough to recover the twelve that went in, so nothing was thrown away in the middle. Ask for every response of that kind the ports admit. Those responses compose, they undo, and they come in a continuous supply, which makes them dials.

Count them. Twelve, one for each reading the ports hold, and no dial among the twelve leaves all twelve readings alone. Turn any one of them by any amount and some port reads differently afterward. That is the condition that welds the dials to the ports instead of to machinery standing behind them. Every direction of response does something. There is nothing left over for a thirteenth to act on.

Twelve dials, and how many forces there can be is settled by how those twelve fall apart into independent pieces.

25.7 The swap and the center

One move of the twelve ports came out of chapter fourteen indifferent to order: the swap that trades every port for its opposite. Do it before any of the sixty rotations or after, and the ports finish in the same arrangement either way. Twelve ports can be shuffled 479,001,600 ways, and of the 479,001,599 that move something, the swap is the only one that gives the same result in either order with all sixty rotations.

Collect, in any group, the moves that commute with every move in the group. That collection is the **center** of the group. Chapter nineteen used the same word for the questions inside an algebra that commute with every other question, which is where records live, and the property is identical, commuting with everything in sight; the difference is what is being commuted, an algebra of questions there and a group of moves here.

Groups differ enormously in how much of themselves lives in the center. Settings of the single amplifier dial all commute with each other, so that group is its own center, every last move of it. Turns of a solid object in three dimensions go the other way: the book on the table showed that a quarter turn and a flip disagree about order, and if you work through the possibilities the only turn that commutes with every other one is the turn that does nothing. Its center holds a single move. The move is nothing happening, zero dials wide.

Go back to the clause chapter four spent a section on. Two parties who cannot look a rate up anywhere outside the room have to manufacture one out of what both of them hold. The same clause binds the twelve ports harder, since nothing anywhere holds a preferred labeling of them. Any change in how the readings are labeled has to be a change the box itself can perform.

Take that concretely. The twelve ports carry no names of their own. Numbering them is something an observer does on a piece of paper, a second observer numbers them differently, and the wiring is indifferent to both. What the box can do is turn its own dials. So the relabelings with any standing are the ones the dials produce, and applying one is a three-step operation built out of moves the box has: turn the dials to the new labeling, do whatever was going to be done, turn them back.

Feed a central move into that. It commutes with the relabeling, so the relabeling and its undoing slide together and cancel. The move comes back out untouched. Every change of description leaves the center exactly where it was.

The sixty rotations of the wiring are changes of description. Each of them is an inside job by that same clause. So the center has to sit inside the part of the twelve readings that all sixty rotations leave fixed.

That part is easy to find, because chapter fourteen showed the sixty rotations carry every port to every other port. A set of twelve readings that survives all sixty unchanged has to read the same at every port. One number, repeated twelve times, and one number is one dial.

The center of the twelve is at most one dial wide.

25.8 One, three, eight

A continuous group of this kind comes apart into its center and a remainder with no center in it at all. The center is one dial or none. The remainder is eleven dials or twelve.

A group that cannot be broken into independent pieces is called **simple**, and the sizes a simple group is allowed to have are a fixed list rather than a matter of taste: three, eight, ten, fourteen, fifteen, twenty-one, and on upward. Below twelve, the list holds exactly three entries. There is no simple group of size four, or five, or six, or seven, or nine, or eleven, or twelve.

So the count of forces is a sum to be hit exactly. The parts come off that list, the total to be reached is twelve less whatever sits in the center, and nothing may be left over at the end. Two targets, then, twelve and eleven. Take a pencil to both, using threes, eights and tens, as many of each as you like.

Suppose the center is nothing at all. Then twelve dials have to be assembled out of pieces of size three, eight and ten. Ten leaves two to find, and the list starts at three. Eight leaves four, and four is not on the list. Threes run three, six, nine, twelve, and the fourth one lands exactly. Three plus three plus three plus three is the only way to do it: four copies of the smallest simple group there is.

Check whether the wiring has room for four copies. Pairing every port with the port opposite it gives the six axes of chapter fourteen. Each axis carries two readings, splitting into the part that is the same at both ends and the part that is equal and opposite at the two ends: six even readings and six odd ones. Among the six even readings sits the uniform one, the same number everywhere, and taking it out leaves five. The six odd readings fall into two separate blocks of three. So the twelve come apart as one, three, three and five, and no rotation of the wiring mixes any block with any other.

Four blocks of three are needed. Two exist. The rotations cannot get around that by shuffling four copies among themselves either, because a shuffle of four things would require the sixty rotations to fall into four families of fifteen, and no fifteen of the sixty close up into a group of their own.

So the center is exactly one dial, the remainder is eleven, and eleven has to be built out of three, eight and ten. Ten leaves one, and one is smaller than anything on offer. Two eights overshoot to sixteen. Threes run three, six, nine and then jump to twelve, stepping over eleven without touching it. Eight leaves three, and three is the first entry on the list. Three plus eight, and no other way at all.

$$1 + 3 + 8 = 12$$

One abelian dial, a three-dial piece and an eight-dial piece, with nothing left over and nothing borrowed. Two sums were tried. The largest number that came up in trying them was sixteen.

Every simple group in the catalogue died at the same step, and all of them died before a single particle was looked at. A simple group's center is zero dials wide, by the definition of simple. This wiring hands over one central dial that no relabeling can touch, and it hands it over before anybody asks what the forces act on. Howard Georgi and Sheldon Glashow proposed in 1974 that the three forces sit inside one larger simple symmetry. A carrier holding a central dial has nowhere to put them.

The allocation is fixed too, and not by size alone. The uniform reading, the one thing all sixty rotations leave alone, is the central dial. One of the two odd blocks of three carries the three-dial piece. The remaining block of three and the block of five carry the eight-dial piece between them, and they carry it because the eight dials of the three-slot group turn among themselves the same way those eight readings do under the rotations, which is a stronger match than eight equals three plus five.

Three pieces, and the world has names for them. The single abelian dial is the phase that chapter twelve's relabelings turned at each port, and its field is the one whose potential falls off as seven thirty-sixths, one ninetieth, minus seven one-hundred-and-eightieths and minus one eighteenth. That is electromagnetism. The three-dial piece is the weak force, which is why beta decay happens and why the Sun burns slowly enough to be worth living near. The eight-dial piece is the strong force. Its eight dials are the eight gluons that hold a proton together.

Look at the full list of what went into that. Twelve ports and thirty seams, which came out of chapter fourteen's counting. Sixty rotations, which came out of the same wiring. The requirement that every relabeling be an inside job, which is chapter four. The splitting of twelve readings into blocks of one, three, three and five, which is arithmetic on the same sixty rotations. And the list of sizes a simple group is allowed to have, which was settled by Wilhelm Killing in four papers between 1888 and 1890, and made rigorous by Élie Cartan in his thesis of 1894, for reasons that had nothing whatever to do with forces.

The count consulted no mass, no coupling strength, no decay rate and no accelerator run. Every input to it was a whole number obtained by counting something on a wiring diagram or looking something up in a classification finished before the electron was discovered. The forces come out one, three and eight.

Which opens a gap, because a force with nothing to push on is a dial wired to nothing. Each of the three pieces turns something. The machine as counted holds nothing for them to turn. Whatever carries those charges has to fit inside the same twelve readings the dials were counted from, since there is nowhere else on the carrier to put it, and has to fit without disturbing the count that has just been made.

The catalogue of things that do exist makes that look difficult. The electron carries one unit of electric charge. The down quark carries one third of a unit and the up quark carries two thirds, and every quark carries one of three color charges besides, and the whole arrangement is repeated three times over at rising masses. Thirds, in a world assembled out of a table of whole numbers. And three of everything.

Chapter 26

Why This Exact List of Matter, and Why Three Times Over?

Two physicists at the Massachusetts Institute of Technology filled a metal cavity with sulfur hexafluoride in 1973 and waited for the gas to sing.

Sulfur hexafluoride is heavy and chemically dead, which is why electrical engineers pump it into switchgear to keep the contacts from arcing. H. Frederick Dylla and John G. King wanted it for a different property. They wanted a gas whose molecules they could count.

Across the cavity they ran an alternating electric field. If every molecule carried even the smallest excess of charge, the field would shove the whole body of gas one way and then the other in time with itself, and a gas shoved back and forth a few hundred times a second is a sound. They mounted a calibrated microphone in the cavity wall, switched on the field, and listened.

The cavity stayed quiet.

The microphone put a ceiling on what it had failed to hear. The net charge on one molecule of sulfur hexafluoride came in below two parts in ten billion billion of the charge on a single electron. That molecule carries 146 units of atomic mass, so per unit of mass the difference between the charge on a proton and the charge on an electron sits below one part in a thousand billion billion. Hydrogen, which is one proton and one electron and nothing else, is electrically neutral to better than one part in a billion billion.

Look at what has to be true for that. A proton is two up quarks and a down quark. The up quark carries two thirds of a unit of electric charge and the down quark carries minus one third. Two thirds plus two thirds minus one third is one. An electron contains no quarks. The strong force that binds the proton's insides passes through an electron without noticing it. The electron's charge is minus one, which is to say exactly three times the down quark's. The exactness holds past every decimal place anyone has reached.

Nobody ordered that. Write the same physics down with the down quark at minus 0.333333 and the electron at minus 1.000000 and every equation goes through unchanged. What changes is the sky: a mismatch of one part in a billion billion would leave two hydrogen atoms pushing each other apart about as hard as their gravity pulls them together, and a cloud of the stuff would blow itself apart instead of collapsing into a star.

So the charges are locked to each other, across species that have nothing else in common, to a precision whose absence a microphone in Cambridge, Massachusetts was able to hear. Two questions follow. Why does matter come with exactly these charges and no others, and why does the whole list arrive three times over, in copies that differ in nothing except how heavy they are?

26.1 Three colors, two weak slots

Chapter twenty-five counted the forces and came back with twelve directions, split as one plus three plus eight. The three belongs to the weak force and the eight to the strong force. Those two numbers say how many things each force shuffles.

The single abelian factor carries **hypercharge**. What that dial reads on a species, combined with its position in a weak pair, gives electric charge. Electromagnetism emerges from that combination.

A force that shuffles some number of slots has a fixed stock of independent moves, the square of the number of slots minus one, which chapter twenty-five counted on its fingers. Set that against the blocks: the three-dimensional block matches a force shuffling two slots, the eight-dimensional block matches a force shuffling three, and the next size up, fifteen, overshoots while three is spoken for.

Two color slots would give a different matter pattern. Their color and anticolor representations are equivalent, and two quarks can form a color-neutral combination. Three quarks cannot: reversing every color vector changes the sign of their product, so it cannot be invariant. Three color slots instead admit both quark–antiquark pairs and an antisymmetric three-quark combination. Those are the color structures used for mesons and baryons. Matching these representations to observed particles is a physical input.

Representing these symmetries by three complex color slots and two weak slots is an additional matter assumption. The construction also assumes fermionic antisymmetry and a primitive charge grid. The following count derives the roster within that model; identifying its slots and fields with physical matter requires separate evidence.

Each slot carries a hypercharge tag. Two conditions within this model fix the tags. The three color tags and the two weak tags, added with their multiplicities, have to come to zero. The primitive grid fixes the charge normalization. Those requirements leave one answer and its mirror image: each color slot tagged minus one third, each weak slot tagged plus one half.

Check the balance. Three times minus one third is minus one, two times one half is one, and they cancel.

26.2 Choose two, choose four

Take the five slots and build things out of them by picking a handful at a time.

One rule governs the picking, the rule that makes matter behave like matter. Write a pick as an ordered list of slot names, and demand that swapping any two names flips the sign of the whole object. That is **anti-symmetry**. It has an immediate consequence: pick the same slot twice and swapping the two copies changes nothing while also flipping the sign, so the object equals minus itself, so the object is zero. Every slot appears at most once. Order stops mattering, since any reordering costs a sign and nothing else. What survives is the bare question of which slots are in and which are out.

The number of ways to choose two things from five is the number of pairs you can make: five choices for the first, four for the second, and a division by two because each pair got counted in both orders. Ten. The number of ways to choose four from five is easier, because choosing four is the same as leaving one out. There are five slots to leave out. Five. Numbers of this kind are called **binomial coefficients**. For cases this small you can write the choices down and count them.

Ten and five. Fifteen.

Fifteen is the number of states in one generation of matter. The charges come out with it. The hypercharge of a combination is the sum of the tags of the slots in it. The strong and weak forces act on whichever of their own slots are present.

Slots picked	Ways	Hypercharge	What it is
two color	3	minus two thirds	anti-up quark, one per color
one color, one weak	6	plus one sixth	up and down quarks, three colors each
two weak	1	plus one	positron
three color, one weak	2	minus one half	electron and its neutrino
two color, two weak	3	plus one third	anti-down quark, one per color

Every hypercharge in that column is the sum of the tags. Minus a third and minus a third is minus two thirds. Minus a third and a half is a sixth. Three color tags and one weak tag is minus one plus a half, or minus a half. The rows add to three plus six plus one plus two plus three, which is fifteen.

Electric charge is one more addition. Where a pick includes exactly one of the two weak slots, the weak force can rotate that choice into the other, so those states come in pairs. The two members of a pair sit half a step either side of the hypercharge tag. A sixth plus a half is two thirds, the up quark. A sixth minus a half is minus one third, the down quark. Minus a half plus a half is zero, the neutrino. Minus a half minus a half is minus one, the electron. The unpaired rows carry their tag unchanged: minus two thirds, plus one third, plus one.

That is the roster. Up and down quarks in three colors, the electron, its neutrino, and the antiparticles of the ones that need them, with every charge in the Standard Model's table reproduced, out of choosing two or four from five slots and adding up tags.

Two further counts fall out of the same table. The quark pair appears once per color and the lepton pair once, so a generation carries four weak pairs. A spin-zero field with hypercharge plus one half supplies three mass-coupling channels. The quark pair couples to the anti-up through that field; the quark pair couples to the anti-down, and the lepton pair to the positron, through its conjugate with hypercharge minus one half. These charge cancellations use the same left-handed convention as the roster. The field and its coupling strengths are supplied.

26.3 Dirac's holes

Three of the five rows above are antiparticles. In 1928 Paul Dirac wrote an equation for the electron that respected relativity. It came with solutions of negative energy that nobody wanted. He tried for a while to identify the holes in that sea with protons, and in a paper submitted to the Royal Society on 29 May 1931 he gave that up and proposed instead a particle nobody had seen: the mass of an electron, the opposite charge. Carl Anderson photographed one at Caltech on 2 August 1932, a track in a cloud chamber curving the wrong way through a magnetic field, and the name came from an editor at *Physical Review*, who proposed it when Anderson sent the discovery paper in: positron. The pattern turned out to be general. Every species of matter has a partner with the same mass and every charge reversed, and the partner is as ordinary as the original: hospitals make positrons to order out of fluorine-18, and the pair of flashes from each annihilation is what a scanner photographs. Writing a species as its antiparticle is therefore a change of notation rather than a change of content, which is why the roster above lists an anti-up quark where a textbook table lists a right-handed up quark, and the reason for preferring one of those descriptions is cobalt.

26.4 Wu's cobalt

Hold your hands out in front of you, palms down. They are the same object twice. No amount of turning either one will make it into the other, because a mirror sits between them. A screw thread has the same property, and so does a particle: give it a direction of travel and a sense of spin, and either the spin turns the way a right-handed screw turns as it advances, or it turns the other way. The two cases are called right-handed and left-handed, a mirror swaps them, and the property is **chirality**.

A symmetry is the claim that some part of a description was never physical. Mirror symmetry would say that left and right are a labeling convention, and that nothing in an experiment distinguishes a world from its reflection. Until 1956 every physicist assumed it, because electricity and the strong force obey it exactly and nobody had looked anywhere else.

Chien-Shiung Wu was born in Liuhe, near Shanghai, in 1912, took her doctorate at Berkeley in 1940 under Ernest Lawrence, and worked on uranium separation for the Manhattan Project at Columbia, where she stayed. She was the person other physicists went to about beta decay. In 1956 Tsung-Dao Lee and Chen Ning Yang worked out that the evidence for mirror symmetry in the weak force amounted to nothing whatever, since no experiment had ever tested it, and they brought Wu a list of things to try. She picked the decay of cobalt-60 and cancelled a trip to Geneva and the Far East that she had planned with her husband, on the grounds that somebody else would think of it.

The experiment needs the cobalt nuclei lined up, which needs them cold. On 4 June 1956 Wu telephoned Ernest Ambler at the National Bureau of Standards in Washington, whose group could reach three thousandths of a degree above absolute zero by adiabatic demagnetization, and asked him to collaborate. Several months went into getting the apparatus to work. The first successful run began at four minutes past noon on 27 December 1956.

A cobalt-60 nucleus decays by throwing out an electron. With the nuclei aligned by a magnetic field, the question is whether the electrons prefer the spin direction or the opposite one. In a mirror the spin direction survives, because a spin is a circulation rather than an arrow, while the direction the electron flies in is reversed. So a world obeying mirror symmetry has to emit equal numbers each way. Any asymmetry at all kills the symmetry outright.

The electrons came out against the spin. Reverse the magnetic field and the count dropped, then climbed back to baseline over about six minutes as the sample warmed and the alignment was lost. Wu, Ambler, R. W. Hayward, D. D. Hoppes and R. P. Hudson published it in *Physical Review* volume 105, page 1413, in 1957. Lee and Yang took the Nobel Prize in Physics that year. Wu took the first Wolf Prize in Physics, in 1978.

What her cobalt established is that the weak force acts on left-handed particles and declines to act on right-handed ones. Which is why the roster above is written the way it is: put every species into its left-handed form, converting the right-handed ones into their left-handed antiparticles as you go, and the list becomes a list of things the weak force either pairs up or leaves alone. The quark pair and the lepton pair are the pairs. The anti-up, the anti-down and the positron are the singles. A list of that shape differs from its own mirror image, and a list that differs from its own mirror image is called chiral.

Chirality is the symmetry the world conspicuously lacks. Every other structural fact about matter says that some distinction was never there; this one says that a distinction nobody put in is physical. It is why the roster has to be rewritten all in one handedness before it can be counted at all.

26.5 One thousand and twenty-four lists

The five slots generate more than the fifteen states above. Pick none of them, pick one, pick three, pick all five: there are thirty-two picks in all, and they group into components according to how many color slots and how many weak slots each one uses. Drop the empty pick and the pick that takes everything, and ten components are left, each with its charges attached. Any subset of those ten is a possible content for matter: two to the tenth possible lists, or 1,024.

Each of the 1,024 gets tested against two demands. It has to be chiral, since the mirror is what Wu's cobalt says the world distinguishes, and it has to make four separate sums come out at zero.

Of 1,024 lists, exactly two survive.

The two survivors are the even picks and the odd picks: the objects built from an even number of slots, and the objects built from an odd number. Charge conjugation, which swaps every particle for its antiparticle, carries each of the two onto the other, so they are one list and its mirror image, and a world taking either one gets the same physics with matter and antimatter labeled the other way round. The even one is the two-slot picks and the four-slot picks: ten plus five, and the roster above.

The search asked each list for two things, that it be chiral and that its four sums vanish, and slot count was not among them. What came back divides by how many slots a pick uses: all even, or all odd. The rule doing the dividing is the sign rule the picks were built with, that swapping two labels flips the sign of the whole object, and that minus sign is the one matter carries when two of its particles change places. The sign came back attached to the answer.

26.6 Four sums and a count

Chapter twelve proved that a force works if and only if its charge is conserved, tight enough that violating it leaves no theory at all rather than a damaged one. The quantum version of that proof comes with a bill. The bill is arithmetic.

Certain sums, taken over every species on the list at once, have to come out at exactly zero. When one of them does not, the conservation law the force depends on fails and the theory stops predicting anything. Sums of this kind are called **anomalies**. A matter list either satisfies all of them or describes no possible world. Every entry contributes, so the list has to be checked whole. A species cannot be added or dropped on its own.

Five conditions apply to a list like this one. Take the first.

The input is every species the strong force touches, tagged with its hypercharge: the quark pair, which has two weak members and so counts twice, the anti-up, and the anti-down. The operation adds their hypercharges, with color triplets and antitriplets entering the same way. One sixth counted twice is one third. The anti-up brings minus two thirds. The anti-down brings plus one third. One third minus two thirds plus one third is zero.

The second is the harder one, because hypercharge enters three times over. The input is all fifteen states with their hypercharges cubed, each multiplied by how many states carry it. The quark pair supplies six states at a sixth, and a sixth cubed is one part in 216, so six of them give one thirty-sixth. The anti-up supplies three states at minus two thirds, and minus two thirds cubed is minus eight twenty-sevenths, so three of them give minus eight ninths. The anti-down supplies three at one third, giving one ninth. The lepton pair supplies two at minus one half, giving minus one quarter. The positron supplies one at plus one, giving one. Put all five over thirty-six and the numerators are one, minus thirty-two, four, minus nine and thirty-six, which add to zero.

Nothing on that list was arranged to make it come out. The tags were fixed by a balance condition and a grid, the multiplicities were fixed by counting ways to pick slots out of five, and five cubes over a common thirty-six come to nothing.

The three conditions left over are cheaper. Adding the plain hypercharges of all fifteen states gives one, minus two, one, minus one and one, which is zero. Adding the hypercharges of the weak pairs gives three color copies at a sixth, a half in total, against the lepton pair's minus a half. The last condition is a count rather than a sum: a world with an odd number of weak pairs has no consistent weak force at all, which Edward Witten proved in 1982, and this generation carries four.

Five conditions. A list assembled by choosing slots and adding tags meets every one. In the way physics is normally taught these conditions arrive after the fact, as a check on charges that were read off a detector, and their passing is one of the things that make the Standard Model feel like something somebody got away with. Here they are the sieve, and 1,022 of the 1,024 lists die on them.

26.7 A clock with six positions

The sums fix the charges relative to one another. The unit they are counted in comes from somewhere else.

Chapter four's clock face bundles three and fifteen together and declares them one thing, because for a clock's purposes the difference between them does not exist. Do that to a group of symmetry moves and you get a **quotient group**: find the moves that change nothing observable, bundle every one of them in with doing nothing at all, and what is left over is the group that does the physical work. The clock bundles the multiples of twelve. The question here is which combinations of gauge moves this world bundles.

Chapter twenty-five's center supplies the moves to try. The center of a group is the moves that commute with everything, and for the strong force it is the move that advances all three color slots together by a third of a full turn, for the weak force it is the half turn on both weak slots, and for hypercharge it is every rotation of the dial. Combine one from each and you get a move that might do nothing, settled by trying it on the roster.

Take the quark pair, which sits in three colors, in a weak pair, at hypercharge one sixth. Apply the color move: a third of a turn. Apply the weak move: half a turn. Apply the hypercharge move, under which a species turns through a fraction of a full turn equal to its hypercharge, so a hypercharge of one sixth turns through one sixth of a turn. Add the three angles. A third plus a half plus a sixth is two sixths plus three sixths plus one sixth, or six sixths, which is one complete turn. The quark pair comes back exactly where it started.

Try the anti-up, which takes two color slots and no weak slot and sits at hypercharge minus two thirds. Each color slot it holds turns a third, so the object turns two thirds, and the hypercharge move contributes minus four sixths, which is the same two thirds backward. They cancel. The lepton pair takes all three color slots, a whole turn by itself, and adds half a turn from the weak move against minus a half from hypercharge. The positron takes both weak slots, which is another whole turn, and one more turn from its hypercharge of plus one. All fifteen states come home.

So there is a combination of gauge moves that does nothing whatever to any piece of matter that exists, and repeating it generates a family of exactly six such combinations, counting the one that does nothing to start with. Six moves that nothing in the world can detect. By the argument chapter four made about permitted relabelings, they get bundled in with the identity, and the group that does the physical work is the quotient.

Which is where the sixths come from. The bundling holds together only if the hypercharge dial advances in steps that fit that six-element pattern. A species whose hypercharge misses a whole multiple of one sixth turns through an angle that spoils the cancellation above, so it cannot exist without breaking a bundle that is a fact about which moves the world can tell apart. Every hypercharge in nature is a whole number of sixths. Which sixth is tied to how many color slots the species carries. That tie is where the electric charges land: anything carrying color comes out on a third, and anything without color comes out on a whole number.

That is the lock. The down quark and the electron are separated by everything. They are counted in the same unit, because the unit belongs to the wiring rather than to either of them. The down quark stands at minus one third, which is minus two sixths, and the electron stands at minus one, which is minus six sixths, and three times two is six, exactly and leaving zero room for a discrepancy in the twentieth decimal place. The proton, being two ups and a down, lands on plus one for the same reason. The microphone in the cavity had nothing to hear.

26.8 Three at the least, five at the most

That settles the charges. The whole of it then arrives three times over.

The muon is a heavier electron and nothing else: same charge, same weak behavior, same everything except a mass two hundred and seven times larger and a lifetime of two microseconds. The tau is heavier again. Behind each of them sits a full copy of the fifteen-state roster, quarks included. Isidor Rabi supplied the standard phrasing on being told about the muon, “who ordered that”, and it holds for the third copy as well.

The lower bound comes from requiring the quark-mixing mechanism to distinguish matter from antimatter.

Charge conjugation swaps every particle for its antiparticle. Combine it with a mirror reflection and you have an operation turning any process into its opposite-and-reflected version, and the question is whether the two run at the same rate. On 27 July 1964 James Christenson, James Cronin, Val Fitch and René Turlay published the answer from the Brookhaven accelerator: among the long-lived neutral kaons in their beam, 45 events decayed the way the symmetry forbids, about one in five hundred. The decays distinguish matter from antimatter. Explaining the cosmic excess of matter requires a further dynamical account.

In the Standard Model’s quark-mixing mechanism, the weak force couples different quark families through a square table with one row and one column per family. Distinguishing matter from antimatter through this mechanism requires a complex phase that no relabeling of the fields removes. Counting the phases that survive relabeling is arithmetic on the size of the table:

$$\frac{(N-1)(N-2)}{2}$$

Here N is the number of families, and the expression counts the quark-mixing phases left after field rephasings. Put N equal to one or two and the answer is zero. Put N equal to three and it is one. With fewer than three families, this mechanism supplies no such phase. Makoto Kobayashi and Toshihide Maskawa ran that reasoning in 1973, when three quarks were known and a fourth was a rumor, and concluded that this mechanism needed six quarks.

The ceiling comes from asymptotic freedom. Put a charge in a vacuum and the vacuum responds: pairs flicker in and out around it, orient themselves and screen it, so the charge you measure from far away is smaller than the charge close up. The strong force does the reverse, because its own carriers carry the charge they transmit, which spreads the charge outward instead of hiding it and leaves the force feebler at short range than at long. David Gross, David Politzer and Frank Wilczek worked this out in 1973. It is why quarks rattle around almost freely inside a proton while being impossible to pull out of one.

Species of matter push the other way, back toward ordinary screening, and each family adds species. Requiring asymptotic freedom of the weak interaction is an additional assumption. With one spin-zero doublet, its tally starts at twenty-two thirds, each family costs four thirds, and the doublet costs a sixth. Requiring a positive total puts the number of families below five and three eighths.

Which leaves three, four or five, and the shape of the problem is familiar to anybody who has tried to get a week into cabin luggage: a floor set by what the trip needs, a ceiling set by what the case holds, and nothing in either constraint that picks a number.

The declared family model assigns generations to one complete carrier band that distinguishes all rotations. Chapter fifteen's twelve directions split into uncuttable pieces of sizes one, three, three and five. The one-dimensional piece cannot distinguish rotations; the others can. Within the assumed window of three through five, the complete eligible bands therefore have sizes three, three and five. There is no four-dimensional band. Mixtures or repeated copies of bands fall outside this model.

The declared Laplacian cost prices the three eligible bands at five minus the square root of five, six, and five plus the square root of five: 2.7639, 6 and 7.2361. Its minimum selects a band of size three. Interpreting that band as physical generations is a separate assumption.

An independent measurement counts light neutrino species with the usual weak coupling. Through the 1990s the four detectors at CERN's

Large Electron-Positron collider measured the Z boson's invisible decay rate and divided by the rate for a single species. The full data set gives 2.9840 plus or minus 0.0082. It excludes a fourth such light species; it does not exclude a neutrino too heavy for the Z to produce.

Every entry on that roster is a set of charges. A charge is an answer: what the strong force gets back when it asks, what the weak force gets back, what the dial reads. Ask an up quark what it is, as opposed to what it answers to, and five slots and fifteen states say nothing at all.

Chapter 27

What Is a Particle, What Is a Wave, and Why Does It Look Like Both?

In April 1925, in a laboratory on West Street in Manhattan, a bottle of liquid air blew apart next to an electron-scattering experiment Clinton Davisson had been running for years and that Lester Germer had joined in 1924. The bottle stood near a vacuum tube containing a nickel target. The target was hot. Air rushed in and the nickel oxidized.

They spent the following weeks baking the oxide off, heating the target for long stretches to drive the damage out, and then started the run again. The experiment was simple to describe: fire electrons at the nickel, count how many come back at each angle. Before the accident the counts had traced a smooth hill. Afterward the same apparatus gave sharp peaks at particular angles and very little in between.

Nothing about the electron gun had changed. What had changed was the metal. The prolonged heating had recrystallized the target, and where the nickel had been a mass of tiny crystals pointing every way, it had become a handful of large ones. A handful of large crystals is a diffraction grating. They rebuilt the apparatus around a single crystal. The strongest peak came at an accelerating voltage of 54 volts and an angle of 50 degrees, and the atomic spacing of nickel converts that angle into a wavelength of 0.165 nanometers, against the 0.167 nanometers Louis de Broglie's rule of 1924 assigns an electron of that speed.

Nickel settled one half of the matter. The other half was put on film in 1989 at the Hitachi Advanced Research Laboratory, where Akira Tonomura's group sent electrons past a charged wire that bends two paths back together, which is a two-slit arrangement with no slits in it. They ran the beam so faint that the instrument held one electron at a time. Each one landed as a single dot at a definite place on a counting detector. It took about twenty minutes of accumulating dots for the stripes to appear.

An electron lands at one spot. The spots it chooses over many landings are the spots a wave makes bright.

The argument about how one object manages both has been running since 1905, when Einstein published a paper in the *Annalen der Physik* proposing that light is emitted and absorbed in lumps of a definite size, at a moment when every optical instrument in Europe worked because light is a wave. A century of textbooks has handled the tension by declaring the object to have two natures and moving on. The question the nickel put, and the charged wire put again sixty-four years later, is how a spreading pattern and a single count can be one thing happening.

27.1 One word, two objects

A value assigned at every place. That is one of the two things physics calls a field, and chapter twenty-five built one out of a table of who touches whom: a number at every port, falling off with hop distance in four exact fractions, and no space anywhere in the calculation.

The other one starts with a room.

Take a region and ask what can be answered inside it without leaving. Chapter nineteen made a question into an operation. A collection of operations you can compose, scale and add is an algebra. So attach to each region the algebra of everything answerable there, with its composition rule. Regions that contain other regions carry longer lists, because anything answerable in the small one is answerable in the large one. The small algebra sits inside the big one exactly.

The list is longer than it sounds. Everything a thermometer, a photographic plate, a photomultiplier and a wristwatch inside that region could between them establish is on it, along with every combination of those readings, and nothing whose answer would require somebody to go outside and look.

Take two regions far enough apart that no light ray leaves one and reaches the other while the questions are being put. Every operation on one side then commutes with every operation on the other. Alice asks and Bob asks, in either order, and the two orders give the same answers, so there is no fact about who went first for anybody to establish. Chapter twenty used that as a step in the Bell argument, where it is the statement that neither of them can signal to the other.

An assignment of an algebra to every region, with those two properties, is what the word field names in quantum field theory. There is no arrow at a point anywhere in it.

On the carrier that structure is exact. Two observer regions with no seam in common sit as exactly the two tensor factors of the joint description. Chapter twenty built the tensor product out of a deck of cards: four suits, thirteen ranks, dimensions multiplying, and most joint descriptions refusing to come apart into a fact about the suit and a fact about the rank. Two separated regions are the suit and the rank. Each one's questions are questions about its own factor, the two lists commute across the gap, and the pair's description is assembled by the operation that multiplies dimensions.

Everything in that construction is countable. The regions are finite collections of patches, the lists of questions are finite, and the commuting across the gap is arithmetic on the wiring rather than the limit of a sequence of approximations, so a continuum is never called for and never used. Chapter twenty-five's field is a value at every place. This one is a list of questions per region. Physics runs both under one word. The second is the one the Standard Model is written in.

27.2 Why the iron bar does not slosh

Warm one end of an iron bar and the warmth crawls along and stays, arriving late at the far end and never overshooting it. Chapter nine's repair move is that same equation written on a graph: my reading minus the average of my neighbors', a sixtieth at a time. Diffusion by itself produces no waves at all.

A reversible wave model adds a second quantity that the first one feeds. A pendulum has two: displacement sets the rate at which velocity changes, and velocity sets the rate at which displacement changes. The pair circles instead of settling. The seam operator can supply the stiffness of such a pair. Choosing that reversible evolution is an additional assumption: conservation of transported load alone does not turn diffusive repair into a wave.

Fix a direction of travel and a wavelength, and the wavelength fixes a wavenumber: how many crests fit into a set length. The amplitudes available to a wave going that way are the ones perpendicular to it. In three dimensions the perpendicular directions form a plane. Two of them. Two is a rank in chapter fifteen's sense, the number of independent directions a table of numbers contains, rather than a count of anything anybody has been out and observed. Chapter seventeen reached the same two by subtracting the direction of travel from the three directions of an observer's rest space.

Write A for the amplitude in that perpendicular plane, and write the Greek letter lambda, capitalized, for the stiffness the wiring supplies at a given wavenumber k, which is the number saying how hard the carrier pushes back on a mismatch of that wavelength. Two quantities feeding each other give one equation.

$$A'' + \Lambda(k)\, A = 0$$

The double prime is the rate of change of the rate of change, in time, so that is the equation of a mass on a spring with the wiring supplying the spring, and its solutions oscillate at a frequency whose square is the stiffness.

The stiffness is computed from the complete thirty-seam operator. At long wavelengths it approaches the square of the wavenumber. Its first correction subtracts a twentieth of the fourth power times the square of the seam spacing. There is also an exact bound on the full operator: the difference from the square of the wavenumber never exceeds that correction in magnitude. At zero wavenumber the stiffness is zero. The limiting wave speed comes from the quadratic normalization and the chosen evolution, not from the zero alone.

This controls the whole evolving field. Put the seam-shaped operator on one ordinary three-dimensional space, choose a common time parameter, and shrink its spacing. The paired fields converge to Maxwell's equations, with an error bound proportional to the square of the spacing for sufficiently smooth initial fields. Every field of finite total energy converges, even without that smoothness. A prescribed current can be included, and conservation of its charge preserves Gauss's law.

The space and the reversible evolution are inputs to this construction. It does not turn the observer's event records into positions, identify a repair count with laboratory time, or identify its paired amplitudes with readings of electric and magnetic fields. That requires a map from the bounded observer's local state, ports, readback, records and feedback into the field family. The continuum theorem specifies what that map must reach.

A finite software observer can read its own field without keeping a spare copy. It remembers a port's value, averages that port with an adjacent value, and reads the average. Twice the average minus the remembered value recovers the hidden value. Writing the recovered pair back lets the next measurement start from the same state. Applied to forty-two declared potential coordinates, this procedure supplies the records that drive a sourced Maxwell update. Its reconstructed fields satisfy the same finite action, including the charged paths. Exact memory and writable classical ports are assumptions. Counting these feedback cycles does not establish a physical clock.

The recorded potentials also define fields inside a solid built by joining the carrier's twenty triangular faces to its centre. This adds an explicit volume to the surface description. A geometric interpolation preserves magnetic flux closure and Faraday induction. Energy exposes a further condition: adding up the field energy throughout that volume gives a different action from counting its stored coordinates, and interpolation between time steps adds a separate correction. Both differences are calculated on the recorded history. The reconstructed fields have nonzero source-equation residuals under this geometric action. A volume field map therefore exists under the supplied geometry. This interpolated history and a stationary volume history are different constructions. This finite solid has no proved connection to the shrinking-spacing field family above.

The volume can also carry its own evolution. Its geometric action gives equations for every edge, including the edges running through the interior. Starting from a field that satisfies Gauss's law makes the evolving field keep that constraint. An exact bound controls its field energy. A software observer reads and restores fifty-five potential coordinates and uses the decoded records to advance the field. This gives a recorded stationary volume history under the chosen geometry and evolution rule.

The same geometric action has thirty independent radiative modes. With the usual quantum rule supplied, these become thirty quantum oscillators. Their occupation numbers form an infinite Hilbert space, and a self-adjoint Hamiltonian evolves its states unitarily. This is the free radiative sector; it does not quantize a charged interacting system or obtain quantum probabilities from classical readback.

Dynamical matter can be included through a separately supplied charged scalar field. Its interpolation changes with the electromagnetic potential as gauge symmetry requires. Both fields follow one stationary action. Positive conserved energy ensures global classical solutions of the finite equations. A short numerical evolution starts with opposite charges near the centre and boundary. Five software patches record, probe and restore the symmetry-reduced state, and each advance reads those decoded records. Independent reconstruction checks every finite field equation. Exact interval arithmetic encloses the same solution throughout two units of model time. Replaying the observer records certifies all five positions and five velocities at its eighty-one decoded checkpoints within 1.0001×10^{-10} of that solution. Temporary probe values and clock readouts have separate errors. The geometry, matter law and couplings are inputs.

A real scalar sector gives a controlled continuum of evolving matter. Its electromagnetic fields vanish, while its quartic self-interaction remains. Subdivide the solid without making the tetrahedra progressively flatter. For a smooth continuum solution and a compatible projection of its initial fields onto the finite mesh, the resulting motion approaches the continuum nonlinear wave equation. The error in the field and its velocity decreases with the cell size on a fixed smooth time interval. The same convergence rate also holds for complex matter in a prescribed uniform magnetic field, with initial data projected using its magnetic energy. The background remains fixed; the estimate does not yet include its response to the matter. Both results compare trajectories within the supplied geometry and matter model.

The changing configuration also provides a model clock. Its kinetic metric and supplied conserved energy assign a duration along regular segments with positive kinetic energy, independently of the labels used to order the samples. Applied to decoded records, this approximates the action's elapsed time without reading its timestamps. The construction excludes turning points and equilibria. Physical units and a comparison with laboratory clocks require additional evidence.

The full interacting action has a Hilbert-space description under a supplied quantum prescription. Removing redundant gauge descriptions gives a complete positive kinetic metric. Choosing its volume measure and Laplace–Beltrami operator, the curved-space version of the ordinary Laplacian, fixes a Hamiltonian with a unique self-adjoint extension. It evolves neutral states unitarily. An explicit normalized initial state gives exactly calculated matter observables. Its physical preparation and the identification of the calculated fields with physical observer records require separate evidence.

The classical charged state also supplies position and momentum parameters for quantum preparation, after removing the redundant gauge coordinates correctly. Gaussian packets retain all fifty-six configuration directions. Averaging over the remaining common phase produces neutral states, with an exact positive normalization formula. Their scalar radius can be calculated. The classical centers label these prepared states; they do not describe their subsequent quantum motion.

A trial history on the full fifty-six-dimensional configuration space changes the centered state's phase. Its distance from the exact quantum evolution is at most one tenth in Hilbert norm for times of magnitude at most 2^{-55} in the supplied units. The trial's configuration probabilities stay fixed. This certifies a tiny interval within the finite model; it supplies no practical quantum-field simulation.

A separate comparison theorem states what a useful effective quantum description must control. Preparation error, the difference between the full Hamiltonians, detector mismatch and probability outside the chosen energy window together bound the error in measured probabilities. Coupling to omitted states matters even when the retained matrix entries agree exactly. Applying that theorem to these interacting fields requires calculated bounds for one common pair of models.

27.3 Two questions, two orders

Amplitudes can cancel and probabilities cannot. That is chapter nineteen's difference and the reason there are dark bands at all. Cancellation needs a sign. A pattern that slides as one path is lengthened needs more than a sign, because it needs something that turns continuously, and the amplitude turns: it carries an angle, and the arithmetic of turning angles is the arithmetic of the square root of minus one.

The suspicion that the turning is bookkeeping is reasonable enough. Turning arrows multiply conveniently, and physicists like conveniences, and somewhere underneath there might be real numbers doing the work.

The carrier settles it on a two-by-two table.

The machine hands over two yes-or-no questions from behind one public reading, both real, both built out of labels an actual run produced, and the two fail to commute. It hands over two states as well, and every real question in the entire web scores those two states identically: put any of them to either state and the same odds come back. The states are different states. Nothing real tells them apart.

Build the missing question out of what is there. Start with the even split, which is the do-nothing operation halved and which as a description gives fifty-fifty odds to every question in sight. Take the two questions, call them P and Q, do them in one order and then in the other, and subtract. That is chapter nineteen's commutator. Multiply that difference by the square root of minus one and by two divided by the square root of three. Subtract the result from the even split, and call what comes out Y.

$$Y = \frac{1}{2}I - \frac{2\sqrt{3}}{3}\, i\,(QP - PQ)$$

I is the operation that does nothing, P and Q are the two real questions the carrier supplied, and i is the square root of minus one.

Y squares to itself, making it a question in chapter nineteen's sense, with a yes, a no and no third possibility. Put it to the two states that nothing real could separate. It answers yes on one of them with certainty and no on the other with certainty.

The coefficient was not fitted to make that happen. The commutator of those two particular questions has off-diagonal entry exactly the square root of three over four, and two over the square root of three scales that entry to exactly one half, which is the value that makes the result square to itself. Two real questions and one subtraction produce the direction that neither of them held.

And the direction is unreachable from below. Close the real questions under everything real processing does to them: average them, scale them, take complements, push them through any real operation of the kind a physical process performs on a description. Every element of that closure hands the two states the same number, and Y hands them different ones, so no amount of coarse-graining can manufacture it.

Three questions then fix everything. The record question the run supplied, its rotated partner, and Y between them pin down any description of a two-outcome system whose odds add up to one, leaving nothing about it undetermined.

The direction that was missing is the phase. Ernst Stueckelberg tried the other route in 1960, writing quantum theory in a real space in *Helvetica Physica Acta*, and made it work by adding an operator that squares to minus one and commutes with every observable, which is the imaginary unit with a different label on the jar.

27.4 The leftover, named

Twelve observers wired into an icosahedron, one record out of thirty flipped, two thousand and forty-eight starting states and sixteen schedules apiece: chapter six ran all of it. Every run converged. Every run left exactly one unit of disagreement sitting in the system, movable to any location and removable from none, with no edge of the world to push it off.

That leftover satisfies three conditions. The three conditions are what a **particle** is.

It is a pattern in the way record content moves from port to port, a property of the transport rule rather than a lump occupying a spot. Every observer whose patch overlaps yours reads the same pattern in the overlap, so it survives comparison and is not an artifact of one observer's labels. And it survives a regrade: change the resolution of the records, finer or coarser, and the pattern is there with its identity intact. Fluctuations of the repair process fail that last condition, because the faster bands decay faster and wash out under regrading.

Two checks make it quantitative, and both of them are taken against the vacuum, which is the state the machine arrives at when every repair available has been performed and no record has been written anywhere. Chapter six's undamaged run landed on it from all two thousand and forty-eight starting configurations under every schedule.

Does the pattern propagate: expand the response around that state to second order, remove the redundancy, and see whether what survives obeys the oscillator equation with a positive coefficient, which is the condition for a mode to carry energy rather than drain it. And does it register as one thing: every quantity built from a pattern has its weight spread across a range of energies, and a particle contributes a spike of zero width at one energy with nothing at the energies beside it. That spike is what a detector reports as an invariant mass. A pattern whose weight smears across a range instead of spiking at a point is several things rather than one. The smooth background lying under the spike is the weight belonging to two or more of the things at once.

27.5 Five places a pattern can sit

One direction, three and eight in the part of the twelve-dimensional response that does the transporting, with nothing else arithmetically available. Fifteen states across three families in the part that gets transported. Chapters twenty-five and twenty-six between them leave five places where a pattern can hold together, and no sixth.

The single direction is the one in which all twelve ports move together, so no rotation of the carrier touches it. On the selected abelian Maxwell action branch, the curvature has no nonabelian self-interaction term. With the declared gauge redundancy and field constraints, two transverse modes survive. They have the polarization structure of a photon. Identifying them with measured light also requires the physical field, propagation and detector maps; the dimension of the invariant band does not select those maps or the action by itself.

The three and the eight do not commute. Composing two directions in one of those blocks gives a third, so the curvature picks up a term quadratic in the field itself and the modes carry the charge of the field they are made of. Two crossing light beams pass through each other untouched, and two crossing color fields pull on each other. Pull two such patterns apart and the response between them holds at constant strength instead of thinning with distance, so the stored energy climbs without bound as the separation grows, and long before the separation is large it costs less to make a fresh pair than to keep pulling. These modes propagate and never register as isolated spikes, because the spike belongs to the neutral combination and the colored piece has none of its own. Confinement is one sentence about a bracket.

The massive carriers start out in the same bands as the massless ones, so whatever separates them comes from what those bands are read against. The vacuum carries a vector of definite size, 246.22 billion electron-volts. The moves that leave the direction of that vector alone form a two-parameter family; the moves that leave the vector itself alone form a one-parameter family; and the response reads the vector. Three of the four electroweak directions move it and one does not. A mode along a direction that moves the vector costs energy however long its wavelength, and a constant added to the stiffness destroys the exact zero: the frequency at zero wavenumber stops being zero and the spike leaves the origin. That constant, converted into the units a detector uses, is a mass, and the Z sits at 91.1876 billion electron-volts. The one direction that leaves the vector alone stays massless. It is the photon. Electromagnetism and the weak force are one band read against a vacuum that distinguishes them. The bookkeeping closes too: each direction the vacuum was displaced along stops being a pattern of its own and becomes the third way of waving that a massive mode has and a photon lacks. Three massless modes with two polarizations each, plus three displaced directions, is three massive modes with three polarizations each. Nine before and nine after.

Matter lives in the part of the response that says what is transported rather than how. Chapter twenty-six built it as a pair of rank-fifteen projectors conjugate to one another, graded by whether the port pattern is odd or even, and fifteen states across three families is forty-five states on one observer's screen. Those projectors are orthogonal, which is chapter nineteen's condition on exclusive alternatives: questions that never overlap, so a yes to one is a no to the other. A matter pattern occupies one of the forty-five states and no second pattern occupies the same one. That exclusivity is the exclusion principle, and why matter stacks into shells and atoms and solids, and why a laser will pile arbitrarily many photons into a single mode while an atom refuses to put two electrons into one orbital.

The fifth type sets the vector the other four are read against. Three of the types transport, one is transported, and this one is the pattern of the vector-selecting field itself. Its excitation is a change in the size of the vector rather than in its direction. A change in size costs energy and relaxes back, which is what it takes to produce a spike. It carries no polarization index whatever, because a magnitude has no direction for a rotation to mix. That is what spin zero means. The two CERN collaborations of chapter four announced this one on 4 July 2012, and selecting the vector is its job.

A sixth type would need somewhere to sit. The response has a part that transports and a part that is transported and no third part. The transporting part splits into the central direction and the non-central ones, and the non-central ones split by whether they move the vacuum vector. The vector-selecting pattern is itself an excitation. Every slot is occupied, no slot is left over, and the splitting of a twelve-dimensional response into one plus three plus eight admits no alternative.

27.6 When the record gets written

Questions that refuse to commute, and a center where the answers commute, survive being read twice, and survive being copied to a neighbor: chapter nineteen supplied both halves of the answer and left them lying apart. Pinching carries a question from the first into the second by deleting every comparison that only made sense between two bins.

Between commits, no record exists of which route a pattern took. What exists is the spreading mismatch of the wave, with a phase and a frequency. Every question about it lives in the part of the algebra that refuses to commute. Amplitudes for different routes add, opposite amplitudes cancel, and cancellation is what a dark band is made of.

At a commit a record is written. A record lives in the center. Everything there commutes with everything else, the answers survive reading and copying, and they are ordinary classical numbers: this one, here, count of one.

The electron crossed the tube on West Street with no record written anywhere of which crystal plane it scattered from, and arrived at the collector where a record was written and a count went up by one. The pattern of counts is the wave. Each count is the particle. Nothing about the electron differed between the crossing and the arrival except how much of it got written down.

The same reading settles what happens when somebody tries to watch. Put an instrument in the apparatus that succeeds in telling you which route was taken and the bands go. Delicacy buys nothing. No version of the instrument has ever been built gentle enough to report the route and leave the pattern standing, because reporting the route is writing a record, and a record is a commit. What the instrument deletes is every comparison between the two routes, which is where the cancellation lived. An instrument that fails to determine the route deletes nothing and the bands survive, however violently it fails.

Duality is a statement about when the record is written. The older question, which of two natures the thing has, carries an unstated variable, and the variable is the depth at which the reading is being taken. Fix the depth and the ambiguity goes. Nothing whatever happens to the electron at the plate. A record gets written, and a record holds a number.

27.7 The term that cannot be written

In 1919 Ernest Rutherford published four papers in the *London, Edinburgh and Dublin Philosophical Magazine* on what happens when alpha particles are fired into light gases. In nitrogen he saw scintillations at distances no alpha particle could travel, identified them as hydrogen nuclei knocked out of the nitrogen nucleus, and gave the thing its name the following year. Proton.

Chapter one told the other half. Grand unification put the eight, the three and the one inside a single simple symmetry, and the fractional charges and the family structure fell into place, and the bill came with the carriers. Building three blocks into one simple group forces extra directions into the list of carriers. The smallest group with room for all three holds twenty-four carrier directions, which is the twelve the world has and twelve more. Every one of the extra twelve carries color and weak charge at once. A carrier of that kind couples a quark directly to a lepton, so a proton comes apart into a positron and a neutral pion.

The list of carrier directions on the twelve-port response is the sum of the blocks and nothing besides: eight color directions carrying no weak charge, three weak directions carrying no color, and one carrying neither. A simple group is compelled to mix its directions into one another, because its own internal moves carry any direction to any other, which is what being simple amounts to. A product of blocks has no move connecting one block to another, so its list stays block by block. The direction a quark-to-lepton carrier would have to occupy is absent from a twelve-dimensional list whose splitting is fixed. So the carrier that would make a proton decay the grand-unified way, a gauge boson turning a quark into a lepton, cannot be assembled out of the directions that exist. That route is closed exactly, with no small coupling in it and no heavy mass sitting in a denominator to make it slow.

The companion embarrassment goes the same way. A monopole of the unified kind is a knot in the field configuration, formed when a large symmetry is squeezed down to a smaller one, and whether such knots exist is settled by asking which loops in the space of ways to do the squeezing can be shrunk to a point without leaving that space. Whether a loop can be shrunk to nothing is a homotopy question. The loops that refuse are what count the knots. The carrier emits the product of blocks from the first moment there is a group at all, with no earlier stage at which it was simple, so the squeezing never happens and the loops are never drawn.

The fifty thousand tons of ultrapure water under the mountain in Japan have been running since April 1996. A search through 450 kiloton-years of data, which is the mass of the water multiplied by the years it was under observation, turned up nothing of the right shape. The floor that search puts under the decay sits at 2.4 times ten to the thirty-fourth years. Every proton Rutherford knocked out of a nitrogen nucleus in 1919 has the same list of things it can turn into that it had that afternoon. The list is empty.

Five entries, and not one of them says how hard anything pulls. Twelve directions, thirty seams and a census of what can hold together leave the photon on the list and the forty-five states of matter on the list, and say nothing whatever about the grip of the first on the second. That grip is a number near one part in 137. It sets the size of every atom in your hand.

Chapter 28

Why 1/137, and Why Is There a Limit to What the Universe Can Remember?

Wolfgang Pauli went into the Rotkreuzspital in Zurich in December 1958 with pancreatic cancer. His assistant Charles Enz came to visit him there, and Pauli asked whether he had noticed the number on the door of the room. It was 137. He died in that room on the fifteenth of the month.

The number had been at him for most of his working life. Take one electron and one photon and ask how strongly the first couples to the second. Write the answer so that every unit cancels out of it: no meters, no seconds, no kilograms, nothing that depends on anybody's choice of ruler. What is left is a bare number close to one part in 137. Arnold Sommerfeld pulled it out of the fine splitting of the lines in the hydrogen spectrum in 1916, where it appears squared, as a splitting about one part in nineteen thousand of the gap it sits inside. That is where the name fine-structure constant comes from. The alpha that stands for it in every equation is Sommerfeld's letter.

Arthur Eddington thought he could derive it. He had organized the 1919 eclipse expedition that measured starlight bending around the sun, he wrote the book that taught a generation of English speakers what relativity was, and by the end of the 1920s he was arguing that the reciprocal of this constant had to be a whole number, and that the whole number was 136, which he obtained by counting the independent components of an algebra. Then the measurements drifted toward 137. He revised the count, produced 137, and his colleagues started calling him Arthur Adding-One. The book he was folding all of it into, *Fundamental Theory*, appeared in 1946, two years after his death, and the derivation in it persuaded nobody.

Feynman put the number in *QED: The Strange Theory of Light and Matter* in 1985 and called it one of the greatest damn mysteries of physics, a magic number that comes to us with no understanding by man. He added that all good theoretical physicists put it up on their wall and worry about it.

What they are worrying about is one number that quantum electrodynamics does not set and cannot set. Every other quantity in that theory is computed. This one is supplied from outside. Once it is supplied it fixes the strength of every chemical bond, the rate at which an excited atom drops back down and throws off a photon, and the color of everything you have ever seen. A theory that computes the electron's magnetic moment to twelve places takes its central constant from a laboratory in Illinois.

That constant is measured to ten significant figures. Take a single electron, hold it in a Penning trap for months, and read off its magnetic moment: Xing Fan and his colleagues at Northwestern published that measurement in 2023 to thirteen parts in a hundred thousand billion. Turning it into the constant requires the theoretical side to be worked out to the same depth, which took the evaluation of all 12,672 Feynman diagrams that contribute at tenth order, a job finished by Tatsumi Aoyama, Masashi Hayakawa, Toichiro Kinoshita and Makiko Nio. The recommended reciprocal that comes out of the whole enterprise is 137.035999177, with an uncertainty of twenty-one in the last two places, which is one and a half parts in ten billion. On the line that fits on a coffee mug it is one of the numbers somebody measured and typed in.

Two questions, and one answer covers both of them. Where does that number come from, and why is there a ceiling on how much the universe can hold? Both fall to one move: take an object, read it from the outside and from the inside, and require the two readings to agree. The object is one cell of an observer's screen, and then, at the other end of every scale there is, the whole horizon.

28.1 A number with no units in it

The speed of light stopped being a measurement in 1983, when the meter was redefined to make it 299,792,458 meters per second exactly. Those digits are a fact about the second and the meter. Pick a different unit of length and every one of them changes. Nothing whatever has happened to light.

A pure number has no such escape. 137.035999177 is what it is in any unit system, on any planet, for any species that gets as far as building a spectrometer, and the same goes for the 1836.15 that the proton outweighs the electron by. There is nothing in either of them that anybody chose. The electron in the innermost orbit of a hydrogen atom travels at one 137th of the speed of light. That is the same number again, the reason atoms are the size they are relative to the light they emit.

The strength of electromagnetism depends on how closely you look. A charge sitting in the vacuum is surrounded by pairs of particles flickering in and out, each pair oriented so as to screen the charge a little, so an experiment probing from far away sees a weakened version and an experiment that gets in closer sees more of the bare charge. The coupling carries a scale with it. Read at rest, at the far end of the range where the probing photon has essentially no energy of its own, the reciprocal is 137.036. Read at the energy of the Z boson, ninety-one billion electronvolts, the same coupling comes out near 128. Colliders watched electrons and positrons scatter off each other at rising energies and read the coupling climbing as the collisions sharpened, so the sliding is measured rather than inferred. Say which energy a fine-structure constant was read at, or the number means nothing. The 137 value is the one at the far end. It is the one Pauli wanted.

28.2 The balance point

Chapter twenty-three showed that an outsider's finite record access is controlled by the boundary rather than the hidden interior. Chapter twenty-four supplies a gravitational area unit from Newton's constant. The numerical branch assumes a physical screen-cell map joining those two constructions and measures the assigned cell area in that unit. Call the resulting pure number the **grain**. The finite wiring does not produce the physical area map or a ruler by itself.

Look first at where the grain would sit if nothing were ever observed.

A cap of screen is the region inside a circle drawn on it, the way a constellation is a region of sky, with an inside and an edge like any other bounded thing. Take one and count its entropy the way chapter twenty-three counted it, in two parts. There is what the cap holds inside it, and there is what the cut around its edge contributes, and the total is the sum of the two. Ask for the arrangement in which that accounting looks the same at every magnification: the total is to the inside as the inside is to the edge. Write x for the ratio of the total to the inside. Then the inside divided by the edge is also x, so the edge is the inside divided by x, and the total is the inside plus the inside divided by x. Divide through by the inside and what is left is x equals one plus one over x.

Which is the instruction from chapter two, the one with the calculator: divide one by your number, add one, repeat. Its fixed point is 1.6180339887, the golden ratio, reached here from an entropy balance on a screen rather than from a sequence of fractions on a phone. The same number sets the coordinates of the twelve corners of the carrier, a property of the icosahedron that was known long before anybody was looking for a coupling constant, and not something this argument arranged.

A screen sitting exactly on that value is self-similar all the way down. Every cap has the same accounting as every other cap at every magnification, which means the entropy is flat everywhere across it. No cap is distinguishable from any other by anything either of them holds. An entropy that is flat everywhere has no gradients in it. A gradient is the only thing a record can be written into. Chapter sixteen's arrow runs down a difference, chapter twenty-two's heat will not move between two things at one temperature, and chapter nine's repair fires only where two readings disagree. Take away every difference and there is no repair, no commit, no record, and nobody in a position to hold one. A universe sitting exactly on the golden ratio is empty of observers. The small amount by which this one misses it is why there is anything to remember at all.

The margin is thin. This screen sits above the balance point by less than one percent of its own grain. Everything that has ever been written down anywhere, every fossil, every measurement of the electron's magnetic moment, every memory anybody has of anything, fits inside that fraction of a percent.

28.3 Two readings of one cell

So the grain sits a little above the balance point. The question is what fixes the distance.

The proposed outside reading begins by assigning the cell a physical area. Its excess above the balance point becomes a pure number after a supplied map converts a boundary profile into a weight per unit area. The edge is treated as a profile smeared across the boundary rather than a drawn line, and its declared normalization brings in the square root of pi, 1.772453851. The finite screen does not select that physical profile map.

The proposed inside reading assigns the same cell to the smallest electromagnetic act available to an observer. That assignment requires a physical gauge carrier, a scale and an operational readout. Authenticated record provenance alone does not identify a cell with a photon exchange.

The physical branch identifies the geometric and electromagnetic readings as two descriptions of one operational resolution. This is a same-carrier hypothesis. It becomes a theorem about Nature only if the screen cell, electromagnetic act and readout are constructed on one realized observer history.

After that same-carrier map is supplied, chapter thirteen's rule about consistent readings sets the two values equal:

$$P = \varphi + \alpha\sqrt{\pi}$$

Here P is the assigned screen-cell area in units of the selected area scale. The Greek letter phi is the golden ratio, 1.618033989, used as the declared entropy balance point. Alpha is the electromagnetic coupling transported to that grain, and the square root of pi is the supplied boundary normalization. The equation defines the conditional comparison branch.

The unknown appears on both sides. Inside the declared map, a trial grain fixes a candidate common coupling at the cell scale, and a specified electroweak transport returns an electromagnetic coupling. Feed in a grain and get a grain back. The fixed-point certificate proves uniqueness for this map; it does not source-select the map or its physical attachments.

28.4 The last three operations

Take a calculator again.

The chain from a trial grain down to a coupling is the part a machine does. The number it hands back for a grain of 1.630968209 is 137.036, with the digits past the third deciding the last place. The coupling is its reciprocal, 0.0072973523.

Multiply by the boundary normalization, 1.772453851, and you get 0.012934220.

Add the entropy balance point, 1.618033989, and you get 1.630968209.

Which is what was fed in, to the last of the nine decimal places quoted. The equation closes on itself on the back of an envelope, and the whole electromagnetic term, the entire distance between a universe with observers in it and a universe with none, is 0.012934220, about eight parts in a thousand of the grain.

Run it backwards if you prefer. Subtract 1.618033989 from 1.630968209 and you have 0.012934220. Divide that by 1.772453851 and you have 0.0072973523. Take the reciprocal and 137.036 comes back out, which is the strength of electromagnetism read off the area of one cell of screen.

The obvious suspicion about arithmetic like that is the one Eddington earned, and the way to settle it is to count the freedom. A formula with one adjustable exponent in it can be made to hit 137.035999177, and so can a formula with two adjustable anythings, which is why people have been hitting it since the 1930s. There is nothing to adjust here. The golden ratio is forced by the entropy balance and has no dial on it. The square root of pi comes from the shape of the boundary profile, and a Gaussian normalization is whatever it is. The coupling is whatever the forward chain returns for the grain it was handed. Change the boundary factor from 1.772453851 to a round 1.77 and the loop returns 1.630950 instead of 1.630968, and the equation stops closing in the fifth decimal place.

The second test is what else the number is on the hook for. An expression that hits one target and produces nothing further is a coincidence with decimals on it, and the literature since the 1930s is full of them. The grain is the coordinate the rest of the numbers are read from: the single coupling the three forces share at the cell scale, which comes out at one part in 24.32, the running masses of the quarks, the scale at which the weak force lives. Move the grain in the ninth decimal place and every one of those moves with it.

28.5 Two endpoints instead of one number

Chapter two's other root, minus 0.618034, is a fixed point nothing ever arrives at: start a calculator anywhere near it and the numbers flee. This one is approached, and the rate can be bounded.

Take two trial grains a small distance apart and run both through the chain. What comes back is two grains at most 0.0724 of that distance apart. A map that pulls its inputs closer together like that is a **contraction**, and a contraction has exactly one fixed point and drags everything toward it. Seed it with 1, a guess wrong by more than a third. After one pass the answer is inside 0.0457 of the destination, after two inside 0.0033, after three inside 0.00024, and after ten passes inside two parts in a million million. Twelve digits, from a deliberately terrible starting guess, in ten passes, with nothing measured entering at any of them.

Arriving at a number and proving one are different operations, and a chain of this length rounds at every step. Anybody who has written software has met the small version: ask a computer for 0.1 plus 0.2 and it answers 0.30000000000000004, which is the machine telling the truth about what it is holding: a tenth has no exact form in binary and never did. Ten thousand operations later, the digits on the screen are a report about the arithmetic rather than about the world.

The repair is to stop carrying numbers and start carrying pairs of them. Every quantity becomes two endpoints, a low one and a high one, with the true value somewhere between. Every operation rounds the low end downward and the high end upward, so the bracket can widen and can never lie. Add the bracket that runs from one to two to the bracket that runs from three to five and the answer is the bracket from four to seven. Multiply two brackets and you get a bracket containing every product of anything in the first with anything in the second. This is called **interval arithmetic**. It turns a computation into a proof. Feed an interval of grains into the chain, and if what comes back is an interval strictly inside the one you fed in, then a fixed point sits in there as a theorem, and the root it holds is certified rather than merely computed. Sampling the slope of the chain at points along it suggests it pulls its inputs together by a factor of about 0.003 a pass. The interval run proves 0.0724, which is the worse figure and which holds at every point of the domain at once, so 0.0724 is the number the argument is allowed to use. The proof that the interval maps inside itself takes 203 seconds of processor time, which is a strange price for a fact about the strength of electromagnetism.

28.6 The quark cloud

Solve the same equation for the coupling instead of the grain. Subtract the balance point from the grain, divide by the square root of pi, and read the result at the point where laboratories read it.

The certified value is 137.035660. The measured value is 137.035999. They differ by two and a half parts per million.

The two numbers 137.036 and 137.035660 are different objects. The 137.036 that the chain consumes is the coupling read at the cell. The 137.035660 is the same coupling read one transport step further out, where an experimenter reads it, and the two differ in the fourth decimal place for that reason.

Transport to the laboratory coupling includes the hadronic contribution: the effect of quarks and their interactions on the photon response. This term requires independent physical information and a consistent convention for separating it from the other contributions. Its value and uncertainty must be propagated through the same calculation as the source map.

The narrow certified enclosure proves that numerical error in the fixed-point solve cannot account for the difference between 137.035660 and 137.035999. It does not identify the difference as a physical quark contribution. That identification requires a calculation connecting the same carrier, scale and readout to the laboratory endpoint. Choosing a correction from the gap would use the target to reproduce itself.

The root is therefore a precise diagnostic for the declared map. A physical prediction must combine an independently justified source map with independently determined transport and an uncertainty small enough to distinguish its output from alternatives.

28.7 The horizon's budget

The same move works on the largest object an observer has.

Look far enough out and the expansion carries a galaxy away faster than light can close the distance, so its signals arrive at no point of your future, ever. The sphere at that distance is your horizon. Gary Gibbons and Stephen Hawking showed in 1977 that it carries a temperature, a couple of thousandths of a billionth of a billionth of a billionth of a degree. Chapter twenty-three supplies a finite boundary-capacity candidate. Reading that candidate as the entropy or physical area of this horizon is an additional identification, not a consequence of counting records.

Counting the records that survive is a different count from counting the ones that can be written. Take a channel with five symbols arranged in a ring, where each symbol can be confused with either of its two neighbors and with nothing else. How many can you use with zero chance of being misunderstood? Two. Number them one to five around the ring and send only one and three. Those two are not neighbors, so nothing can turn either into the other, and there is no room for a third: two sits next to both of them, four sits next to the three, five sits next to the one, and the ring holds nothing else. **Zero-error capacity** counts the messages that stay distinguishable whatever the channel does to them, the count that matters when a record has to survive being read back by somebody else.

The proposed capacity branch asks whether the horizon can be read two ways: from outside by its geometric area, and from inside by the durable public records observers can write, keep, tell apart and read back. Equating those readings is a physical hypothesis. It requires one realized horizon, a map from correctable-record capacity to its entropy and area, and a calibrated density law connecting event counts to spacetime volume. The authenticated causal order supplies none of those maps by itself.

Inside that conditional branch, the arithmetic takes one input, the grain, and one correction. A shipping container holds slightly less than its stated volume, because the bracing that keeps the load from shifting takes up room. Every shared cut on the screen is assigned a small uniform reserve in the same way: the grain divided by twenty-four, chapter twenty-five's twelve dials counted twice, once out and once back. The reserve is 0.067957 against a budget of 281.07, and because the candidate capacity is an exponential of the budget, those seven hundredths move it by six and a half percent, from 3.5321 times ten to the 122nd before the reserve to 3.3001 times ten to the 122nd after it.

The weighted Planck expansion coordinate is 3.3129 times ten to the 122nd, four parts in a thousand above that candidate. This numerical proximity is a comparison on the supplied horizon-record branch; it does not prove that either finite count is the Universe's storage or volume.

If the same physical branch identifies that capacity with de Sitter horizon entropy, then the dimensionless relation is $\Lambda\ell_\star^2 = 3\pi/N_\star$. A time-independent selected capacity would give the constant-dark-energy equation of state $w = -1$. Neither the identification nor time-independence follows from the finite causal poset, so a measured evolution of w would falsify this physical capacity branch rather than the source-derived order or rank-three carrier. Five thousand and twenty robotic fibers on the four-meter Mayall telescope at Kitt Peak, repointing themselves to a few microns every twenty minutes, are collecting the spectra of tens of millions of galaxies that test such constant-dark-energy models.

28.8 One value, and why there is one

Take those two numbers away from the machinery for a moment and look at what they normally are.

Alpha is a dial. Quantum electrodynamics is a consistent theory at any setting of it, and nothing inside the theory prefers one setting to another.

You measure it, you enter it, and everything downstream follows. The cosmological constant is worse, because it enters the field equations as a term you are permitted to add with any coefficient at all, and the one estimate that tries to compute it from the vacuum overshoots the sky by a hundred and twenty orders of magnitude. The two govern nothing in common. One sets the size of an atom. The other sets how fast the universe comes apart.

This chapter studies a single conditional move on both of them. Read a cell of screen from the outside and from the inside, and set the two readings equal. Read the horizon from the outside and from the inside, and set those equal. Both use chapter thirteen's rule about seams, but only after the compared readings have been shown to refer to the same physical carrier. That last attachment is a premise of the numerical branch.

That demand needs something to hold before it can be made at all. It needs the world to have an inside to be read from, and it needs what the inside holds to be a description of the thing the inside is part of. A structure carrying no account of itself has no second reading to set against the first. The equation cannot be written, and the number stays free.

On the fully attached branch the constants are wired together. The grain P enters the candidate common-force coupling, running that down the range returns alpha, and alpha appears back inside P. The grain also enters a budget, the budget enters a candidate capacity N, and the separately identified de Sitter capacity fixes the dimensionless cosmological term through three pi divided by N. P is a resolution and N a capacity; converting either into a measured scale consumes the independent physical carrier and unit maps.

A loop of that shape is chapter two's equation with the world inside it. The unknown stands on both sides. The finite certificates prove uniqueness only inside their stated maps and feasible sets; they do not prove that Nature inhabits every physical attachment used in the loop. If those attachments are realized, the branch has one selected solution rather than a continuously fitted dial. Without them, the numbers remain comparison candidates rather than derived observables.

Which is the part worth sitting with, because nobody went looking for it. Self-reference turned up in chapter two to answer why there is anything at all, and it turns up again in chapter thirty-six to take the beginning away. The mathematical point is narrower and still striking: once the named maps are supplied, self-consistency can turn a continuously adjustable parameter into a unique fixed point. Establishing those maps from the source-derived causal poset is what would turn the comparison into a physical derivation.

What that is worth is what the rest of the chapter said it was worth, and no more. The declared coupling closure returns 137.035660 against a laboratory reading of 137.035999. The candidate capacity lands four parts in a thousand from the weighted expansion coordinate. These are retrospective numerical comparisons until the physical carrier maps are supplied.

Feynman's went up on the wall because nothing inside the theory could reach it. The difference here is that the theory has an inside, and the inside is what reaches it.

Chapter 29

Why Do the Masses Land Where They Do?

Yoshio Koide was an associate professor at Shizuoka Women's University at the end of 1981, working on composite models of quarks and leptons. Models of that kind put something smaller underneath the particles on the roster, so that the shape of the roster becomes a consequence of how few pieces there are to build from. The field was busy. In 1979 Haim Harari and Michael Shupe published the same proposal four pages apart in the same issue of *Physics Letters B*, without either knowing what the other was doing: every quark and every lepton built from triples of two objects, one carrying a third of the electron's charge and one carrying none. Harari called them rishons. Nobody has ever seen one. One line of arithmetic came out of Koide's model and stayed.

Take the masses of the three charged leptons, the electron, the muon and the tau. Add them up. Then take the square root of each of the three masses, add the three square roots together, and square that total. Divide the first number by the second.

The answer is two thirds.

The three masses have no business producing a number like that. The muon is two hundred and seven times the electron and the tau is three and a half thousand times the electron, and a quantity assembled out of three separately measured masses is supposed to come out as a decimal with nothing in it. Koide was in a position to notice, because his model had given him a reason to compute exactly that combination.

Two of the three masses were known to many digits by 1981. The tau was the trouble. It had been found at Stanford in the mid-seventies, it dies in less than a picosecond, and its mass had to be reconstructed from what came out of the decay rather than weighed. The world average sat at 1784.2 million electron-volts, give or take 3.2. Koide's relation, given the electron and the muon and asked for the third, returned 1776.97.

The gap is 7.2 million electron-volts, and physics reports a gap in units of the experiment's own quoted uncertainty. A measurement is a claim about a range, which is how chapter four read two different Higgs announcements as one result. The half-width of that range has a symbol, the Greek letter sigma, and a distance between two numbers gets quoted as a multiple of it. Koide's tau mass sat 2.25 sigma from the measured one. Gaps that size open and close as the measurements improve. The word discovery is reserved for five, which is the threshold ATLAS and CMS cleared in July 2012.

The unit in that comparison belongs to somebody. Sigma is whatever the experiment declared its own uncertainty to be, so a distance in sigma is a distance measured with a ruler that a particular group of people manufactured in a particular year, and rulers get replaced.

In the spring of 1992 a group running the Beijing Spectrometer, a detector on China's electron-positron collider, went at the tau mass from a different direction. Below a certain beam energy you cannot make a pair of taus at all and the production rate is zero. Above it the rate climbs. The corner between the two is the mass, twice over. They set the beam using the standing world average of 1784, took twelve measurements across the threshold, and found the corner at 1776.9 million electron-volts, 7.2 below where the average had put it, with an uncertainty smaller by a factor of seven.

Koide's 1776.97 had not moved. The world average had walked 7.2 million electron-volts and arrived on it.

Something holds the three masses at two thirds to three parts in a million. It is a triangle.

29.1 Three corners, two numbers

Chapter twenty-six put the three families of matter on the three corners of an oriented face of the icosahedron, which is where the count of three came from. The electron, the muon and the tau sit one to a corner.

An oriented face has one rotation left in it. Rotate the corners round the face: one goes to two, two goes to three, three goes to one. Do that three times and everything is back where it started. Of the sixty rotations of the icosahedron, that turn, its repeat and leaving the solid where it stands are the only three that keep both the face and its sense of circulation.

Take three identical rooms arranged in a ring, each joined to its two neighbors by identical doorways, and put a heater in one of them. Wait, then measure the temperature in all three. Doing that for each room in turn gives you nine numbers: how warm room one gets when room one is heated, how warm room two gets when room one is heated, and so on across and down. Nine numbers, and seven of them are redundant, because the ring has no room that differs from any other room. There is a number for how much heat stays where it was put and a number for how much crosses a doorway. The whole table is those two numbers arranged in a pattern.

The face works the same way. Push at one corner and read what comes back at all three, which is again a three-by-three table.

A response that respects the rotation cannot tell corner one from corner two, because the rotation carries the first onto the second and the response has no way of objecting. So the second row of the table is the first row shifted one place along, and the third row is the second shifted again. Nine numbers collapse into two: one for a corner's response to itself, written a, and one for its response to a neighbor. The neighbor entry carries a size and an angle, because a response can arrive turned as well as scaled, and chapter nineteen's amplitudes are that kind of number. Call the size b and call the angle the twist.

A table whose every row is the row above it shifted one step round the cycle is called a **circulant**.

Chapter fifteen took a twelve-by-twelve grid of repair weights and hunted for the grid's own directions, the ones it merely scales without swinging them round, and called the scale factors eigenvalues. Finding those for a general grid means solving an equation whose degree is the width of the grid. A circulant hands them over for nothing, because its own directions are fixed by the cycle before anybody says what a and b are: they are the three ways of stepping a phase round the triangle in equal parts.

The three scale factors come out as a, plus twice the size b, multiplied by a cosine. The three cosines are the cosine of the twist, the cosine of the twist plus a hundred and twenty degrees, and the cosine of the twist plus two hundred and forty.

Three cosines, a hundred and twenty degrees apart. The face has fixed their spacing and left the starting angle free.

29.2 Two lines about cosines

Draw three arrows of length one from a single point, a hundred and twenty degrees apart. They are the spokes of an equilateral triangle drawn from its center. They add up to nothing, because there is no direction for the sum to point. The cosine of an angle is the shadow one of those arrows casts on a chosen axis, and the shadow of a sum is the sum of the shadows.

Three cosines a hundred and twenty degrees apart sum to zero.

For the squares, use the fact that the square of a cosine is half of one plus half the cosine of the doubled angle. Double the three angles and you get the twist doubled, that plus two hundred and forty degrees, and that plus four hundred and eighty. Four hundred and eighty degrees is a hundred and twenty with a full turn discarded, so the doubled angles are three directions a hundred and twenty degrees apart, and by the line above their cosines sum to zero. Three halves of one, plus half of nothing.

Three cosines a hundred and twenty degrees apart have squares summing to three halves.

Add the three eigenvalues. The three copies of a give three a. The cosine terms cancel by the first line. Total three a, whatever the twist happens to be.

Add their squares. Each square is a squared, plus twice a times twice b times a cosine, plus four b squared times the square of a cosine. The middle terms cancel by the first line. The last terms come to four b squared times three halves, which is six b squared. Total three a squared plus six b squared.

The proposed mass readout squares the three eigenvalues, with an overall scale that fixes the units and cancels out of any ratio. Restricting to the positive chamber makes those eigenvalues the positive square roots used in Koide's relation. This defines a mathematical mass model; identifying it with the observed leptons requires a physical attachment.

Koide's ratio, written Q, is the sum of the eigenvalues' squares divided by the square of their sum. The two lines above have evaluated both of those.

$$Q = \frac{m_e + m_\mu + m_\tau}{\left(\sqrt{m_e} + \sqrt{m_\mu} + \sqrt{m_\tau}\right)^2} = \frac{1}{3} + \frac{2}{3}\left(\frac{b}{a}\right)^2$$

On the left are the three charged lepton masses, measured in laboratories. On the right is one number, the size of a corner's response to its neighbor divided by the size of its response to itself. Everything else has cancelled, the twist included, which appears nowhere on the right and therefore cannot be tuned to help.

Q equals two thirds exactly when b over a is one over the square root of two.

The left side of that equation is trapped, for any three positive masses anybody cares to write down. Set all three equal and it gives one third, which the right side agrees with by sending b to zero. Let one mass run away from the other two and it climbs toward one, without arriving. So every possible triple of masses in the universe lands somewhere in the range from one third to one. Two thirds is the exact midpoint of that range. The three charged leptons, whose masses cover a factor of three and a half thousand, sit on the midpoint.

Two pieces make up the response. There is the direction in which all three corners move together, which carries weight three a squared, and there is the two-dimensional plane in which they move against one another, which carries weight six b squared. Setting b over a to one over the square root of two sets those two weights equal. One condition on the response of a single face, and Koide's two thirds is that condition written out in masses.

The twist gets a small allowance rather than a value. At balance, the three eigenvalues stay positive as long as the twist is within fifteen degrees of one of the three spokes, and anywhere inside that band the ratio sits at two thirds without moving.

Take the three square roots of the masses as the three coordinates of a point in space. There is a line through that space along which all three coordinates are equal, the line where the three leptons would weigh the same. The balance says the point stands at exactly forty-five degrees from that line, and forty-five degrees is why the relation looks so peculiar written in masses. A clean statement about square roots turns into an agreement to five decimal places once everything in it has been squared.

29.3 Seventy-two electron-volts

Feed in the two masses known best. The electron is 0.51099895 million electron-volts and the muon is 105.6583755 million electron-volts. Requiring the balance to hold turns the relation into a quadratic in the square root of the third mass. A quadratic has two roots. One of them is 3.3174 million electron-volts, which would put the third charged lepton between the electron and the muon rather than above both of them. The tau is the heaviest of the three, which is why it can decay into a muon and a muon cannot decay into it, so the upper root is the one.

Chapter twenty-eight turned a computation into a proof by carrying intervals rather than points. The same machinery applies here: give the electron and the muon their measured widths, do the arithmetic on the widths, and what comes back is an interval guaranteed to contain the answer.

It runs from 1776.968991 to 1776.969063 million electron-volts, centered on 1776.969027.

That window is 72 electron-volts wide. The measured tau mass is 1776.93 million electron-volts, give or take 0.09, and 0.09 million electron-volts is ninety thousand electron-volts, so the uncertainty on the measurement is twelve hundred and fifty times the width of the window it is being compared against. The center of the window is 0.039 million electron-volts from the measured value, which is 0.43 sigma.

Fed the measured tau mass, Koide's ratio comes out at 0.66666446.

The sharpest tau measurements are threshold scans of the kind the Beijing Spectrometer ran, reaching ninety thousand electron-volts. What limits them is how precisely a beam energy can be calibrated at the point where a pair of taus becomes possible, and that is where the next three decimal places are.

A measured central value more than three standard uncertainties from 1776.969027 destroys the balance condition. There is no free parameter left to absorb the move: the twist cancelled, the scale cancelled, and the one remaining quantity was set to one over the square root of two by the requirement that the two pieces of the response weigh the same. At an uncertainty of 0.09 million electron-volts, three of them is 0.27, so a central value below 1776.70 or above 1777.24 ends it.

29.4 Thirty-one axes

In June 1963 Nicola Cabibbo published three pages in *Physical Review Letters* under the title "Unitary Symmetry and Leptonic Decays". Decays that changed strangeness were running slower than decays that did not, by a consistent factor, and the theory of the day had no room for a consistent factor. Cabibbo's move was to let the weak interaction couple to a rotated combination of the down quark and the strange quark, turned by a single angle. One number, fitted from decay rates, and the discrepancy went away.

The strength of the transition from a down quark to a strange one is measured at 0.2250. The angle whose sine that is comes to 13.0029 degrees. Rates go as the square of that strength, so a decay that changes strangeness runs at about a twentieth of the rate of one that does not. That factor of twenty is the suppression sitting in the data he was accounting for.

An icosahedron has axes of three kinds. Six of them run through pairs of opposite corners. A fifth of a turn about one of those leaves the solid looking untouched. Ten run through pairs of opposite faces, at a third of a turn. Fifteen run through pairs of opposite edge midpoints, at half a turn. Six and ten and fifteen is thirty-one.

Take the angle between every pair of those axes. Four hundred and sixty-five pairs, one angle each, and the smallest nonzero angle anywhere in the list is 20.9052 degrees.

13.0029 is not in the list. The reason it cannot be got at is that 20.9052 is the floor. The smallest angle the geometry has to offer is sixty percent larger than the one the quarks require, and relabeling which axis is which does nothing, because the floor is a property of the solid rather than of the labeling. The angle that sets quark mixing is not an axis angle of this geometry.

The list has an end, which is what makes an empty search a result: four hundred and sixty-five angles, written out in full and read down to the last one, at an arithmetic cost below what a phone spends drawing its own screen once.

Koide's own model reached past the leptons to the quarks. The quark side did not hold. Two structures built out of entirely different material, a set of subparticles and a twenty-sided solid, keep the charged leptons and stop at the quarks.

On the selected matter representation, the wiring fixes charge ratios and color multiplicity. For the lepton model, equal weights of the two response sectors are an additional balance condition. Combined with the measured electron and muon masses, that condition gives the 72-electron-volt tau interval and a precise comparison that can fail. It does not hand over the angle by which the weak interaction turns a down quark into a strange quark: 20.9052 against 13.0029 proves that the angle is no axis angle of this solid. Had 13.0029 turned up anywhere on that list, the charges and the family count would have been worth less, because a solid that yields whatever angle is asked of it has yielded nothing.

The entries end up on the same line. The Standard Model action combines symmetry constraints, supplied field content and measured parameters. Reading it as a reconstruction means identifying which inputs a proposed deeper theory explains and which require additional premises.

Chapter 30

What Is a Lagrangian, and Why Is It the Last Thing?

In 1788 the widow Desaint published a book on mechanics in Paris that contained no pictures, and its author put that on the first page as though it were the attraction. Joseph-Louis Lagrange had come down from Berlin the year before. The Académie sent the manuscript to a committee of Laplace, Cousin, Legendre and Condorcet, who approved it, and Legendre read the proofs.

No diagrams will be found in this work, the preface opens. The methods set out in it, Lagrange goes on, call for neither constructions nor geometrical or mechanical reasoning, but only algebraic operations subject to a regular and uniform procedure.

Mechanics until then was drawn. Newton's *Principia* argues in figures: an arc, a tangent, a shaded triangle, a limit taken as two points slide together, and the reader follows by looking. For a century after 1687 doing mechanics meant getting the picture right first, and the pictures were different for a pendulum, a spinning top and a planet, so each of them was a separate accomplishment. What Lagrange had found was that somebody who cannot draw at all obtains every one of those results by writing down a single function of the system and then turning a handle, and that the handle turns the same way in all three cases.

The line working physicists write down when somebody asks what the world is made of, the one that fits on a mug, is an object of the same kind: a few terms wide, with a subject folded into each one. Two questions come with an object like that. What sort of thing is it, that a whole subject folds into one line at all? And what does the folding cost, since a book of mechanics with the figures taken out has thrown something away, whatever else it has gained.

30.1 Proof of a supreme being

On 15 April 1744, before the Académie Royale des Sciences in Paris, Pierre-Louis Moreau de Maupertuis read a paper called "Accord de différentes lois de la nature qui avaient jusqu'ici paru incompatibles", agreement between different laws of nature which had until then seemed incompatible. The laws were the ones for how light bends and how it bounces. He proposed that a single quantity, which he named the action, comes out as small as it can be whenever anything in nature changes.

Two years later in Berlin, running Frederick's academy, he published "Les Loix du mouvement et du repos déduites d'un principe métaphysique". There the principle acquired a second job. Maupertuis presented it as a demonstration of a supreme being, and ranked it above the older proofs drawn from the beauty and the order of the world, on the ground that an equation is harder to argue with than a wonder.

The inference is the obvious one: a rule that picks the smallest total out of every total available looks like an economy. An economy has a treasurer.

Maupertuis had earned the standing to say such things by going out and getting a number. In 1736 he led an expedition to the Torne valley in Swedish Lapland to measure the length of a degree of the meridian close to the Arctic Circle. The party came home in August 1737 with a degree longer than the one measured in France, which meant the Earth is flattened at the poles, which meant Newton was right and the Cassinis were wrong. Robert Levrac-Tournières painted him afterward in a fur hat and reindeer skins with one hand pressed down on a globe, squashing it out of round. Voltaire, an admirer at that stage, addressed him as the flattener of the world and of Cassini.

The rest went badly. In 1751 Samuel König claimed that Leibniz had had the least-action principle first and produced a letter to prove it, and the Berlin academy, with Maupertuis presiding, declared the letter a forgery. Voltaire took König's side and wrote the *Diatribe du docteur Akakia, médecin du Pape*, in which a papal physician works through the published proposals of a self-styled academy president, most of them quoted from what Maupertuis had actually printed. Frederick had the pamphlet burned by the public executioner on Christmas Eve 1752, with the king and Voltaire both standing at the fire. Maupertuis left Berlin and died in Basel in 1759.

His principle survived him exactly as he stated it, and so did his reading of it. A rule that scores a whole path and then selects one looks as though the path was compared against paths that never happened.

30.2 A ball, described twice

Throw a ball across a room. It leaves your hand at one place and arrives at another a second later.

Here is the ordinary description. At each moment the ball has a position and a velocity. Gravity pulls down, which changes the velocity a little, which changes the position a little, and repeating that as finely as you like traces out the arc. The description looks forward one step at a time and needs nothing except what is true where the ball is.

Here is the other one. Draw every arc you like between those two points taking the same second: the true one, one that goes twice as high and comes down in a rush, one that dawdles halfway across and then sprints, one shaped like a staircase. Score each of them with a single number. The arc the ball flies is picked out by what its score does when you nudge it.

The scoring rule has two parts. The future appears in neither.

The first part is one function. At any instant, two things about the ball matter: where it is, and how fast it is going. Feed both into a single function and it hands back a number. For a thrown ball, that function is the part that grows with speed less the part that grows with height.

Both parts are countable for a ball you could actually throw. The height part is the **potential energy**, what the ball's position owes: the mass times the strength of gravity times the height. Gravity at the Earth's surface pulls at 9.81 meters per second per second, and rounding that to ten costs two percent and buys arithmetic you can do standing up, so a one-kilogram ball held a meter off the floor owes ten joules. The speed part is the **kinetic energy**, which is what the motion holds: half the mass times the square of the speed. That same ball moving at two meters per second holds two joules, and at four meters per second it holds eight, because the square is where the growth is. The function is the second of those less the first, joule for joule.

The second part is a running total. The number the function hands back is per second, so run along the arc from the first moment to the last and keep a running total of it, the way a taxi meter accumulates while the wheels turn. One arc in, one number out. Every arc on the list gets a total this way, including the ridiculous ones.

Selection uses slopes. Stand on a hillside and step one pace in any direction: your height changes by an amount proportional to the size of the step. Stand on the floor of the valley and step one pace: your height barely changes, because the ground under you is level, and the change you do get shrinks with the square of the step rather than with the step. The bottom of a valley is where the slope is zero.

The arc a thrown ball flies is where the slope is zero. Push the middle of it up by a hair, holding both ends where they are, and the running total does not change. Push it down by a hair and the running total does not change. Every neighboring arc scores the same to first order. The score is flat there in the way the valley floor is flat.

Take the crudest version of that and check it by hand. Three moments and two hops: a one-kilogram ball starts on the floor, ends on the floor one second later, and the only freedom is how high it is at the halfway mark. Raising the midpoint costs in the speed part, because the ball has to climb further and fall further in the same second. It pays in the height part, because the ball spends the middle of the flight higher up. Each hop lasts half a second and there are two of them, so anything holding steady through both contributes its own value times one second to the running total, and the halves cancel out of the arithmetic before it starts.

A peak of one meter. The ball climbs a meter in half a second, so it travels at two meters per second and holds two joules of motion, and its height averages half a meter across each hop, which owes five joules. Two less five is minus three.

A peak of one and a half meters. Three meters per second, four and a half joules of motion, average height three quarters of a meter, seven and a half joules owed. Minus three again.

A peak of one and a quarter. Two and a half meters per second, three and an eighth joules of motion, average height five eighths of a meter, six and a quarter joules owed. Minus three and an eighth.

The arc that overshoots and the arc that undershoots score the same number. The one between them scores below both, by an eighth of a joule-second in either direction. A single line says why, with h standing for the height of the halfway mark in meters and the total coming out in joule-seconds:

$$\text{score} \;=\; 2h^2 \;-\; 5h$$

The cost grows with the square of the midpoint height and the payment grows in proportion to it, so the cost overtakes. At one midpoint height and no other, raising the peak by a hair costs exactly what it pays. There the total has slope zero. That is the height the rule selects, one and a quarter meters, where a ball airborne for a second peaks: it leaves the floor at five meters per second, sheds ten of those every second, and stops climbing after half of one. Three moments is the coarsest account of a flight with any freedom left in it, and lands on the height exactly. Cut the second into four hops instead of two and the corners of the selected shape land on a parabola, and into forty and they land on the same one.

Look at what had to be in your hand to do any of that. All three sums used the landing. Fixing the speed on each hop took the ball being back on the floor at one second, and without the finish there was nothing to add up. The ball has the start. It leaves your hand with a place and a speed and no information whatever about where it comes down, and the rule says it flies the arc whose total sits at the bottom of a parabola whose shape depends on where it comes down. Something in that description knows the ending before it happens. It is not the ball. Maupertuis put a treasurer in the gap, which at least has the merit of being a mechanism.

The one function of where you are and how fast you are going is the **Lagrangian**. Its running total along a path is the **action**, which is the word chapter twelve used for a single number scored over a whole arrangement. This is the same object scored along a history instead of across a wiring. The rule that selects the path where wiggling changes nothing is **stationary action**.

Everything about a system is in its Lagrangian. Hand somebody that function and every quantity the system has gets its equation of motion, by the same handle-turning in each case: take the function's sensitivity to how fast the quantity is changing, ask how that sensitivity changes as the run proceeds, and set it equal to the function's sensitivity to the quantity itself. One line goes in and the entire behavior comes out. **A Lagrangian is a compression**: the thing you keep when you throw away the histories, chosen so that the histories can be regenerated from it.

Watch it come apart once. The Lagrangian of a thrown ball has two pieces, the speed piece and the height piece. Its sensitivity to how fast the ball is going comes from the speed piece and is the mass times the velocity, which is the ball's momentum. Its sensitivity to where the ball is comes from the height piece and is the mass times the strength of gravity, pointing down. Set the rate of change of the first equal to the second and out drops the statement that the ball's momentum gains mass times gravity downward every second, which was Newton's second law and which nobody wrote into the function. Two pieces of arithmetic and the handle, and the same handle applied to a spinning top gives its equations too.

30.3 Göttingen, 1918

The second thing a Lagrangian compresses is the reason anything is conserved.

Emmy Noether arrived at Göttingen in the spring of 1915 at the invitation of Hilbert and Klein, and worked there for years with no salary and no post. The faculty blocked her habilitation. Her lecture courses appeared in the catalogue under Hilbert's name, with herself listed as the person who would actually be giving them. When she had the result that carries her name she could not read it to the Göttingen Scientific Society, not being a member of it; Klein read it out for her on 26 July 1918. It went into the society's proceedings under the title "Invariante Variationsprobleme", pages 235 to 257.

In April 1933 the Law for the Restoration of the Professional Civil Service removed her from the university. She went to Bryn Mawr College in

Pennsylvania, and once a week to Princeton to lecture. In the spring of 1935 she had surgery for a tumor, the operation went well, and four days afterward she developed a fever and lost consciousness. She died on 14 April 1935, at fifty-three.

Chapter five gave the correspondence and chapter eighteen used it: every continuous symmetry of the laws comes with exactly one conserved quantity, determined by the symmetry, and momentum and energy are what you get from the laws not caring where and not caring when. The machine that turns the one into the other is a recipe with three steps.

Take the Lagrangian. Find a change you can make to it continuously, by an amount you can dial down to nothing, which leaves its value alone: slide the whole system a hair sideways, turn it by a small angle, advance every clock by the same interval, add the same small amount to a phase everywhere. Each of those is a direction of change with a size attached to it.

Then, for every quantity the system has, take the Lagrangian's sensitivity to how fast that quantity is changing. That sensitivity is the momentum belonging to the quantity, and falls out of the function rather than being defined off to one side. Multiply each sensitivity by how far the symmetry moves its own quantity, and add the products over the whole list. Advancing the clock takes one more move, since shifting the clock shifts the running total too: subtract the function's own value times the interval.

The number you get does not change along any history the Lagrangian selects. One symmetry in, one number out, by a recipe that does not care which symmetry it was fed.

Slide sideways and the number is momentum. Turn and it is angular momentum. Advance the clocks and it is energy. Add the same phase everywhere and it is electric charge, arriving by a second road. Chapter twelve reached it from the other end, by showing that a force whose charge leaks stops being a force at all. The recipe here starts from the same phase and hands back the quantity you can total up.

The same statement holds on a discrete chain with no limit taken anywhere. A run is a list of states with one step between each neighboring pair, and a symmetry is a change that carries each state to another while leaving each step's contribution to the running total alone. Take the contribution's sensitivity to the change across one step and it comes out the same across every step of the run. Segment by segment, the number holds, and there is no continuum anywhere in the argument.

Bookkeeping in this world was never a separate law added on top of the dynamics. It is a property of the line, extracted by a fixed recipe. A world has as many conservation laws as its line has continuous symmetries.

30.4 A product of small numbers

None of that touches Maupertuis's reading. The recipe is exact. A rule that scores whole paths and selects the flat one continues to look like a comparison, which needs the losers to be somewhere in order to lose.

Start again from the smallest assumption a world can be given.

Something is in some configuration. There is a rule saying what it does next. The rule has three properties. It is local, so the next step depends only on the configuration it is standing in. It is memoryless, so the rule does not consult how that configuration was arrived at. And it is strictly positive, so nothing is flatly forbidden: every configuration it could move to has some chance, however small.

That is the whole of the input. There is no action in it, no Lagrangian, no scoring, and nothing that looks at a history.

Ask about histories anyway, since a history is a thing that has happened and can be asked about. Take a list of configurations, one after another, from a start to a finish. What are the odds of that particular list? Each step carries odds of its own and the steps do not consult each other, so the odds of the list are the odds of the first step times the odds of the second times the odds of the third, all the way to the end.

A product of a thousand numbers each below one is a number nobody can read. Chapter sixteen introduced the operation that fixes it: a logarithm converts multiplying into adding. Take the logarithm of each step's odds, flip the sign so the contributions come out positive, and add them along the history instead of multiplying. Written out, with the odds of the first step called p one, the odds of the second p two, and the history running to n steps:

$$-\log(p_1\, p_2 \cdots p_n) \;=\; (-\log p_1) \;+\; (-\log p_2) \;+\; \cdots \;+\; (-\log p_n)$$

On the left is one number belonging to a whole history. Check it on the smallest case: two steps, each a coin flip, so the history has odds of one in four, which is two of chapter sixteen's yes-or-no questions, one from each step. On the right are n numbers, each belonging to a single step and worked out from the two configurations that step ran between, none of them consulting any other step, and the equals sign in the middle is the line of algebra that takes the foresight out.

Look at what that produces. It is a running total along a path, one contribution per step, and each contribution depends on where its step started and where it ended and on nothing else anywhere in the history. That is the shape of the thing the thrown ball was scored with, arrived at from a rule that has never heard of it.

A history with a large total took improbable steps. A history with the smallest total is the most probable history there is. And a history whose total does not move when you jiggle one of its interior configurations, holding the ends, is sitting on the valley floor in that direction, which is what it means to be locally most likely. **Least action and maximum probability are two readouts of one thing.**

Run the total backwards. Give each history a weight that shrinks as its total grows, one over the exponential of the total, and out comes the odds of that history again, because exponentiating undoes the logarithm that built the total. Nothing has to be divided by anything to bring the weights to one across all the histories, since the odds added to one before any logarithm was taken.

Then suppose somebody hands you a different running total, and a different overall scale to multiply it by, and their combination also reproduces the same odds. Their scale times their total equals the total above plus a constant, at every history, and there is no other freedom anywhere in the problem. You may slide the zero, and you may change the units. Everything else is forced.

A world whose step rule is local, memoryless and strictly positive has an action, then, whether or not anybody writes one down.

Nothing in that argument looks ahead, because the step rule has nothing to look ahead with. The endpoint turns up in the statement of the principle because the statement is about a completed history. A completed history has a finish in it the same way a receipt has a total printed at the bottom. A machine repairing locally, with no view past its own neighbors, produces a selection that looks global, because that is what taking the logarithm of a product of local factors does.

Maupertuis had found something real. The quantity a world comes out smallest on is improbability. A chain of local steps runs that tally by itself, without a manager and without a comparison ever being performed.

30.5 Plateau's wire frames

Joseph Plateau lost the sight of both eyes by the end of 1843, at forty-two. Ghent promoted him to full professor a few months later, and for the next four decades his wife Fanny, and afterwards his son and son-in-law, did the seeing for him. They dipped wire frames of every shape into soapy water mixed with glycerine, drew them out, and described aloud exactly what surface had formed. In 1873 he published nearly a thousand pages on the statics of liquids held together by molecular forces alone, every word of it dictated. Jean Taylor proved his rules about how the films meet each other in 1976, a hundred and three years afterward, as consequences of minimizing area.

Dip a bent wire in soap solution and the film stretched across it is the surface of least area that wire will hold. It looks like the answer to a question about all possible surfaces. The film has no way of finding out what any other surface would have cost. Each patch of it pulls on the patches around it, the pulling is settled locally, patch against neighbor, and when no patch is being pulled harder one way than another the film stops moving. Where it stops has the least area available. Nothing in the film has ever evaluated an area.

The settling is also invisible, which is most of why the film looks clever. By the time the frame is steady in your hand the film has finished, so nobody has ever watched a soap film choose. What Plateau's family described to him across forty years was the finished surface, every time.

30.6 The line

Physics as it is normally taught reaches a Lagrangian in the first week and spends the rest of the time taking it apart. The spacetime is there before the pen touches the page, the group is written down because the detectors reported it, the fields are listed because they were found, and the numbers are entered because laboratories measured them.

The line has four terms in it, written schematically, each standing for a block that opens out further.

$$\mathcal{L} = \mathcal{L}_{\text{carry}} + \mathcal{L}_{\text{matter}} + \mathcal{L}_{\text{couple}} + \mathcal{L}_{\text{select}}$$

The representation table supports a local action on specified flat spacetime with a spin convention. Gauge fields, anticommuting fermions and the Higgs field enter one expression, with all three kinds of Yukawa coupling. Gauge identities hold even when the relabeling varies from place to place: its derivatives cancel. The kinetic choice, Higgs terms and coefficients are supplied; connecting this action to the observers' repair records remains a separate requirement.

The script L on the left is a Lagrangian in the sense taught above, with one change of scale. Where the thrown ball had a position and a speed, this one takes a field's value at a place and how fast that value is changing, and its running total runs over all of space and all of time rather than along a single arc.

The first term is the carriers. Chapter twenty-five split the twelve dials of the wiring into a central one, a block of three and a block of eight, with nothing else arithmetically available. This term holds the energy stored in what those dials do around a loop. Chapter twelve's loop readings are the field strength, and squaring them and adding over the arrangement is the term.

The second term is matter, carrying chapter twenty-six's fifteen states, three families over. Inside it sits a comparison between a reading here and a reading next door, taken after the reading next door has been run through the dictionary sitting on the seam between them, which is chapter twelve's connection written with derivatives instead of exchange rates. That comparison is the entire content of the phrase "the force acts on matter".

The third term holds the allowed couplings: the channels through which the vector-selecting field touches matter. The representation scan constrains these channels. Their strengths are separate parameters of the local action; the charged-lepton comparison in chapter twenty-nine does not determine the complete physical coupling matrices.

The fourth term belongs to the vector-selecting field itself, chapter twenty-seven's single spin-zero type, which carries no polarization index because a magnitude has no direction for a rotation to mix, and whose vacuum vector has a size of 246.22 billion electron-volts.

Alongside the four sits gravity's term, chapter twenty-four's curvature. On the named physical-scale branch, an independently calibrated screen cell is identified with Newton's area, 2.61 times ten to the minus seventy square meters; the finite screen count does not supply that meter-valued identification. Beside it is the constant term, invisible to null contractions, which is why the cosmological constant remains undetermined by the local null-tomography calculation.

The prologue promised a distinction between reconstructed structure and the physical assumptions behind each symbol. Chapter fifteen supplies a continuous three-dimensional carrier and finite populations selected by its distances. Chapter twenty-five constrains the local gauge Lie algebra under complete reversible response and internal transport; chapter twenty-seven classifies field candidates under its stated interfaces. These results organize the ingredients of the line. Their common physical realization, the interaction coefficients and the assignment of laboratory units require their own justification. A four-dimensional space of possible event coordinates does not by itself provide a physical clock or make recorded dependencies into signals.

The proposed face-to-family identification supplies three copies of the matter grammar. The local action and its gauge identities are checked at that declared multiplicity; they do not establish its physical realization.

One charged scalar component also has a consistent classical reduction of this action, with both diagonal gauge fields aligned by their kinetic weights. A local execution on 64 prepared spatial sites records a phase disturbance, the resulting current and electric feedback. A gauge transformation of the same preparation leaves the invariant readings unchanged. This calculation uses specified field laws and model time. It supplies a finite dynamical example, without identifying its charged field with a measured particle or computing a quantum history.

30.7 What a compression costs

A compression is allowed to be lossy, and the useful ones generally are.

A finite system can justify a smooth field description through a controlled approximation. On prepared golden source addresses, one positive nearest-neighbor tensor action gives classical waves and, after canonical quantization, a full oscillator Fock space. The comparison theory is a massive scalar field in a cube with fixed boundary values. Refining the actual source addresses makes smeared classical and coherent quantum readouts converge.

The same calculation resolves a compact disturbance at a detector confined to a separate slab. At 233 source coordinates per axis, the total probability error stays below 0.050055 while the induced signal exceeds 0.137754 throughout model time from 0.95 to 1. Both spatial localization errors and the omitted spectral comparison modes are controlled. Tensor factorization makes the calculation small; it is not a full observer-network execution. The action, protected address records, quantization and model time remain specified inputs. Their connection to physical repair dynamics and to the count clock remains a further task.

The second loss is sharper, and can be written down exactly.

Take the line and add a term chosen so that it comes out zero at every configuration the world realizes. On a chain, the realized configurations sit at the corners. A term that vanishes at every corner adds nothing to the score of any realized history. So every realized history keeps its old score. Swap an interior configuration for another the world can realize, holding the ends, and the answer comes back what it was: the same run is the most probable run, and sits at the bottom against every single-step swap.

The momentum changes. Momentum is the line's sensitivity to how fast a quantity is changing. A term pinned to zero at every corner tilts as it passes through, in a direction no realized step ever moves. So the two lines assign different momenta to the same motion, and therefore different energies, since energy is what you get by trading the speeds for the momenta.

The freedom is one number. The added term bends with speed by any positive amount you like, and its bend with position is then forced, the two summing to one particular number the step odds themselves set. Under that constraint the most probable run stays at the bottom when an interior configuration slides continuously rather than jumping corner to corner, and the momentum stays a strictly increasing, invertible function of speed. Each setting is a different line carrying a different energy. Nothing in any history of this chain distinguishes them from each other. The histories are the same histories, and they were the only evidence there was.

So the line is the best compression anybody has written down. A compression is not the thing compressed. Lagrange knew what he had taken out, which is why he mentioned it in the first sentence.

Hand the line to somebody with a computer and enough patience and out comes the arc of a thrown ball to as many decimal places as the ball has, the spectrum of hydrogen, the lifetime of a muon and the rate at which the sun burns. Then put the ball in their hand and push it, and watch it go. The line scores the path it takes. It says nothing whatever about why a pushed thing moves.

Chapter 31

Why Does Anything Move?

The OPH reading in this chapter is conditional on a physical carrier, action, clock, ruler and source attachment. The finite seam graph supplies exact currents and shell-count precursors. Its read-from dependency cone is not a physical signal cone, and its rank-three carrier does not by itself establish the dimension or volume growth of space.

Benjamin Robins hung a broad iron plate from a frame in 1742, screwed a thick sheet of wood across the front of it, and fired a musket into the wood from twenty-five feet away.

The plate swung back. A narrow ribbon ran from the plate through a clamp gripped just tightly enough to hold, so the swing dragged a length of ribbon out through the slot and left it lying there. Robins picked up the ribbon and measured it. Its length is the chord of the arc the plate traveled, the chord gives the speed the plate left with, and the speed the plate left with gives the speed of a ball nobody had ever seen move: sixteen hundred and seventy feet per second, for a three-quarter-inch ball out of a forty-five-inch barrel. The same ball measured at seventy-five feet from the muzzle came in at fifteen hundred and fifty, and at a hundred and twenty-five feet at fourteen hundred and twenty-five. The difference is what the air took.

New Principles of Gunnery came out the same year and turned artillery into a subject with mathematics in it. Euler translated it into German in 1745 and added notes until the translation ran to roughly five times the length of the book it was translating.

What Robins built is a piece of arithmetic with a musket pointed at it. The ball goes into the wood and stops being a ball. Whatever it was carrying does not stop: the plate leaves with all of it, and a length of ribbon on the floor is a measurement of a quantity nobody in the room can see, trusted because the books balance.

Three things have to hold for that ribbon to mean anything. The quantity has to survive a collision that destroys the shape of the ball, the shape of the wood and most of the geometry of the problem. It has to arrive at the plate in full, with no fraction of it lost in the splintering. And it has to be the same quantity in a ball made of lead as in a plate made of iron, since the two are being added together. Robins had all three from firing the gun and derived none of them. The ribbon measures a quantity whose entire definition is that it comes out even.

Twenty-nine chapters have built a world and none of them has said why a thing that gets pushed goes. Four words carry every account of it, and a reader owns all four before opening any book: momentum, inertia, force, acceleration. Robins' ribbon measures the first of them, which a collision hands over whether or not anybody has defined it.

31.1 A chain of readings

Take a history of the kind chapter thirty scored: a list of readings, one per event, from a start to a finish. Chapter thirty's rule walks that list and adds one contribution per step. Each contribution depends on the two readings at the ends of its own step and on nothing else anywhere in the history.

For a pattern crossing the seam network with nothing pulling at it, the contribution is half the square of the change across the step, weighted by what the pattern costs to hold together. That square is the one the brass balls settled in chapter eighteen's tray of clay, and the half in front of it is the half that stands in front of chapter thirty's kinetic energy.

Here is a history with five readings in it, carried by a pattern with a standing cost of two units, chosen because two cancels the half and keeps the arithmetic in whole numbers.

0, 3, 6, 9, 12

The steps are 3, 3, 3 and 3, their squares are 9, 9, 9 and 9, and half the standing cost is one, so the history scores 36.

Slide the whole thing. Add five to every reading and the list reads 5, 8, 11, 14, 17. Every step is 3, every square is 9, the total is 36 again, and nothing in the arithmetic noticed. That indifference is the laws not caring where you are, on a list of five numbers you can check with a pencil, and chapter fifteen met the same thing in the wiring: shifting all the readings together is the one direction repair leaves exactly where it found it, and the one direction no observer inside the arrangement can read.

Chapter thirty's recipe converts an indifference of that kind into a number. Take one step's contribution and ask how sensitive it is to how fast the reading is changing across that step. A square's sensitivity is twice the step, since widening a square by a hair lays a fresh strip along two of its sides. The half in front of the square cancels that two and the standing cost multiplies what is left, so the answer is the standing cost times the step. Then weight it by how far the symmetry moves the reading. A slide moves every reading by the same amount, so the weight is one everywhere, and with one reading per event there is nothing to add up. Every step of the history therefore carries a number of its own. On a pattern costing two it is twice the step.

On 0, 3, 6, 9, 12 the segments carry 6, 6, 6 and 6.

Move one reading and watch what breaks. Change the second reading from 3 to 4 and leave the rest alone: 0, 4, 6, 9, 12. The steps are 4, 2, 3 and 3, the squares are 16, 4, 9 and 9, and the total is 38. The segments carry 8, 4, 6 and 6.

Two things went wrong at the same place. The score went up by two. The segment numbers stopped agreeing across the junction at the moved reading, where an 8 comes in and a 4 goes out. One move cures both: put the reading back at 3, which is the midpoint of the 0 and the 6 on either side of it.

The midpoint is chapter nine's repair, arriving from a direction chapter nine never looked in. A history that repair has nothing to do to is one whose every interior reading sits at the midpoint of its two neighbors. That is exactly the condition under which the number carried by the segment coming in equals the number carried by the segment going out. Where the two disagree, a repair is owed, and it is owed at precisely the record where they disagree.

So the number is the same on every segment of an undisturbed chain. Whatever one segment hands over the next one receives, no segment in the middle keeps a share, and the chain has nowhere to put any that went missing. A quantity that behaves like that along a chain is a **current**. This one is four numbers on four segments, computed from a list of five readings by taking each difference and multiplying it by the standing cost.

A constant total says less than that. A total can hold while the quantity behind it disappears at one end of the graph and turns up at the other. An audit of the total would find nothing to object to. The chain rules that out by saying where the quantity was in between. It crossed every junction on the route, in order, one segment at a time. Any place where the number changes is a place with a repair owed at it. Chapter nine's bound is the same graph-local count run the other way: read-from dependence advances through declared seams. Interpreting that bound as physical propagation needs the causal, ruler and clock attachments.

On the declared physical-action branch, the number is **momentum**. Chapter five named it as what you get from the laws not caring where you are, chapter thirty supplied the machine that turns the one into the other, and this chain is the machine running on an object small enough to check by hand. Chapter eighteen listed the same quantity as the second, third and fourth of a region's four numbers, the drift of its records across the seam network along the rank-three carrier. The physical identification and mass-shell relation consume the attachments named at the start of this chapter.

Two patterns meeting have one chain between them at the seam where they touch. The segment on that seam carries one number. What leaves the ball arrives at the plate because the same segment is the last one of the ball's chain and the first one of the plate's, and a segment carries one number rather than two. Robins' ribbon is a reading of that number, taken at the far end of the plate's chain, some fraction of a second after the wood stopped splintering.

31.2 Why a heavy thing is hard to push

Chapter eighteen built mass out of the heap of water John Scott Russell chased along the canal: a bounded set of records that comes back after every round of repair, because flattening it would open wider disagreements around its edge than it settles in the middle. Repair never finishes with an arrangement like that. The rate it goes on demanding while the records stay put is the standing cost. A balance reads that cost off as a mass, the cost divided by the square of the speed of light, a conversion that has stood at exactly 89,875,517,873,681,764 joules to the kilogram since the meter was redefined in 1983 and that has no error bar on it anywhere, because nobody measured it: somebody squared 299,792,458.

Such an arrangement holds its shape and not its contents. The records carrying it at one event are not the records carrying it at the next. It survives by being re-established out of whatever is there, event after event, across every seam on its boundary.

So ask what a change of drift costs. Redirecting the pattern means re-establishing it in a different place relative to its neighbors, so every seam on the boundary has to be renegotiated instead of reconfirmed. The seams that have to be renegotiated are the same seams, and the work each one takes is the same work, that holding the pattern in place demands at every event anyway. A pattern that costs twice as much to hold costs twice as much to redirect, because it is one bill with two headings. That is **inertia**, and explains a fact that otherwise has to be assumed: the price of a given change of drift is proportional to the standing cost and to nothing else about the pattern.

The proportionality is exact. A measurement can contradict it. If the price of redirecting a pattern depended on anything besides what the pattern costs to hold, then two lumps carrying the same standing cost and different insides would need different pushes to change their drift by the same amount, and a balance that put them at the same weight would be reporting on a different quantity from the one a push reports on. Everything anybody has ever weighed has been weighed on the assumption that those are one quantity.

Inertia settles the undisturbed case too. With nothing pulling at the seams, every interior reading sits at its midpoint, the segment numbers agree the whole way along, and the advance per event is whatever it was at the start. A pattern with nothing to repair goes on going. The going costs nothing to maintain, because the maintenance is the standing cost and that is paid whether the pattern travels or sits.

The standing cost turns up in one other place. Chapter twenty-four's field equation takes what a region's records demand in the way of work as its source, and hands back the shape of the neighborhood around it. So the number that fixes how much shape a lump makes is the number that fixes what it costs to redirect the lump. The mass that resists a push and the mass that responds to gravity are one entry in one ledger, read twice.

One of those readings is a price and the other is a source, they come off the same entry, and their equality needs no principle to keep it upright. It has one anyway, called the **equivalence principle**. The two hollow cylinders of chapter twenty-four, one of a titanium alloy and one of a platinum alloy, went into orbit to make the two readings disagree, and across five months of usable free-fall data they failed to disagree by as much as a part in a thousand million million.

31.3 Eleven, three, one

Chapter twenty-five put one unit of source on one port of the twelve-port wiring and solved for what the other eleven read. Over a common denominator of a hundred and eighty the twelve values are thirty-five at the source, two at each of the five ports one seam out, minus seven at each of the five ports two seams out, and minus ten at the port directly opposite.

A pattern sitting on the wiring cannot respond to those values. Add the same amount to all twelve and every seam sees what it saw before, so the readings are pinned only up to a common shift. A quantity pinned only up to a shift has nothing in it to push with.

What survives the shift is the drop from one port to the port next door. Over sixty rather than a hundred and eighty, those drops are eleven from the source to the first ring, three from the first ring to the second, and one from the second to the far port. Steep near the source, gentle far from it.

Stand on the source port and step outward and a unit of charge drops eleven sixtieths on the first hop, three on the second, one on the third. Stand on the far port and step inward and it climbs the same three amounts in the reverse order, a sixtieth, then three, then eleven, and the climb is the part somebody has to pay for. A port in the middle of that sits on five seams at once, and each seam has its own drop across it: a first-ring port reads a rise of eleven sixtieths toward the source, a fall of three sixtieths on each of the two seams leading outward, and nothing at all on the two that join it to the rest of its own ring.

The fall per hop is the **gradient** of the potential. The force on a unit of charge is the gradient across the seam it is sitting on. A force is a difference between what one port reads and what its neighbor reads. The difference is the renegotiation the next event owes.

31.4 The size of a shell

Follow the outflow rather than the readings.

Five seams leave the source port, one to each member of the first ring, and each carries the drop across it, eleven sixtieths. Five elevens is fifty-five, so eleven twelfths of a unit crosses the first surface you can draw around the source.

Ten seams join the first ring to the second, two from each of the five, and each carries three sixtieths. Ten threes is thirty, so a half crosses the second surface.

Five seams join the second ring to the far port, each carrying one sixtieth, so a twelfth crosses the third.

The seams inside a ring carry nothing whatever, since both ends of them read the same number. The entire outflow therefore crosses the seams that lead outward, and eleven twelfths, a half and a twelfth are the whole of it.

Check them against what is enclosed. Chapter twenty-five's source arrived with its own balance, a twelfth subtracted at every port including its own, so the source holds eleven twelfths and every other port holds minus a twelfth. Eleven twelfths is what crosses the first surface. Eleven twelfths less the five twelfths held by the first ring is a half, which is what crosses the second. A half less another five twelfths is a twelfth, which is what crosses the third, and the far port's own twelfth brings the total to nothing.

The amount crossing a closed surface is the **flux** through it. The flux is fixed by what is inside and takes no interest at all in which surface you drew. Nothing in that depends on the influence being an electric one. Any quantity whose total outflow is fixed by its source and which has to cross the seams between here and there obeys the same count, which is why two forces with nothing else in common carry the same exponent, and why the same exponent was on the books for gravity and for electricity long before anybody could say what either of them was.

Distance enters only as a count. The strength on one seam is the flux divided by the number of seams in the shell it crosses. The flux is the same across every shell, so the strength falls off exactly as fast as the shells grow and nothing else about the arrangement enters.

How fast a shell grows is fixed by how many directions there are to grow in. A container twice as wide in every direction holds eight times as much, because there are three directions available to be twice as wide in. A ball's content grows as the cube of its radius when the carrier has three independent directions in it, and as the fourth power when it has four. A shell is what you get by growing the ball's radius by a hair, so it grows one power more slowly than the ball it bounds: three directions in the ball, two in the shell, and the strength falls as the square of the distance.

Check that on two shells. The twelve ports run out of room at three hops, the far port being the end of the walk, so take the count on a seam network that keeps going. Stand two hops from the source and count the seams in the shell around you, then stand four hops out and count again. The far shell is twice as wide in each of the two directions a shell has when the carrier has three, so it holds four times as many seams. The flux crossing the two is the same number, so each far seam carries a quarter of what a near seam carries. Hand the carrier a fourth direction and the shell picks up a third direction of its own, the far shell holds eight seams for every one the near shell holds, and each of them carries an eighth.

The count runs on two numbers. Both are shorter as letters. Let r be how many hops out from the source you have gone, and d the number of independent directions the carrier gives anything to spread into. The shell at r has d minus one directions in it, its seam count grows as r raised to that power, and has the same flux to share out however far out it sits.

$$\text{strength} \;\propto\; \frac{1}{r^{\,d-1}}$$

The curly sign is a proportionality, which holds the two sides in step while the source's flux sets the scale, and everything distance does to a force is sitting in that exponent.

Put d = 3 in. Three minus one is two, the strength falls as one over r squared, and the quarter you counted between two hops and four hops comes back out. Put d = 4 in. Four minus one is three, the strength falls as one over r cubed, and the same step outward costs an eighth. Nothing in either substitution asked what the source was made of, or whether the influence crossing the seams was gravity or charge.

The exponent is the number of directions in the carrier, less one.

Chapter fifteen fixes the carrier rank rather than the physical dimension. Twelve ports leave twelve shadows in the band that outlives every other band, all 144 comparisons between those shadows come out as 1, minus 1, or one over the square root of five with a sign in front, and the count of independent rows in that table is three. The inverse-square exponent follows only when a physical realization also supplies three-dimensional shell growth, the source law and equidistribution. The finite graph result is a precursor to that conditional law.

Chapter twenty-five put that exponent at two out to the fifteenth decimal place. Edwin Williams, James Faller and Henry Hill measured it at Wesleyan University in 1971, on an apparatus built out of five concentric icosahedral shells, twelve corners and thirty edges apiece. Four megahertz at ten kilovolts peak to peak went across the outer two, and a battery-powered amplifier sitting inside the innermost shell listened for any trace of that signal appearing across the inner two. It heard nothing. A carrier whose surviving band held four directions instead of three would put the exponent at three rather than two. An instrument that sensitive could not have missed a whole integer.

31.5 Mass times acceleration

A pattern's advance per event is its step. The rate at which that advance changes is the difference between one step and the next, and a chain of readings shows it directly: the change in the step across a record is twice the amount by which that record falls short of the midpoint of its two neighbors. A record sitting at the midpoint has equal steps on both sides and an unchanging advance. A record sitting one unit above the midpoint leaves with a step two units shorter than the one it arrived on.

That is **acceleration**, the rate at which a pattern's advance changes relative to its neighbors. On the chain it is a difference of differences, with no limit taken anywhere and nothing smooth required.

The history with the moved reading has one in it. In 0, 4, 6, 9, 12 the second reading sits one unit above the midpoint of the 0 and the 6 on either side of it, and its steps run 4 and then 2. The step drops by two across that record and the segment number drops by four, from the 8 coming in to the 4 going out, which is the standing cost of two multiplied by the change in the step. That record is accelerating, and also the record where the segment numbers disagreed and the score could be brought down. A pattern accelerates at exactly the records where its history has a repair outstanding.

Put every piece on one chain. A pattern with a standing cost, crossing a wiring that has a potential on it, scores each step twice over: half the standing cost times the square of the step, less what the potential reads at the port the record is sitting on. Ask, at one interior record, that the score not move when the record is nudged, which is chapter thirty's condition for the history the world runs. One line of arithmetic comes back. The standing cost, multiplied by the change in the step across that record, equals the fall of the potential across that hop.

Read the line in the words its pieces arrived with. On the left, a standing cost multiplied by the rate an advance is changing: mass times acceleration. The same left-hand side is the change in the segment number from the incoming segment to the outgoing one, which is a change of momentum across one event. On the right, the gradient of chapter twenty-five's potential: force.

Written with one letter for each piece, it is the line that opens every mechanics course.

$$F = ma$$

F is the drop in the potential across one hop, m is the standing cost of holding the pattern together, and a is the rate at which the pattern's advance changes relative to its neighbors. Nothing in it was measured in a laboratory and typed in. The mass is what a self-re-asserting pattern demands while its records go nowhere, the acceleration is a difference of differences along a chain of readings, the force is the fall of chapter twenty-five's exact potential across one hop, and the equals sign is chapter thirty's condition that the score not move when a record is nudged.

The left-hand side is a rate. A rate here is counted against repair events, at whatever pace a region's own state sets them, which was chapter twenty-one's answer to what a clock is made of. Every textbook writes that side as a change of momentum per unit of time, and not one of them says what the time is made of.

Divide both sides by the standing cost. It goes from the left, and when the potential is gravity's it goes from the right as well, because the entry that priced the redirection is the entry that put the shape there. What is left is an acceleration with nothing about the pattern in it, which is why a lump of anything at a given place changes its advance at the same rate as a lump of anything else.

Robins fired one ball and it crossed the room by one route. The plate swung once, the ribbon came out at one length, and every part of that has an account behind it: the number handed over at the wood, the price of turning the iron, the fall the ball rode down on the way. One thing about it has no account. The ball took a path, and it took one of them. Nothing that has been built says why it did not take several.

Chapter 32

Why Was Every Open Problem the Same Problem?

In September 2006 Lee Smolin published *The Trouble with Physics*. The most useful page in it sits near the front, before the argument starts. He sets out five problems and calls them the five great problems in theoretical physics. Combine general relativity and quantum theory into a single theory of nature. Resolve the trouble in the foundations of quantum mechanics. Find out whether the particles and the forces are aspects of one thing. Explain how the free constants of the Standard Model came to have the values they have. Account for dark matter and dark energy, or work out how gravity is modified at galactic scales instead.

A physicist can spend forty years on any one of those five and never need to open the literature of the other four. They have separate conferences, separate journals, separate graduate courses and separate senior figures who do not attend one another's talks. Somebody working on the measurement problem and somebody working on the cosmological constant share about as much technical vocabulary as a cardiologist and a bridge engineer.

OPH approaches these questions through one observer architecture. Records, local agreement and bounded access recur in its constructions of quantum probabilities, geometry and matter. Their common starting point creates opportunities to connect the problems, while each physical conclusion retains its own assumptions.

Chapter one's comparisons showed several ways to reconstruct parts of the physical description. The question here is what the observer route contributes. Removing an assumption is useful only when a replacement explains the relevant physics. A successful solution must reach quantities that somebody can measure.

32.1 Ten words Bell wanted banned

In August 1990 John Bell published eight pages in *Physics World* listing ten words that, he wrote, have no place in a formulation with any pretension to physical precision. System. Apparatus. Environment. Microscopic.

Macroscopic. Reversible. Irreversible. Observable. Information. Measurement. The worst of all, he added, is the last one.

His target was a joint. The theory as John von Neumann set it out in 1932 contains two kinds of change. One is smooth and deterministic and runs whenever nobody is looking. The other is a jump: it happens when a measurement is made, it produces one outcome out of several, and the odds come from Born's footnote. Nothing in the theory says where the first kind stops and the second begins, and von Neumann showed that the joint slides anywhere along the chain from the instrument through the amplifier to the eye of the physicist without changing a single prediction. Fifty-eight years later Bell was pointing out that a fundamental theory containing a movable part, whose position no experiment fixes, is an unfinished theory.

The assumption under the joint is older than the theory and it sits inside the word Bell most wanted removed. To measure is to find out a value that was there beforehand. To disturb is to spoil a value that was there beforehand. Both take for granted a world stocked with definite values and experiments as encounters with them, and the measurement problem is the question of how one encounter manages to be smooth and a jump at once.

In the finite reconstruction, an observer has an algebra of questions and a commuting record layer. The probability theorem concerns assignments to all quantum effects, including weighted questions, that are normalized and additive whenever the sum is another allowed effect. Under those hypotheses, the assignment is the trace against a unique state in every finite dimension. Additivity only on sharp projectors needs the familiar dimension restriction; agreement on a finite set of observed questions is weaker still.

An instrument connects those probabilities to records. A nonselective readout can discard distinctions in the reduced description while a larger reversible system retains them. The formal representation does not choose a physical instrument or explain which outcome an observer experiences. Those identifications are additional parts of the physical account.

32.2 Loschmidt's objection

In 1876 Josef Loschmidt put an objection to Ludwig Boltzmann that has never been met on its own terms. Take any mechanical process, run it, then reverse every velocity in it. The reversed motion obeys the same equations exactly, so a quantity that rose during the first run falls during the second, and no theorem about time-symmetric mechanics can deliver a time-asymmetric conclusion. Every derivation that appears to do it has smuggled the asymmetry in somewhere.

Where it gets smuggled in is a statement about how everything started: an arrangement of the rare, low-entropy kind, fourteen billion years ago, asserted by people who were not there and defended on the ground that nothing else does the job. David Albert named it the past hypothesis in 2000.

The assumption sits one line above the fix. The elementary dynamics is blind to direction, so a direction has to be imported from outside the dynamics, and the only place left to import one from is the boundary.

The finite repair construction makes its direction assumptions explicit. A stochastic kernel preserves a supplied reference distribution. With strictly positive initial, reference and output laws, the average exponential of minus entropy production is one, and mean entropy production equals the fall in relative entropy. Detailed balance supplies the stronger relation between forward and reverse transition weights. These are statements about a declared probability model.

A commit can retain its inputs in a reversible record, so commits alone do not evade Loschmidt's objection. Physical irreversibility requires the channel, boundary conditions and discarded information to be identified. Thermal heat bounds additionally require the reservoir and energy assumptions of chapter twenty-two. The finite inequality connects to the physical arrow only through those bridges.

32.3 A hundred and twenty orders of magnitude

Add up the zero-point energies of the fields of empty space, cut the sum off at the shortest length anybody takes seriously, and compare the total against what the sky reports, which is a cosmological constant of about 1.1 times ten to the minus fifty-second per square meter. The two disagree by roughly a hundred and twenty orders of magnitude, the largest gap between a calculation and a measurement in the history of the subject.

The assumption is that empty space carries an energy density, that the density sits in a spacetime which was there first, and that gravity weighs it the way a balance weighs a sack of flour.

The local null-probe route uses light rays. A Lorentz-invariant vacuum stress is a number times the metric, and the metric evaluated twice on a light-ray direction is zero. That route is therefore blind to the metric-proportional term. It derives the local Einstein relation only up to one cosmological constant; it does not set the constant to zero or determine its value.

The proposed global branch supplies a candidate for the missing term. It identifies correctable public-record capacity with the entropy of one realized de Sitter horizon and imports the physical area and scale calibration. Under those assumptions the candidate count is 3.3001 times ten to the hundred and twenty-second, against a weighted expansion coordinate of 3.3129 times ten to the hundred and twenty-second. The finite record count alone fixes no cosmological constant.

32.4 An ensemble of one

In 1953 Fred Hoyle turned up at the Kellogg Radiation Laboratory at Caltech and told the nuclear physicists working there that the carbon-12 nucleus had to have an excited state at around 7.68 million electron-volts, on the ground that stars manufacture carbon and have no other way of doing it. Ward Whaling, who had helped build the laboratory's magnetic spectrometer, put his group on it: a nitrogen-14 target, a beam of deuterons, and a scan across the alpha particles that came off. The state is there, at 7.68 million electron-volts give or take 0.03. The measurement was published in *Physical Review* the same year.

Shift that level by a fraction of a percent and the reaction that builds carbon inside stars stops working, which makes that level the first exhibit in every account of a universe tuned for its occupants. Take any of the twenty-six measured numbers of the Standard Model, change it on paper and work out what the resulting world does, and for several of them the answer is a world with no chemistry in it. The constants of this one look, from that angle, like settings somebody dialed in.

An argument of that shape needs both a collection of candidates and a probability measure on it. String compactifications supply examples with different particle spectra and parameters. Counting many candidates alone does not determine the probability of a hospitable world; that depends on their physical content and the weights assigned to them.

The numerical proposal reads two quantities twice, once from the finite structure and once through a supplied physical observer map. If each pair is proved to refer to the same carrier, consistency gives an equation with the quantity on both sides. The declared map is a contraction, so it has one fixed point in its feasible set. Its coupling coordinate is 137.035660 against the measured 137.035999. This proves uniqueness inside the named map, not that the source-derived causal poset selects every physical attachment or excludes every other architecture.

32.5 Two parts in ten to the seventeenth

The Higgs boson was measured at 125.20 billion electron-volts, with an uncertainty of 0.11. The scale at which gravity becomes as strong as everything else sits around 1.2 times ten to the nineteenth billion electron-volts, seventeen orders of magnitude higher up. In a theory where a scalar's mass collects contributions from every scale in between, that mass belongs near the top of the range and it sits at the bottom of it. The standard repair was a symmetry pairing every particle with a partner whose contributions cancel against its own. The Large Hadron Collider spent the whole of its second run looking. The floor under the partner that theory assigns to the gluon sits above 2.3 trillion electron-volts.

The assumption is the top of the range. A hierarchy problem is a comparison. A comparison needs two scales, one of which was written into the theory by hand as the place where the description gives out.

The hierarchy branch begins with a named physical-scale identification: a screen cell is assigned Newton's area, 2.61 times ten to the minus seventy square meters. The finite screen count does not derive that meter-valued calibration. Conditional on it and on the branch's coupling and matching premises, the weak scale is obtained from the same solve. The ratio between the two is the exponential of minus two pi divided by four times the coupling at a cell, and that coupling is 0.0411, one part in 24.32. Work the exponential and out comes two parts in ten to the seventeenth, which is the size of the scalar's vacuum vector, 246.22 billion electron-volts, measured against the calibrated cell scale.

A gap of seventeen orders of magnitude is what one number near a twenty-fourth does when it lands in an exponent. Within the calibrated branch, the exponential relation does not introduce a second fitted hierarchy parameter at this step; the physical-scale and matching inputs remain explicit.

32.6 The groups that died in 1894

In 1974 Howard Georgi and Sheldon Glashow proposed that the eight, the three and the one of the Standard Model sit inside a single simple symmetry. The thirds of quark charge fell into place, the three force strengths appeared to converge at high energy, and the proton acquired a finite lifetime. Fifty thousand tons of ultrapure water under a mountain in Japan have been watching for the decay since April 1996 and have seen nothing of the right shape, which puts the floor under the lifetime at 2.4 times ten to the thirty-fourth years.

The assumption is in the choosing. A symmetry group gets picked by the person writing the theory and imposed on the world from outside, which is a move available only where there is an outside to impose it from.

The clause that removes it arrived with two firms keeping books in different currencies. They need an exchange rate, there is no market outside the room to look one up in, so the rate has to be computed out of what the two of them hold between them. Applied to the twelve ports of a patch, the same clause says that every relabeling of the readings has to be a move the patch can perform on itself, since no venue anywhere holds a preferred labeling of twelve ports. The relabelings with any standing are the ones the dials produce.

That requirement decides the group. A central move commutes with every relabeling, so a relabeling and its undoing slide together and cancel and the move comes back untouched, which puts the center inside the part of the twelve readings that all sixty rotations of the wiring leave alone. Those sixty rotations carry every port to every other port, so the surviving part is one number repeated twelve times: one dial, and at most one. A simple group has a center zero dials wide, by the definition of simple. Twelve dials come apart as one plus eleven, eleven splits as three plus eight and in no other way, and a simple group has nowhere to stand.

What survives is the arrangement nobody would have picked for its looks. One dial that no rotation touches, a block of three and a block of eight, which are electromagnetism, the weak force and the strong force, in exactly the proportions the Standard Model was written down with because that is what the detectors had reported. A wiring diagram hands the product over, and it hands it over before anybody asks what the three of them are supposed to be acting on.

So the beautiful groups died of a clause about exchange rates. The catalogue of sizes a simple group is permitted to have was finished by Wilhelm Killing in four papers between 1888 and 1890 and made rigorous in Élie Cartan's thesis of 1894, twenty-six years before the proton had a name.

32.7 A proof one step long

On 27 November 1997 Juan Maldacena posted a paper showing that a theory of gravity in a particular curved spacetime is the same theory as a field theory without gravity living on that spacetime's boundary at infinity. Everything in the interior has a counterpart on the surface. The interior is the surface's information reorganized into an extra dimension. In 2006 Shinsei Ryu and Tadashi Takayanagi supplied the geometric dictionary entry: the entanglement entropy of a region of that boundary is the area of the smallest interior surface hanging on the region's edge, divided by four times Newton's constant. It is the most productive thing anybody has done with the idea that a region's memory scales with its surface.

The assumption is that the inside of a region is fixed by what is written on its edge. Stating that carefully takes three pieces: a region with an inside, a summary that lives on the edge, and the claim that two insides carrying the same summary are the same inside. Everything depends on which summary. The first one anybody reaches for is the edge itself, because the edge is where the information was supposed to be. Set it up that way and the line going in reads: two insides with the same edge are the same inside. The line coming out reads: two insides with the same edge are the same inside. A proof checker handed both lines gets from one to the other in a single step. The step is the one that says a thing equals itself.

The claim has content when the summary is built out of something other than the edge and shown, on its own ground, to fix the inside. Then a pair of insides agreeing along the edge and differing behind it would break the claim. Whether the network holds such a pair is settled by its seams.

Here the edge is a collar. A band of patches one patch thick, with an interior on either side of it, and both interiors keeping records, which makes both of them observers. The surface where the information sits is inside the network, between two parties comparing notes across it.

Two theorems come with that arrangement. Finitely many patches, finitely many readings each, and a local check on every seam demanding that its two readings answer each other under the dictionary sitting there: a setting of all the patches meeting every check at once is a legal word, the legal words are exactly the globally consistent states, and an object of that shape is a constraint code, of the same family as the one Richard Hamming built in 1950 out of four data bits and three checks. The second theorem is a negative one: a minimum-cut formula does not come with the code. A list of checks fixes which configurations are legal and says nothing whatever about how far apart two legal configurations sit, or where the cheapest bottleneck through the wiring runs, and both of those depend on which port is joined to which port. An area formula for the entropy of a region is a computation on one particular connector, one seam at a time.

A region's memory is set by its surface because everything anybody can learn about the region arrives through the ports on that surface, which chapter thirteen got out of a wiring diagram with no gravity in the argument. What drops out of the account is the outside boundary. The far side of any collar in this world is another interior with somebody in it, keeping its own records and repairing its own seams.

32.8 Eight short answers

The other famous questions went the same way.

Why are there three families of matter rather than one or seven? Because a physical difference between matter and antimatter needs at least three, asymptotic freedom permits at most five and three eighths, and the twenty triangular faces of the wiring carry a threefold turn whose orbit supplies exactly three.

Why do the charges come in sixths? Because six combinations of gauge moves do nothing whatever to any piece of matter, so they get bundled in with the move that does nothing at all, and the bundle holds only if every hypercharge advances in whole multiples of one sixth. That is the lock that leaves hydrogen neutral to better than one part in a billion billion.

Why does the proton refuse to decay the grand-unified way? Because the carrier that route needs, one direction coupling color to weak charge at once, is missing from a twelve-dimensional list that splits into eight, three and one, and a missing direction cannot be made small or slow: that route is closed exactly.

Why is the fine-structure constant near one part in 137? A declared screen map has one fixed point, with diagnostic coupling 137.035660 against the measured 137.035999. Connecting its two readings to one physical carrier and independently determining the transport are necessary for a physical comparison.

What does the circulant model say about Koide's relation? In its positive chamber, equal weights of the two response sectors give two thirds. With measured electron and muon masses as inputs, this balance condition gives a tau window seventy-two electron-volts wide, centered at 1776.969027 million electron-volts against a measured 1776.93. Positivity alone does not select the balance.

Why does light have two polarizations? Because the central band carries four components, removing the redundancy takes two of them away, and what is left is transverse and two wide.

Why does the position carrier have three directions? Because the twelve readings of a patch split into blocks that fade at different rates under repair, and the slowest block, holding 0.9539 of what it carries per pass against nine tenths and 0.8794 for the others, has rank three. Its interpretation as physical space remains conditional.

Why does the Lorentz module have a limiting cone? Because its determinant has one positive and three negative directions and the celestial sphere is its projective null boundary. Identifying that cone with physical signal propagation, and calibrating its speed, require the separate causal, ruler and clock receipts.

On the proposed dark-sector branch, missing-mass phenomenology is represented by a repair charge on the collars rather than by an additional particle. The charge persists only if later accepted commits do not write the support needed to clear it; the map from that conditional record defect to a galactic profile, geometry and acceleration law remains a physical premise.

32.9 The empty column

The observer constructions supply a set of conditional answers. Each row states both the mathematical contribution and the physical inputs it consumes.

The problem	Observer construction	Physical requirement
Measurement	Effect-additive probabilities and a commuting record layer	An instrument and an outcome interpretation

The problem	Observer construction	Physical requirement
The arrow of time	Relative-entropy contraction for a reference-preserving channel	Dynamics, boundary conditions and a physical clock
The cosmological constant	Null data leave one metric term free; a horizon model assigns capacity	A common entropy, area and scale identification
Fine-tuning	Unique fixed points of declared maps	Independent map selection and physical readouts
The hierarchy	An exponential relation on a calibrated branch	Physical scale, coupling and matching inputs
Grand unification	Response assumptions exclude a simple local gauge Lie algebra	Realized gauge and matter interfaces
Holographic duality	Finite overlaps define a constraint code	Geometry and an entropy law for the physical connector

The connection between the rows is the bounded observer architecture. Twelve ports and thirty seams supply a common finite construction. Physical carrier, scale, stress, clock and horizon maps identify what its mathematical objects measure. The strength of the proposed unification is tested by whether those maps belong to one physical realization.

There is a test that separates the outcomes. Ask what a solution would have to contain. The finite construction can remove a misplaced mechanism or replace a free parameter with a conditional fixed-point equation. A physical result additionally needs its named bridge to instruments. For the cosmological constant, the null argument explains why local light-ray data leave the metric term undetermined. The horizon-capacity proposal supplies a value only if its record, entropy, area and scale identifications hold on one physical branch.

Which leaves the picture everybody starts with, the one with a container, a clock running at one rate for everybody, and objects sitting at positions inside it. That picture is the same assumption in domestic form, accurate to more decimal places than any instrument in a house can resolve. Somebody assumed a container and a clock and got the tides out of it, and the precession of the equinoxes, and the mass of the Sun worked out from how long a planet takes to go round it. That is two hundred years of correct answers out of a picture whose opening sentence puts the world inside a box that was never there.

Chapter 33

Why Was Newton Right for Two Hundred Years?

Yaakov Fein, Markus Arndt and six colleagues sent molecules down a two-meter tube in a laboratory at the University of Vienna in 2019 and collected what came out the far end. The molecules were tailored oligoporphyrins, up to two thousand atoms apiece, weighing more than twenty-five thousand atomic mass units, which is roughly a small protein. What arrived at the detector was a pattern of light and dark bands. The wavelength that produced it was 53 femtometers, five orders of magnitude smaller than the molecules doing the interfering.

Nothing heavier has ever been caught doing that.

The natural reading is that the bands hold up to some weight and give out above it, and that a kitchen chair misses the cutoff by a margin nobody would bother writing down. The natural reading is wrong. An experiment run fifteen years earlier in the same city says why.

In 2004 Lucia Hackermüller, Klaus Hornberger, Björn Brezger, Anton Zeilinger and Markus Arndt put C70 fullerenes through an interferometer, seventy carbon atoms each, and heated them on the way in with a multi-pass laser. Every photon a molecule absorbed raised its internal temperature by about 140 kelvin. The beam entered the interferometer at temperatures approaching three thousand. A molecule that hot glows. The bands weakened as the molecules got hotter, and the weakening matched, quantity for quantity, the rate at which the emitted infrared light was carrying off which route through the gratings the molecule had taken.

Nobody read that light. There was no detector aimed at it, no experimenter waiting on it, and no measurement in any sense a laboratory notebook would recognize. The molecule wrote its own position into the room, in a handful of infrared photons. The writing was enough. Weight was never the variable. The pattern ended when a record got written.

The chair in your kitchen is in the same business and has never once come out in bands. A person can go a whole life in a kitchen without wondering why. Why does a chair have a definite place, when a molecule of two thousand atoms does not have to? Why does everything any instrument has ever written down come out as ordinary numbers, obeying ordinary arithmetic, with none of the refusals that run through everything underneath? Why does space look smooth? And why did a description of space and time that put both of them in by hand, unexplained, run for two hundred and eighteen years with almost nothing to show against it?

33.1 Sixteen copies

Take a fact worth sixteen yes-or-no questions: enough to fix where something is to one part in sixty-five thousand, along one direction. Write it into one place and it is a record that a determined liar could rewrite by guessing sixteen bits, which comes off once in 65,536 tries.

Write it into sixteen places instead.

The liar has to produce all sixteen copies without holding the original, because a single copy that disagrees with its neighbors exposes the whole attempt. Take 65,536, the odds against faking one copy, and multiply it by 65,536 fifteen more times, once for every copy after the first. That is two hundred and fifty-six bits of blind guessing, and comes off about once in 1.16 times ten to the seventy-seventh.

Look at what the sixteenth copy contributed to the world's knowledge. Nothing. It says what the first copy says, to the bit. The world learned sixteen bits when the first copy was written, and it holds sixteen bits after all sixteen exist, and the entire effect of the other fifteen is on what it would cost to lie.

A chair is not held by sixteen copies. Every photon that leaves it carries where it was, every air molecule that comes off its surface leaves with the same information, and the copies are being made continuously, in numbers that make sixteen look like a rounding error. That copying is chapter nineteen's pinching, performed by the room rather than by an experimenter, and decoherence is its name.

So definiteness belongs to the record rather than to the chair. There is a record of where the chair is, written into every photon that has left it since you last looked at it, no two copies disagreeing, and the odds on any alternative pinned so far below one part in ten to the seventy-seventh that no experiment will ever separate them from zero. Definiteness is what that redundancy looks like from inside, to an observer who is one more copy.

The public half comes free with it. Chapter four's overlap rule says that where two observers can compare, they have to agree, and the redundancy is what makes the agreement cheap: you and everybody else in the kitchen are reading different copies of the same sixteen bits, so nobody has to go and check with anybody. Four people walk into a room and the chair is in the same place for all four, which is what the word objective is describing, and the reason it holds is that all four are downstream of a record copied more times before they arrived than any of them could count in a lifetime.

Chapter eleven has the name for it. A settled state that the record fixes, rather than the path taken to it, is an observation-determined normal form. At kitchen resolution the record is the photons and the settled state is where the chair is. The room hands it to all four of them without any of them having seen a single repair.

It is also being paid for. Chapter twenty-two put a price on a committed record: at three hundred kelvin, 2.87 divided by ten to the twenty-first joules for every yes-or-no question the commit settles, expelled into the room as heat. The chair sits at a definite place because the room is buying it a definite place, bit by bit, without pause, at a rate your kitchen absorbs without anybody noticing the bill.

33.2 This or that, here or there

Chapter nineteen found where records live. The questions that commute with everything else in an observer's algebra collect into its center. A record has to be there, because a record is exactly a question that nothing else the observer can ask will disturb.

Take an inventory of that center once the copying has done its work. A finite list of alternatives. This one or that one, here or there, one or two. Every pair of them can be asked together, in either order, with no cost to either answer. Probabilities that add the way probabilities added before 1925.

That is an ordinary classical event algebra, the kind George Boole set out in 1854 in a book about the laws of thought, where every question has an answer, the answers combine with and, or and not, and nothing depends on the order you ask in. It is the whole of what any instrument can print. A kitchen scale showing 1.24 kilograms is one alternative from a finite list, written into a place where nothing else the observer can ask will move it, and the same goes for a pointer, a photographic plate, a click in a counter and the sentence a physicist writes in a notebook afterward.

Everything an observer can write down was always going to look like ordinary numbers, because the writing is what makes it ordinary. Physics before 1925 was an accurate theory of the center, put together by people whose instruments could not reach a question living anywhere else.

33.3 The traffic runs one way

A physicist who works only with pointer readings is working inside the center, and everything that person writes down is true. Whether it is the whole of what there is gets settled by the maps between the two layers. Those maps are lopsided.

Upward, every classical record is one of the operators. Take a reading off your kitchen scale and find the question in the algebra that asks it: add two readings and the questions add, multiply them and the questions multiply, swap yes for no and the question flips with it. A record about one region lands inside that region's own algebra, carrying the odds it had. Nothing is lost and nothing is approximated. Records are the operators that quarrel with nothing.

Downward there is no such map. The reason is arithmetic rather than difficulty.

A classical readout assigns a number to a question. Suppose some rule did that for every question in one of chapter nineteen's blocks, thirteen alternatives across, and suppose it added the way the questions add and multiplied the way they multiply. Numbers multiply the same way round in either order, so the rule would have to hand the same number to a product taken one way as to the product taken the other, and hand their difference zero. Those differences, taken over every pair in the block, fill out everything in it except the overall scale. A rule that sends all of them to zero has one option left: report the sum of the diagonal, divided by thirteen, which is an average.

Then take one question inside the block and watch the average fail. The question says yes to one direction out of thirteen, so its average is one thirteenth. Ask it twice and you have asked it once, which means the square of the question is the question, so the average of the square is one thirteenth as well. Square one thirteenth yourself. A rule that respected products would have to hand the square that number, one part in a hundred and sixty-nine, and the average hands it one part in thirteen: same block, same question, and a factor of thirteen between the two answers.

So there is no structure-preserving map from the operator layer down to the classical one, in any block bigger than one alternative across, and the obstruction is blunt: a readout has to assign a number, and a thirteen-by-thirteen block of operators has no numbers in it to be assigned. What exists downward is the average. It keeps every total right, keeps every probability between zero and one, carries each record across untouched, and mangles multiplication.

The everyday world is therefore one exact copy plus one lossy average, and multiplication is what the average loses. That settles which of the two layers is holding the other up. A classical account lifts into the operator account entire, products included, and lands inside it as a working part. Send the traffic the other way and the best map available gets the product of a question with itself wrong. No amount of care in the averaging repairs it, because the failure sits in what a number can do rather than in how the average was taken. A classical world could contain no operator layer at all. An operator layer contains a classical world exactly, in its center, and hands it out to anybody who writes anything down.

33.4 A world that cannot flow

There is a second obstruction, the one that decides whether a classical world could ever have been the ground floor.

A record takes one of finitely many values. A change of the record layer that preserves its structure has to send alternatives to alternatives, which is to say it shuffles the labels. A finite list has only finitely many shuffles. So consider a continuous family of them, one shuffle for every real number, starting from the shuffle that does nothing: the kind of family that any dynamics, any symmetry with a dial on it, any smooth motion whatsoever would have to supply.

Such a family cannot move. To carry a label off itself it would have to pass through values in between, and between two alternatives there is nothing: a record whose value is partway from heads to tails is not a record. The parameter runs over the real numbers, which come in one connected piece, and the labels sit apart from one another with gaps that no continuous path crosses. A path that starts at the label it started at stays there.

Every continuous one-parameter group of purely classical public symmetries is the identity. The proof is the connectedness of the real numbers on one side and the gaps between the labels on the other. There is nothing else in it.

Try to write down the rate of change of a record and the same wall arrives from the other direction. A rate of change is a difference divided by a shrinking interval, and needs the quantity to take values arbitrarily close to the one it has. A record takes the value heads, and the values near heads are heads. Every derivative of it is zero, every equation of motion for it says nothing changes, and the only motion the layer supports is a jump from one label to another with no time spent in between.

Chapter twenty-one pulled a clock out of a state and found the flow living on the part of the algebra that refuses to commute, reversible, supplied by whatever the observer holds. The connectedness argument reaches the same place from the classical side. A purely classical world cannot flow. It can be relabeled in steps, and between the steps it does nothing whatever.

Which is why the classical layer is a readout rather than a foundation. Whatever is flowing is flowing underneath, in the part where order matters. A classical dynamics is a sequence of photographs of that flow, taken at a resolution where the photographs join up smoothly enough to differentiate. Newton wrote down a continuous dynamics for positions and momenta and it worked for two centuries. What moved in it was the layer underneath, and not the records anybody wrote down.

33.5 The finest length anybody has measured

Run a hand along your kitchen counter. It is smooth under your fingers, while chapter fifteen built an abstract Euclidean carrier completion out of comparison counts that are not smooth. Identifying that completion with physical space requires its own realization and gluing receipts. Whatever an observer holds is a finite list of whole numbers taken after a finite number of passes. Refinement has no completed final stage. Descriptions get finer and no finite description arrives at its completion.

The instruments say the same thing from the other side. In Louisiana and in Washington State there are two L-shaped machines with arms four kilometers long, built to catch gravitational waves, and what they detect is a change in the length of one arm of about a meter divided by ten to the nineteenth. A proton is about 1.7 of a meter divided by ten to the fifteenth across, so what those machines resolve is under one ten-thousandth of the width of a proton. That is the finest length measurement human beings have made.

It lands at a finite resolution, like every measurement before it. The next instrument will land at a finer one, and the one after that finer again, and the sequence of resolutions has no last term in it, which is what it means for smooth space to be a limit rather than a place. The counter feels smooth because every instrument you could put on it stops short of the limit by less than that instrument can register.

A position fixed to one part in 65,536 takes sixteen yes-or-no questions, each one halving the stretch the answer can be in. Do that along the edge of the counter. Halve a meter ten times and you have cut it by 1,024, which is a thousand for envelope purposes. Do six lots of ten and you are at a meter divided by ten to the eighteenth. Three more halvings supply a factor of eight.

Sixty-three questions therefore place a point on your counter to about a meter divided by ten to the nineteenth, which is the wobble those four-kilometer arms register. Going from where a chair is to the finest length anybody has measured costs four times the questions and nothing else.

Ask a sixty-fourth and what you hold is one bit longer and a whole number. Ask sixty-three more, a second copy of everything between your counter and Louisiana, a longer whole number, finite, exactly as it was at the first question, and with as many questions left to ask as there were before anybody asked one. The smooth counter is where those answers pile up. Every answer anybody holds about it is a finite list of whole numbers.

Calculus works on the world for that reason and no other. A derivative is a limit of differences, so writing one down commits you to differences taken at every scale. The world supplies them down to whatever depth the apparatus reaches and no further. The answers agree because the disagreement is parked below the last decimal place any instrument has produced, which is exactly where an abstraction hides its leak until the day somebody builds a better instrument.

33.6 Twenty-two and a half meters

The *Principia* was published on 5 July 1687 with a scholium near the front saying that absolute, true and mathematical time, of itself and from its own nature, flows equably without regard to anything external. Absolute space got the same treatment in the sentence after, similar and immovable of its own nature, answerable to nothing outside. Chapter one called those two sentences the load-bearing assumptions of the book, put in by hand by a man who had just explained that they were too obvious to need stating.

They ran for two hundred and eighteen years. A wrong assumption at the base of a theory does not usually take two centuries to show itself.

In 1960 Robert Pound and Glen Rebka took the tower of the Jefferson Physical Laboratory at Harvard, 22.5 meters of it, put an iron-57 source near the roof and an absorber in the basement, and watched what happened to 14.4 kilo-electronvolt gamma rays on the way down. The prediction they were testing was that the frequency arriving at the bottom would be higher than the frequency leaving the top by about 2.5 parts in ten to the fifteenth, because a clock in the basement runs slower than a clock on the roof. Over ten days of counting they measured 2.56 parts in ten to the fifteenth, give or take 0.25.

That is the size of Newton's error about absolute time, measured across a building. Two and a half parts in a thousand million million. To see it at all, Pound and Rebka needed a nuclear transition, a technique Rudolf Mössbauer had published two years earlier, and six thermocouples watching the two ends of the shaft, because one degree Celsius of difference between the source and the absorber shifts the frequency by about as much as the whole height of the building does.

Set that against the instrument. The best timekeeper of the eighteenth century was John Harrison's fourth attempt at a sea watch, which sailed from Portsmouth in November 1761 and reached Jamaica on 19 January 1762 having lost 5.1 seconds. Over two months at sea, 5.1 seconds is an error of roughly one part in a million, and the shift Pound and Rebka went after is a few hundred million times smaller than the error in the finest clock of its century. Newton's absolute time was safe from every clock built in the two hundred years after him, and safe by a margin that no amount of care with brass and jewels could have closed.

The sky had the better instruments. It took a century and a half of them to turn up a single discrepancy. Urbain Le Verrier went through the transits of Mercury from 1697 to 1848, subtracted the pull of every other planet, and reported in 1859 that the orbit's closest point was creeping around the sun about 38 arcseconds per century faster than Newton's law allowed. Simon Newcomb redid the subtraction in 1882 and made it 43. Forty-three arcseconds per century, out of 574 that the whole orbit turns through against the fixed stars in the same time. All 574 of them come to a sixth of a degree. That is the one leftover in the solar system no further Newtonian bookkeeping ever absorbed.

The density of records does the same work on the other side of the account. Everything a human body handles is copied into its surroundings so many times over that the odds on any alternative place for it sit below one part in ten to the seventy-seventh and keep falling, so the indefiniteness Newton left out of his description is as invisible from a kitchen as the difference between the clock upstairs and the clock downstairs. Speeds far below the speed of light, and records copied past counting: those are the two conditions a person lives under, and under both of them the corrections sit in the fifteenth decimal place, nine places past anything Newton's contemporaries could read.

The scholium was an excellent local approximation, and Newton wrote it down as a definition of what time is, of itself and from its own nature. A quantity introduced that way has no route back to being corrected, because there is nothing it was ever answerable to. That is chapter one's error in its purest form, committed by the man who was right about everything else for two hundred years.

33.7 What the abstraction is an abstraction of

The prologue put the everyday picture on the table: a three-dimensional container, time running inside it at one rate for everybody, objects at positions in it, physics describing how the objects move. It named that picture a leaky abstraction, in Joel Spolsky's sense of a simplification that holds until the thing underneath does something the simplification has no words for, and listed six places the leaks come through.

Thirty-two chapters later, the everyday container is the commuting, redundant, coarse-grained readout of a finite network of observers repairing their disagreements. Commuting, because the questions that survive being written down are the ones that refuse nothing, and that is the center. Redundant, because a fact written in enough places has odds no experiment can lift off zero or lower off one. Coarse-grained, because the only downward map is an average, and the average discards the order in which questions are asked. And a readout, because the layer it reads has no motion of its own and never did.

Every one of the prologue's six leaks is a place where the reading is taken at a resolution the abstraction has no words for: a cabin that gives no sign of its own motion, an orbiting clock running fast by forty-five microseconds a day and slow by seven because a rate belongs to whoever holds it, a needle whose bands survive until a route gets written down, two laboratories 1.3 kilometers apart with no separate answers to have been carrying, a black hole that cannot destroy the record of what fell into it, and a gravitational entropy ceiling controlled by a surface. The finite observer model supplies a boundary-record count; matching it to physical entropy and area needs the separate screen calibration.

There is nothing wrong with the container picture. It is what the world looks like from inside, at the resolution a body operates at. It is not where the world starts.

Throw a ball across the kitchen. Its path is a pattern following a gradient, re-established at each place in the comparison count; its mass is the standing cost of holding that pattern together against the seams it touches; its position is written into the room by every photon that leaves it; and it lands in your other hand with a definite place and a definite speed, both of them readings taken off a layer that has neither. Newton's account of that throw is accurate to every decimal place your hand can feel. Thirty-two chapters went in underneath it, and not one of them has said what the ball is.

Chapter 34

What Is Anything Made Of?

Antoine Lavoisier paid for his laboratory out of tax collection. He had bought a share in the Ferme générale, the private company holding the right to collect the crown's duties, and that is what killed him: on 8 May 1794 he and twenty-seven other tax farmers were tried in a morning, driven to the Place de la Révolution and guillotined, Lavoisier fourth in the line.

Five years earlier he had printed in Paris the *Traité élémentaire de chimie*. In its first volume there is a table headed Tableau des substances simples. Thirty-three rows. Everything he judged could not be taken apart any further, sorted into four groups, with the name he was proposing for each one printed beside the older name it was replacing.

Twenty-three of those thirty-three are chemical elements and hang on the wall of every school laboratory today. He got them by taking things apart and weighing what came out, which nobody had done at that scale before, and for a list drawn up in 1789 that is a great deal of it left standing: oxygen, nitrogen, hydrogen, carbon, sulfur, phosphorus, and seventeen metals running from antimony to zinc.

Two rows sit above the others, in the group he headed as the substances that may be regarded as the elements of bodies. Lumière and calorique. Light and heat, entered as materials, on the same footing as gold, in the first two lines of the sheet.

Caloric is the fluid chapter twenty-two met, running downhill out of hot bodies into cold ones and doing work on the way. It could be neither created nor destroyed, and Lavoisier and Laplace had measured it in 1783 by the weight of ice a guinea pig could melt. In 1798 Benjamin Thompson, running the arsenal at Munich, set a cannon barrel in a box of water, fitted the boring machine with a deliberately blunted bit, and put two horses on it. The water boiled in two and a half hours, and would boil again the next day out of the same brass, which lost no weight and no capacity for heat. Whatever comes out of one lump of metal without limit, while the metal loses nothing, was never a fluid. Heat is a motion of the parts. The second row of the most careful table in eighteenth-century chemistry was a process wearing a substance's clothes.

Catch the ball. Pick up anything else at all: a beam of light, the ache in a knee. What is any of it made of? The table was drawn up to answer that. No list drawn up since has answered it all the way down.

34.1 Ten entries

The repair rule and its response metric define an abstract continuous three-dimensional carrier. Twelve readings sit on a patch, and the rule that repairs them cannot tell one port from another, so the twelve directions fall into four bands that no rule respecting the wiring can mix, of sizes one, three, three and five. The band of size one carries no comparison, so nobody inside can read it. The other three fade at rates of their own, and a disturbance in the fastest is down to half in 5.4 passes while the slowest takes 14.7. The quickest question a patch can put about itself runs three seams out and three back, six repairs, by which time most of the fast content has drained, so every observer is a slow ear.

The slow band has three directions, and the integer comparison records are dense in its continuous Euclidean completion. Joined to an independent real axis, this gives a finite-dimensional ambient carrier with one positive sign, three negative signs and a two-sphere of null directions. The authenticated direct-parent order separately computes source height as the exact longest parent-chain length ending at each event and proves strict increase along generated ancestry. One explicit positive global scale on source height enters only through event placement. This is not an intrinsic four-dimensional geometry of the event poset. The six whole numbers attached to opposite port pairs are candidate spatial readbacks, not physical locations, and scaled source height is not a physical clock.

Time is three objects sharing one word. The first is order, a partial order on committed repairs: one commit precedes another when the later one certifies that it read a register version the earlier one wrote. Among the four hundred and thirty-five pairs of seams in the twelve-observer wiring, a hundred and twenty may compete through a shared observer and three hundred and fifteen are structurally disjoint. Those are possibilities fixed by topology. Which pairs are actually ordered is fixed by the read-from provenance of a particular history. The second is direction, which is the irreversibility of a commit: seven arrangements of a seam go into the move and one comes out, and what separated them was in the arrangement the move overwrote. The third is flow, which is what a state does to its own algebra, and the entire supply of rate. Flow is reversible, so it carries no arrow. It belongs to whoever holds the state. There is no timeline for events to be strung along. There is a web of repairs with a direction on each authenticated strand. Each observer's rate is a property of what that observer is holding.

Mass is a standing cost. A bounded arrangement of records that keeps re-asserting itself against the seams around it has to be paid for on every cycle, because flattening it would open larger disagreements at its edge than it closes inside, so repair works on it at every event and gets nowhere with it. The size of that bill is the mass. It resists a push and it bends the geometry around it. Those are one entry read twice. A proton comes to 1.0072764665789 atomic mass units on every scale that has ever been put under one, the uncertainty confined to the last two of those digits, and not one of the records holding it together stays put long enough to be the proton.

Energy is a rate: how fast a region's records are worked through by repair, counted against the clock that region's own state manufactures. Three properties classical physics had to assume come out of that one sentence. The rate never goes negative, because outstanding disagreement is a sum of squares and its floor is total agreement. Rates of separated regions add, because two regions with no seam between them have nothing to repair across the gap. And the books balance, because a repair takes off one reading exactly what it puts on the other, and no move on the list writes one half of an entry.

Motion is transport of a pattern, and nothing travels. The pattern is re-established at one place in the comparison count after another, each place holding it for a cycle and handing it on, the way a wave crosses a lake while the water goes up and down and stays where it was. What advances is an arrangement. The material carrying the arrangement is different material at every step. Momentum is what the arrangement carries while this is going on, conserved because the repair rule does not care which seam it is applied at.

Force is a steepness. Put one unit of source on the twelve ports and the potential reads one value at the source, a second at every port a hop away, a third at two hops, and a fourth at the port directly opposite, and the drops between those four are eleven sixtieths, three sixtieths and one sixtieth. Eleven sixtieths of the fall happens between the source and its nearest neighbors, and everything past them, out to the far side, is worth the remaining four sixtieths, computed off a list of which port touches which and nothing else. A pattern sitting in a fall like that gets re-established slightly off-center on each cycle, toward the steep side, and how far off-center is the force.

Matter is a defect that no order of repair removes, together with a record saying it is there. Flip one record out of thirty on twelve wired observers and the machine converges from all two thousand and forty-eight starting states under every schedule and leaves exactly one unit of disagreement behind. Repair in another order and it sits somewhere else, at the same total. A defect of that kind is a pattern in the way content moves rather than a lump occupying a spot. It reads the same in every overlap, and survives being regraded to a coarser or finer resolution. The record is the other half of the entry: some observer has to have committed the fact that the defect is there. That writing is a separate event from the defect it is about.

Charge is a number you pick up going round a loop that will not go away. Each observer's labels are its own business, and the translations sitting on the seams are the only thing anybody can check; chain them round a closed loop and what comes back is a translation from your own conventions into your own conventions. Relabeling cannot reach it, because relabelings leave round trips where they found them. Nor can repair flush it out: repair works one seam at a time, and no single seam holds it. Twelve observers have nineteen independent loops for such numbers to sit in, and what sits there has to balance at every port and be paid for in energy.

Light is the transverse mode of the repair current. Mismatch leaving a port is transported rather than deleted, so the mismatch and the current carrying it drive each other in a circle instead of settling, and the pair travels. The mode starts with four amplitudes, two of which are a choice of labels rather than anything anybody can read; strip those and what is left is transverse, a plane of rank two. That is why light has two polarizations and never three. At zero wavenumber the stiffness the wiring supplies is exactly zero, so a disturbance of unlimited wavelength costs nothing whatever, and that exact zero is why every long wave goes at one speed.

A record is a commitment. It is an irreversible write into the part of the algebra where everything commutes with everything, which is what makes it readable twice with the same answer, copyable to a neighbor, and proof against being disturbed by the reading. The flow does nothing to it: the price of commuting with everything in sight is having no clock of your own. Of the ten entries this is the only one that is not a rate or a count, and the one the other nine are written in.

34.2 Rates, counts and one commitment

Read those ten down the page and ask what kind of object each one is. A rank. A relation, a loss and a flow. A cost per cycle. A rate. A re-establishment. A gradient. A residue with a note attached to it. A loop reading. A mode and the rank of a plane. A write.

There is nothing underneath them. None of the ten is a stuff that the other nine are made out of, and there is no eleventh entry held in reserve that the ten are the behavior of. An inventory of this kind is supposed to contain at least one substance. Lavoisier's contained seventeen metals you could pick up and put down again. This one contains a rank, a rate, a residue and a write.

Take the most solid item available and walk down through it. A proton is a standing cost, the cost is a rate of repair, a repair is one comparison between two readings held at the ends of a seam, and a reading is a number an observer holds about a neighbor it cannot see. Four steps. At no step does a material turn up that the numbers are numbers of.

Lavoisier's table had thirty-three rows and thirty-one of them were things you could put in a jar. The two you could not put in a jar were the two he placed at the very top, as the most elementary items on the list, and they are the two he got wrong. The ten entries above finish the correction he was not in a position to make: the jar is empty all the way down, and the inventory is complete anyway.

The world is made of accountancy.

The road to that sentence ran through a bell in an Essex church tower, a bead held in a laser in a dish of water in Canberra, and a nickel target in a laboratory on West Street, and every step of it was a count of something.

Stuff is what it feels like. The feeling has an account too. You are made of the same entries. The hand you put on a table is a set of patterns with a standing cost, the pressure you feel is a rate, the table's refusal to let the hand through is one pattern declining to occupy a state another pattern is holding, and your conviction that you have touched something solid is a commitment written into the part of you where everything commutes. There is nowhere in that arrangement to stand and look at the accountancy from outside, because the eye, the ruler and the place to stand would each have to be built out of more of it.

34.3 The shape of space

Cosmology puts a question to the sky that sounds like the most concrete question anybody could ask. Is space finite or infinite? Does it curve away like the surface of a ball, so that a straight line eventually comes home, or run flat forever, or close on itself the way a doughnut does, joined up in a direction nobody can point at? There are instruments aimed at it and a number attached to it. The number comes back flat to within a fraction of a percent.

Chapter fifteen gives the question a mathematical starting point. Under its stated repair rule and response metric, integer comparison records are dense in a continuous three-dimensional Euclidean carrier. A minimum-separation rule selects finite populations that cover bounded regions, with three-dimensional bounds on their density. Connecting nearby selected records gives a spatial transport graph without inserting a Cartesian grid. The region and resolution are supplied, and neither the selection rule nor an edge's physical travel time follows from the availability of its endpoint records.

Local geometry does not settle global shape. A small patch of a flat surface cannot tell you whether the surface extends forever or wraps around. The same distinction applies to three-dimensional space. Curvature, overall topology and total extent are different questions. The local carrier alone selects none of them. A cosmological model that joins the patches and identifies their physical events can make those questions precise, and observations can test its answers.

A shape at an instant also needs a choice of simultaneous events. Authenticated record dependencies give a partial order: one event precedes another when the later event genuinely consumes its record. That order alone supplies no common clock or preferred slice through the whole universe. Once a physical geometry and a slicing are specified, the shape of such a slice is a meaningful property of the model. Its dependence on that choice is part of the answer.

The instruments constrain this physical interpretation. A real axis joined to the continuous three-dimensional carrier gives a four-dimensional Lorentzian space of candidate event coordinates. Source height can enter an event placement, but its scale is not a calibrated clock. The carrier's dimension does not determine which recorded dependencies represent signals. A claim that the events themselves sample spacetime volume additionally requires interval counts, divided by a calibrated density, to converge to that volume. Event count is distinct from public-record capacity.

An effective field description has a direct construction on the selected spatial population. Weight each point by its nearest region's volume and supply a positive coupling between nearby field values. One stationary action then gives scalar motion whose detector readings approach those of a continuum wave. Before boundary effects arrive, the approximation error vanishes under the stated refinement conditions. The wave's finite propagation speed belongs to that continuum equation; a tiny instantaneous tail on the finite graph is tested against the detector error. The action and its clock interpretation remain assumptions about the physical source.

Exact agreement between event order and cone order is a stronger, separate route. A controlled order approximation can also suffice when its error vanishes with refinement. Neither route identifies a physical source merely by choosing event coordinates. Compatible readbacks and clocks connect an effective geometry to what observers can measure.

A concrete record population also gives an effective causal spacetime. Conservative source records place distinct sites throughout a bounded three-dimensional window. At each layer, every site reads all sites within a declared radius on the preceding layer, including itself. As the read radius shrinks more slowly than the gaps between points, the order approaches the light-cone order of flat spacetime. Equally weighted event counts approach four-dimensional volume, and comparable pairs inside an interior causal diamond approach a fraction of one tenth. The population and read rule are specified. Their physical selection and agreement with the scalar action's propagation require a common realization.

The event counts also recover a clock without reading the layer labels. Choose fixed timelike intervals whose causal diamonds lie inside the spatial window. Compare their counts and take the fourth root, using one interval as the reference. In the continuum limit this gives the ratio of proper durations. The observer must be able to read the interval records. That access and the reference unit are explicit; the readout does not select the microscopic law or a laboratory second.

The shape of the cone also has a precise conditional answer. Suppose some nonzero influence is possible and its allowed displacements form a closed convex cone containing no complete line: limits and positive combinations are allowed, but a nonzero displacement and its reverse cannot both occur. If the source's rotations and every boost along one axis preserve these possible influences, the cone must be the Lorentz cone, up to its time orientation. The symmetry must act on actual preparations, signals and readouts. A Lorentz transformation of a coordinate diagram alone does not establish that physical premise.

On a source-selected placement, the edge-speed bound gives the forward implication, while equal-height spatial separation and strict spacelikeness of increasing-height incomparable pairs give the reverse implication and hence exact order agreement. Antisymmetry then derives event separation, while the two-way equivalence gives exact interval preservation. None of this makes source height a clock, supplies physical event coordinates, determines intrinsic dimension or constructs a physical spacetime continuum.

A second theorem places supplied geometric and stress tensors on the same finite events. Nine balances along fixed algebraic source directions, together with symmetry, conservation and connectedness, give balance on every null direction and an Einstein-shaped tensor identity with one constant metric term. The dependency order does no work in that tensor calculation, and no dependency link chooses a direction, a finite-difference operator or a tensor. Those are algebraic labels rather than observed signals. A common manifoldlike refinement, count-to-volume calibration, small-region geometry and a same-family tensor-curvature identification with the smooth Einstein tensor, or the independently premised continuum small-ball/null-balance route, are required before this identity can be read as the smooth Einstein equation. Scalar-curvature convergence alone is only a diagnostic.

The same distinction matters when the question turns from shape to origin. In the standard cosmological account, running the scale factor backwards leads toward a hotter, denser era. Continuing a classical model to zero scale can produce a singular boundary. That extrapolation uses the model's time coordinate and equations; identifying its boundary with a first physical event requires control beyond the point where those equations cease to describe the physics.

The reconstruction starts from records held inside finite patches. The hot, dense era is supported by those observations. A first moment is a stronger claim than the success of that reconstruction over the range where it can be tested. The local carrier and authenticated event order do not decide whether such a boundary exists.

Chapter thirty-six returns to the question of a beginning. Here the distinction is enough: a continuous local carrier gives geometry a starting point, while global shape and temporal extent depend on how physical events realize it.

34.4 The Link Trainer

In 1929, in his family's organ factory in Binghamton, New York, Edwin Link built a stubby wooden aircraft that went nowhere. Organ bellows underneath it, driven by an electric pump, pitched and rolled the box while the pilot inside worked the controls, and the instruments on the panel moved the way instruments move in the air. The first customers were amusement parks. In the spring of 1934 the Army Air Corps was carrying the mail and losing aircraft to weather, and Link flew in to a demonstration in conditions nobody else would fly in. They ordered six, at three thousand five hundred dollars each.

That box is a simulation. The reason has nothing to do with the bellows. It is a simulation because flying exists. There are aircraft, in the air, doing the thing the box imitates. The box is pointed at them. Take the aircraft away and the trainer does not become more authentic; it becomes a wooden cabinet that pitches and rolls. Imitation is a relation with two ends, something standing in for something else, and with one end missing the word has nothing to grip.

So put the question to the list. Which of the ten entries is standing in for something else? Where is the original that a repair rate is a copy of, and what is a loop residue an imitation of? Chapter two ran two rules on a string of letters, and the strings that came out were not descriptions of realer strings kept in a warehouse somewhere; you ran the rules, and what came out is what there is. Every entry above has that character. Nothing on the sheet points outward at an original.

Outward is the load-bearing word, because a second question hides inside the same objection and that one has an answer. A structure can hold a description of itself while nothing in it stands in for anything, since standing in takes two objects and self-description has one. The description has a subject. The subject is the thing holding the description. Chapter thirty-six is where that gets demanded and priced. The reason the simulation argument finds nothing to grip here is the reason chapter twenty-eight can compute a constant rather than measure it: there is no second copy for the first one to be a copy of, so the accounts have to settle inside.

The other half of the objection is the word computed, used as though it were a synonym for fake. A thing that is computed is a thing that happens.

When a repair runs at a seam, a disagreement that was there has gone afterwards, the total is where it was, and seven possible pasts have been discarded, which is an event with consequences rather than a picture of one. The same holds for a magnetic field and for the taste of coffee. Both are computed in the only sense of the word this world supplies, and both occur. There is no clock outside the arrangement stepping it along, either: the eighty-one thousand nine hundred and twenty patches of chapter ten worked every seam between them into agreement with no tick anywhere in the machinery, each region advancing at the rate its own state manufactures. There is no next frame, and nothing anywhere that would be producing one.

34.5 The reader

Every entry above came apart into something more basic than itself. One item has not.

Look at what the ten have in common besides being counts. A rate is counted against somebody's clock. A rank is the rank of somebody's table of comparisons. A distance is a count of comparisons, and a comparison takes two parties that hold something to compare. A record is a write into somebody's center. Charge is what a loop reads on the way round, and a loop is a walk from one observer to the next. Every quantity on the sheet is a reading, and a reading takes something that reads.

That something has not been dissolved. The reason is structural: the dissolving was performed with it. Chapter three asked what the job of holding a description requires and got three plain items, a bounded view, somewhere to write, and a reading that makes a difference, all three of them met by a thermostat screwed to the wall of a classroom in Whitewater, Wisconsin. Chapter nineteen took the same object and gave it four: a local algebra of questions, a record algebra sitting in the center of it, an interface that reads and updates the same way every time, and enough of a state to fix the odds on the next question. Nothing in the chapters since has explained either version away, because every one of them was built out of it.

Which is what the first page meant by putting observers first, arrived at from the far end of the argument. The stage was assembled out of what bounded record-keepers can consistently write down about each other, in that order, and the ten entries are the assembly. Take the record-keepers out and there is nothing to take them out of.

Every number quoted in the last thirty-three chapters was obtained by observers standing inside the thing they were measuring, with instruments assembled out of the ten entries. Lavoisier weighed the elements on a balance made of them. Nobody since has found a way to step outside and check.

Chapter 35

Why These Laws and Not Others?

On 18 June 1858 a package reached Down House in Kent, and Charles Darwin opened it and found twenty pages of his own theory written by somebody he had never met. Alfred Russel Wallace had drafted the essay on Ternate, in the Moluccas, between bouts of fever, called it "On the Tendency of Varieties to Depart Indefinitely from the Original Type", and asked that it be passed to Charles Lyell if it seemed worth anything. Darwin had been assembling the same argument for twenty years, through barnacles and pigeon fanciers and a filing system of small stubborn facts, and he wrote to Lyell that day. Your words have come true with a vengeance. If Wallace had my manuscript sketch written out in 1842 he could not have made a better short abstract. All my originality, whatever it may amount to, will be smashed.

Ten days later his son Charles Waring Darwin, eighteen months old, died of scarlet fever. On 1 July, Charles Lyell and Joseph Hooker took both accounts to the Linnean Society of London and put them before the meeting, where the society's secretary, John Joseph Bennett, read them aloud. Darwin was at Down with his family, burying the child. Wallace was in the Malay Archipelago and learned that the meeting had happened months afterward, in a letter. The reading drew no discussion.

Thomas Bell, the society's president, reviewed the year in his address the following May. The year which has passed, he said, has not indeed been marked by any of those striking discoveries which at once revolutionize the department of science on which they bear. Bell had been in the chair on the evening of 1 July.

What Bennett read aloud was a procedure for producing the appearance of design with nothing in it that chooses: variation happens, some variants leave more descendants, and the arithmetic does the rest. It has an application in physics that has nothing to do with finches. Chapter one counted the numbers physics takes from laboratories and enters by hand, nineteen of them or twenty-six depending on how neutrinos are handled, and the question underneath that list is why the world runs on these rules with these numbers instead of some others. Two answers are usually on offer. They are brute facts and there is nothing to say about them, or they were drawn from an enormous collection of alternatives and we see this set because the others have nobody in them to look. Both answers end the conversation somewhere the reader cannot go. The first says there is nothing behind the numbers and does not say how it knows. The second needs a shelf with the other worlds standing on it. A shelf has to be somewhere.

35.1 What a description is for

In 1950 Vitaly Ginzburg and Lev Landau wrote down an account of superconductivity that mentions no electrons at all. They introduced one quantity standing for how much of the metal has gone superconducting at each place, expanded the energy in powers of it, and read off the two lengths a superconductor has: how far the superconducting state can vary over, and how deep a magnetic field soaks in before the metal expels it. The account worked near the temperature where superconductivity sets in and nowhere else. The constants in it were fixed by measurement, because nothing in the argument could produce them.

In 1957 John Bardeen, Leon Cooper and Robert Schrieffer explained superconductivity from underneath: electrons pair up through their coupling to the vibrations of the lattice, and the pairs move as single objects. Two years later Lev Gor'kov showed that the 1950 account is what the 1957 theory becomes near that temperature. The constants nobody had been able to calculate came out in terms of the pairing. The 1950 account did not become wrong when its reasons arrived. It stopped being the bottom and stayed in the textbooks, because near the transition it is the easier of the two to compute with and gives the same answers.

That is an **effective description**: correct inside a range, carrying constants it does not generate, and usually able to say where it will fail. Ginzburg and Landau's account announces its own ceiling, since the expansion it rests on is only good while the superconducting quantity is small, which is to say near the transition and not far below it. Physics is made of descriptions like that and there is no defeat in having one. The Standard Model is an effective description of the same kind. The complaint chapter one lodged against it was about direction rather than count. Numbers go in and predictions come out. No arrangement of it sends a number the other way.

35.2 The planet that was easier to move

Chapter thirty-three left one leftover in the solar system: 43 seconds of arc per century in Mercury's orbit that no amount of Newtonian bookkeeping supplied, about a hundredth of a degree in a human lifetime. It cost the nineteenth century an entire planet. The man who proposed the planet had form.

Urbain Le Verrier had found Neptune on paper in 1846, by taking the anomalies in the orbit of Uranus and computing where an unseen body would have to sit, and the telescope at Berlin found it that night within a degree of the position he gave. So he proposed another one, inside Mercury's orbit, hot and small and hard to see against the sun. A country doctor at Orgères-en-Beauce named Edmond Lescarbault wrote to say he had watched a dark spot cross the face of the sun in March of that year. Le Verrier travelled out to interview him, came back convinced, and the planet was named Vulcan.

Vulcan was never there. Each failed search moved it rather than killing it: smaller, closer in, differently tilted, and in one version broken into a belt of bodies each too small to be seen, which no observation could exclude. Nothing in the proposal forbade those moves, because a planet comes with a mass and an orbit and both are dials. During the total eclipse of 29 July 1878 two experienced American observers, James Craig Watson at Separation in Wyoming and Lewis Swift near Denver, both reported a reddish object near the sun. The two positions did not agree with each other. Nobody else at that eclipse saw anything. Eclipse expeditions went on looking through 1908.

In November 1915 Einstein computed the same 43 seconds of arc from field equations with nothing adjustable anywhere in them.

The history illustrates why a fit needs an independent test. Adjusting an unseen planet's mass and orbit could accommodate a perihelion discrepancy; the resulting model also had to agree with observations elsewhere. Changing the planet to a belt changed the proposal being tested. Chapter one counted parameters, but a count alone does not decide between models. What matters is which observations fixed the inputs and which consequences can fail independently.

35.3 What an assumption costs

In 1978 Jorma Rissanen published a paper called "Modeling by shortest data description". Its rule for choosing between accounts of the same data is to add two lengths: what it takes to write the account down, and what it takes to write down the data once you have the account. Take whichever account makes the two together smaller. Rissanen's name for that sum is the description length. An explanation is a compression. A rule that costs more to state than it saves has explained nothing, whatever else it may have done.

The inverse electromagnetic coupling read at rest is 137.035999177. The declared closure in chapter twenty-eight returns the diagnostic 137.035660, about two and a half parts per million away. The exact root and its physical interpretation are separate questions: an independent carrier identification and transport calculation must connect the map to that measurement. The comparison becomes explanatory when its physical inputs and outputs are accounted for together.

The package from Ternate is the same accounting seen from the other end. What Darwin had been accumulating for twenty years was data: island species, barnacle anatomy, fossil beds, the breeding records of pigeon fanciers who had made fantails and pouters and tumblers out of the rock dove, geographical ranges that made no sense on any other reading. Stored as itself, that is a library. The sentence that variants which leave more descendants become more common adds no fact to the library. It shortens it, by removing the need to hold each adaptation as a separate item. That shortening is why two men at opposite ends of the world wrote the same sentence out of two entirely different collections of animals.

35.4 Reading the collection as a sieve

String compactifications give candidate low-energy worlds by choosing an internal geometry, bundles and other data. A specific construction can then be tested against a specified target. This is a comparison between mathematical models before it is a comparison with Nature.

The OPH target also has premises. The response theorem constrains the local gauge symmetry; its global form requires compatible representation data. Matter content and family counting consume the finite menu and additional assumptions explained in chapters twenty-five and twenty-six. Together these define structural tests for a candidate, with physical predictions requiring their own realization.

A useful example is Vincent Bouchard and Ron Donagi's construction, published in 2006. Heterotic strings are compactified on a Calabi-Yau threefold with a noncontractible loop and a suitable gauge bundle. Transport around that loop breaks a larger gauge symmetry to the visible one. The resulting supersymmetric spectrum has three generations and no exotic matter, with regions containing zero, one or two conjugate Higgs-doublet pairs. The one-pair region is the structural benchmark used here.

On its specified smooth one-Higgs region, the pre-completion count used in the comparison is seventy-one complex directions: eleven for size, eleven for shape and forty-nine for bundle data. That is a hundred and forty-two real directions. Extra completion conditions could cut this space; the count does not describe a fully stabilized physical vacuum.

The declared comparison uses five real readouts. Its executable proxy consumes none of the candidate's moduli, so its derivative with respect to them is zero. That proves the proxy selects no point. A smooth five-output map on the whole hundred-and-forty-two-dimensional space has derivative nullity at least a hundred and thirty-seven by dimension alone. Neither statement computes the physical response of the string construction. A physical comparison requires the missing map from its parameters to the measured quantities.

35.5 Testing a specified world

A specified vacuum can be ruled out when its physical predictions disagree with data. Moving to another candidate changes the model being tested. The existence of alternatives neither saves a rejected candidate nor makes a successful one describe Nature automatically. OPH faces the same requirement: connect its construction to measurements and accept the consequences of that comparison.

The OPH numerical branch proposes a different route. Chapter twenty-eight assigns one screen cell an outside area reading and an inside electromagnetic reading. A physical screen map, gauge carrier, scale and operational readout must first show that they are readings of one quantity. Conditional on that same-carrier identification, setting them equal gives an equation whose unknown appears on both sides.

The equation has exactly one fixed point in its declared feasible set. Feed it two trial grains some distance apart and it hands back two grains at most 0.0724 of that distance apart, so the passes squeeze rather than spread. The corresponding perturbative endpoint is 137.035660, against the measured 137.035999177; completing the transport requires the separately measured hadronic contribution. The contraction proves uniqueness for the declared map. It does not prove that Nature realizes the map or rule out a different physical architecture.

35.6 Scoring a rule

Chapter nine derived the repair move from three requirements. Every move that fails one of them is a rule somebody could have written down, and rules that get written down compete.

Three quantities, each of them a count of something. The first is schedule stability. Take a starting arrangement, run a rule until it stops under one order of repairs, run it again under a different order, and ask whether the two settled readings agree; do that across many starting arrangements and many orders, and the fraction that agree is a number between zero and one. Chapter six's twelve observers ran from every one of two thousand and forty-eight states under sixteen schedules and landed every time on the same arrangement, which is that number coming out at 1.

The second is observer yield: how many observers a rule produces. A region counts as one when its record reads the same at the end of a stretch of time as it did at the beginning, and when what it holds carries information about what its boundary does next. Both halves are measurable, the first as how far the reading has drifted and the second as how much the record tells you about the neighbors' future. Count the regions that qualify and take the average over the running of the world.

The third is the cost of writing the rule down, in Rissanen's sense, and enters with a minus sign. A rule's score is the first two with positive weights, minus the third.

The three pull against each other. A rule that repairs nothing scores a perfect 1 on stability, because every order of doing nothing agrees with every other order of doing nothing. It is about as cheap to write down as a rule can be, and yields not one region holding a record about anything, so the second term deletes it. A rule that buys the second term by copying down everything it ever sees pays for the copies in the third. The rule chapter nine derived, which moves two disagreeing readings to their midpoint and touches nothing else, is where the three pulls balance.

The rules are competing for one finite thing, which is the capacity of the structure to repair itself. A rule's share is the fraction of that capacity being spent its way. A rule scoring above the average grows its share at a rate proportional to how far above the average it sits, and one below shrinks the same way. That is **replicator dynamics**, arithmetic for the sentence that fitter types spread. Nothing in it involves universes having children.

Write f for a rule's score, put a bar over it for the average of the scores across the rules holding capacity, weighted by their shares, and let t measure how far the reallocation has run. Ask how that average moves as the shares move, and the answer is one line.

$$\frac{d\bar{f}}{dt} = \mathrm{Var}(f)$$

The left side is the rate at which the average score changes. The right side is the variance of the scores: the average of the squared distances of the individual scores from the average score. So the average climbs at exactly the rate at which the survivors disagree with each other, and since a squared distance is never negative, the average never falls.

Two rules make it concrete. Scores of 3 and 1, half the capacity each, and an average of 2. The first gains half a share per unit of t and the second loses the same. The variance is 1, and the average is climbing at 1. As the first takes over, the spread closes and the climb stops, because the mean rises only on disagreement and by then there is none.

What survives that is a rule whose answers do not depend on the order anybody works in, which is thick with regions that hold stable predictive records, and which is cheap to state. Chapter six measured the first of those on twelve observers, and chapter nine derived the move that supplies it.

35.7 An ensemble of one

The anthropic argument runs as follows: there is a large collection of worlds with different settings, the great majority are barren, ours is one of the few in which anybody could be around to take a reading, and no further explanation of the settings is needed. As selection reasoning it is sound. It requires the collection.

Take the phrase "the largest prime number". It has the grammar of a name and there is nothing at the end of it, and Euclid's argument shows why: take any finite list of primes, multiply them together, add one, and the result is divisible by none of them, so no finite list is all of them. The phrase does not pick out a number that turned out to have some other property. It picks out nothing.

The phrase "a universe with a different fine-structure constant" is excluded only inside the fully attached closure branch. On that branch the physical area and electromagnetic readings refer to one carrier, and their equality selects one fixed point. Without the same-carrier map, the finite equation classifies a candidate rather than every possible world.

Within the declared map the feasible fixed-point set has one member, so that branch supplies no continuous parameter for an anthropic filter to scan. This is a statement about the map's solution set. It is not a proof that the space of physical architectures contains one world.

Running from inside has a price. The capacity being competed for is the structure's own, the observers being counted include whoever is doing the counting, and the rule that takes the whole share is the rule the judges are running on. Darwin's argument reached Down House in an envelope from Ternate, carried halfway around the world by a post office, so there was a morning before it arrived and a morning after. There is no morning before this one. Nothing was posted and nothing was opened, and whatever holds the arrangement up is somewhere inside it.

Chapter 36

Why Does the Loop Have No Beginning?

The room was emptying. It was the afternoon of 7 September 1930, the last day of a three-day conference on the epistemology of the exact sciences that Kurt Reidemeister had organized at the University of Königsberg, and the closing item was a roundtable on the foundations of mathematics with Hans Hahn, Rudolf Carnap, Arend Heyting, John von Neumann, Arnold Scholz and Reidemeister himself at the table. Everyone had said their piece over the previous two days: Carnap for Russell's programme, Heyting for Brouwer's, von Neumann for Hilbert's. The proceedings went into the journal *Erkenntnis* the following year, volume 2, pages 87 to 190, with the roundtable transcript at pages 135 to 151, so it is possible to read what was said and in what order.

Near the end, a twenty-four-year-old Viennese logician remarked that one can give examples of statements of arithmetic that are true and that no proof inside the system reaches. Kurt Gödel had come to Königsberg to present a completeness theorem, the result he had built his reputation on, and he left having mentioned in passing the result that closed the question the three days had been about.

The discussion moved on. Almost nobody in the room reacted. One person did: von Neumann cornered him afterward, wanted the construction, went home and thought about it for two months, and wrote to Gödel on 20 November with a further consequence that Gödel had reached first.

The next morning the city of Königsberg made David Hilbert an honorary citizen. He was sixty-eight, he had been born there, he had retired from Göttingen that year, and he addressed the Society of German Natural Scientists and Physicians on knowledge of nature and logic. He closed with six words that were later cut into his gravestone: we must know, we will know. A recording of the ending survives, 373 words read for local radio and running four minutes. The answer had been supplied the previous afternoon, in a smaller room, at a table of seven, one of whom was listening.

What Gödel had was a construction. Hand it a system that can talk about its own formulas and it hands back a sentence about that system, assembled out of the system's own material and pointing at nothing outside it. A world made of observers comparing records leaves a bill that no amount of further comparing pays. It has to settle that bill the same way, out of its own material. Something has to be doing the comparing. Where did the network come from, what is it running on, and what started it?

36.1 A program that reads its own text

Every programmer has wanted the same tool at some point. Something you can point at a piece of code that tells you whether the code finishes or runs forever, and answers by deciding rather than by waiting, correctly, on any program you hand it.

Suppose you have it. Call it the oracle. It takes a program's text along with an input, answers finishes or runs forever, and is never wrong.

Then write the following, which takes about four lines. It reads a piece of program text. It asks the oracle what that program does when it is fed its own text as its input. If the oracle says finishes, it goes into an infinite loop. If the oracle says runs forever, it stops immediately.

Hand that program its own text.

If it finishes, the oracle must have answered runs forever, since that is the only answer that makes it stop, and the oracle was wrong. If it runs forever, the oracle must have answered finishes, and the oracle was wrong the other way. There is no third case, so there is no oracle. Alan Turing published that argument in 1936.

Look at what did the work, because the contradiction is the disposable part. The four-line program is one whose behavior is the opposite of what is predicted about it. Hand somebody the transformation "do the opposite of the prediction" and the construction hands back a program satisfying it. Feeding a description into the thing it describes is a way of solving an equation. The equation is chapter two's: return what you were given.

The calculator in chapter two hunted for a number the operation left alone and found 1.6180339887. The same demand, made of descriptions instead of numbers, has the same kind of answer: a description the transformation returns unchanged. That is a fixed point, and self-reference is the construction that builds one. Chapter two's warning travels with it, because a fixed point can be a place nothing ever arrives at.

36.2 Four rows and a diagonal

The move underneath the oracle is older than computers and fits on an envelope. Georg Cantor published it in 1891, four pages in the first yearbook of the German mathematical society, of which he was the first president. It runs on a numbered list.

Write four strings of four bits, one per line.

row 1: 1 1 0 0 row 2: 0 0 1 0 row 3: 1 1 1 0 row 4: 0 1 0 1

Walk the diagonal: the first bit of row one, the second bit of row two, the third of row three, the fourth of row four. That reads 1, 0, 1, 1. Flip every bit you landed on and you have built 0, 1, 0, 0.

Check it against the list. It differs from row one in the first place, which is where you flipped row one's own bit. It differs from row two in the second place, from row three in the third, from row four in the fourth. Four rows, four disagreements, one per row, each one forced by where the bit was taken from.

Nothing in the walk cared that the list was four long. Run it down a numbered list of strings that never end, one string per whole number, and what you build differs from the seventeenth string in the seventeenth place and from the millionth in the millionth, so it sits on no line of the list. A minute of bit flipping. No table of any size holds every string.

Turing's four lines are that walk with programs on the list and the oracle's prediction as the bit. Gödel's sentence is the identical construction pointed at proof instead of at halting. Number the formulas, so that a statement about numbers is also a statement about formulas, and build the sentence whose content is that it has no proof in the system. If the system proves it, the system proves a falsehood. If the system does not, the sentence says something true that no proof reaches. The lever was never a pun on the word this.

The numbering is the engineering and the self-reference is what falls out of it. Give every symbol a number. A formula is a list of symbols, so code the list by prime powers: symbols numbered three and one give two cubed times three, which is twenty-four, and twenty-four factors into primes one way only, so the code hands back three and one. Proofs are lists of formulas and get their numbers the same way, so the relation between a proof and what it proves is arithmetic, and "no number codes a proof of the formula coded by twenty-four" is a statement about numbers that a system able to do arithmetic can write down.

Getting a sentence to carry its own code with no regress is Quine's device, and works in English with no numbers in it at all. Take a phrase, write it once inside quotation marks, then write it again outside them.

"is twenty words long when appended to its own quotation" is twenty words long when appended to its own quotation.

Count. The quoted phrase is ten words and the phrase after it is the same ten, so the sentence is twenty words long, which is what it says about itself. It does that without the word this and without a second sentence propping it up.

Gödel's sentence is built that way. It gets called a paradox. It is not one. The liar sentence, the one saying that it is false, has no consistent reading: true makes it false, false makes it true, and the machinery seizes. Take Gödel's instead and suppose the system proves it. Then a proof of it exists, so what the sentence says is false, and the system can read that proof and derive the opposite of what it just proved. It contradicts itself, which is chapter two's disease, buying everything and ruling nothing out. So a system that contradicts nothing does not prove that sentence, which is what that sentence says about itself. The sentence is true.

That is what the table of seven was handed: a consistent system with a truth inside it that its own proofs do not reach. Hilbert wanted consistent and complete, and complete was the half that went.

36.3 Two hands and a sheet of paper

In January 1948 M. C. Escher printed a lithograph, 28.2 by 33.2 centimeters, showing a sheet of paper lying on a surface. Two shirt cuffs are drawn on the sheet in flat gray pencil. Out of each cuff a hand rises off the paper in full modeling and shadow, and each hand holds a pencil, and each pencil is drawing the cuff of the other.

Chapter six borrowed the impossible staircase, which this artist put into a lithograph twelve years later. The staircase fails at the closing: every individual step up is unobjectionable and the loop cannot be assembled. *Drawing Hands* fails nowhere. Every line in it is a line an artist could draw. Both hands are drawn and both hands are drawing, and cover any part of it you like and what is left is consistent. What the picture lacks is a ground floor. Neither hand is the real one. There is no corner of the composition to stand in and ask which was drawn first.

Douglas Hofstadter named the arrangement in *Gödel, Escher, Bach* in 1979 and spent seven hundred pages on it. A **strange loop** is a hierarchy that climbs through its levels and arrives back at the level it started from, with no level underneath holding the rest of them up. The staircase is what a strange loop looks like when it fails. The hands are what one looks like when it holds. Everything remaining in this chapter is the second picture with something else in the cuffs.

36.4 The description and the thing described

Chapter eleven put a name on what a network of observers holds in common. The record map takes the full state and returns the part that survives comparison, which is every seam entry and nothing that only one observer could have known. That list has a fiber, the set of consistent states carrying it, and when the fiber holds exactly one member, that member is the observable normal form. Objective reality is that state, up to changes nothing can see.

Every object in that sentence is a description or an operation on one. The record list is a description. The normal form is what the description specifies. And the operation running from the one to the other, take the description and return the world it names, is defined on any description at all, including a description of everything.

So point it at everything. Take the whole structure, records included, as one object, and apply the operation to it. What comes back is what that object's own description specifies. The demand is that the two agree.

The universe is the observable normal form of its own description.

One line covers the calculator, the four-line program, the sentence carrying its own code and the demand just made of the whole structure, and the same unknown stands on both sides of it.

$$x = f(x)$$

Here x is whatever is being solved for and f is the operation applied to it. On the calculator, x was a number and f was divide one by it and add one. Applied to everything, x is the whole structure with its records and f is take the description and return the world it specifies. Self-reference is that equation, and closure is that equation with the universe in it.

Chapter eleven's three verdicts apply to that description the way they apply to any other. The fiber could be empty. A description with an empty fiber names no consistent state, which is the reading a witness produces when no arrangement of the world satisfies the testimony. It could hold several, in which case more than one structure answers to the description and the description is of none of them in particular. Closure is the middle verdict, taken at the top: the fiber holds exactly one member, and the member is the structure the description came out of.

That single requirement is severe, in chapter two's sense, and does most of the work of existing. Let a structure's description specify something other than that structure, and the description is of the other one, so the structure it came out of is not described at all, which is a way of failing to be anything rather than a way of being wrong. A structure whose observers could never recover its architecture from their records fails the same demand from the other side, because then the description its observers hold is not a description of it, and the operation returns something else.

Nothing outside supplies the description. Asking what switched the universe on assumes an outside to switch it on from. The demand that picks out this object is the demand that there is no outside. Anybody asking has to supply the outside themselves. Every outside anybody has supplied has needed one of its own.

There is a possible dividend in that which nobody went looking for. Chapter twenty-eight writes the grain and a candidate coupling in terms of each other, and a separate conditional branch joins grain, record capacity and the cosmological term. The declared finite maps have unique fixed points in their feasible sets. Turning those points into measured constants requires a source-selected physical screen, gauge transport, horizon-record identification and unit calibration. Closure removes a dial inside an inhabited map; it does not prove that every required physical map is inhabited.

36.5 Three orders

The word loop invites a picture of something traveling backward into its own past. Three different orderings are in play. Every paradox that gets built out of the loop comes from confusing two of them.

The first is the fixed point. It contains no order whatsoever. It is a relation among the parts of one structure, holding or failing to hold. Nothing steps it, nothing iterates it, and there is no count of how many times it has been applied, because an equation is not a process. A loop with no beginning is what an equation looks like when nothing solves it first.

The second is an observer's record order: the chain of committed records that observer holds. It supplies informational precedence and an irreversible direction for accepted writes. Memories and clock readings can be recorded in that chain, but the chain is not a physical clock. Elapsed time, rate and agreement with signal causality require a realized instrument plus the separate causal, ruler and clock maps.

The third is the descent of repair: the order on authorized updates given by the disagreement count, which falls strictly at every accepted step and cannot fall forever. It is what proves the repairs terminate and what lets two schedules be compared. A repair step is not a second on anybody's clock. Two incomparable schedules do not make two parallel histories. Two repairs at opposite ends of the network, with no seam joining the patches that made them, stand in no order whatsoever: no fact settles which came first, and the question never enters anybody's records because there is no record it could enter.

Set the three side by side and the paradox has nowhere to live. First they work out the architecture, later they build it, is a statement about the second order, holding inside the records of particular observers. It says nothing about the first, because the first has no first. The recovered specification, the built machine and the inhabited structure are three readings of one object, and only an observer's own chain of records puts them in an order.

36.6 A thousand machines and one machine

Run the network on one computer. One process, one memory, one loop that picks a seam, checks it, repairs it if the two sides disagree, and goes round again. It ends where chapter ten said it ends. The endpoint does not depend on which seam it picked first.

Then run the same network across a thousand computers, which is what anybody does in practice with eighty-two thousand patches. Cut the patches into a thousand pieces, one per machine. Each machine holds its own patches plus a copy of the seam data along its border, refreshed by messages that arrive late and out of order. Machines take checkpoints. One dies and restarts from its last checkpoint, replaying updates it had applied before it fell over. A supervisor notices another machine falling behind and re-cuts the entire partition to move work off it.

Anybody who has run a large computation knows what the second version looks like from the operator's chair. The logs differ. The timings differ. Two runs of the identical setup interleave their operations differently. Neither interleaving matches anything the single machine does.

Whether the two are the same universe takes one bookkeeping move to settle: say exactly what counts as the state of the world and what is scaffolding around it. Keep the authoritative record each patch holds. Quotient by the changes nothing can see, which chapter eleven's group names. Throw away the message queues, the retry counters, the labels saying which machine owns which patch, and any wall clock the machines happen to be reading. What is left is a state of the plain single-machine network. Every readout an observer inside can perform is computed from that.

Then classify what a distributed run is permitted to do, and it comes to three things.

A commit, which projects onto a legal repair path of the single-machine network. Something disagreed, a patch fixed it, and the fix corresponds to a repair the one machine could have made.

A stutter, which projects onto nothing at all. Delivering a message, refreshing a border copy, taking a checkpoint, preparing a transaction, aborting one, starting a machine, stopping a machine, replaying an update that has been applied once, and every byte of metadata describing how the work was cut up. The projected state before and after is the same state.

A rollback to an earlier committed state, carrying the name of the committed state it returns to.

That exhausts the list. The rest is an induction with three cases. Chapter ten's result is that the endpoint of the repair process is constant across everything reachable: if legal repairs get you from one state to another, both have the same endpoint. A commit moves along legal repairs, so the endpoint is unchanged. A stutter moves nowhere, so the endpoint is unchanged. A rollback lands on a state whose endpoint was the same one. Start the distributed run and the single-machine run from the same initial state, and after every event in the run, including the last, the endpoint is the same.

So the thousand-machine run settles into the state the one-machine run settles into. Every observer readout that goes through the projection agrees.

Partition, worker count, schedule, restart history and repartition metadata appear in no observable.

That is a symmetry, in chapter five's sense and in no weaker one. Name the change: divide the work differently. Name the invariant it leaves alone: everything anybody can measure. Chapter five also supplied the deflation. It applies here at full strength. A symmetry is the claim that part of your description was never physical, and how the work was divided was never physical.

Nothing else in the architecture moves as much. Rotating the carrier moves twelve ports. A gauge change moves a convention chosen separately in every place. This one moves the entire execution, and goes uncounted as a symmetry because it looks like a fact about implementation rather than a fact about the world, which is what the labeling of the twelve ports and the choice of gauge in every place both looked like until somebody counted them.

A rendering loop has a frame number, and every object in the scene shares it, which is what makes it readable: ask any two objects what frame it is and they agree. Ask which step a given patch is on here, and two runs that agree on every measurement anybody can take come back with different numbers. A quantity that differs between runs nobody can tell apart belongs to the bookkeeping. No frame, no tick, no rendering loop, and no place to stand and watch the next frame get drawn. What there is instead is a state picked out by the constraints and arrived at by descent from wherever the system happened to be. The descent has no rate, because a rate would be a step count.

So a description of the universe, running inside the universe, is a distributed presentation of it. Its partition and its schedule reach no observable. It settles where the thing it is running inside settles.

36.7 Five cards

Chapter six's observers have no clock, no leader and no queue. Can a thing built like that do anything besides agree?

Deal five cards face up in a row and number the places one to five. A shuffle is a rule saying where the card in each place goes. Call the plain one C: every card moves one place along, and the card in place five comes round to the front.

Two more shuffles, each of which is C wearing different clothes. Shuffle A sends place one to two, two to four, four to five, five to three, and three back to one. Shuffle B sends one to four, four to three, three to five, five to two, and two back to one. Trace either of them and you visit all five places once and return to where you began, which is exactly what C does. They differ in the route and in nothing else.

Chapter nineteen measured how far two operations are from being swappable by doing them in both orders and subtracting. In a group there is nothing to subtract, so you cancel instead: undo one, undo the other, then do each in turn. If the two get along, every doing cancels its undoing and the deck is as you found it. Try it with these two. Perform, in this order, A backwards, B backwards, A, and B.

Read the row off. Every card has moved one place along and the last has come round to the front. Four shuffles that were supposed to cancel have delivered C.

Five pieces of card on a table are enough to check that identity.

Write a program as a list of instructions, where an instruction reads one input bit and says: if that bit is one, do this shuffle, otherwise do that one. Run the instructions in order on the deck. Say the program computes a function of the input bits when the deck ends up cycled by C exactly on the inputs where the function is true, and untouched on the others.

A single instruction computes a single input bit. Building everything else out of that is the identity above, used once per gate. Given a program for one statement and a program for another, assemble four blocks in the order the identity wants them: the first in A's clothing run backwards, the second in B's clothing run backwards, the first in A's clothing, the second in B's clothing. If either statement is false, its two blocks do nothing and the surviving pair cancels, so the deck is untouched. If both are true, the four blocks are the four shuffles above and the deck comes out cycled, by a route that depends on the clothes and can be steered to any cycle you want by choosing them, which is what lets the step be applied again to its own output. That is the AND gate, and with the complement it is every gate there is. David Barrington published the construction in 1989 in the *Journal of Computer and System Sciences*.

Every function of bits is a tree of gates, so every function of bits is a fixed list of shuffles of five cards. The deck never holds more than five things in a row. The price sits in the length: four blocks per level of the tree, so a formula of depth ten compiles into a word of about a million shuffles, and depth is expensive.

Those five cards are hardware. Chapter fourteen counted the rotations of the twenty-faced carrier, got sixty, and found five families of half turns that those sixty rotations shuffle among themselves. Sixty permutations of five objects is the group that leaves the fifth-degree equation without a solution formula, and the carrier's own group of moves. The deck is the carrier. The shuffles are things the carrier does to itself.

The property making the construction work is the property Galois found. The leftover from those four steps, the thing that came out as C instead of as nothing, is itself a move of the group. Collect every leftover the group produces that way and run the same operation on the collection. In a group with the shape a formula in radicals requires, the leftovers shrink at every round and die out. In this group of sixty they never shrink at all: the leftovers are the whole group again, round after round. The five-card identity is that refusal to shrink, written out on one case. The absence of a formula for the quintic and the presence of universal computation are one property of one group, read from two ends.

Compile a function into a federation of observers instead of into a deck: one patch per gate, each patch holding a few registers, each patch checking the seams it touches and repairing its own registers and nothing else. The input arrives as patches that hold their bits and accept nothing else. Set every other register in the network to anything at all. Let the patches repair in any order, with no clock and no leader appearing anywhere in the argument. The network settles, and the value of the function is sitting in a designated register of the settled state, from every starting state, and any two settled states of that network agree there whatever order the repairs ran in.

The small cases can be checked on every input rather than on a sample. A full adder's sum bit, the three-input circuit that adds two bits and a carry and sits underneath every binary addition anybody has ever performed, comes out right on all eight inputs from every starting state. So do the two bit-mixing functions called choose and majority, which take three inputs each and sit inside the hash function that Bitcoin's ledger is built on. A two-input NAND, the gate that answers false only when both its inputs are true, runs end to end on both engines, the deck and the federation, and they agree on all four inputs.

Patch-local repair, with nothing global anywhere inside it, computes every function of bits.

36.8 The hardware

Set the two results next to each other.

The first says the division of labor is invisible from inside. A description of the world, running inside the world, has a partition, a worker count, a schedule and a restart history, and not one of those reaches any observable. The second says the world's local repair is a general-purpose computer: the same rules that make two neighbors agree about a seam will, wired up, evaluate any function you can write down.

Together they close the loop. Observers inside the structure read their records, recover the architecture that produced them, and build it, in their own record time and at their own expense. What they build emits records of the kind they recovered it from. There is no step at which the built thing and the inhabited thing are two objects, because the only differences between them were partition, schedule and worker count, and those are the coordinates the first result deletes. The hardware the observers reverse engineered is the hardware they are running on.

Which leaves the observers themselves, who have been doing all the work and getting none of the explanation. A loop that computes has pieces. Some of those pieces spend everything they have on continuing to be pieces: holding a boundary, keeping a record, repairing themselves faster than they come apart. Every rule in the architecture is indifferent to which patterns survive. Some of the patterns are not.

Chapter 37

Why Does Anything Try to Persist?

The rabbits came from the market and their urine was wrong.

Claude Bernard tells the story in a book he published in 1865, and the detail he leads with is the one that was sitting in front of him: the urine in the tray was clear, and it was acid. Rabbits eat grass. A grass eater's urine is cloudy and alkaline, and it had been cloudy and alkaline in every rabbit that had ever come through the laboratory. These animals were producing the urine of a cat.

He guessed they had not been fed for some time. He gave them grass, and within hours the urine went cloudy and alkaline. He took the grass away. It turned clear and acid again. Then he fed rabbits cold boiled beef, which a rabbit will eat if there is nothing else on offer, and for as long as the beef kept arriving the urine stayed clear and acid, exactly as it had while the animals were starving.

Fasting and a meat diet produce the same rabbit. A rabbit that has stopped eating is a carnivore. The meat it is eating is the rabbit.

The urine is the smallest part of this. What the fasting animal was defending is the composition of the fluid its cells sit in: the sugar in it, the salt, the water, the acid, the heat. Cut the supply from outside and those quantities hold. They hold because the liver dismantles the animal in order to hold them. Bernard proved that the liver manufactures sugar rather than merely storing it by washing a freshly removed dog's liver until no sugar could be found in it, leaving it overnight, and finding it full of sugar in the morning. That was 1855. He had pure glycogen out of liver tissue by 1857.

Repair happens by itself. Two readings disagree, an observer works out a move and commits it, the disagreement drops, and nothing anywhere wants anything: chapter six ran two thousand and forty-eight universes to a standstill on exactly that, and no observer in any of them was trying to get anywhere. A rabbit interrupts that pattern. It holds a set of numbers fixed at its own expense and goes on holding them until it has nothing left to spend. Why does anything do that, what exactly is it spending, and what does the spending buy that nothing else buys?

37.1 The internal environment

Bernard had a name for that fluid. He had been using it since 1854 and he set it out in the 1865 book, *Introduction to the Study of Experimental Medicine*: the internal environment, the *milieu intérieur*. His claim about it was blunt. A complex animal does not live in the air or the sea. It lives inside a fluid it manufactures and defends. The liberty of a warm-blooded animal to walk out into weather that would kill a fish is bought entirely by how tightly that fluid is held.

In lectures published in 1878, the year he died, he compressed the program into one line: the constancy of the internal environment is the condition of a free and independent life. A creature whose insides track its outsides is at the mercy of its outsides. Every degree of independence from the weather is paid for by an equal amount of work done indoors. Bernard's rabbits were buying independence from having eaten. The currency was their own tissue.

The explanation available at the time was a vital principle: living matter has a property, dead matter lacks it, and the property is why the numbers hold. Bernard spent a career refusing that. His position was that the constancy is produced by machinery, that a physiologist's job is to find the machinery, and that the machinery obeys the same determinism as a furnace or a barometer.

The word everybody uses arrived sixty-one years after the book. In May 1926 a volume was published in Paris for Charles Richet's jubilee, edited by A. Pettit, in an edition of five hundred copies. Walter Cannon, professor of physiology at Harvard, contributed a paper to it called "Physiological regulation of normal states: some tentative postulates concerning biological homeostatics". In that paper he gave the steady states of the body's fluid matrix the name **homeostasis**.

Cannon had come to the subject by watching the thing fail. He went to France with Base Hospital No. 5, a unit of Harvard clinicians, moved forward to a casualty clearing station run by the British, and spent the war on wound shock: men whose blood pressure fell and kept falling, whose blood turned acid, and who died with their injuries dressed and their wounds clean. He carried on the work in William Bayliss's laboratory in London and at the American laboratory in Dijon, and published *Traumatic Shock* in 1923. The list of quantities he assembled afterward as the ones a body holds steady, temperature and water and salt and sugar and oxygen and blood calcium, is a list of the readings he had watched go.

Cannon explained the choice six years later in *The Wisdom of the Body*. Equilibrium was taken: it had acquired an exact meaning for simple physico-chemical states in closed systems where known forces balance, and the coordinated business of brain and nerves and heart and lungs and kidneys holding a mammal's blood where it belongs is a different kind of thing.

"The word does not imply something set and immobile, a stagnation," he wrote. It means a condition that varies and comes out roughly constant anyway.

37.2 A symmetry with a payer

Chapter five settled what a symmetry is: a change that leaves a named invariant alone, which makes it the claim that part of your description was never physical.

Homeostasis has that exact shape. The shape is easy to fill in. Name the invariant: the concentration of sugar in the rabbit's internal fluid. Name the change: everything that happens to a rabbit across a day, being fed, not being fed, running, sitting in a cold room. The reading comes out the same, and it goes on coming out the same while the animal producing it is being eaten to produce it.

The symmetries of the earlier chapters arrive free of charge. The laws not caring where you stand costs nobody anything: move the apparatus a meter to the left and the same numbers come out. Move a rabbit from fed to fasting and the same number comes out because a liver was taken apart to produce it.

So there are two ways for a reading to be invariant, two different physical situations wearing one word. In the first, the invariance is a property of the world's laws. The observer has nothing to do with it. In the second, the invariance is manufactured by a bounded observer out of its own capacity, continuously, for as long as it can afford the outlay. What the vital principle was tracking is the outlay, which in Bernard's rabbits was paid in liver.

The outlay has to be continuous because seams do not sit quiet. An observer's neighbors are repairing as well, every commit any of them makes is irreversible in the sense chapter sixteen gave that word, and the entries on a shared boundary get rewritten whether or not the observer on the other side of it wanted them rewritten. A pattern that spends nothing inherits whatever its neighbors' repairs happen to leave on its boundary. Holding an invariant against that traffic is a rate rather than an act, which is why the liver does not finish, and why a mammal that stops paying cools toward the temperature of the room it is lying in.

Chapters eighteen and twenty-one already priced that rate. A region's energy is what drives its own clock, and what drives that clock is the repair running through its records. So the outlay here is the same quantity a calorimeter reads rather than a metaphor borrowed from biology, and eating is how an animal settles it.

37.3 What a living thing is

Chapter eleven divided a network's state into what any neighbor can check and what only one observer holds. What an observer publishes is the entries on its seams. The fiber of a reading is the set of consistent worlds that produce exactly those entries. A reading whose fiber is empty describes nothing that could have happened.

A bounded observer persists when the reading on its seams keeps having a consistent extension. At every point along its own record time there is a world that carries what it publishes. It is spending to keep that true.

Bounded homeostatic persistence is the whole definition of a living thing. There is no ingredient anywhere in the phrase. Bounded, because an observer has a finite interface and a finite budget. Homeostatic, because the invariance on its seams is manufactured rather than handed over. Persistence, because the claim is about the reading holding across record time rather than about any single moment. A rabbit is alive on Tuesday because it was alive on Monday and paid for the difference.

Chapter five's invariants survived something done to a system from outside: a rotation, a shift, a relabeling. This one survives the system's own running. A rabbit is whatever a rabbit's metabolism leaves alone while it takes everything else apart.

The rabbit makes the point about fibers by itself. Across a week without food almost everything about its interior changes: the liver empties, the fat goes, the mass falls by a fraction anybody can measure on a kitchen scale. What it publishes holds. The animal is traveling an enormous distance inside the set of states that produce one reading. The traveling is what the reading costs.

Stone persists too. Nothing inside a stone is spending to make that happen. Leave a stone alone and its readings sit where they sit. Leave a rabbit alone for a week and its readings walk off, because holding them where Bernard found them was work, and the work stopped. When the entries on an observer's seams stop having any consistent extension, the fiber is empty, and repair does what repair does: the surrounding patches redistribute until the record closes again, with the boundary of the thing that was doing the spending drawn somewhere else. Every atom is where it was a minute earlier. The reading is what stopped.

37.4 Two budgets, and what a policy is

Chapter nine found two kinds of repair and separated them by who pays. In the first, an observer rewrites its own patch so that its reading matches what the seam says: the capacity spent is the observer's own, nothing outside the patch changes, and the boundary record comes out of the transaction saying what it said going in. In the second, the boundary record itself is rewritten so that both sides read something new, and the capacity spent sits on the overlap, owned by neither observer alone.

A bounded thing that persists is choosing between those two, at every event, out of a finite budget. The rule it chooses by is its **policy**, and running a policy is the whole of behavior. There is nothing else in the word. An organism, a firm, a thermostat and a bacterium each run one: how much to spend bringing themselves into line with what the world reports, and how much to spend on changing what the world reports.

In 1972 Howard Berg and Douglas Brown built a microscope that chased bacteria. Rather than hold the field of view fixed and watch a cell swim out of it, the instrument moved to keep one *Escherichia coli* centered and recorded how far it had to move, which gave them the three-dimensional track of a single cell. The results went into *Nature* that year, volume 239, pages 500 to 504.

A cell swims in nearly straight lines, which they called runs, interrupted by brief episodes of jittering in place, which they called twiddles, after which it sets off in a direction unrelated to the one it came in on. In plain water that is a random walk. Put a gradient of something edible across the chamber and exactly one thing changes: a run that happens to be pointed up the gradient goes on longer. Runs pointed the wrong way come out at ordinary length. The cell has no steering at all. It has a brake. It declines to apply the brake while things are improving.

A bacterium cannot read a gradient across its own body, because the difference in concentration between its two ends is far below the noise of molecules arriving at random. Robert Macnab and Daniel Koshland showed what it does instead, in the *Proceedings of the National Academy of Sciences* in September 1972. They built an apparatus that mixed a cell suspension abruptly out of one uniform concentration into another, so that the population experienced a change in time with no gradient in space anywhere in the vessel. The cells responded anyway. A sudden drop set them tumbling, a sudden rise sent them swimming smoothly, and what the bacterium is reading is therefore its own record: what the seam says against what the seam was saying a moment ago.

Both budgets are open to that cell, and it uses both. It can retune its own detectors, and it does: after a step up in attractant, the receptors are chemically modified until the same concentration stops counting as news and the tumbling rate comes back to where it started. That is capacity spent inside the boundary. The sugar in the chamber is precisely where it was. Or it can swim, which puts its boundary against a different piece of the world, so that the entry on that boundary comes out different because the boundary is elsewhere.

Swimming is expensive in the only currency the cell has. The flagellar motor is turned by ions falling across the membrane, and that gradient is the same one the cell draws on to build everything else it builds, so a run is charged to the account that would otherwise have bought the next protein.

A company faces the identical pair. It can work on the product, which is capacity spent inside the boundary and leaves what the market says untouched on the day. Or it can buy entries in the shared account that customers, competitors and regulators all read from, which is what an advertising campaign is. Both are expensive and both are ordinary, and the two come out of different accounts: a firm that does more of the first does not thereby obtain any of the second.

A bacterium declining to tumble and a company buying billboards are the same move at different scales under different budgets.

37.5 Four states

Strip the arrangement to the smallest thing on which the two budgets can be told apart at all.

Take an observer with two registers, each holding one bit. One of them sits on the overlap: it is the shared record, the entry a neighbor can read. The other is private. What it holds is a flag, set when the observer's own state fails to match what the shared record says and clear when it matches. Four states in total, of which two are consistent, being the two values of the shared record with the flag clear.

Two moves exist. The private move clears the flag and touches the shared record not at all, which is self-repair in its smallest possible form: the observer bringing itself into line with what the seam says. The network move writes the shared record, so that what both sides read comes out different.

A rule for picking moves, step after step, possibly at random, is a scheduler. Constrain it three ways, all of them constraints the repair law carries anyway: it never increases the mismatch, it writes at most one register per step, and it spends at most one unit of capacity per step. Two sub-families inside that class do different things. A self-update-only scheduler never writes the shared register. A network-update-only scheduler never writes the private one.

Start in a consistent state, with the shared record holding one of its two values. Ask for the other one: the state in which the shared record holds the opposite value and the flag is clear.

Three facts hold about that target, on a system with four states in it.

A self-update-only scheduler never arrives. The failure has nothing to do with budget or patience. From every consistent start, over any number of steps, for every member of that sub-family, the probability of arrival is zero.

An explicit network-update-only scheduler arrives with probability one.

The third fact prices the other two. Any scheduler in the class, whatever its rule, that has a positive probability of arriving writes the shared register somewhere along the way. A positive chance of getting there costs at least one shared write. There is no arrangement of the private budget that supplies it.

The reason for the first of the three is one line. It is chapter five's word doing the work. A private move leaves the shared register alone. Compose any number of moves that each leave a quantity alone and the quantity is left alone, so the value of the shared register is an invariant of the whole self-update-only sub-family, and every state that sub-family can reach carries the value it started with. The size of the budget never appears in that argument, which is why enlarging the budget does nothing to the conclusion. The private register can be cleared, set, cleared again and interrogated for as long as anybody has patience, and the neighbor reading the seam sees the entry it saw at the beginning.

The two budgets do not buy the same things. The demonstration fits on four states, which matters more than a large example would, because a system with four states has nowhere for an inefficiency to hide. The gap between the budgets is not a shortfall in the private one that a larger private budget would cover. There is a class of outcome that self-repair does not reach at any size. The only thing that reaches it is expenditure on the record two observers hold jointly.

Every bounded system that persists therefore runs into that class. Bernard's rabbit can convert its own body into blood sugar for days and hold the internal environment at the value he measured, and the entry on the seam between the rabbit and the room, which is that there is nothing in the room to eat, comes through the whole performance unaltered. A bacterium that has finished retuning its receptors reads zero mismatch while sitting in a patch of water with no food in it. Perfect internal adjustment and starvation are compatible states. A policy that spends only on the first arrives reliably at the second.

The company gets the same result with the accounts written out. A firm can spend a decade making its product better and arrive at a market that has heard nothing about it, because what the market holds is an entry the firm never wrote to, and that entry carries the value it carried before. Which is why the workshop and the billboard both exist, and why the division between them is fought over every year in every firm that has one.

37.6 The entry is one entry

An entry on a seam is one entry. It is read by both patches, owned by neither, and holds a single value at a time.

Two observers spending on the same entry are spending against each other. Whatever stands written there when the spending stops is what both of them have to live with. Each brings a finite budget to the argument. A bacterium in a gradient is in that position with respect to every other bacterium in the same drop of water, which is why the gradient it is climbing is partly a record of what the others have taken out of it.

That is a competition. It is settled by which pattern goes on having a consistent extension afterwards, and was running on the thirty seams of a twelve-patch net long before anything with a flagellum turned up to enter it.

Chapter 38

Why Is a Bacterium Freer Than an Atom?

Every morning somebody in a laboratory lifts twelve glass flasks off a shaker and moves one hundredth of the liquid in each of them into a fresh flask of the same clear solution. The used flasks go into the autoclave. The new ones go back onto the shaker at 37 degrees Celsius for the next twenty-four hours. The job takes a few minutes, and somebody has done it every day since 24 February 1988.

Each flask holds ten milliliters. The entire experiment, all twelve populations of it, comes to a hundred and twenty milliliters, which is less than a glass of wine.

What is in the flasks is *Escherichia coli*. On the first day all twelve held the descendants of a single cell. Six of them were founded from that cell's line and six from a variant of it differing by one change that does nothing in this medium except make its colonies come up a different color on one kind of indicator plate, which is how a contaminated flask gets caught before it ruins anything. The solution carries twenty-five milligrams of glucose in every liter and nothing else these cells can eat. It also carries citrate, about twelve times as much of it as glucose by count of molecules, which is in the recipe to keep iron in solution. *E. coli* cannot use citrate as a carbon source when oxygen is present. Handing a bacterium a tube of citrate and watching whether it grows is one of the routine ways of telling *E. coli* apart from the things that look like it under a microscope, so the inability belongs to the species rather than to these twelve flasks. The test is on the identification key, and a strain that passes it gets filed as something other than *E. coli*.

The daily transfer carries over one percent. Overnight that one percent eats the glucose and grows back a hundredfold, which is six and two thirds doublings, so a day in the flask is six and two thirds generations and a year is roughly two thousand four hundred.

Nobody in that laboratory picks which cells get transferred. The pipette takes a hundredth of whatever is in the flask and cannot tell one cell from another, and the person holding it has no way of knowing which lineage is which until long afterward, when somebody sequences something. Yet the flask holds a different population every year. The differences run in a consistent direction. Something settles which lineages are in there. It is not in the room.

There is a second question in that flask, about one cell rather than about a population. A bacterium has options. It can put the next minute's energy into dividing or into repairing itself, it can swim toward the sugar or sit, it can carry the machinery for eating something and switch it off. A hydrogen atom in the same flask has nothing of the kind. The atom came first and has fewer moving parts, and less room to be otherwise than the bacterium does, which runs backwards from the way the world is usually described.

38.1 Three things, and no fourth

Something copies itself imperfectly. The imperfection is passed to the copies. Some versions end up with more copies than others.

That is the whole mechanism. Each of the three parts is visible in the flask. Copying is imperfect: over thirty thousand generations, each of the twelve populations tested billions of mutations, which is what happens when a population of hundreds of millions divides six and two thirds times a day for decades. The imperfection is heritable: a daughter cell gets its mother's copy of the genome, errors and all. And the versions do not fare alike, because the glucose in a fresh flask is gone within a few hours, and a lineage that got through more divisions before it ran out is a larger fraction of what is sitting there when the pipette arrives in the morning. A larger fraction of the flask is a larger fraction of the hundredth that gets moved.

The name for those three conditions running together is **natural selection**. There is no fourth item in which anybody decides anything. Nothing in the flask is aiming at a result, no lineage is trying to win, and the only agency in the building is a plastic tip that samples without discrimination. The sorting has finished before anybody arrives for the morning transfer.

"Fitter" is a bookkeeping word here and carries none of its usual freight. A fitter lineage holds a larger share of tomorrow's flask than of today's. That exhausts the meaning. The word predicts nothing about the day after tomorrow, because the flask the lineage will be competing in is the flask it is in the middle of changing.

38.2 The freezer

Richard Lenski set the twelve flasks going at the University of California, Irvine, with two thousand generations in mind, which at six and two thirds a day is about ten months of transfers. He moved the populations to Michigan State in 1991 and they carried on there. The organism he picked reproduces without sex, so every lineage is an unmixed line of descent and no gene ever arrives sideways from a neighbor. It also survives being frozen.

Every five hundred generations, which comes to every seventy-five days, the part of each population that does not go into the fresh flask gets a dose of glycerol and goes into a freezer at minus 80 degrees Celsius. The cells come through it alive. Thaw a vial and the population resumes, from that day, with the mutations it had accumulated by then and none of the ones it acquired afterward.

Twelve populations, one ancestor, and a shelf of vials sampled every seventy-five days for decades. It is a fossil record whose fossils are not dead, which makes this the only evolutionary experiment anybody can run twice.

The freezer earned its keep early in 2020, when the populations had passed seventy-three thousand generations and the laboratory shut. The flasks went into the freezer. In September the experiment came back out of it and carried on from where it had stopped. The rest of the time the freezer is doing something stranger. Thaw a vial from before an event and run the population forward again, and whether that event had to happen becomes a question somebody can answer by counting.

The one quantity this experiment reports comes out of the freezer as well. To find out how much better a population has got, thaw the ancestor, mix the two in equal parts in a fresh flask, and plate a sample at the start and again a day later. The ancestor's colonies come up one color and the competitor's come up the other, so counting red against white on a plate gives the ratio before and the ratio after, and the change in that ratio over one day is what the word fitness denotes here. Every measurement of it is a competition against cells that were frozen in an earlier year and thawed for the purpose.

Michael Wiser, Noah Ribeck and Richard Lenski ran that measurement across all twelve populations through fifty thousand generations and published the trajectory in *Science* on 13 December 2013. Mean fitness increases without bound, consistent with a power law, with no sign of leveling off after twenty years of it. Chapter thirty-five's argument said that mean fitness never decreases in a population sorted this way, and that the climb stops once the survivors stop disagreeing. A flask never runs out of disagreement, because copying keeps making more of it. Twelve flasks on a shaker in Michigan are where somebody went and measured it.

38.3 Citrate

For more than thirty thousand generations, the citrate sat in all twelve flasks, twelve times more abundant than the food, untouched. Then one of the twelve, the population known as Ara-3, started eating it. By generation 31,500 there were cells in that flask growing on citrate in air, which is the thing the diagnostic test says *E. coli* does not do. The population got larger, because there was suddenly far more carbon available to it, and it got more diverse, because the cells that could not use citrate were doing something else for a living and kept doing it.

Zachary Blount, Christina Borland and Richard Lenski published the result on 10 June 2008, in a paper called "Historical contingency and the evolution of a key innovation in an experimental population of *Escherichia coli*". It was Lenski's inaugural article for the National Academy of Sciences, the paper a newly elected member writes on a subject of their own choosing. Most of it is taken up with the experiment Blount ran after the announcement.

Two explanations were available for why one population out of twelve did this after thirty thousand generations and the other eleven did not. Either the mutation required is astronomically rare, in which case any of the twelve could have got it on any day and this one was lucky, or the mutation is ordinary and works only in a cell that carries some other change first, in which case the population had to build up to it.

Those two stories make different predictions about the freezer, and the freezer answers. Blount thawed vials from across that population's history and grew enormous numbers of cells from each of them. From the ancestor, not one citrate-eater among 8.4 trillion cells. From sixty clones sampled across the first fifteen thousand generations, none among nine trillion. From clones taken later, citrate-eating appeared repeatedly and significantly more often than chance allows, which puts the arrival of the enabling change at around generation twenty thousand. That change left the cells no more error-prone in general. It made this one step reachable.

Run the population from the early part of its own past and it cannot get to citrate. Run it from a later part of the same past and it can, over and over. What a lineage is able to become is set by what that lineage has done. The set gets bigger as the history gets longer.

38.4 One budget with two halves

Chapter thirty-seven left a living thing as a pattern that keeps its public reading extendable at its own expense, and behavior as a policy over the two ways of spending: on itself, or on the record it shares with a neighbor.

A pattern persists by being re-asserted into the record every cycle, and a record costs capacity, and the capacity is finite and shared. Two patterns whose re-assertions cannot both be extended cannot both persist. Which of them does is settled by which one keeps having an extension. Nothing about food or territory is needed to get this started. The scarcity is in the bookkeeping, and would be there in a universe with unlimited everything else.

Chapter thirty-five ran that argument on laws, and got a filter with no chooser in it: what survives is what closes, and the share held by the descriptions that close does not decrease. This is the same argument one level down, with organisms where the laws were. Selection at the level of physics and selection at the level of bacteria are one filter applied at two grain sizes. The finer one inherits its shape from the coarser.

A pattern can spend on either side of the split. The two sides buy different things.

Spend everything privately and you get homeostasis: an interior in excellent order, every internal reading consistent, and an overlap you have conceded to whatever else is spending out there. That is a stable arrangement right up until the neighborhood's reading stops extending yours, at which point the beautifully maintained interior is a private fiction with no public correlate, and the pattern is finished, in perfect health.

Spend everything on the overlap and you get domination: the neighborhood rewritten in your terms, every seam carrying your version, and an interior that has gone unrepaired for as long as the spending has run. The reading you are imposing has less and less behind it. A reading with nothing behind it fails the first check that reaches for the state underneath.

So fitting in and crowding out are one expenditure seen from the two sides of the split. Nothing that persists optimizes for either alone. The cells in those flasks do both, all night, and the boundary between the two runs through individual molecules. Refolding a protein and proofreading a copy of the genome are interior work. Eating the glucose is overlap work, because what a lineage does to the flask is what every lineage in it, its own descendants included, has to live in tomorrow. Pushing a proton out through the membrane is both at once: the cell has kept its own interior in range and made the flask more acidic for everybody. The citrate-eaters are the cleanest case, because the mutation is in the cell and the consequence is in the flask: they opened a carbon source nobody was using, and the whole population, competitors included, got bigger.

The harder result in chapter thirty-seven was that some outcomes are reachable only by spending shared capacity, and that no amount of private self-repair arrives at them at any budget. The flasks are what that looks like when it runs for decades. A lineage cannot reach citrate by getting better at being itself. It gets there by changing what the flask contains, which is a change to everybody's record, and then living in the flask it changed.

Different lineages settle on different policies, in flasks prepared from the same recipe on the same bench. Paul Sniegowski, Philip Gerrish and Richard Lenski reported in *Nature* on 12 June 1997 that lineages which copy themselves less accurately had arisen on their own in some of the twelve populations and spread. Sloppier copying is a strange thing to spend an interior budget on, because most mutations are damaging and the lineage pays for every one of them. It spreads anyway, because a sloppier copier reaches the useful change sooner, and in a flask where everything is sampled once a day, sooner is the whole of the argument. Twelve flasks, one ancestor, and the populations do not agree about how carefully to be themselves.

38.5 What a hydrogen atom cannot do

Ask how much room something has to be otherwise, and count it at each level.

Chapter fourteen counted the ports: twelve, with thirty seams between them, on the declared carrier branch. Chapter fifteen found a rank-three candidate position band, with the other bands fading faster; its physical-space interpretation remains a separate receipt. Chapter sixteen found an informational arrow: one way round, because a commit discards what it resolved. Identifying that arrow with physical causality in an early universe is another step.

Those three were settled with no history available to settle them. There was no record, nothing had happened, and the conditions had exactly one solution. A universe with eleven ports never lost a competition to a universe with twelve. Eleven ports does not survive the counting, so there was never an eleven-port world for the twelve-port one to beat.

The early universe is therefore the most constrained thing there has ever been. Every question about it that has an answer has a forced answer. The forcing comes from its age rather than from its simplicity. Nothing had been written down.

Then things start getting written down. Chapter eleven's result was that the normal form is unique given the boundary data, which is to say that different records on the boundary settle into different worlds. Boundary records are made, one commit at a time. A commit is a fact that the closure conditions did not supply and could not have supplied. Every one of them enters as an input to what comes next. So the fraction of what a thing is that traces to its own history rather than to the conditions that permitted anything at all rises with every cycle, and it rises fastest in the patterns that are making the most records.

Look at a hydrogen atom with that in hand. Its ground state is one state. Two hydrogen atoms in it are the same object in the strong sense chapter twenty-seven needed for the exclusion argument: there is no record anywhere that distinguishes them, and the absence is not a limitation of anybody's instruments. It has no policy over the direction split. Holding that pattern together costs what chapter eighteen said it costs, a standing bill re-asserted every cycle, and the bill has exactly one line on it and no way to move an amount from one line to another. Put the atom in a field and its response is fixed by the same conditions that fixed the number of directions the field can point in. A hydrogen atom has no strategy available to it. There is no record in it that could carry one.

Two *E. coli* cells in the same flask at generation 31,500, both of them descended from the cell that went into a flask in February 1988, differ. The difference sits in their records, it is heritable, and one of them can eat something the other cannot. That lineage carries a policy and spends it every cycle: divide or repair, swim or sit, keep an unused piece of machinery running or shut it down and save the cost. Each of those is a choice about which half of the budget to spend on, and each of them is available only because of a long run of commits in that lineage's past that could have gone otherwise.

Freedom, if the word is going to mean anything measurable, means room to be otherwise, and room to be otherwise accumulates. It is the residue of history rather than a late arrival with brains. There is more of it the further down the tree you go.

38.6 The view from inside

Chapter six wired twelve observers into a closed surface and ran every one of their two thousand and forty-eight states through sixteen schedules apiece. Every run ended in the same arrangement. Nothing chose that arrangement and nobody voted on it. It was the one configuration that satisfied every seam at once. The observers arrived at it by repairing whatever was in front of them.

Chapter eleven gave that its general form. A reading has a fiber, which is the set of states consistent with it. A reading whose fiber is empty belongs to no world at all. Nothing rejects such a reading. There is nowhere for it to be. Rejection would need a judge, and having nowhere to be needs nobody.

Seen from outside, the twelve observers are a solve: a system with a unique consistent state settling into it. There is no outside. The only available view is the one from within, held by the patterns that came through, and from there it looks like something else entirely. Some patterns are here and others are not. The ones here are the ones whose readings kept having an extension. The missing ones ran out of world to be consistent with.

That is selection, and needs no agent, because differential persistence is the whole requirement. Lenski's laboratory has a pipette that cannot tell one cell from another. The universe contains no sampling device of any kind.

So the filter and the fixed point are one object seen from two positions. One of the two positions is not available to anybody. Selection is what a fixed point looks like from the inside, when the patterns competing to be in it are the ones doing the looking.

Twelve flasks, a hundred and twenty milliliters, one ancestor, and a freezer full of its own past. Nothing in that arrangement wants anything.

The same is true of the arrangement the flasks are sitting in. One of the patterns that came through the filter has hands, and a laboratory, and a habit of writing things down, and is holding this book.

Chapter 39

Does It Feel Like Anything, and What Is Left to Decide?

The first big interdisciplinary conference on consciousness was organized by an anesthesiologist.

Stuart Hameroff worked in anesthesia at the University of Arizona Medical Center, and in April 1994 he convened a meeting in Tucson called Toward a Scientific Basis for Consciousness, which put philosophers, neuroscientists and psychologists in the same rooms for a week. He spent his working days putting patients under and waking them again. The meeting was about what consciousness is.

The talk people were repeating in the corridors came from a philosopher a year out of his doctorate, holding a fellowship in a philosophy, neuroscience and psychology program at Washington University in St. Louis. David Chalmers opened by splitting the subject in two. The split was a list.

How a system discriminates between things around it, categorizes them and reacts. How it brings information together from separate places. How it reports its own states in words. How it gets access to its own internal states at all. How attention gets focused on one thing rather than another. How behavior gets deliberately controlled. What separates being awake from being asleep.

Seven items, and he called them the easy problems of consciousness, with a warning attached to the adjective: getting the details right, he wrote, will probably take a century or two of difficult empirical work. They count as easy because in every case anybody can say what an answer would look like. It would be a mechanism. Specify the machinery that retrieves information about internal states and makes it available for verbal report, and reportability has been explained.

Then the other one. When our cognitive systems engage in visual and auditory information-processing, he wrote, we have visual or auditory experience: the quality of deep blue, the sensation of middle C. Thomas Nagel had supplied the standard phrase for it twenty years before, in the October 1974 issue of the *Philosophical Review*: an organism is conscious if there is something it is like to be that organism. Why physical processing should give rise to any of that seems objectively unreasonable, Chalmers wrote, and yet it does. The talk went into the *Journal of Consciousness Studies* the following year, volume 2, pages 200 to 219, under the title "Facing Up to the Problem of Consciousness". The phrase the hard problem became the standard name for that question.

Descartes had a physical world made of extended stuff and a thinking thing that occupies no place at all, and his trouble was traffic between the two: how a decision taken in something with no location moves an arm that has one. He put the meeting at the pineal gland, on the grounds that every other part of the brain comes in pairs and that one does not, so it was the only place the signals from two eyes and two ears could be brought together into one impression. Every version since has kept the shape and dropped the gland. There is an account of the world written in the third person, there is what it is like to be one of the things in it, and nobody has a rule that turns either into the other. The name for that is the **mind-body problem**.

Chapter three was blunt about the word observer. An observer is any bounded system that keeps records, a thermostat is one, and consciousness had nothing to do with the five items on the list. It also made a promise about the conscious ones: when they arrive, they will not arrive as a sixth item.

One of these bounded record-keepers is holding this book. There is something it is like to be it. Where in the architecture does that sit? And if the network settles into the same arrangement whatever order anybody repairs in, what is there for that reader to decide?

39.1 A creature molecule for molecule identical to you

Chalmers put the sharpest form of the first question two years later, in a book called *The Conscious Mind*. Consider, he wrote, his zombie twin: a creature molecule for molecule identical to me, lacking conscious experience entirely. Same cells, same firing, same words coming out of its mouth, and nobody inside.

The argument runs three lines. Such a creature is conceivable, and Chalmers set the standard in those words: it certainly seems that a coherent situation is described, and I can discern no contradiction in the description. Whatever is conceivable in that sense is possible. And if a world physically identical to this one can lack experience, then experience is something over and above the physical facts, and physics by itself does not reach it.

The twin writes the paper about zombies. It sits down, works through the conceivability argument, publishes it in a journal and answers the objections. There is nothing it is like to be the author. Chalmers grants that without flinching, because the twin does everything he does.

Look at the operation the argument asks for, one step at a time. Take the complete physical description of a person. Remove the interior. Hand back what is left and call it the same physical situation.

Everything rests on the first step, the step nobody pauses on, because taking a complete physical description sounds like the part you get for free.

39.2 The mechanical properties of the ether

Chapter one's crash log had an entry for the substance light was supposed to wave in, rigid enough to carry a wave at the speed light travels and insubstantial enough for a planet to cross without losing speed. The nineteenth century held both properties rather than drop the idea that a wave needs something to wave in. The interferometer in the Cleveland basement went looking for the Earth's motion through it in 1887 and found nothing, and forty years of better instruments found nothing again.

Nobody ever caught the ether not being there. Einstein took the null result as the law rather than as the disappointment, and after that every question the ether had been introduced to answer had an answer without it. Its stiffness explained nothing, because nothing needed explaining by a stiffness. Its density had no consequence anywhere. The ether was made redundant, which is a more complete failure than going undetected, because a thing nobody detects may have effects that need accounting for.

Take the sentence "what are the mechanical properties of the ether?" It has a subject and a verb and it never got an answer. It stopped being a question, because one of its words stopped naming anything. A question with a term that names nothing does not have a false answer either.

Two things can happen to a question, then. It gets answered. The question stands with a fact attached to it. Or one of its terms is shown to name nothing. The question goes away and stops consuming effort. The second is called **dissolving** the question. It happens rarely, and from the outside it looks the same as a refusal to answer.

The hard problem asks how experience arises from the physical facts, where the physical facts are meant as an inventory belonging to nobody: a list of what there is, compiled from no position. Go looking for that list and it fails to turn up, one entry at a time.

Distance counts comparisons. There is no comparison without two parties each holding something. A rate needs a clock, and chapter twenty-one left no clock that is not somebody's. Charge is read off a loop that runs through one observer after another. A record sits in the center of one particular algebra of questions. Even the leftover of exactly one, sitting on a twelve-patch net with one flipped edge, is a quantity that every observer in that net agrees about, which is a different thing from a quantity nobody holds.

Objective, in the sense chapter eleven fixed, means settled between parties who each have a boundary and a budget, with no further list standing behind theirs for it to approximate.

So the hard problem asks how somebody's readings arise from a list that is nobody's reading. One of its terms names nothing. It goes the way the ether question went.

The zombie is the same move at the size of one person. Chapter eleven said exactly what an observer's interior is: the part the record map deletes. Alice's coin. Real, hers to consult at any time, and off the shared list because there is no check she and a second party could run on it together. The deletion is what makes the list shared.

Take the interior out and what is left is a seam with one side missing. The entries on that seam were entries about a relation between two things that each held something. One of them holds nothing. The behavior does not survive the operation unchanged, because the behavior was the boundary traffic of a thing with an inside. The zombie ends where the ether ended, with nothing left to do.

39.3 Coordinates an observer reads on itself

Chapter thirty-seven got an observer down to two registers, which is the smallest arrangement in which its two budgets come apart. A neighbor reads the first one. The second holds one bit saying whether the observer matches what the seam says about it.

What that private register holds is the observer's reading of its own relation to the world. The observer's next move is taken on the strength of it. Nothing outside can consult it, and no repair anywhere in the network reaches it.

A system carrying readings of its own update state is doing something a photographic plate does not do. It divides itself into parts and reads the division. The reading goes to its own repair policy and nowhere else. Give it more registers and the division gets finer. A stable cell of that division is a coordinate the observer can locate itself along and repair itself with.

Redness is one such coordinate. An ache in a knee is one. The state of having understood something and the state of being about to be wrong about something are two more, and people report both with the confidence they report colors with and with the same inability to hand the thing over.

Reading such a coordinate is the whole of having it. There is no second copy of the ache, kept elsewhere, for the reading to be a reading of, and no fact about whether the one you have is the genuine article.

Consciousness is owned boundary repair, read from the inside.

Owned, because the capacity being spent belongs to the observer and comes out of its own budget. Boundary, because what is being held is the interface between it and everything else. Repair, because the holding is a rate rather than an act, since the neighbors keep writing and the entries keep moving. Read from the inside, because the observer's records include readings of its own state, and those readings are the half of the record the map deletes: real by the same standard Alice's coin is real, and unpublishable by construction.

Nothing in that sentence adds to chapter three's list. A boundary, a place to write, updating, comparing, and sticking around between comparisons. Five items, and no sixth. A thermostat has all five and reads exactly one coordinate on itself, which is why nobody wonders what it is like to be one. What separates a person from it is how many coordinates, how finely the division runs, and how much of the repair policy is computed off them. The difference is one of degree with an enormous number sitting in the middle of it. The number is biology's to measure.

Which cells implement which coordinate, why activity in one part of the visual cortex reads as color while activity a few centimeters away reads as motion, what a stroke takes away when it takes a coordinate away, how many coordinates a mouse reads on itself, and at which point in an anesthesiologist's induction a person stops reading any, are questions for people with electrodes and scanners.

No instrument reads a coordinate from outside, and none could be built. A coordinate readable from outside would be a seam entry, and would have sat on the shared list from the beginning.

39.4 Three cells down the diagonal

Ask somebody what red is like and everything comes back except the thing you asked for. A wavelength. The name the paint shop prints on the can. Both of those cross the gap between two people intact. Neither answers the question. The word for what failed to cross is a **quale**, plural qualia, meaning how a particular experience is for the thing undergoing it, as against the wavelength, the firing pattern or the word.

Every quantity in the preceding chapters is somebody's readout, taken at a bounded interface. A readout comes in two parts. Chapter nineteen wrote an observer's questions as tables of numbers, so a readout that tells three things apart is three rows by three columns. Nine cells. Three of them lie down the diagonal, one for each alternative, and those three commute with each other, which is what makes a record a record: readable without disturbance, and the same again in a neighbor's notebook. The other six sit off the diagonal. They are what makes the order of two questions matter, and chapter twenty-one found the observer's entire supply of rate in them, because the flow leaves the record layer where it finds it and moves only the part that refuses to be answered alongside itself.

Do that count at three other sizes, on the back of an envelope, because the numbers are small enough. A thermostat tells two things apart: four cells, two down the diagonal, two off it, and two is the only size at which the two parts come out equal. Take it to ten alternatives. Ten rows by ten columns is a hundred cells, the diagonal takes ten of them, so ninety are off it. Take it to a hundred. Ten thousand cells, a hundred on the diagonal, nine thousand nine hundred off. That one step, ten alternatives up to a hundred, multiplied the publishable part by ten and multiplied the rest by a hundred and ten.

One line covers every size, with N standing for the number of alternatives a readout tells apart.

$$\text{cells on the diagonal} \;=\; N, \qquad \text{cells off it} \;=\; N^2 - N$$

The diagonal count keeps pace with N and the other one climbs with N squared, so the share of a readout that can be handed to anybody is one cell in N. Push it to a thousand alternatives and work that share out: a million cells, and the thousand sitting on the diagonal are the whole of what leaves the observer.

A quale is the part of a readout that never gets committed, held by the observer whose interface it is. Which cells count as the diagonal is fixed by that observer's own record layer, so the division is taken in its terms and nobody else's.

The reason it does not travel is the reason nothing uncommitted travels. Communicating a quale would mean copying it, and chapter twenty settled what copies: a duplicator preserves overlaps and squares them in the same move, so it works at overlap zero and at overlap one and nowhere in between, and two distinct committed records sit at overlap zero. What is copyable is the record layer, exactly and only, and a quale never entered it. So the wavelength duplicates and the redness does not. The gap between the two of them is a fact about copying rather than a mystery about matter.

Look at what none of that required. A theory that begins with a stage and puts matter on it has to add experience later. The addition is where the hard problem lives, because nothing about a stage says what would switch the lights on or when. Nothing was added here. An observer is a bounded interface that reads, compares and writes, and a structure with no observer anywhere in it holds no records and has nothing that could be true of it. Experience is the material the rest was built out of rather than a late achievement of complicated matter. The question with content in it is why some of it grows elaborate enough to fill a room in Tucson.

39.5 Then nothing you do matters

Somebody is sitting up at night working out whether to say the true thing or the convenient one. Three results say the sitting up changes nothing.

Chapter six started a twelve-patch network in every one of its two thousand and forty-eight distinguishable states, ran sixteen repair schedules on each of them, and ended in the same arrangement every time. Chapter ten ran the same process on eighty-two thousand patches, where 102,415 disagreements came down to none under every order the repairs were run in. Chapter thirty-six showed that a run cut across a thousand machines settles where the single-machine run settles, and that the partition, the worker count, the schedule and the restart history reach no observable anywhere.

The endpoint does not depend on where the system started. It does not depend on the order anybody repaired in. There is no clock outside it, so there is not even a sense in which the endpoint is waiting its turn. So the person sitting up at night is a subsystem executing a delay, everything between here and the endpoint is theater, and the arrangement the whole thing lands in carries the same entries either way.

This is a harder version of fatalism than the one usually argued about. Ordinary determinism concedes that the future depends on the initial condition, which at least leaves the past doing some work. Here the initial condition washes out as well. Two thousand and forty-eight starting states, one endpoint.

39.6 Who writes the boundary

Chapter six gave twelve observers a coin each and thirty seam records saying whether two coins matched. The repairs inside that net move coins and leave the records standing, so where the coins stop is a function of the records handed over before anybody moved, from the first repair to the last.

Chapter eleven's theorem carries a qualifier. The fatalism above works by dropping it. The theorem returns one endpoint given the boundary data, and a region's boundary data is the entries on the seams where it meets what it is not, which the patches inside can read and cannot set. Regions that start anywhere at all with the same edge entries finish in the same place. Regions with different edge entries finish apart, with nothing to choose between the two endpoints, since both satisfy every record inside.

Drop the words "given the boundary data" and out comes the fatalism above. Keep them and boundary records vary, so "the endpoint" does not pick out anything until somebody has said which entries are sitting on the edge.

Chapter six checked that on twelve patches. Thirty entries, all satisfiable, and the twelve coins have exactly one setting up to the flip nothing can see.

Change one entry out of the thirty, from same to different, and the number of consistent settings drops to zero and what the machine stops on carries a leftover of exactly one. One record, and the coins stop somewhere else.

So where do entries come from? The patches inside a region read its edge and cannot set it, so the write happens at the seam itself, between the two observers who share it. Chapter thirty-seven named the move. A private self-update spends the observer's own capacity, clears its own mismatch, and leaves the shared entry saying what it said going in. A network update writes the shared entry. Each side then reads a different value. On the four-state system where the two can first be told apart, no self-update-only scheduler ever reaches the state where the shared entry has moved, at any budget, over any number of steps. A network update reaches it with probability one, and any scheme with a positive chance of getting there writes the shared register on the way.

The shared entries are the boundary data. The move that writes them is the move no quantity of private effort substitutes for.

Say what those writes are in a person's vocabulary.

What to admit, which is what you let across your boundary from outside and allow to change your state. What to emit, which is what you write onto the seams you are one side of. What to repair, which is where a finite budget actually goes. What to ignore, which costs nothing to decide and shows up in the record as an entry left standing. What to confess, which is putting onto the shared record something only you were holding. What to protect, which is spending your own capacity so that somebody else's reading keeps having a consistent extension. And what to amplify, which is writing an entry onto seams that had no opinion about it until you arrived.

Those seven are the boundary data. They are the input the theorem takes, and they are the reason it returns a different world on different days.

So free will is self-reading participation in repair. An observer that reads coordinates on itself, acts on the readings, and writes the results onto seams it shares with others is a place where the boundary record gets set. The convergence does not run around such a thing and it does not suspend itself on request. It runs through. The entry it reads on the way past is the entry that observer wrote.

39.7 Pavia, 524

Boethius was consul in 510 and master of the offices under Theodoric, and in 522 his two sons held the consulship together, which is the top of a Roman career of that period. He then defended the senator Albinus, who was accused of writing to the emperor Justin in Constantinople against Theodoric's rule. He was taken to Pavia, held there, and executed in 524.

In the prison he wrote the *Consolation of Philosophy*. Its fifth and last book is about the question above.

He puts it in the vocabulary of divine foreknowledge. If the whole order is possessed at once, entire, by something that does not take it a piece at a time, then it is settled, and a settled order leaves nothing for anybody inside it to decide.

The answer has two parts and both survive translation out of the vocabulary.

The first is a distinction between two necessities. One is simple: men are necessarily mortal. The other is conditional: if you know that someone is walking, he is necessarily walking. The second attaches to the act only while the condition holds, imported by the knowing rather than produced by the walking. Then the test question, which Boethius puts directly: does the act of vision add any necessity to the things you see before your eyes? Watching a man cross a courtyard does not make him cross it.

The second part is his definition of eternity: the possession of endless life whole and perfect at a single moment. That is chapter thirty-six's first order, written down in a cell in Pavia. A structure with no ordering inside it is not a very long time. It is no time at all, nothing in it is earlier than anything else, and nothing in it gets to be the reason for anything else by having come first.

Fourteen centuries later the requirement itself was taken apart, in eleven pages. What most people mean by freedom is that you are responsible for what you did only if you could have done otherwise. Harry Frankfurt attacked that sentence in the *Journal of Philosophy* of 4 December 1969, in a paper called "Alternate Possibilities and Moral Responsibility", pages 829 to 839.

His case is a piece of machinery. Somebody, call him Black, wants Jones to do a particular thing, and is prepared to step in and make him do it if Jones shows any sign of deciding otherwise. Jones decides to do it on his own. Black watches and never moves. Jones could not have done otherwise, because Black stood by with the means to prevent it. Jones is responsible anyway, because what he did came out of his own deliberation and Black played no part.

What carries the responsibility is therefore the structure of the will that produced the act and the record of the act itself. Whether some other setting of the boundary was reachable is a question about the space of records. Which entry got written, out of which observer's own repair policy, is a question about this record. This record is what everything downstream is computed from.

39.8 It was going to happen anyway

The fatalism above has a second life as an excuse.

Somebody writes an entry onto a shared seam and leaves the repair bill with whoever is on the other side. Asked about it afterward, the sentence available is that the network converges regardless, that the endpoint was going to be the endpoint, and that nothing turned on the write.

The structure blocks that, and it blocks it with arithmetic rather than with disapproval.

Chapter six's flipped edge is the clean case, because the whole net can be checked. Two of its nineteen independent loops carry a residue, no setting of the twelve coins satisfies all thirty records, and the machine converges anyway, from all two thousand and forty-eight starts and under all sixteen schedules, onto an arrangement carrying a leftover of exactly one. Convergence is guaranteed and so is the size of the leftover.

Where it sits is a different matter. Run the repairs in another order and the leftover ends up on a different part of the net. It survives any amount of working harder, and it can be placed, and which patch is carrying it when the repairs stop is settled by who repaired what, in which order, and which way each one pushed the disagreement it found.

That trajectory is in the record. Chapter sixteen made a commit irreversible by arithmetic: seven arrangements at a seam average down to one, and what told them apart is gone from the state and written into the order of commits. So a record holds more than its endpoint. It holds the descent. The descent tells apart a patch that absorbed a mismatch at its own expense from a patch that wrote the mismatch onto a neighbor and let the neighbor pay for it.

The leftover comes to one either way. The record that produced it names which patch absorbed the mismatch and which one exported it. The naming is no moral gloss applied afterward by somebody keeping accounts. It is the entries, in order, which is the only material anybody downstream has to compute from.

The excuse also needs the endpoint to be the kind of thing that could have caused something, and chapter thirty-six wrote the endpoint down. Take the whole structure, take the record it holds, look up the world that record names, and demand that what comes back is the structure you started with.

$$\mathrm{world}(R(W)) \;=\; W$$

W is the whole structure, R is chapter eleven's record map, which returns the part of a state that survives comparison, and world is the lookup running from a record to the consistent state that record names. The unknown sits on both sides, which makes this a fixed point rather than a recipe.

A fixed point has no first solver. Nothing evaluates one side and arrives at the other, because an equation contains no step and nothing that goes first. So “the endpoint was going to be the endpoint” names a solution and uses it as a cause. Nothing in a solved equation pushes.

What an observer writes lands in the record R returns. The entries it puts on the seams it touches are part of the record the lookup gets applied to, which makes them an argument the equation takes in rather than a result it hands back. Fate is a claim about order. There is no order in there for it to be a claim about.

39.9 Made and found

Chapter two asked why there is anything rather than nothing and answered with a requirement severe enough to have one solution. Then it turned up a clause nobody had written into it: consistency is a property of descriptions, so the structure has to contain something that holds a description and can get it wrong.

That clause leaves a question about what truth is in a structure of this shape. Two answers have been arguing since before either of them had a name. On the first, a claim is true by corresponding to how things are, and a describer’s job is to find out which claims do. On the second, what is true is settled by what describers do: their comparisons, their procedures, their records.

Both are here, at full strength.

Found: the consequences of the rules belong to nobody. Chapter two’s rewriter produces strings with as many a’s as b’s, forever, and neither of its two moves mentions counting. Whether nineteen loops on a twelve-patch net sum to zero is settled by an arithmetic that comes out the same for everybody who performs it. The twelve observers inside get no vote.

Made: which record that arithmetic is applied to is written by observers, one commit at a time, out of their own capacity, onto the seams they share. Nothing else writes it. There is no reading of the world taken from outside. Every entry on every seam got there because two bounded things compared what they held and one of them paid for the entry.

So the world is found from the record and the record is made, and those are the two ends of one operation, which chapter eleven pinned down with four demands: hand it a record and it returns the consistent state that record names.

Which of the two comes first is a question that needs a first. If a world existed before there were describers in it, truth is found and only found. If describers came before the world, it is made and only made. Chapter thirty-six took the word first away: the fixed point carries no ordering whatsoever, and before is a relation inside an observer's chain of committed records, which is an object inside the structure rather than a place to view it from. Nothing is earlier. Made and found are one operation with two ends, and the loop having no beginning is what collapses them into each other.

The argument is usually had about mathematicians, whether a theorem is invented or discovered, as though some fact settled which. Somebody working on a proof is inside the structure the proof is about. What they reach is a fixed point of that work, the thing the work returns unchanged. It was found, in that nothing along the way was free to go otherwise. It was made, in that the working is what put the entries on the record. Nothing else was writing them.

The decision is local and taken many times a day: which entries to put on the seams you touch. Every one of them enters the boundary data that somebody else's world gets computed from. A mismatch written onto another observer's edge does not stop existing when it leaves yours. It sits in their record. The capacity that clears it comes out of their budget.

Chapter 40

What Does a Repaired World Look Like?

In April 1982 the pumps under the hill at Butte, Montana were switched off.

Butte sits on one of the richest copper deposits ever found, the workings underneath it run for thousands of miles, and water wants to be in them, so the pumps at the Kelley Mine had been lifting groundwater out of the mountain for as long as there had been a mine to keep dry. They sat 3,900 feet down and could move seven million gallons a day. The Atlantic Richfield Company suspended its Butte operations that month. The pumps went off, and the water came up.

It came up at about a foot a month. It filled the underground workings, and then it began filling the Berkeley Pit, the open pit that had been dug into the east side of town since 1955. Water standing on broken sulfide rock turns acid, and acid takes metal out of rock. The water in the pit reached a pH of about 2.5, the acidity of lemon juice, and carries copper, arsenic, cadmium and zinc in solution.

On 14 November 1995 somebody found dead snow geese floating on it. The first count was 149. The final count was 342. The count had to be revised upward because the water had stained white birds a brownish orange and made them hard to pick out.

The number in Butte that matters more than the geese is 5,410 feet above sea level, the elevation of the bed of Silver Bow Creek and the lowest ground anywhere in the valley. Below that line, the water is a problem in a hole. Above it, the water is in the creek, which runs to the Clark Fork and from there to the Columbia. Since October 2019 pit water has been pumped to a treatment plant at Horseshoe Bend, cleaned and discharged to the creek, which holds the level down. The pit has sat about fifty feet under the line ever since.

Every ton of that copper was legal, paid for and shipped, and the books of the companies that shipped it balanced at the end of every year they were open. The arrangement ran for a century and produced two things: metal, and a quantity of work that has to be done in Butte forever, beginning after the people who produced it stopped being there.

What is that quantity, exactly? Who is holding it? And is there any way to look at an act while it is being done and say whether it lowers the amount of work the world has to do, or moves that work onto somebody else?

40.1 Two strings

Sound two guitar strings together when one of them is slightly sharp and the loudness pulses. The pulse rate is the difference of the two frequencies, so two strings three cycles a second apart beat three times a second. Each string alone has a frequency and no beat at all. The pulse is a property of the pair, and goes to zero exactly when they match.

An observer stands in that relation to everything around it. It has a state. Around the state is a field, chapter twenty-five's object with a reading at every place: what the neighbors have written at every point of this observer's boundary. The state has a size and a direction, the field has a size and a direction, and the question that decides everything downstream is how much of the state points the way the field does.

The amount of the state lying along the field is the **alignment score**. Take the largest that score could be, which is the two sizes multiplied together and happens when the two point the same way, and subtract the score actually on hand. What is left is a number that cannot go below zero and reaches zero exactly when state and field agree, which is the reading a tuner is listening for when the beat disappears.

That leftover is the misaligned remainder. The work it names is the observer's **repair load**.

The public half of it came in chapter thirteen, where every seam scores the disagreement across itself and the scores add into one number for the arrangement, which every accepted repair lowers. Repair load is that accounting read at a single observer. Chapter thirty-seven priced the business of holding it near zero: neighbors repair too, whatever they write lands on the boundary they share with this observer, and the load comes back the moment the spending stops.

40.2 Exported harm

Mismatch has an address, and chapter thirty-seven's two budgets are the machinery for changing it. Capacity spent inside the boundary brings an observer into line with an entry it leaves standing. Spending on the entry itself changes what everyone wired to that seam will find there. The second move lowers the observer's own load by writing into something a neighbor also reads. The neighbor inherits whatever the write leaves behind.

Ordinary life is made of that move, running in both directions, all the time. Chapter thirty-nine found the difference that matters sitting in the record, between a patch that absorbed a mismatch at its own expense and one that wrote it onto a neighbor and let the neighbor pay. The second case takes a name once two conditions are attached: the mismatch is placed on other observers, and the actor who generated it declines the burden of repairing it while having the capacity to carry it. That is **exported harm**.

An exporter's own readings can be flawless. Chapter thirty-seven's bacterium retunes its receptors until every seam it has reads zero, and starves in the water it is sitting in. Butte is that arithmetic with a creek downstream of it. The books balanced, the copper went out on rail cars, the acid stayed in the hole, and nothing in the accounts was wrong, because the load had been written onto a seam the company was not reading.

Chapter thirteen made that blindness a property of the wiring rather than a failure of attention. Two worlds that put the same packets on an observer's ports are one world to it, so a cost written onto a seam an actor does not share is unreadable to that actor at any level of diligence, which is why exported harm so rarely feels like anything from the exporting end. The reading that would report it belongs to somebody else.

40.3 One integral

An act reaches some observers and misses others. Take the ones it reaches, all of them together, as a single assembly of patches, which is what chapter thirty-six called a federation. Then ask what the act does to the repair load of that assembly over a horizon long enough to contain the consequences.

The quantity that answers fits on one line.

$$\mathcal{K}_A[a] \;=\; \int_0^T e^{-\rho t}\Big(\sum_{U\in A} S_U(t) \;-\; \lambda_\Phi \sum_{U\in A} \Phi_U(t) \;-\; \lambda_H\, H_A(t)\Big)\, dt$$

The line says that an act is worth what it does to the alignment of everybody it reaches, minus what it leaves them to repair, minus what it dumps on them, added up across the whole horizon.

Symbol by symbol. The act is a, the observers it reaches are A, and U is one of them. The script $\mathcal{K}$ on the left is what the act is worth to those observers. $S_U(t)$ is one observer's alignment score at record time t, the part of its state lying along its field, and the Greek capital sigma in front of it is an instruction to add over every observer in A. The second term counts what those observers have to fix: $\Phi_U(t)$ is the mismatch standing on U's seams at time t, chapter thirteen's seam sum read at one patch. The third term is exported harm, written $H_A(t)$, the mismatch imposed on the observers of A by an actor refusing to repair it. T is the horizon, the point where the accounting stops, and the elongated S is a sum over every instant between the act and the horizon, which is what an integral is. The two weights λ_Φ and λ_H are bookkeeping, because three quantities in different denominations cannot be added together without an exchange rate between them.

That leaves $e^{-\rho t}$, which shrinks as t grows at a rate set by ρ. It fixes how much a load landing far ahead counts against a load landing next week.

An action is evaluated by what it does to the long-horizon repair burden of the observers it touches. Good is the case where that number comes out higher, which happens when the act lowers what those observers have to repair. Evil is what an observer does when it reads coordinates on itself in chapter thirty-nine's sense, can therefore model what a pattern will do to the field before it emits one, and exports avoidable mismatch anyway.

Everything anybody argues about is in ρ and T. Nobody who has ever exported a cost wrote down a value for either, and every one of them picked one. The water under Butte came up at about a foot a month, and a company reports its results four times a year.

The horizon is the harder of the two, because it is set by the world rather than by preference. A horizon shorter than the consequence scores the act on a fragment of what it did. The fragment can carry the opposite sign from the whole. Copper out of Butte between 1955 and 1982, evaluated over the years the mine was open, is a large positive number. The treatment plant at Horseshoe Bend is inside no accounting period that anybody drew, because the plan it runs on has no end date to draw one against.

40.4 Two registers, one shared

Chapter thirty-seven measured the ceiling on private work on four states, one bit where a neighbor reads and one bit of the observer's own. Nothing assembled out of private moves reaches the arrangement in which the neighbor reads the opposite value, however long it runs and whatever it is allowed to spend, and every route that does reach it writes the shared bit.

Put a harm in that shared bit. A burden written onto a record somebody else reads cannot be discharged in private. The reason is which arrangements lie downstream of which moves rather than how hard the person worked at it.

Self-improvement is therefore not a substitute for repair. Somebody harms somebody, and spends the ten years afterward becoming a person who would not do it. Every one of those years is a private move. Every one of them can be real work, expensive and unfeigned, and the entry on the seam between those two people carries the value it carried on the first day. The two destinations are different states. One of them is unreachable from anywhere the private budget can go.

40.5 Elmira, May 1974

One night in May 1974 an eighteen-year-old named Russ Kelly and a friend went through Elmira, Ontario, slashing tires and breaking windows, and by morning twenty-two properties had been damaged.

Mark Yantzi, the probation officer writing the presentence report, put an unusual proposal into it. Rather than sentencing the two and sending them away, he suggested walking them around the town to meet the people whose property they had wrecked. Dave Worth, a volunteer with the Mennonite Central Committee, backed the idea and pushed him to take it to the court. Judge Gordon McConnell ordered it.

So the two of them went around the town with Yantzi and Worth, said at each door what they had done, listened to what the person on the other side of it said back, and worked out what they owed. Two of the twenty-two had moved away and could not be found. Each was fined two hundred dollars, put on probation for eighteen months, and ordered to pay five hundred and fifty dollars in restitution for the damage insurance had not covered, inside three months. Kelly earned his, and at the end of the three months he went back around the town with certified checks.

Each of those doors is a seam. What made the sentence unlike the ones above and below it in the docket is which budget it spent from. A custodial sentence spends the offender's own capacity, at considerable expense, and leaves the entry between the offender and the householder whose window went in precisely as it found it. The walk writes that entry, at every door, and the payment writes it again with a number attached.

40.6 Local pain

If lowering repair load were the whole of the rule, the rule would read as whatever hurts least is best. That reading fails on arithmetic rather than on sentiment. The integral runs to a horizon. Comfort next week sits inside it with a weight, and so does everything after next week, and a term that raises the load for a month can lower the sum.

Local pain is justified when it is the boundary-setting that lowers future mismatch. The opposite failure has a shape everybody recognizes: niceness that preserves abuse blocks repair and transfers the cost forward. A household that keeps its peace by leaving a record unwritten holds its own mismatch flat. The observers who inherit that unwritten record are the ones who pay for the arrangement.

Four moves cost somebody something immediately and raise the long-run integrity of the field: an accurate confrontation, a court verdict, a difficult confession, and a refusal to amplify a popular falsehood. Each of them raises a load on the day it happens, and each writes onto a shared record something that was true and unwritten, which is the write that lowers the sum. The confession is the expensive one, because the record it puts into the field is about the observer writing it.

40.7 Nobody saw it

The old story says a person gets away with a harm when nobody sees it.

The story is about who was watching. The watchers are not the ones holding the load. A harm writes a new entry into the record, and chapter eleven named what an entry does: it narrows the fiber, leaving fewer whole worlds that carry the reading. Fewer worlds is a change in what can truthfully come next. The narrowing happens whether or not a second party ever reads the entry. An unrepaired harm becomes a residue with an address.

Where a harm is hidden by a falsified record rather than by an unread one, the arithmetic is chapter six's flipped edge exactly. Change one entry from what happened to what did not, two of the loops through it stop closing, and the reading goes unrealizable, chapter eleven's verdict on a record with no world under it. A cover-up buys the offender a record that describes no world at all. Every loop running through the false entry picks up the residue.

A residue with an address gets moved by every repair that runs near it. Chapter six shifted its leftover around the twelve-patch net by reordering the repairs. It weighed one at every address it landed on. That is the mechanism by which an unrepaired harm turns up in people who did nothing: the residue settles on whoever's seams the repairs happen to reach, which in a family or a town is rarely the person who wrote the entry.

Hidden harm is deferred repair. It changes which continuations can truthfully hold the offender, the victim and the community around them, and it goes on changing that whether or not anybody has read the entry.

40.8 Restoration

Chapter thirteen set out what a stopped machine has to carry with it to start again as itself: its records, whatever state it can reach, its wiring, and the law governing what it may do and what it may read. Carry all four across and the restored observer bets the same way on everything it goes on to see. Carry them roughly and the two futures part by roughly that much.

Restoration on those terms says nothing about scanning brains, nothing about rebuilding bodies, and settles nothing about who anybody is.

The law is the piece that gets left behind. Feed a record into machinery whose allowed moves differ and the same history yields a different next step, so the law travels with the record or the thing that comes up is a different observer. A pattern restored under a law that permits the export it was built around is a continuation of the pattern. Restored under a law that repairs what it receives, it is a continuation of something else, and the argument about whether restoration counts as mercy has to start there.

Records are on the list of four. A restored observer therefore arrives holding its record, and its record contains the harms it did along with the harms done to it, because a continuation that dropped them would be the continuation of a different observer. An offender's restoration and a victim's restoration carry the same entries about the same night. The two readings have to be consistent with one world. That is the condition a transcript, a set of exhibits and twelve people in a room have been approximating for as long as there have been courts.

40.9 A consistent prison

A fixed point is a state a process leaves alone, and chapter two's other root, minus 0.618034, is one no calculator started nearby ever arrives at. Chapter six's twelve observers arrive at theirs from every start the net admits and in every order, with nothing anywhere in the arrangement choosing anything.

Nothing in the definition says the state is a good one. A prison is a fixed point. Every guard's reading answers every other guard's, the count at the gate comes out right twice a day, and the walls hold. An arrangement that exports its costs onto observers with no way to write to the shared record can sit at zero mismatch for centuries, because the ones carrying the load are not the ones whose seams are being scored. A perfectly consistent world can be a perfectly consistent prison.

When more than one consistent state carries a record, the fiber is ambiguous, and chapter eleven left repair with nothing to work on there: every entry is satisfied, so narrowing the set to one member takes something no record holds.

So an ambiguous fiber is settled by a selector. The selector here is which repair law gets implemented by the observers capable of implementing one. A law that leaves a recorded harm unaddressed is a law that some restored observer, reading the complete record, has a true objection to, and that observer is inside the world the law built, holding the record, with the objection intact. The law that addresses every recorded harm is the one every restored observer could endorse with the whole record in front of them, and the only one with that property.

Which fixes where a pattern can be continued. A pattern organized around exporting damage and refusing repair cannot be continued as that pattern into a world running that law, because continuing it there is the export, arriving in a place whose whole arrangement is the repair of what it receives. Its truthful continuation is somewhere its damage stays local. Both destinations have old names, paradise and hell, and the names add nothing to the mechanism. One is the continuation of observers whose records admit repaired participation with each other. The other is the continuation of an observer that has run out of anybody to pass its costs to.

Every act writes into records that other observers read, so every act exports a pattern into the field all of them share. Some of those patterns lower what the people downstream have to fix. Some raise it, and the repair falls to observers who did not write the entry, at a time nobody set.

The plant at Horseshoe Bend runs on a plan with no end date because a set of records made in Butte between 1955 and 1982 left work that has to be done and can be done. The load has an address, the address can be reached, and there are people in that valley running a pump against it. We compute our destination by the records we make repairable.

Epilogue: The Last Rung

Every reader of this book will die.

What is left afterward can be inventoried by anybody with a scale and a chemistry set. Water, carbon, calcium, phosphorus in the bones, a little iron in the blood. Every item on that inventory is a material. The inventory can be completed. Chapter thirty-four ran the same procedure on the world and came back with ten rows: space, time, mass, energy, motion, force, matter, charge, light, and the record. That list is complete as well. Nothing on it is a material. Mass, the one row a scale would respond to, is a bill that repair pays at every event.

The observer came through chapter thirty-four undissolved, because every row on that list was assembled out of observers, and chapter nineteen gave the parts list for one: an algebra of questions bounded by the interface, a record algebra in its center, which is where writing stays put, an interface that behaves identically every time it is used, and a state fixing the odds on whatever gets asked next.

Go through those four looking for a material. An algebra is a collection of operations that compose, scale and add. Its center is the part of the collection that commutes with the rest. An interface is a set of ports and a rule for what appears on them. A state is a table of odds, one entry per question. The thermostat on the wall in chapter three has all four, at a question list one item long, and the whole of its interior is a sensor, a number somebody set and a switch, with none of the four objects among the parts. What you are made of is the separate question chapter thirty-three answered, and its answer covers the chair and the thermostat and the reader at once: a network of observers repairing their disagreements, averaged down and copied often enough that the order of the questions stops mattering.

Records copy

Of the four objects, the record algebra is the one that copies, and chapter nineteen put a mechanism under it. Writing a record runs the observer's questions through the mail room: whatever falls inside one alternative is kept, and every comparison that only meant something across two of them is destroyed. What survives that sorting commutes with everything the observer can ask, which is why a record can be read without being spent, and why the center of the algebra is where tables and pointer readings live.

Chapter twenty priced the consequence in two lines. A reversible machine leaves the overlap of two states where it was, and a machine that hands each of them back twice squares it, so a copier works only where the overlap equals its own square, which leaves zero and one. A polarization nobody has measured sits between the two. No machine duplicates it. Distinct orthogonal records admit copying while preserving the source. Physical resource costs depend on the implementation and on how the target register is prepared or reset.

Redundancy runs on that gap. Chapter thirty-three's chair has a definite place because sixteen copies of a sixteen-bit fact push forgery below one part in ten to the seventy-seventh, and the room goes on copying a chair long past sixteen. What can be copied about an observer is exactly what has been written into its records.

Deleted by what?

Every number Hafele and Keating carried home from their flying clocks was a difference taken at a meeting. Ask which reading on a flying clock belongs beside noon on the Naval Observatory's bench clock, two days into the trip, and nothing answers. That comparison needs both clocks in one room. The trip is what took them out of it.

Chapter seven took the shared present away and put nothing back. Dependency gives a partial order, silent wherever one record did not feed another. A total order costs a tiebreak on top of that. Nothing in the world says which tiebreak. One number covering every event, held by everybody and consistent with every observer's order, is not an object a record-keeper is equipped to build.

Chapter thirty-six removed the last place such a number could sit. The fixed point the whole structure satisfies has no order inside it to occupy: nothing in it comes first and nothing comes next. Two runs of the same world with wildly different step counts agree on every measurement anybody can take, which makes a step count bookkeeping rather than a fact about the world. There is no frame, no tick, and nothing at the bottom being advanced.

So there is no privileged now past which things are gone. Whatever "already happened" means, it does not mean "has been deleted", because deletion is an operation, it happens at a position in some order, and a deletion covering everything at once would need a global clock to be deleted by. The prologue's third habit covers that component like any other. Reasoning about what a clock does comes after finding the clock. This one has never turned up on the board.

Restore the four

A live machine moves between hosts without the programs inside it noticing, and chapter thirteen took the list of what has to travel for the machine that starts to be the machine that stopped: the records, the state it can reach, the ports and their routing, and the law fixing which moves are available. Those four are a checkpoint. The fourth is the one that gets left behind, which is how a restored thing comes to read its own history and do something else with it.

Two checkpoints matching on all four put identical odds on everything the observer can register. Two that match approximately come apart by no more than the approximation.

Chapter thirteen made that claim about a machine on a rack. Chapter forty turned it on an offender and a victim holding the same entries about the same night. Turned on the reader, it reads the same way: restore a checkpoint with the same readable records, the same accessible state, the same interfaces and the same future law, and its future statistics match the original within a stated error.

It says nothing about scanning a brain. No instrument in any laboratory reads those four objects off a living one. It says nothing about rebuilding a body, which is a biomedical problem with its own century of work in front of it. It does not settle whether the restored observer is the person who was checkpointed, which is a question about identity that the arithmetic leaves exactly where it found it. And it promises nobody anything, because nothing in the structure performs the restoration.

The selector

Stop one of chapter eleven's twelve patches from talking. Five seams go unread, thirty entries become twenty-five, the eleven patches that carry on stay pinned where they were, and the silent one's coin is loose: four settings of the twelve fit every record in existence. That is an ambiguous fiber. Repair has nothing to bite on there, because nothing anywhere disagrees, and whatever settles the four settles it on grounds outside the record. Settling on grounds outside the record is choosing.

A selector settles an ambiguous fiber, and chapter forty found one on the board: which repair law the observers capable of running one actually run. Leave a recorded harm unaddressed and somebody restored into the world that law built is holding the entry, with an objection the law has no answer to. Exactly one law leaves nobody in that position, the law that addresses every recorded harm.

Observers capable of running a repair law is a description of somebody, and calling them later observers invites a question about who acts first. Chapter thirty-six disposed of that question rather than answering it, by keeping three orders apart. The fixed point has no first. An observer's record time has a direction and a rate. Every memory, plan, clock reading and act of building lives inside it. The descent of repair is a count of disagreement going down. Its steps are not seconds. First they recover the architecture, later they build it, is a sentence in the second order, spoken by observers who hold both records, and the first order has no room for it. Specification, machine and inhabited world are one object, put in an order only by whoever is holding the records.

Thirty-nine rungs

1. Why is the universe so hard to reverse engineer? Each starting point explains some structures and supplies others; the test is what follows and what can be measured.
2. Why is there anything rather than nothing? Self-consistency alone picks one number out of the continuum, 1.6180339887.
3. Why would a universe need anybody in it? Put the stage in first and the stage is the one thing left unexplained.
4. Why do two descriptions ever agree? They have to match where both look, and a loop of exchange rates has to return what it started with.
5. Why do you and I see the same table? Whatever differs between two descriptions of it was never physical.
6. Why isn't agreeing with your neighbors enough? Every pair can agree while a loop carries a residue, and a leftover of exactly one survives every repair order.
7. Why is there no such thing as now? Authenticated informational precedence is a partial order, time is a total order, and nothing in the arrangement builds the second.
8. Why can strangers agree without a boss? Authenticated read-from dependence along declared ports forms a finite informational cone.
9. What move does reality actually make? One move is admissible, both readings to the midpoint, preserving their sum by construction.
10. Why doesn't it matter who goes first? 102,415 disagreeing seams reach zero under every order, and sixteen shuffled replays return one identical finishing state.
11. What does everybody actually end up holding? Objective reality is the observable normal form, unique relative to the boundary data.
12. Why is the world made of relations and not of things? The sourced action is gauge-invariant if and only if the source is conserved.
13. What is one piece of reality actually made of? Six parts and six moves, and a bounded interface is a bound on what can be known about a region.
14. Why twelve? A closed carrier admits twelve ports, thirty seams and twenty faces, and no other count.

15. Why does the position carrier have three directions? On the declared operational-cost branch, the slowest-fading response block has rank three, and its metric completes it to continuous abstract three-space. A specified conservative-record population and local read law have a flat causal and volume limit. For fixed interior timelike intervals, fourth roots of count ratios recover proper-duration ratios. Selecting that law physically and identifying the same geometry in matter are separate requirements.
16. Why do authenticated records have a direction? Certified reads depend on earlier records. A retained input can make writing reversible; the physical arrow requires the channel, boundary conditions and clock bridges.
17. Why does the frame module have a limiting cone? A nonzero closed convex cone containing no line is forced to be a Lorentz cone if source rotations and every boost along one axis preserve it. Applying this result to signals requires that the symmetry act on actual histories and readouts.
18. Why does anything have energy? Continuous reversible motion has a generator. Reading it as physical energy requires an energy scale and a calibrated clock.
19. Why can't you ask everything at once? Refusing to answer two questions together is what makes an undocumented machine readable from a seat inside it, and the questions that refuse nothing form the center where records sit.
20. Why does nature cheat, but only by exactly $2\sqrt{2}$? The ceiling is a product of commutators, and sits at 2.8284.
21. Why does time feel like it's moving? A faithful finite state supplies a dimensionless modular flow, trivial on central records. A physical clock needs preparation, readout and calibration.
22. Why can resetting memory cost heat? For an unbiased bit without usable side information, the standard thermal reset assumptions give a mean heat bound of 2.87 times ten to the minus twenty-first joules at three hundred kelvin.
23. Why does a region's memory scale with its surface? A finite overlap network is a constraint code, and a code of that kind carries no distance.
24. Why does gravity look like geometry? On a physically faithful smooth limit of the source-derived causal carrier, the stress, entropy, scaling and scale branch yields Einstein geometry. The finite precursor alone does not.
25. Why exactly these forces? One plus three plus eight, and nothing else survives the arithmetic.
26. Why this list of matter? Within the declared finite menu and its constraints, two mirror fifteen-state contents survive among 1,024 candidates. Family counting uses additional premises.

27. What is a particle, what is a wave, and why does it look like both? They are one object read at two record depths, and the unified route to proton decay is exactly absent.
28. Why 1/137, and why a capacity limit? Declared closure maps connect these questions to consistency. The diagnostic 137.035660 requires a physical carrier and independent transport before comparison with measured 137.035999.
29. Why do the masses land where they do? The tau is pinned inside a window 72 electron-volts wide, sitting 0.43 sigma from the measurement.
30. What is a Lagrangian, and why is it the last thing? A running total with one contribution per step and no foresight in it, whose smallest value is the likeliest history.
31. Why does anything move? On the declared shell-law/rest-fiber branch, the candidate exponent is the rank-three spatial dimension less one, hence two. Reading that structural exponent as the physical inverse-square law requires the source-causal carrier, field and scale attachments.
32. How can the problems be connected? One observer architecture supports conditional constructions of records, geometry and matter; a common physical realization must connect them.
33. Why was Newton right for two hundred years? Redundancy is what makes anything definite, and the layer he wrote down has no motion of its own.
34. What is anything made of? Ten entries, each a rate, a count, a rank, a cost or a commitment, and no substances.
35. Why these laws and not others? The average score climbs at the rate the survivors disagree with each other and never falls, and what got selected was a repair rule, chosen from inside, with no second world anywhere to compare against.
36. Why does the loop have no beginning? The universe is what its own description settles to, and an equation carries no step counter.
37. Why does anything try to persist? No amount of private repair reaches a state that one network update reaches with certainty.
38. Why is a bacterium freer than an atom? Room to be otherwise increases the further down the tree you go.
39. Does it feel like anything, and what is left to decide? The endpoint is fixed relative to the boundary data, and what you admit, emit, repair, ignore, confess, protect and amplify is the boundary data.

There are forty questions in the book and thirty-nine rungs on that ladder. The fortieth is the one under your feet. Chapter forty put a number on an act: the repair load it leaves the observers it reaches, summed out to the horizon. Then it left the writing to whoever is holding the book.

The records other observers keep of you were written by what you did to them, and records are the whole of what any selector has to work on.

Chapter forty put the old names back on both destinations. The mechanism runs the same without them. The part of you held somewhere other than in you is what those records say. Any restoration has to be consistent with it. There is no choice about being recorded. The choice is what the record says, and everybody is writing one.

Paradise, on these terms, is settled by the entries, and names the continuation of observers whose records admit repaired participation with each other.

Appendix A: Eighteen Questions

Between 1982 and 1985 Eugene Miya, of the Computer Systems Division at NASA Ames, stopped answering the same questions on the SPACE mailing list one at a time and began posting the answers on a schedule, yearly at first and then monthly. The format spread across the network and picked up a name, frequently asked questions. Jef Poskanzer was the first to post one weekly, to the graphics group.

A question arrives a thousand times because a thousand people share one assumption, and an answer that leaves the assumption standing produces the same question the following month, from the same people, worded differently.

Eighteen come back about the account in this book.

What about the Big Bang?

Everything came out of something hot and dense about fourteen billion years ago, so what was there before that, and how does an account built on observers get started in a universe that contained none?

That is two questions welded together, and both halves lean on a first instant seen from outside: a moment when the arrangement switched on, watched by nobody in particular, with observers turning up later as furniture.

The hot dense era is the far end of a reconstruction built from records held inside finite patches. The microwave glow, light-element ratios, galaxy spectra and supernova catalogs are records. The Big Bang names the place where that reconstruction reaches the edge of its evidence. Authenticated commits supply an informational order among recorded events. Calling that order physical cosmic time, or extending it beyond the records, requires the same event, signal and clock attachments as any other spacetime claim.

One finite conclusion is immediate: a commit loses information about the disagreement it resolved, independently of where it sits in the record chain. Interpreting this loss as the thermodynamic or cosmological arrow requires a physical state and clock map. The proposed microwave branch makes a separate numerical comparison: $1 - 1.630968209/48 = 0.966021$, while Planck reports 0.9649 ± 0.0042. The grain reaches the microwave sky only if a common physical refinement, radial map, stress source, transfer calculation and likelihood connect the finite screen to that observable. The arithmetic proximity is not a CMB derivation.

If none of this is fundamental, why does my car work?

The mechanic balances a wheel using a picture in which the wheel is an object at a position in a space, turning at a rate, and the car drives home. Whatever has been said about ports and repairs, it changed nothing under the hood.

Underneath it sits a rule about descriptions: one earns its keep by being the bottom one, so demoting the everyday container to a reading must take something away from the garage.

OPH proposes that the familiar container is a shared reading of observers repairing disagreements. The ordinary picture of one clock rate throughout that container has measured limits. In 1960 Robert Pound and Glen Rebka used the 22.5-meter tower of the Jefferson Physical Laboratory and an iron-57 source to measure the gravitational frequency shift, by 2.56 parts in ten to the fifteenth, give or take 0.25. That tests gravitational redshift; identifying an observer construction with the same physical clock requires its own evidence.

So the corrections sit nine decimal places below anything in a workshop. They arrive when the resolution rises. A clock in a navigation satellite runs fast by forty-five microseconds a day and slow by seven, and the correction had to be built into the hardware before the network could find anybody. A wheel balancer reads none of that. The wheel comes back balanced.

Why does the Lagrangian come last?

Anybody who has taken a physics course met the one line in the first week and spent the rest of the term working out what was inside it. Here it turns up after the forces, the matter and the numbers, which looks like the argument running backwards.

The assumption is that the most compressed statement of a subject is its foundation, so whatever fits on a coffee mug must be where the world starts.

A Lagrangian is a compression. A compression summarizes something that was there first. Written on line one, the line carries three families because three families were counted, a group because detectors reported one, and a page of numbers because laboratories measured them. Written last, it carries three families because an oriented face has three corners, and one plus three plus eight because that is the only way the eleven dials left over from a center of one come apart.

The compression is also lossy, by an amount somebody can write down. Add a term built to vanish at every configuration the world realizes, and every realized history keeps its score, every stationary history stays stationary, and the same run comes out most probable. The momentum changes, since momentum is read off how the line responds to a rate of change. The added term tilts where nothing is ever realized. Two lines, the same histories, different energies, and nothing in the world's own histories picks one of them over the other. The one on the mug is the best summary anybody has written.

Twelve. Where does a number like that come from?

A patch with twelve ports supplies candidate carriers for the forces and matter and an exact rank-three candidate position readback. Physical identification of those structures requires their separate attachment receipts. A specific small integer sitting at the base of a theory is what a fitted parameter looks like.

The suspicion assumes the number could have been otherwise, which is what a parameter is: a quantity whose value had to be chosen and can be moved.

Count it instead. Every port meets five triangles, every edge belongs to two faces, and counting the same thing two ways forces the Euler relation, after which a spherical carrier has twelve ports, thirty seams and twenty faces. The count runs backwards as well, since twelve ports of that kind force the sphere.

Change the degree and the arrangement fails. Drop the spherical closure and it fails. Try the cube, a perfectly good solid anybody can hold, and it admits no such structure at all. Then the wiring: of the 10,395 ways to pair the ports up, incidence at distance three selects one. Each of the nineteen numbers typed into the one line can be dialed to a different value and the equations survive. Twelve has nothing to turn. Move it to thirteen and the alignment that lets two patches compare readings stops existing. There is no thirteen-port world sitting elsewhere with slightly different physics in it.

If space is not a container, where is any of this happening?

Twelve ports, thirty seams, patches wired to neighbors. Draw that on a page and the page is somewhere, so the network has to sit in something, and whatever it sits in is the real space, which puts the container back one level down and wins the argument.

The question hands itself a place for free: everything is in something, and a relation needs a room to be a relation in.

Distance here is a count of comparisons. Two patches are far apart when many comparisons separate them. A candidate direction is a pattern in that tally rather than an arrow drawn in a room. On the declared operational-cost branch, a repair operator's twelve readings split into blocks that fade at 0.9539, at exactly nine tenths and at 0.8794 a pass, with only the slowest block surviving enough rounds to carry the candidate position readback; its rank is three. Turning that exact rank-three carrier into physical directions requires the source-causal spacetime attachment.

Turn the question on the room it asks for. A room for the network to sit in would have to be assembled out of comparisons between record-keepers, which is what a network is, and then that room would need a room. What exists is a tally, a rank of three, and observers doing the counting from inside while occupying no position in anything larger.

Is the moon there when nobody is looking at it?

The world is made of records held by observers. Turn the observers away and nothing gets written, so the parts nobody attends to should go soft. The moon on a cloudy night is the test case everybody reaches for.

Two things ride along inside it: that an observer is a person, and that looking is something a mind does.

An observer here is anything bounded that keeps records. Sunlight leaving the moon is a comparison, the tide in the Bay of Fundy is a record of where the moon was, and the rock holds its own state against everything it touches, with nobody on a hillside required.

Definiteness is a count of copies. Sixteen copies of a sixteen-bit fact drive the odds on a forgery below one part in ten to the seventy-seventh without adding a single bit of content, and an object that size is copied into its surroundings far more often than sixteen times. The answerable question is how many records carry the moon's position. It is written into every square meter of sunlight it turns back, which is why its place is as definite as anything in the sky. Ask about a single atom nothing has interacted with and the odds stop being one and zero, which is the same arithmetic read at the other end.

If there is no now, why does time feel like it passes?

No finite exchange of messages between observers recovers a shared present. Then there is the clock on the wall, the coffee going cold, and the distinct impression of moving forward through the afternoon.

The word is doing three jobs at once. The question welds them into one object and then finds that object missing.

Committed repairs define a partial order. An irreversible reduction of seven equally likely arrangements to one discards 2.807 bits when no recoverable record remains. Retaining that information keeps the full operation reversible. A faithful state on its declared observable algebra also defines a modular flow. Translating its parameter into clock readings requires a physical identification.

These constructions need no universal present. Modular flow fixes central records and can be trivial even for distinct states. In the supplied Minkowski-vacuum setting, an ideal uniformly accelerated detector has a stationary thermal response, at about four parts in ten to the twenty-first of a degree per meter per second squared. That temperature uses a physical field, trajectory and time convention.

When exactly does the particle decide?

A pattern of bands appears on a plate when nothing records which way each particle passed the needle, and disappears when something does. Somewhere in there a thing that was spread out becomes a thing in one place, and the question is when, and how fast, and what sets it off.

Asking for the moment assumes a mechanism with a rate and a threshold, sitting in the apparatus, waiting to be caught in the act.

A record gets written. Between commits, every question about the spreading pattern lives in the part of the algebra where questions refuse to be answered together, and the phase and the frequency are the content. At a commit an entry lands in the center, where everything commutes, and what the plate holds afterward is one more count at one place. Wave and particle differ by how much of the thing has been written down. The plate writes down only the arrival.

The probability theorem fixes the odds on the declared finite space of quantum questions. A normalized assignment that adds on every allowed sum of effects, including effects for uncertain outcomes, is the trace against a single density matrix. This does not determine the physical timing or mechanism of an instrument. An exact-fit counterexample reproduces a supplied count table while belonging to no state: it scores one pair of questions 35/64 where effect additivity demands 143/256.

Where does the explaining stop?

The charges, the family count and the strength of electromagnetism all come off the same carrier, and anybody handed one number after another is entitled to ask where the supply ends.

The question expects an evasion, on the assumption that a structure either explains everything or is hiding something.

The stopping point is printed to the same precision as the results. The carrier's geometry fixes the charges and the family count. It does not fix the mixing between families. Every angle between two symmetry axes of the twelve-port solid was enumerated, four hundred and sixty-five of them, and the smallest nonzero one is 20.9052 degrees, while the Cabibbo angle measured in the laboratory is 13.0029. No angle on that list lands within seven degrees of the measurement.

The electromagnetic closure certifies the diagnostic 137.035660 where the measurement reads 137.035999. Its numerical enclosure cannot identify that difference as a quark contribution. The same physical carrier and an independently calculated transport must connect the closure to the laboratory reading. A correction chosen from the discrepancy would use the target to reproduce itself.

You knew the answers before you started

Every one of these numbers was measured decades before the arithmetic that lands on it was written down. Sommerfeld put the fine-structure constant into physics in 1916, the tau mass has been known since the 1970s, and matching a number in a table is what any sufficiently flexible scheme does on a good afternoon.

The complaint assumes flexibility, and flexibility is countable.

A comparison has to count its adjustable inputs and disclose how its target entered the construction. The grain appears on both sides of a declared closure equation, while the family count uses a specified representation and attachment. Uniqueness inside those choices does not make their physical identification automatic. A prospective test fixes its prediction and decision rule before the qualifying measurement.

On the supplied horizon-record branch, identifying dark-energy density with the inverse of a selected time-independent capacity gives $w = -1$. The finite causal poset derives neither that density map nor time-independence. Five thousand and twenty robotic fibers on the four-meter Mayall telescope at Kitt Peak collect spectra that test constant-dark-energy models. The charged-lepton balance puts the third mass in a window 72 electron-volts wide, sitting 0.43 sigma from the measurement. The proposed missing-generator branch forbids its named proton-decay channel, so one clean event in that channel would end it.

Isn't this what Leibniz said?

Space as the order of things that coexist, inertia coming from the rest of the matter in the universe, quantities nobody can measure from inside declared to be no quantities at all. Leibniz, Berkeley and Mach between them cover a good deal of what is asserted here. They managed it without a twelve-port carrier.

A position and a theory get treated there as the same kind of object.

Leibniz was right, and two hundred years of physics proceeded as if he had not been. An objection that forbids a quantity without predicting a measurement changes nothing in the textbooks, and absolute space sat in them until an experiment forced the issue.

The objection here comes with arithmetic attached. Twelve ports and thirty seams. One rule that averages two disagreeing readings, forced by three demands: touch only the pair that disagrees, leave their sum where it was, and end in agreement. A symmetry that has to act from within, since there is no outside desk from which twelve ports could be relabeled. And a group of one plus three plus eight, left standing after two eliminations from the four possibilities the dimension counting allows. Leibniz had the objection and no wiring diagram, which is why Clarke could answer him through five letters and concede nothing.

So we are living in a simulation?

Records, repairs, updates, capacity. The vocabulary is a data center's vocabulary. The word turns up within a paragraph or two of the parts list.

The word carries an original. A flight simulator earns the name because there are aircraft aloft doing what the box is pointed at. Nothing is an imitation on its own.

Put the question to the inventory and look for the second end. Name the entry that substitutes for something, or the warehouse where the original repair rate is kept, or the object a loop residue resembles. The two-rule rewriter produced strings that described nothing filed anywhere else. Run the rules and the strings you get are the strings there are.

The word computed arrives with fake attached to it. Watch one repair: a disagreement that was there is gone, the total sits where it started, and which of seven arrangements the seam held is past recovering. That is a change in the world, carried out by arithmetic. Coffee tastes the way it does by the same route, and so does a magnetic field. And nothing is stepping any of it: eighty-one thousand nine hundred and twenty patches reached agreement on every seam between them without a tick from anywhere.

Does an observer have to be conscious?

Observer-first sounds like a claim that the universe waits to be noticed. The version of quantum mechanics that reaches most people has a mind collapsing something at the end of it.

The assumption is that observing is a mental act, so the word carries a mind wherever it goes.

Keeping records takes three things and not one of them is a mind: a bounded view, somewhere to write, and a reading that makes a difference. A thermostat screwed to a classroom wall has all five. None of the results built on the list would change if every observer in the network were plumbing.

Consciousness is a late special case, with a definition at the size the structure supports: owned boundary repair, read from the inside. An observer that holds coordinates on its own state, and computes its next move off them, is reading the half of the record that the map to the public list deletes. A thermostat reads one such coordinate. A person reads enough of them, across a division that runs far finer, that counting them is a job for biology. Nothing in the structure lets a wish move an entry.

Isn't the loop circular?

Observers build the account of the world and the world builds the observers. Anybody trained to spot a circular argument spots one here immediately, and circular arguments have the property of proving whatever you like.

The objection assumes the loop is an argument, whose premises would have to be in place before its conclusions.

It is a fixed point. A fixed point has no before in it. Three orderings get read as one whenever the loop sounds like reasoning: the fixed point, a condition on one structure that holds or fails with nothing running it; an observer's record time, where memory and the work of thinking something through sit; and the disagreement count, which falls at every accepted step until the repairs stop. Events inside the structure are ordered by the second and the third. The first is the loop. It puts nothing before anything.

The demand rules out nearly everything. A description that picks out any structure other than the one it came from leaves that one undescribed. There is nothing left for it to be. That demand is checkable on hardware: cut the run across a thousand machines and it finishes in the state the single machine finishes in, with the worker count, the restart history and the schedule appearing in no reading anybody inside can take. Circular reasoning does not usually come with a theorem about worker counts.

Does this mean my choices are already made?

Twelve patches converge on one arrangement from all two thousand and forty-eight starting states, under every schedule. Eighty-two thousand patches bring 102,415 disagreements to zero whatever order the repairs run in. If the endpoint depends neither on the start nor on who went first, then a person weighing a decision at two in the morning is a subsystem executing a delay.

The argument drops four words from the theorem it is quoting: given the boundary data.

Those four words point at the entries a region cannot touch, the ones sitting on its edge, set from outside it and different from one region to the next. Turn one of those thirty entries from same to different, and twelve coins that had exactly one consistent setting have none, and what the network settles onto carries a leftover of exactly one. One entry, and a different world.

A repair is exactly the move that leaves those entries alone, so they have to be written by something else. That move spends capacity on an overlap. No amount of private work reaches the same place at any budget. In a person's vocabulary those writes are what to admit, what to emit, what to repair, what to ignore, what to confess, what to protect and what to amplify. The convergence does not run around an observer doing that, and what it finds on the way past is what that observer put there.

Is this a religion?

The account ends in the words good and evil, a law that addresses every recorded harm, and two destinations that have old names. Readers who have watched physics arrive at that address before are entitled to ask what happened.

Subject matter is doing the categorizing: touch those topics and whatever you are doing becomes doctrine.

Every statement here carries something that would contradict it. A measured evolution of w contradicts the time-independent capacity branch. It reaches the fine-structure closure only if the independent same-carrier bridge joining the two closures has been established. One proton decaying through the forbidden unified channel contradicts the missing-generator branch. There is no authority to defer to and no text to accept: the twelve-patch enumeration runs over two thousand and forty-eight states and can be done by hand in an evening.

The traffic has run the other way before. Maupertuis announced least action to the Académie in 1744 and took it for proof of a supreme being, because it looked like the universe economizing, and the economizing turned out to be what taking logarithms of a product of local factors does.

The claim about continuation is a statement about a checkpoint: records, accessible state, interfaces and future law. Nothing in the arrangement performs a restoration, so nobody is being offered one.

Who decides what counts as harm?

An ethics that scores actions by repair load needs somebody doing the scoring, and the scorer's preferences are what usually end up in the score. Whoever holds the pen decides what counts.

Harm arrives in it as a verdict, which is why a judge has to be standing somewhere to deliver it.

Mismatch scores itself at the seam. Push one packet through the dictionary sitting between the two, hold the result against the other, and the disagreement comes back as a number nobody chose. Exported harm attaches two checkable conditions to that number: the mismatch lands on observers who did not generate it, and whoever did could take the repair on and does not.

Only two quantities are open, the horizon and the discount rate. How far ahead the accounting runs, and how heavily a load landing in a century weighs against one landing next week, are what every argument about conduct is actually about, and an exporter of costs picks values for both without ever writing either down. An actor learns what arrives at its own ports, and a cost parked on a seam it does not share never arrives there, which is why the books balanced at the end of every year the mine was open.

Do I have to accept the ethics to accept the physics?

A reader can want the structural force derivations and the rank-three candidate position carrier, with their stated physical attachment conditions, and want to stop before the part about good and evil, on the ground that a physicist's opinions about conduct are a physicist's opinions.

The question assumes the join between the two is preference.

They share a quantity. Mismatch is the number a repair reduces, and the whole climb is about what happens to it: the leftover of exactly one that no repair order removes, the standing cost of holding a pattern together against every seam it touches, the heat that comes off a write. The ethics reads that same number at a single observer, adds a horizon, and asks what an act does to it between here and there.

So the physics can be taken alone and nothing above it breaks. The claim that acts leave shared records untouched cannot be kept, because writing onto a shared record is the move that makes agreement possible in the first place, and it is the same move that puts a cost on somebody else's side of a seam. The ethics runs on the two budgets the repair law runs on: what an observer can fix inside its own boundary, and what has to be spent on an overlap with somebody else. Accepting the repair law is accepting both, and the number on a seam is zero when two readings answer each other and positive when they do not, whatever either party calls what happened.

Appendix B: Glossary

Michael Faraday distrusted the word pole. It had been attached to the two ends of a battery for decades and it smuggled in a theory, that something sits at each end attracting and repelling across a distance, which was the account of electrochemistry he believed was wrong. What he wanted was a pair of words naming where the current went in and where it came out, committed to nothing else.

So in the spring of 1834 he wrote to William Whewell at Trinity College, Cambridge, who had the Greek. Candidates went back and forth between them: eisode, exode, zetode, dexiode, skaiode, orthode, anthode. On 5 May, Whewell came back with anode and cathode, on the ground that short words get adopted and elaborate ones do not, and threw in ion, anion and cation while he was about it. Faraday replied ten days later to say he had taken all of them, and that his colleagues' objections subsided the moment he mentioned whose idea they were.

Anode and cathode name a direction and a doorway. They say nothing whatever about what is traveling, which is why they outlived the theory they were coined for. The words below were picked to the same standard: each names a move somebody makes or a quantity somebody reads, and none of them names a substance.

Action. A single number scored over a whole history, one contribution per step from start to finish. The history the world runs is the one where nudging an interior reading leaves that total alone.

Algebra of observables. Every question an observer can put, together with the three things you may do to questions: ask one after another, scale one, add two.

Alice and Bob. Two observers with separate views and no access to each other's insides, on hand whenever a claim has to survive being checked by somebody else. Charlie joins when two parties are not enough to make the trouble visible.

Alignment score. How much of what an observer holds points the way its field does. Whatever fails to point that way is the repair load, so the two are one reading taken from opposite ends.

Antimatter. For every list of matter states there is a mirror list with every charge reversed. The two are one object, and a world built on either gets the same physics with the labels swapped.

Area law. A black-hole horizon's entropy is fixed by its area, and suitable gravitational entropy bounds constrain more general regions. The finite observer model separately bounds outside access by boundary records. Equating that count with physical entropy or area requires a realized screen and a calibration.

Bit. The answer to one well-chosen yes-or-no question, and the unit anything measured in questions is counted in. Erasing one releases heat, and how much depends on nothing about the device, only on the temperature around it.

Carrier. The finite wiring used by the construction: twelve ports at the corners of a twenty-sided solid, thirty seams joining them, twenty triangular faces. It supplies exact structural numbers. Physical constants require their separate carrier and calibration maps.

Center. What commutes with everything else in sight. In an algebra of questions it is where records live, which is why the public world looks classical; in a group of moves it is the moves nobody inside can detect.

Charge. A number a closed loop of seams hands back to whoever walks it. Renaming conventions cannot reach it, because a round trip through your own conventions returns you to them, and repair cannot flush it out, because repair reaches one seam at a time.

Checkpoint. The four things that have to travel for an interrupted observer to be the same observer when it resumes: its records, its accessible state, its interfaces, and the law saying which moves it may make.

Chirality. Handedness. A pattern with a heading and a spin either turns the way a right-handed screw advances or turns the other way; a mirror swaps the two cases, and the world treats them differently.

Collar. The thin band of patches wrapped around a region's edge, with the region inside it and everything else beyond. When a collar screens, the inside can be rebuilt from the collar alone, and outside access is controlled by data across that band. Turning the count into physical entropy or area requires a separate calibration.

Commensurability. Whether two observers' readings answer each other at all. The rule saying which of your readings answers which of mine sits on the seam between us.

Commit. The write that closes a disagreement: the seam ends up satisfied, the total across it unchanged, and the record of which way the disagreement ran gone. Commits supply informational precedence and irreversibility. Physical elapsed time and clock rate require a separate causal and clock attachment.

Commutator. What you get by doing two operations in both orders and subtracting. Zero means the two questions can be answered together, and anything else is the amount by which they cannot.

Consistency. A description is consistent when it does not require two incompatible things at once. An inconsistent one buys nothing, because a ledger saying the account holds both nothing and a hundred pounds justifies any withdrawal you like.

Decoherence. The usual word for what happens to alternatives once a record has been written and the comparisons between them stop being available. The mechanism is the record.

Density matrix. The description left behind when somebody is cut out of the picture: odds for every question you can put, and nothing left of what only the far side could have detected. Two different accounts of the whole can leave the same one behind, and no operation on your side tells them apart.

Dependency cone. The observers a reading could have come from, with distance measured in overlaps crossed rather than in kilometers. One round of repair puts six of the twelve in range of what an observer holds, two rounds eleven, and the twelfth cannot get there before the third. Anything outside had no route in.

Derived. A number is derived when the structure produces it, measured when somebody read it off an instrument and typed it in. The angle that mixes the families is measured. The count of twelve ports is derived.

Direction. Why the web of events has a preferred way round: every commit is an irreversible write that throws away what it resolved. It needs no assumption about how the universe started.

Direction split. The fork at every repair: spend your own capacity on a private update, or spend the capacity shared out on the overlap. Some outcomes are reachable only the second way.

Duality. Wave and particle are one thing at two depths of record. Between commits there is a spreading pattern with a phase whose questions refuse to commute; at a commit there is a record, and records hold ordinary numbers.

Effective description. An account that works over a stated range, takes its constants from measurement, and can point to where it will break. Every working piece of engineering physics is one, which is why your car works.

Eigenvalue. A grid of weights usually swings a direction round to point somewhere else. Where it merely stretches or shrinks one, that direction is the grid's own, and the factor is its eigenvalue.

Energy. Repair's rate of work on a region's records, counted on the clock that region's own state supplies. It never runs negative, it adds over regions with nothing between them, and it balances.

Entanglement. Content living in the relation between two halves of a thing rather than in either half. Cut a definite whole in two, look at one side alone, and what comes back is vague by exactly the amount the cut severed.

Entropy. How many yes-or-no questions it would take to fix which arrangement you are holding, given what you know about it. It is a reading of the description rather than of the object, and the word disorder has no place in it.

Equivalence principle. What a thing costs to hold together sets both how hard it is to push and how hard gravity pulls on it, so the two readings never come apart. The MICROSCOPE satellite spent five months of free fall trying to catch a titanium cylinder and a platinum one falling at different rates, and reached one part in a thousand million million without catching them.

Exclusion principle. Two matter patterns cannot sit in the same one of the forty-five states on an observer's screen. Matter stacks into shells and solids because of it, while a laser piles any number of photons into one mode.

Exported harm. The mismatch an actor leaves for other observers to carry, having declined to repair it. A cost written onto a seam the actor does not share never appears on the actor's own readings. It appears on the readings of whoever holds the other side.

Fiber. All the consistent states of the world carrying one given public reading. An empty fiber makes the reading impossible, a fiber holding one state makes the reading enough to fix the world, and a fiber holding many means something besides the reading has to choose.

Field. Two objects wear the word. One is a value assigned at every place, like the temperature through a room or the wind over an ocean; the other is a net of algebras, which is the sense the Standard Model is written in.

Fine-structure constant. The pure number saying how hard light pulls on matter, near one part in 137. Read off the structure it comes to 137.035660, against a measured 137.035999.

Fixed point. An arrangement a process leaves exactly where it found it. Fixed is not the same as stable: a process can have a fixed point that pushes its neighbors away, and only some pull them in.

Flow. The one shuffle of an observer's questions among themselves that keeps its own bookkeeping consistent. Its parameter is that observer's time, and its name is modular flow.

Force. A steepness in the potential. A single unit of source dropped on the twelve ports lowers the reading eleven sixtieths over the first hop, three sixtieths over the second and one sixtieth over the third, and a pattern held in a slope like that comes back each cycle a little further down it.

Formal system. Marks you may write, a starting arrangement of them, and moves that take one arrangement to the next. Nothing in it means anything, which is why the rules run identically in the hands of somebody who knows nothing about the subject.

Free parameter. A number a theory does not fix, which somebody measures and types in. The minimal Standard Model carries nineteen, closer to twenty-six once neutrinos have masses, plus Newton's constant and the cosmological constant.

Frustration. A loop of constraints that no assignment satisfies, however long you wait. Three coins in three rooms, each pair recording that it differs, is the smallest one, and a triangular magnet is the same arrangement in a material.

Gauge. A symmetry whose convention is picked afresh at every place rather than once for the world. Twelve observers each choosing for themselves make twelve of them, and what settles between them is the relation rather than the values.

Generation. One complete list of matter states, fifteen of them, ten from picking two slots out of five and five from picking four. The world runs three copies and no more. Fewer than three leaves matter and antimatter with no physical difference between them, more than five and three eighths costs the strong force its weakening at short range, and the threefold turn carried by the twenty faces has an orbit of exactly three.

Geodesic. Where you end up by going straight ahead and never turning. On a floor it draws a straight line; on a globe, the flight path from London to Tokyo, up over the Arctic.

Graph. Dots and the lines joining them, a dot per observer and a line wherever two of them have compared records. A walk that returns to its starting dot without retracing is a cycle, and the cycles are where the trouble lives.

Group. A collection of changes that is closed, so that two of them give a third from the same collection, and that holds a do-nothing move and an undo for every move. The eight ways of picking a square up and putting it down are one.

Hilbert space. The arrows an algebra of questions acts on, given their lengths and angles by the state. Physics courses start there and the order runs the other way, because what an observer holds is a list of questions.

Holonomy. The difference a round trip makes: set off round a loop, come back, and see what changed. Round a triangle of coin records it is one bit saying whether the coins can be set at all; on twelve observers it is a charge.

Homeostasis. Keeping a bounded set of internal readings where they were against a world that keeps pushing them off, at the observer's own expense. Walter Cannon gave the steady states of the body's fluids that name in 1926.

Inertia. What it costs to change a pattern's drift, proportional to what the pattern costs to hold together and to nothing else about it. Redirecting it means renegotiating every seam on its boundary instead of reconfirming them.

Inside-only clause. A relabeling counts only if the system can carry it out on itself. It reads like bookkeeping and it kills grand unification, because a symmetry acting from within forces a center, and simple groups have none.

Interval. The quantity two observers who disagree about distances and durations both compute the same value for. Its sign sorts every pair of events into three classes: reachable by something slower than light, reachable only by light, reachable by nothing.

Invariant. Whatever a given change of description leaves the same. Naming it comes first and the symmetry second: a rotation preserves length, and without length there is nothing for the rotation to be a symmetry of.

Lagrangian. A single function of position and speed from which every equation of motion follows. It is a compression, arrived at last rather than first, and a compression is allowed to be lossy.

Leaky abstraction. An account that lets you get on with the work while the machinery stays out of sight, good until the machinery does something the account has no words for. Joel Spolsky wrote the rule down in 2002, and the everyday container picture of space and time leaks in at least six places.

Mass. What a pattern costs to keep in existence, charged every cycle, because smoothing it away would open more disagreement around its edge than it would settle inside. A balance is an instrument for reading that bill.

Mismatch. What a seam scores when its two readings fail to answer each other: zero when they agree, positive when they do not. Add it over every seam and one number covers the whole arrangement, which every accepted repair lowers.

Momentum. What one segment of a chain of readings hands to the next, with no segment keeping a share. Shift every reading by the same amount and the arithmetic does not notice; momentum is what that indifference conserves, and a musket ball hands its entire supply to whatever stops it.

Natural selection. Variation, heredity and differential persistence running together. The list closes at three, and nothing on it decides anything.

Net of algebras. An assignment to every region of the questions answerable inside it, where a region contained in another carries the shorter list and two regions too far apart to signal carry lists that commute. This is what quantum field theory means by a field.

No-cloning. An unknown quantum state cannot be copied. Anything written down as a record can be copied freely and as often as you like, which is the half of the theorem most treatments leave out.

Noether's procedure. The machine that turns a continuous symmetry of the scored line into a conserved quantity, one for one. Shifting in space gives momentum, shifting in time gives energy.

Normal form. Where a run of local repairs stops, because no observer has a move left. Nobody has to poll the network to find out: an observer that finds nothing to do on its own seams has the answer.

Observer. Five items: a bounded view, a place to write, a way to update, a way to compare, and enough persistence to be there for the next comparison. A thermostat has all five. In machinery: a local algebra of questions with a record algebra sitting in its center, an interface steady enough to be read from and written to, and state enough to set the odds on whatever gets asked next.

Orbit. Every place the moves in a group can send one thing. The four corners of a square make one orbit under its eight moves, the four edge midpoints make another, and nothing carries a corner into the second set.

Order. A relation between two events saying which depends on which, and only for pairs where an authenticated read-from fact exists. In the twelve-observer wiring, a hundred and twenty seam pairs may compete through a shared observer and three hundred and fifteen are structurally disjoint. The wiring lists possibilities; the certified versions actually read in one history determine its ordered pairs. A clock may then put even incomparable events into a sequence.

Overlap rule. Two accounts have to match where both parties look, and nowhere else. Alice's brown table and Bob's waxy one never have to be reconciled.

Packet. What appears at a port: one reading drawn from a finite list, available to whatever sits on the far side. It is what a patch's state looks like from outside at one interface rather than the state itself.

Particle. Three conditions on a leftover: it is a property of how content moves between ports rather than a lump sitting at one, every overlapping observer reads it the same way in the overlap, and it keeps its identity when the records are regraded finer or coarser. A unit of disagreement that no order of repair removes meets all three.

Patch. A machine small enough to describe completely: bounded state inside, a set of interfaces, a finite list of moves. It is the smallest piece of the world there is.

Pinching. Sorting every question into the alternatives a record distinguishes, keeping the parts that fall inside one of them and deleting every comparison that only meant something across two. This is what writing a record does to an observer's algebra, and decoherence is the usual name for the result.

Port. A single place on a patch where part of its state is readable from outside. Twelve of them, arranged so that each sees what every other sees, is the only count the consistency conditions permit.

Probability. A weight assigned to each yes-or-no question, adding correctly over any complete list of alternatives that exclude one another, rather than a tally of how often something came up over many trials. Additivity by itself forces the odds the world runs on.

Projector. What an asking does to a state: put the same question twice and the second asking changes nothing the first did not. Every yes-or-no question anybody can put to a physical system takes this form.

Quotient. Sweeping up everything that counts as the same into one bundle and working with the bundles. The twelve hours on a clock face are the whole numbers with twelve counting as nothing.

Rank. How many rows of a table of numbers are independent of the rest. The candidate position carrier has three directions because the slowest-fading band of comparisons has rank three. Calling it physical space requires a realization map. The two transverse modes of the separately selected Maxwell action branch are likewise not, by themselves, a photon-particle theorem.

Record. A write into the protected part of a patch's state, the part later moves may not touch. It reads the same twice, it can be copied to a neighbor, and everything an observer holds onto is made of them.

Record map. Takes the full state of a network and hands back the part that survives comparison: every seam entry, and nothing that only one observer could ever have known. What it deletes is exactly what no two parties were in a position to check against each other.

Refinement. Describing something at ever finer resolution with no completed infinite object at the end, the way 3, 3.1, 3.14 describes pi and nobody ever holds pi. The carrier completion is smooth in that mathematical sense. Its identification with physical space is a separate claim.

Relative entropy. How many extra questions per state your description costs against the settled one. Zero when the two match, positive when they do not, and nothing done to both of them afterwards pushes it back up.

Repair. The move the world makes: read what a neighbor exposes, compare it with your own reading, and if the two disagree, go to the midpoint. It preserves the sum of the two readings it touches, which is where conservation comes from.

Repair load. How much of an observer's state fails to lie along the field its neighbors have written on its boundary. It cannot go below zero and it reaches zero exactly when the two agree, which is the reading a tuner is listening for when the beat disappears.

Running coupling. A force strength is one number only if nothing surrounds the charge. Screening puts more of the charge in view the closer an experiment gets, so every strength quoted carries a distance with it.

Schedule independence. Every order of repairs lands on the same final arrangement. That is what objectivity amounts to here, and why no observer has to be told when to move.

Screen. Every reading on every port an observer has facing outward. Nobody looks at it, it points nowhere, and its size is counted in readings rather than measured in meters.

Seam. Two ports routed to each other, which makes it the one place a disagreement can show up at all. Twelve observers wired into a twenty-sided solid have thirty.

Selector. What settles an ambiguous fiber, where every record in existence is already satisfied and repair has nothing left to bite on. Which repair law gets run by the observers capable of running one is a selector.

Spin. A two-outcome measurement with a direction attached. Aim the magnet however you like and the atom answers with one of exactly two deflections, and the aim is the only thing you choose.

Superposition. A system described in terms of questions it was not prepared for, carrying one number for each answer. The numbers are amplitudes, two of them can arrive with opposite signs, and a dark band is where that happens.

Symmetry. A change of description that leaves a named invariant where it was. Finding one deletes a part of your description as unphysical, so it subtracts from the world rather than adding to it.

Temperature. The exchange rate between energy and questions, in joules per bit. Three hundred kelvin sets that rate at 2.87 joules divided by ten to the twenty-first, for each yes-or-no question a record settles.

Tsirelson ceiling. How often two observers who cannot signal each other can agree, at most: 85.36 rounds in every hundred when each picks one of two questions, against the seventy-five a sealed answer sheet allows. The score squares to four plus a product of commutators, which puts the ceiling at 2.8284 rather than at 2. Twelve ports on a closed surface reach 2.3416, over the sealed ceiling and under this one.

Verifier. Anything that runs the check on the exposed readings and settles whether a patch's claim about its own work holds. The inside of the patch never enters it, because the check is arithmetic, and arithmetic is indifferent to whose readings it gets.

Wave. What repair looks like while it is under way. A port that hands its mismatch on rather than deleting it has a current to show for it, the two go on feeding each other instead of settling, and the pair crosses the wiring at the speed one seam per event allows.

Zero-error capacity. How many messages a channel can carry with no chance of confusing any two of them. On a ring of five symbols where each can be confused with its two neighbors, one symbol at a time, the answer is two.

Zombie. A creature identical to a person in every physical detail with nobody inside. Constructing one means deleting an interior and keeping the boundary traffic the interior was one side of, and what the deletion leaves is half a seam.

Appendix C: Sources and Further Reading

The proof assistant Lean has a keyword for a step nobody has done. The keyword is "sorry". A file containing one compiles anyway and announces itself while doing it, which is about as close as mathematics comes to a smoke alarm.

The formal proofs use Lean 4 and Mathlib, the community's library of formalized mathematics, without admitted steps. The papers also give analytic proofs, including the continuum count, clock and cone arguments. They sit with the numerical code, the simulation and the technical papers in the public OPH repository. The numerical work is Python, and one documented command reproduces it from a clean copy. The principal paper there is "Finite Observer Consensus as a Reconstruction Principle". Longer treatments build the mathematics at learn.floatingpragma.io, and the repair dynamics can be watched settling at simulation.floatingpragma.io.

The quoted runs were produced with the finite patch engine OPH-FPE. The protected-consensus archive supports chapters ten through thirteen. It carries the configuration, seed streams, endpoint arrays, full cycle trace and port state, together with a verifier that imports none of the simulator's code and rebuilds the result from the raw arrays.

The signed record-feedback archive probes all 96 carrier-port coordinates while eight carriers commit twelve-coordinate records. The spacetime derivation gives the conservative-record causal and count limit, the clock read from interval counts, and the operational symmetry condition selecting the Lorentz cone. Each statement retains its population, readout and physical-identification assumptions.

The standard physics

Edwin Taylor and John Archibald Wheeler's *Spacetime Physics* (second edition, 1992) builds special relativity out of the invariant interval and holds the Lorentz transformation back to a special topic after chapter three; the authors released it for free download at eftaylor.com. Leonard Susskind and Art Friedman's *Quantum Mechanics: The Theoretical Minimum* (2014) runs alongside Susskind's continuing-studies course at Stanford and starts the reader on a single spin rather than on a wave. Michael Nielsen and Isaac Chuang's *Quantum Computation and Quantum Information* (2000) gives quantum error correction a chapter of its own and no-cloning a box inside the information-theory chapter.

Bruce Schumm's *Deep Down Things* (2004) gets the gauge structure of the Standard Model across to somebody who will never do the field theory. Kip Thorne's *Black Holes and Time Warps* (1994) covers horizons and was written by one of the three men who signed the encyclopedia wager. Ernest Nagel and James Newman's *Gödel's Proof* (1958) does the incompleteness construction in a hundred and eighteen pages. Claude Shannon's "A Mathematical Theory of Communication" ran in the July and October 1948 issues of the Bell System Technical Journal, where the entropy of a message is defined.

The causal-set program begins with Luca Bombelli, Joohan Lee, David Meyer and Rafael Sorkin, "Space-time as a causal set", *Physical Review Letters* 59 (1987), 521. Bombelli and Meyer, "The origin of Lorentzian geometry", *Physics Letters A* 141 (1989), 226, and Graham Brightwell and Ruth Gregory, "Structure of random discrete spacetime", *Physical Review Letters* 66 (1991), 260, develop the order-to-geometry and dimension problem. Seth Major, David Rideout and Sumati Surya, "On recovering continuum topology from a causal set", *Journal of Mathematical Physics* 48 (2007), 032501, treats topology recovery; their "Stable homology as an indicator of manifoldlikeness in causal set theory", *Classical and Quantum Gravity* 26 (2009), 175008, presents a necessary but not sufficient diagnostic. Bombelli, Joe Henson and Sorkin, "Discreteness without symmetry breaking: a theorem", *Modern Physics Letters A* 24 (2009), 2579, gives the Lorentz-symmetry result for Poisson sprinklings. Sumati Surya's "The causal set approach to quantum gravity", *Living Reviews in Relativity* 22 (2019), article 5, gives the modern survey.

The continuum results behind the slogan that causal order fixes conformal geometry are S. W. Hawking, A. R. King and P. J. McCarthy, *Journal of Mathematical Physics* 17 (1976), 174, and David Malament, *Journal of Mathematical Physics* 18 (1977), 1399. They require causal conditions on the continuum; an arbitrary finite order need not have such a limit. OPH's spacetime paper gives a distinct constructive example: conservative record populations with a specified local read law and model clock converge in order and normalized counts to flat four-dimensional spacetime. It proves the one-tenth ordering-fraction limit inside interior causal diamonds and recovers proper-duration ratios from the fourth root of interval-count ratios. The latter uses a common reference interval and complete ancestry access. Physical identification and curved geometry require further hypotheses.

In the exact-embedding route, density uniformity completes the causal-set faithful-embedding condition. Controlled effective approximations can instead allow a causal discrepancy that vanishes with refinement. David Reid, *Physical Review D* 67 (2003), 024034, tests causal-set dimension in conformally flat spacetimes; Lisa Glaser and Surya, *Physical Review D*

88 (2013), 124026, use interval abundance as a locality diagnostic; and Mriganko Roy, Debdeep Sinha and Surya, *Physical Review D* 87 (2013), 044046, give curvature-aware small-diamond geometry.

Dionigi Benincasa and Fay Dowker, *Physical Review Letters* 104 (2010), 181301, construct a causal-set d'Alembertian whose continuum approximation contains the scalar curvature. Fay Dowker and Lisa Glaser, *Classical and Quantum Gravity* 30 (2013), 195016, extend the construction across dimensions. These operators provide a concrete scalar-curvature diagnostic for any source-derived refinement family. They do not reconstruct the Ricci or Einstein tensor: OPH's smooth Einstein promotion additionally requires a same-family tensor-curvature reconstruction or the independently stated continuum small-ball/null-balance identification, as well as a physically identified causal approximation with independently calibrated density and a controlled continuum limit.

Richard Feynman's lecture on the principle of least action, chapter 19 of the second volume of *The Feynman Lectures on Physics* (1964), was delivered to Caltech sophomores and is the shortest good account of where a Lagrangian comes from; Caltech hosts all three volumes online. The measured values in this book come from the Particle Data Group's *Review of Particle Physics*, revised every two years at pdg.lbl.gov. The 2024 edition is S. Navas and others, Physical Review D 110, 030001.

The people

Galois's last letter appeared in the *Revue Encyclopédique* in September 1832 and in Liouville's journal in 1846; Robert Bourgne and Jean-Pierre Azra's critical edition (1962) prints it with its variants. Émilie du Châtelet's *Institutions de physique* was published in Paris in 1740, and her *Principia*, the only complete French one, in 1759. Elizabeth Loftus and John Palmer's car-crash experiment is in the *Journal of Verbal Learning and Verbal Behavior* 13 (1974), 585.

Sadi Carnot's *Réflexions sur la puissance motrice du feu* (1824) is in English in the Dover volume edited by E. Mendoza (1960). Jacob Bekenstein's "Black holes and entropy" is Physical Review D 7, 2333 (1973), and Stephen Hawking's "Black hole explosions?" is Nature 248, 30 (1974). Richard Hamming's codes are in the Bell System Technical Journal 29 (1950), 147, and Rolf Landauer's cost of erasing a bit is in the IBM Journal of Research and Development 5 (1961), 183. John Bell's inequality is *Physics* 1 (1964), 195; the four-setting form is Clauser, Horne, Shimony and Holt, Physical Review Letters 23 (1969), 880; the ceiling on it is Boris Tsirelson, *Letters in Mathematical Physics* 4 (1980), 93; and the loophole-free Delft run is B. Hensen and others, Nature 526 (2015), 682.

The parity experiment is Wu, Ambler, Hayward, Hoppes and Hudson, Physical Review 105, 1413 (1957). Rutherford's four 1919 papers on the collision of alpha particles with light atoms are in volume 37 of the *Philosophical Magazine*, the fourth reporting the anomalous effect in nitrogen. Yoshio Koide's relation is Physical Review D 28, 252 (1983).

Maupertuis read his least-action memoir to the Paris Academy in 1744. Lagrange's *Méchanique analitique* dates from 1788, and the English translation by Auguste Boissonnade and Victor Vagliente from 1997. Emmy Noether's *Invariante Variationsprobleme* is in the Göttingen *Nachrichten* for 1918, and M. A. Tavel's translation appeared in 1971.

The Darwin and Wallace papers were read at the Linnean Society on 1 July 1858 and printed that August. Gödel's paper is in *Monatshefte für Mathematik und Physik* 38 (1931), and in English in Jean van Heijenoort's *From Frege to Gödel* (1967). Claude Bernard's *Introduction à l'étude de la médecine expérimentale* (1865) can be had in Henry Copley Greene's translation. Walter Cannon named homeostasis in the 1926 Paris volume for Charles Richet's jubilee and explained the choice in *The Wisdom of the Body* (1932). Zachary Blount, Christina Borland and Richard Lenski's citrate paper is in the *Proceedings of the National Academy of Sciences* 105 (2008), 7899. Boethius on foreknowledge is book five of *The Consolation of Philosophy*, in Victor Watts's translation for Penguin. Harry Frankfurt's counterexample is in the *Journal of Philosophy* of 4 December 1969.